

Dynamic treatment effects in Education

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Abstract

This paper examines the uncertainty involved in extrapolating education effects measured within short observation windows to lifetime welfare. I document the timing information available in existing studies and develop a model of educational attainment with and without a policy. Calibrated logistic curves describe paths that match retained education inputs, allowing permanent attainment changes and temporary changes in acquisition speed or timing. Additional point observations enter constrained minimum distance, with discrepancies reported when exact matching is unavailable. Two deterministic simulations show how these alternatives change lifetime earnings values and how additional observations can narrow their range. I then apply the model to 38 policy calculations, reconstruct ranges for the marginal value of public funds (MVPF), and propose an ordering rule based on those ranges. Under the specified path and valuation assumptions, the same short-window education evidence can support conflicting welfare judgments even when the estimated effects are treated as known.

1 Introduction

A policy that raises bachelor's degree attainment at age 25 can create a permanent attainment gain, a temporary advantage from earlier graduation, or both. These possibilities have different economic consequences. Earlier completion can generate earnings before the comparison group catches up, while a permanent attainment gain can affect income and tax revenue over a much longer period. Educational costs also depend on when additional schooling occurs. An effect estimated at a particular age therefore answers an important causal question, but does not by itself determine the policy's lifetime value. This distinction matters when education estimates inform long-term earnings projections and comparisons of public spending, as in Hendren and Sprung-Keyser (2020).

This paper asks whether education effects measured within a finite window can rule out conflicting lifetime welfare judgments for the same policy. I take selected causal estimates from existing studies as inputs and construct education paths under explicit outcome conversions and age mappings. Retained reference inputs are matched exactly; additional point estimates are matched exactly when possible and otherwise fit by constrained minimum distance, with residuals disclosed. Within each reconstruction, I hold these inputs, the external comparison profile, path restrictions, and the valuation rule fixed. I then compare earnings net present values (NPVs) and the marginal value of public funds (MVPF), which relates beneficiaries' willingness to pay to net fiscal cost. The replacement passes each new education path through the retained income, tax, and education-cost functions. Transfers and genuinely fixed components remain fixed; education-dependent costs move with the path. The question is whether different paths can place beneficiary welfare above or below

public cost, or change whether a policy finances itself. A comparison with the original welfare calculation can additionally reflect changes in conversion or valuation; the comparison among paths within a reconstruction holds those choices fixed.

How much the reported education effects reveal about lifetime attainment depends on when education is measured. I document this timing information in a review of 459 studies that combines a search of 12 economics and education journals over 2000–2025 with additional welfare-source papers and an author manuscript. Among the 190 studies reporting bachelor’s-or-higher outcomes, 132 report one distinct estimation time or window for that outcome. More directly relevant to welfare extrapolation, 18 of the 31 studies reporting education-based lifetime or career valuation provide only one window for each valued education series. These counts describe the assembled sample rather than all education research. The original authors also discuss dynamics and extrapolation limits. Cohodes and Goodman (2014) interpret persistence and completion patterns as evidence of dropout rather than pure delay, while Angrist et al. (2022) and Schoonover et al. (2026) qualify their projected returns and discuss education timing. Their evidence informs the reconstruction without determining the unobserved continuation.

To connect the observed education effects to their possible continuation, I represent educational attainment for the same target population by two cumulative age profiles: $H_c^0(a)$ without the policy and $H_c^1(a)$ with it. Their difference is the education effect at each age. Both attainment levels are nondecreasing, but the difference can widen, narrow, or reverse sign. Reported effects constrain this difference at the source’s observation ages, over its age windows, or through changes between observations, using the specified conversions. For the numerical exercises, I use logistic continuations anchored at the last unaffected age and restrict their levels, acquisition rates, and empirical support. Census–ACS profiles supply the comparison curves and calibration restrictions, matched to policy geography and birth cohorts in the applications. These population profiles are external calibrations, not counterfactuals identified by the source study’s research design. The paths exactly match the retained reference inputs. Additional selected estimates may have nonzero best-fit residuals; neither unselected source results nor every added point is asserted to be matched exactly.

Two deterministic simulations then examine how education effects and their measurement ages restrict possible lifetime earnings values. The first compares permanent effects on eventual schooling with temporary effects generated by changes in acquisition speed or timing, while retaining the same age-25 measurement rule. In each of the three illustrated scenarios, the measured effect is compatible with both positive and negative education NPVs under a common valuation rule. Some compatible paths reverse the initial education advantage at later ages, so the alternatives extend beyond different durations of a positive effect. These reversals are possibilities within the specified path family. The second exercise holds the generating path fixed and varies the observation ages. In the illustrated comparisons, adding an effect estimate at one additional age can substantially restrict the compatible paths and narrow the resulting NPV range. The reduction depends on when that additional effect is measured. Even when attainment eventually converges, earnings forgone during an earlier delay need not be recovered.

Applying the model to existing policy evaluations, I reconstruct MVPF ranges for 38 policy calculations and propose a rule for ordering the 37 with an original welfare valuation (L1). The reconstructions build on Hendren and Sprung-Keyser (2020) or the original authors’ valuations, replacing education-related earnings and fiscal returns with values implied by compatible education paths. Among these 37 calculations, 29 initial reconstructions (L2) admit positive-cost MVPFs on both sides of one, and 23 admit both beneficiary

losses with public costs and beneficiary gains with self-financing.¹ Single-parameter comparisons show that permanent attainment changes often yield valuations close to L1, whereas matching the same education input through temporary changes in acquisition can produce substantially different lifetime earnings and MVPFs. Additional comparable observations narrow some ranges by excluding continuations allowed by the initial inputs; twelve calculations with L1 support this richer reconstruction (L3), including converted inputs with explicit timing mappings. Seven of these twelve L3 applications select one best-fitting path; five retain exact-match path banks. The cannabis education application also uses a best-fit L3. The ordering uses L3 where available and L2 otherwise, prioritizing higher lower endpoints and then higher upper endpoints among the paths found. Separately, 1,000 specified input repetitions per calculation examine how imprecision in education or accounting inputs changes the paths and their welfare values. These repetitions distinguish archived source draws from constructed perturbations. They address uncertainty in the inputs without supplying observations of the unmeasured education path.

The main contribution is a systematic analysis of how extrapolating finite-window education effects changes lifetime welfare judgments. Representing attainment with and without a policy as H^1 and H^0 , I construct paths that match the same education inputs and show how permanent and temporary effects differ in their implications for earnings and fiscal returns. This makes explicit which parts of a welfare conclusion follow from the selected education evidence and which depend on continuation assumptions. Research on time to degree and dynamic educational choices establishes the importance of when schooling is acquired (Bound et al., 2012; Bárány et al., 2023; Heckman et al., 2016). Work on long-term extrapolation studies how short-term observations can inform later outcomes under additional restrictions (Athey et al., 2026; Obradović, 2026; Fan et al., 2026). I bring this question to the education-based valuations in Hendren and Sprung-Keyser (2020), using the literature review to document the available information and the simulations and policy reconstructions to assess its welfare consequences. Separating uncertainty about continuation from imprecision in estimated inputs also complements research on inference for calibrated parameters and welfare metrics (Cocci and Plagborg-Møller, 2025; Vohra, 2026).

Two verified education paths that satisfy the same restrictions but imply opposing welfare conclusions establish that those restrictions do not resolve the comparison. The sensitivity exercises assess whether such conflicting conclusions remain under alternative calibration, valuation, and search specifications. They cover all 38 calculations and use additional education observations where available. Across the common specifications, 15 calculations continue to admit beneficiary losses with public costs and beneficiary gains with self-financing. These checks support the existence of conflicting judgments under several stated assumptions, but do not validate every assumption or exhaust the feasible paths. A search that finds only favorable paths cannot establish that every compatible path is favorable. The reported extrema are candidate values, no probabilities are assigned to paths, and input-repetition summaries do not establish confidence coverage.

The comparison of alternative education paths also relates to research on lifetime earnings measurement, long-term prediction, and policy choice with incomplete information. Haider and Solon (2006) show that the relationship between current and lifetime earnings varies with age, while Bhuller et al. (2017) document how additional schooling changes the age profile of earnings. These studies examine earnings over the life cycle; the reconstruction here also varies the education path entering the valuation. To forecast beyond experimental

¹Eleven tax calculations require an explicit policy-unit conversion, Early Head Start and Tennessee Promise use disclosed public-data income reconstructions, and algorithmic placement adds a long-run earnings channel; Section 4.1 explains these adjustments. Several calculations share a source study. The remaining application values only the education channel of cannabis legalization and has no whole-policy L1.

follow-up, García et al. (2020) combine experimental and nonexperimental data with economic models to estimate lifetime program benefits and costs. Chen and Ritzwoller (2023) develop efficient estimators combining short-term experimental and longer-term observational outcomes, and Imbens et al. (2025) use the sequential structure of short-term outcomes to address persistent confounding in such data combinations. These approaches clarify how additional data and identifying restrictions support long-term estimates. The present analysis examines which welfare judgments remain unresolved when selected education observations leave room for different attainment continuations. It also relates to sufficient-statistics welfare analysis, which connects reduced-form estimates to welfare through economic restrictions (Chetty, 2009; Kleven, 2021). Projecting lifetime earnings and fiscal returns from an education effect similarly requires a restriction on its continuation. Finally, Manski (2011) studies policy choice when treatment responses are only partly known and no probability distribution is assigned to that uncertainty. This distinction between what the evidence determines and what a decision rule adds also guides the interpretation of the policy ordering in this paper.

The remaining sections move from the available education evidence to path construction, welfare applications, and their interpretation. Section 2 presents the motivating cases and literature statistics, and Section 3 develops the model and simulations. Sections 4 and 5 present the three main applications and the full policy collection. Section 6 discusses assumptions, robustness, and policy implications, and Section 7 concludes.

2 From Observed Education Effects to Lifetime Valuation

Assessing the lifetime value of education policies requires connecting effects estimated within finite observation windows to education paths over the life cycle. Figure 1 shows that educational attainment continues to rise after age 25. Among the 1985–1987 birth cohorts, bachelor’s-or-higher attainment (BA+) rises from 29.34 percent at 25 to 40.80 percent at 35; approximate schooling rises from 13.30 to 13.97 years. The profiles use the full-population American Community Survey (ACS) extract for 2000–2022, with fixed cohort weights.² The cohort comparison in Figure A1 shows that gains in degree attainment and schooling between ages 25 and 35 are larger for later-born groups.

To document the time information used in higher-education research and its role in longer-run interpretation and valuation, I review the studies described in Appendix A. The sample contains 459 papers: 447 from a search of 12 economics journals over 2000–2025, 11 additional welfare-source papers, and one author manuscript. Among 190 papers with BA+ outcomes, 132 report one distinct BA+ estimation time or window (69.5 percent); one window can span several years.³ Separately, 31 papers report education-based lifetime or career valuation. Among these, 18 report one age, follow-up, or stage-specific window per valued education series, and 12 report multiple outcome times.⁴ The following three scholarship studies illustrate the calculations and the dynamic evidence accompanying them.

Angrist et al. (2022) study Susan Thompson Buffett Foundation (STBF) scholarships in a randomized experiment. Among applicants targeting four-year colleges, awards raise bachelor’s degree completion by

²Each cohort contributes ages 16–35 with equal weight at every age. Appendix A.5 gives sample, person-weighting, and education-crosswalk details.

³The journal-only share is 130 of 186 BA+ papers (69.9 percent). Coding distinguishes attained age, individual follow-up, and policy event time.

⁴One further paper values institutional degree flows in policy event time. Appendix A.3 records the valued outcomes and their observation clocks.

8.1 percentage points six years after high school. Combining educational credentials and time enrolled with an ACS earnings model, they project a lifetime earnings gain of \$21,150 per award recipient at a 3 percent discount rate and a marginal value of public funds (MVPF) of 1.75 under hypothetical public funding. The authors examine graduation dynamics and describe the valuation as provisional: observed education histories inform the calculation, while later attainment and earnings require extrapolation.

Cohodes and Goodman (2014) estimate effects of Massachusetts’ Adams Scholarship on college choice, persistence, and completion. Scholarship eligibility lowers the probability of earning a bachelor’s degree within six years of high-school graduation by 2.5 percentage points near the eligibility threshold. The authors interpret the persistence and graduation patterns as evidence of dropout rather than pure delay, and use ACS earnings differences to illustrate lifetime losses for students induced to switch colleges. Hendren and Sprung-Keyser (2020) construct a policy-average MVPF from the scholarship’s enrollment and completion effects, which together imply a 0.018-year reduction in schooling. They project the associated earnings and tax changes, then account for transfers and private and public education costs to obtain an MVPF of 0.72.

Schoonover et al. (2026) estimate the effects of Tennessee Promise in overlapping two-year age groups from 18–19 through 23–24. At ages 23–24, the estimated increase in bachelor’s-or-higher attainment is 2.7 percentage points, statistically significant at the 10% level under permutation inference. The authors combine the education effects with Tennessee-specific returns to schooling to project lifetime income and tax revenue. Their valuation yields positive willingness to pay and negative net public cost, classifying the program as self-financing with an infinite MVPF. They acknowledge that outcomes observed in early adulthood cannot fully distinguish lasting attainment gains from changes in the timing of schooling and labor-market entry.

These studies use different education outcomes and populations, and their authors already examine graduation dynamics or qualify the lifetime projections. The question pursued here is how much those projections can vary when the observed education evidence is retained but its unobserved continuation changes. The next section holds observed education effects fixed and varies age paths within stated economic restrictions. Applying the same valuation rule to each path isolates the role of those paths in the calculation.

3 Education Paths and Lifetime Valuation

A finite observation window leaves the evolution of educational attainment partly unobserved, even when the effects within that window are measured without error. To separate this information problem from sampling uncertainty, I hold a comparison path fixed and construct treated paths consistent with stipulated education effects. In each of the three displayed scenarios, the same age-25 effect admits paths with positive and negative education-channel net present values (NPV) under a common valuation rule. Adding the corresponding age-35 effect reduces the width of the numerical candidate range by 80.4–86.7 percent. These comparisons treat the observed effects as exact and hold the baseline and path restrictions fixed. The first experiment varies the education process while observing it at age 25; the second holds each generating process fixed and varies the observation ages. Together, they show which valuation conclusions follow from the observed effects and which depend on the unobserved schooling path.

3.1 Education Paths, the Baseline, and Path Restrictions

An education effect measured at a particular age describes the attainment difference at that age, while leaving its earlier and later evolution unresolved. For a fixed population in cohort c , let $H_c^0(a)$ and $H_c^1(a)$ denote the

mean of a specified educational attainment outcome at age a without and with the policy, respectively. The outcome may record cumulative college entry, degree attainment, or completed years of schooling.⁵ The numerical illustration below uses years of schooling. For any of these cumulative outcomes, I distinguish the policy effect on the attainment stock from its age- a increment:

$$\Delta H_c(a) \equiv H_c^1(a) - H_c^0(a), \quad \Delta h_c(a) \equiv \Delta H_c(a) - \Delta H_c(a-1). \quad (1)$$

Cumulative attainment satisfies $H_c^j(a) - H_c^j(a-1) \geq 0$ for $j \in \{0, 1\}$ in a fixed population. This restriction does not imply monotonicity of the treatment effect $\Delta H_c(a)$, because the two potential attainment paths can accumulate at different rates. The increment $\Delta h_c(a)$ may therefore be positive, negative, or zero, according to whether attainment accumulates faster, more slowly, or at the same rate under treatment. In particular, $\Delta h_c(a) < 0$ indicates a decline in the treatment-comparison attainment gap; neither level path need decline. Appendix B.1 gives the detailed definitions, and Figure 2 illustrates the two levels, their age-specific gaps, and the change in the gap between adjacent ages.

I represent the treated path with a three-parameter logistic continuation anchored to the comparison level. For a given starting age a_0 , write $\tau_c = a_0 - 1$ and $S_c = H_c^0(\tau_c)$. With $\theta_c = (\Delta K_c, \Delta b_c, \Delta d_c)$, define $L_c(a; \theta_c) = \Lambda[(b_c + \Delta b_c)(a - d_c - \Delta d_c)]$, where $\Lambda(x) = (1 + e^{-x})^{-1}$. The treated path is

$$H_c^1(a; \theta_c, a_0) = \begin{cases} H_c^0(a), & a \leq \tau_c, \\ S_c + (K_c + \Delta K_c - S_c) \frac{L_c(a; \theta_c) - L_c(\tau_c; \theta_c)}{1 - L_c(\tau_c; \theta_c)}, & a > \tau_c \end{cases}. \quad (2)$$

The normalization makes the continuation pass through S_c at τ_c and approach $K_c + \Delta K_c$ at older ages. Thus, the first affected observation reflects acquisition since a common starting point, rather than a level difference imported when switching curves. The parameter K is measured in attainment units, b in inverse age-years, and d in age-years. They govern the plateau and the speed and timing of accumulation conditional on the anchor. I call an education effect permanent when it changes eventual attainment, so that $\lim_{a \rightarrow \infty} \Delta H_c(a) = \Delta K_c \neq 0$. An effect is temporary when the attainment gap eventually vanishes. In the single-parameter exercises, a nonzero K change is permanent, whereas b or d changes with K fixed are temporary.⁶ Holding K , b , and the common anchor fixed, a higher d delays subsequent accumulation: attainment is lower at each finite age after the anchor, and any given level between the starting value and plateau is reached later. The delay varies across attainment levels, so an increase in d does not imply an equally sized delay at every point on the path. These are population-path parameters rather than individual graduation dates. Figure B1 illustrates their roles under the normalization; Appendix B.2 gives the derivatives.

The choice of a_0 reflects when the policy can first affect the specified attainment outcome for the fixed target population. I assume no policy effect or anticipation through τ_c , which justifies the common starting level in Equation (2). This compares the same population under two policy states; it does not require the observed treatment and comparison samples to have identical pre-policy means. In the simulation, the hypothetical policy first affects the population at age 18, making age 17 the anchor. The anchor is therefore determined by exposure timing, not by the age at which the researcher happens to observe an effect.⁷

⁵For binary attainment, $H_c^j(a)$ is a population share and $100\Delta H_c(a)$ is a percentage-point effect. Schooling levels and effects are measured in years. Current enrollment is not cumulative and need not rise with age.

⁶Temporary describes eventual convergence, not disappearance within the valuation horizon. Such an effect may persist beyond age 65 and can have a nonzero lifetime NPV. A permanent effect can also involve changes in acquisition speed or timing.

⁷Observation at 25 does not justify anchoring at 24. The anchor must precede possible effects and anticipation. If exposure ages differ, use a common unaffected age or cohort-specific anchors before aggregating with fixed weights.

For the years-of-schooling simulation, $H_c^0(a)$ is the equally weighted average of fitted ACS profiles for the 1985–1987 birth cohorts, using the full-population data at ages 16–35 introduced in Figure 1. Here c indexes the synthetic three-cohort group. Each cohort is fitted separately, and the fixed one-third weights preserve its contribution at every age; values above age 35 are extrapolated.⁸ I vary H_c^1 while holding this comparison curve fixed. Appendix C, Table C1, and Figure C1 document the construction and fit.

The admissible path class combines the logistic specification with restrictions on attainment levels and annual increments. Both experiments set the first possible effect age to $a_0 = 18$ and value earnings through age 65 and extend education paths through age 72. I require completed schooling under treatment to be nondecreasing, to lie between zero and 25 years, and to increase by no more than one schooling year between adjacent ages, including the transition from age 17 to age 18. The total-parameter domain is $K \in [13.3370, 15.1395]$, $b \in [0.12, 0.65]$, and $d \in [8, 30]$.⁹ Let C collect these maintained conditions. They define which paths enter the comparison; satisfying them does not establish that a particular path is the response to an actual policy.

3.2 What a Single Observed Effect Leaves Unresolved

An education estimate restricts a feature of $\Delta H_c(\cdot)$ rather than revealing the entire path. Let the observation operator $\mathcal{M}$ map a candidate path into the effects represented by a study:

$$\mathbf{m} = \mathcal{M}\{\Delta H_c(\cdot)\}. \quad (3)$$

For attainment measured at one age, the relevant entry of $\mathbf{m}$ is $\Delta H_c(a)$. An average effect over an age window $\mathcal{A}$ instead enters as $\sum_{a \in \mathcal{A}} w_a \Delta H_c(a)$, using the source’s weights.¹⁰ An effect on attainment acquired between ages a_1 and a_2 restricts $\Delta H_c(a_2) - \Delta H_c(a_1)$. These restrictions preserve the source study’s outcome definition and time scale.¹¹ The source-specific mappings for credits, enrollment, and credentials enter the applications in Section 4. For the fixed comparison path and target population, the compatible path set is

$$\mathcal{P}(\mathbf{m}; a_0, C) = \{H_c^1(\cdot; \theta_c, a_0) \in C : \mathcal{M}\{H_c^1(\cdot; \theta_c, a_0) - H_c^0(\cdot)\} = \mathbf{m}\}. \quad (4)$$

An additional observed effect changes the moment restrictions; changing the parameter domain or the path conditions changes C . With one or two exact age-specific effects, the matching equations generally leave two or one free parameter dimensions. Appendix B.3 gives the minimum-distance formulation and the algebra used to solve these equations directly. I impose no probability distribution over compatible paths or their parameters. The resulting NPV and MVPF candidate ranges describe conditional possibilities, without assigning probabilities to a positive NPV or an MVPF above one.¹²

⁸The reference triple averages parameters, whereas H_c^0 averages fitted curves. Their anchored continuations differ by at most 0.000226 schooling years over ages 18–65; zero parameter increments need not imply an exactly zero effect.

⁹The plateau limits are weighted first and 99th percentiles of a historical state-cohort fit bank. Transporting this support to the full-population baseline is a calibration assumption (Appendix C.3). Speed, timing, and annual-increment limits are design restrictions. Binary outcomes require share-consistent bounds instead.

¹⁰An exact 0.1-year effect at 25 imposes $\Delta H_c(25) = 0.1$. An equally weighted two-percentage-point BA effect at ages 23–24 imposes $[\Delta H_c(23) + \Delta H_c(24)]/2 = 0.02$, not a restriction at the midpoint.

¹¹Six years after college entry need not mean age 24. Converting follow-up into age requires entry-age information or source-specific averaging. Policy event time also differs from individual follow-up.

¹²Grid spacing does not assign path probabilities. Here the stipulated effects have no sampling error, so candidate ranges are not confidence intervals. Finite search also does not certify global bounds; Appendix B.7 distinguishes these issues.

The first experiment fixes the observation age at 25 and varies the education process generating that observation. Holding two generating parameters at their reference values, I vary the third over its entire conditional feasible interval. Each scenario supplies its own age-25 effect. I then set aside knowledge of the generating parameters and search the common three-parameter class for paths matching that effect. Only one generating parameter varies at a time, but all three candidate parameters may vary: the observer is not assumed to know which parameter the policy changed. Appendix B.6 specifies the finite grids used to span the intervals.

An education effect observed at age 25 leaves open whether the effect is permanent or temporary. Panel A of Figure 3 compares a permanent increase in completed schooling, a temporary gain from faster accumulation, and a temporary loss from delayed accumulation followed by catch-up, from left to right.¹³ The orange curves describe these stipulated responses. In the middle column, the early advantage largely disappears as the comparison population catches up; in the right column, the treated population initially falls behind and subsequently catches up. The shaded regions summarize the alternative paths found when only the age-25 effect is observed. Within each column, every candidate reproduces that effect and satisfies the same maintained restrictions, yet the subsequent education outcomes can differ. For both positive early effects, the candidates include persistent gains and later reversals of the attainment advantage. For the negative early effect, they include persistent shortfalls, catch-up, and a later attainment advantage. These are possible evolutions of population means; they do not identify particular individuals' graduation or return-to-school decisions. Observing the gap at one age therefore leaves uncertainty about both eventual schooling and how long an advantage or disadvantage lasts. Under the valuation rule introduced next, these differences change the duration of earnings gains or losses and the timing of schooling costs.

3.3 From Education Paths to Net Present Value

Education paths that reproduce the same observed effects can imply different economic values under a common earnings and cost schedule. For college policies without observed earnings effects, Hendren and Sprung-Keyser (2020) project income changes from education outcomes using returns estimated by Zimmerman (2014). The superscripts point and path distinguish the benchmark and its path-based extension. I write both values as functions of the schooling-effect path ΔH_c , holding the observation age a^* fixed. The benchmark uses only the observed gap $\Delta H_c(a^*)$ in completed schooling years per member of the target population.¹⁴ Let $Y_0(a)$ be annual comparison earnings in a common dollar year and r_t^{HSK} the proportional earnings effect per additional schooling year in period t after policy onset. The dollar premium is $\pi_{\text{YS}}^{\text{HSK}}(a) = Y_0(a)r_{a-a_0+1}^{\text{HSK}}$. With first possible effect age a_0 , terminal valuation age A , and discount rate ρ , write the common discount factor as $D_\rho(a; a_0) = (1 + \rho)^{-(a-a_0)}$. The single-effect benchmark is

$$\text{NPV}_c^{\text{point}}(\Delta H_c; a_0, A, \rho) = \sum_{a=a_0}^A D_\rho(a; a_0) \pi_{\text{YS}}^{\text{HSK}}(a) \Delta H_c(a^*). \quad (5)$$

The observation age a^* stays fixed as the valuation age a runs from a_0 to A . The benchmark therefore assigns the same NPV to all paths with the same observed gap. The retained schedule assigns negative returns in the

¹³Only the named parameter moves to its conditional-interval midpoint. Relative to (13.9222, 0.2575, 12.5649), the increments are (0.3160, 0, 0), (0, 0.0030, 0), and (0, 0, 6.4351). The rule is independent of valuations and does not imply equal-sized policy shocks. Table B1 reports all scenario parameters.

¹⁴Appendix B.4 gives schooling conversions and the treatment of age-window averages.

first seven periods and positive returns thereafter.¹⁵ Thus, a single observed education difference generates an income profile that varies with age through earnings, returns, and discounting. Its value depends on using that same education input in periods for which the education path is unobserved.

I extend this benchmark by assigning its signed earnings response to the age at which each part of the attainment gap is acquired. Define the signed completion increment

$$q_c(\tau) = \Delta H_c(\tau) - \Delta H_c(\tau - 1). \quad (6)$$

These increments need not be positive and are not probabilities of individual graduation. They record the difference between the two population paths' acquisition at each age. The net annual earnings effect and its present value are

$$\Delta y_c(a) = Y_0(a) \sum_{\tau} R(a, \tau) q_c(\tau), \quad (7)$$

$$\text{NPV}_c^{\text{path}} = \sum_{a=a_0}^A D_\rho(a; a_0) \Delta y_c(a). \quad (8)$$

For the common simulation, $R(a, \tau) = r^- \mathbf{1}\{\tau - 7 \leq a < \tau\} + r^+ \mathbf{1}\{a \geq \tau\}$, evaluated only at income ages $a \geq a_0$. Here $r^- = -0.0667064939863901$ and $r^+ = 815/7241$. A single completion increment of size m at 25 reproduces the benchmark's negative response at ages 18–24 and positive response from 25 onward. Two increments sum two shifted responses; a smooth path sums increments across all completion ages. This is a maintained transport of an earnings-response template across completion ages. It does not assert that HSK observed everyone graduating at 25 or that each additional schooling year entails seven actual years of enrollment.

Early earnings losses and later gains are already in $\Delta y_c(a)$. No separate opportunity-cost charge is subtracted, and annual changes in completed schooling are not set equal to contemporaneous enrollment. Linearity permits signed increments, including comparison-group catch-up. It also assumes that the same marginal response can be transported across the completion ages and sizes considered, without additional interactions or general-equilibrium effects. These assumptions concern valuation rather than the identification of the source education estimates.

The first experiment uses $a^* = 25$, income ages 18–65, a 3 percent discount rate to age 18, and 2000 dollars. The common baseline remains the fitted 1985–1987 cohorts. Education paths extend through 72 because income at 65 can include a pre-completion response associated with acquisition up to seven years later. The terminal attainment statistic remains the gap at 65. All restrictions, income prices, response coefficients, and discounting are held fixed across candidate paths. Appendix B.4 gives the equivalent stock formula and the continuous-age interpretation. Policy accounts then retain their own income, tax, transfer, and education-cost conventions.

Applying the common path valuation to the candidates in the first experiment produces both positive and negative values in all three displayed scenarios. Table 1 separates each scenario's single-effect benchmark, its known generating-path value, and the values of other paths matching the observed effect. The age-25 candidate NPV ranges, in thousands of 2000 dollars, are $[-1.32, 35.19]$ for the plateau change, $[-12.47, 31.31]$ for

¹⁵The schedule uses approximately -0.0667 in periods 1–7 and 0.1126 thereafter, following the Hendren and Sprung-Keyser replication's conversion of Zimmerman's estimates. It is an illustrative benchmark; their policy accounts also use other education inputs and observed earnings.

the speed change, and $[-26.73, 26.16]$ for the timing change. Finding paths with opposite signs establishes that the stipulated education evidence and maintained restrictions do not determine the NPV sign. The generating-path value exceeds the single-effect benchmark in the plateau and timing examples and is below it in the speed example, so the direction of that comparison is not uniform. Table B2 summarizes the full scans; the complete scenario results remain in the replication files.

The annual curves in Figure 4 show the net earnings response for a single completion batch and for the full attainment path under the same signed template. The illustration combines a higher plateau with changes in acquisition speed and timing; Figure B2 shows the corresponding levels.¹⁶ Its gap is 0.364 years at 25 and 0.263 at 65. The single-effect and full-path NPVs are approximately \$15,390 and \$13,986. Positive and negative portions of each curve are changes in net earnings; they are not separately identified wage premiums and opportunity costs.

The distinction between permanent and temporary effects can also be examined with the education inputs used in the policy applications. Figure B3 in the appendix compares each reference earnings NPV with the value obtained by matching its education input through a plateau, speed, or timing change alone. Table B3 reports the corresponding normalized NPVs and parameter changes, including calculations without a comparable pair. The other two parameter families remain at their calibrated baseline values. The K specification produces a permanent effect; the b and d specifications produce temporary effects as the paths eventually converge. Both axes use each account's monetary units and the same reference-expenditure normalization. These comparisons retain the applications' own valuation schedules, rather than imposing the simulation's common schedule on every policy. Appendix B.5 explains the construction, unit bridges, and differing matched samples. Differences from the reference projection can reflect both continuation and the source-to-path earnings mapping; they are not estimates of which mechanism actually occurred.

3.4 What Additional Observation Ages Reveal

Additional observations can distinguish paths that fit the same early effect, but the information they provide depends on when schooling is measured. The second experiment holds each generating path fixed and changes the observation ages, retaining the same baseline, path restrictions, and valuation rule. I use five positions spanning each parameter's conditional feasible interval and examine every single age and every pair of distinct ages from 18 through 35. The path remains fixed as observations change; its effects at different ages need not be equal. Each added observation restricts the same underlying path set. Appendix B.6 gives the full design and the numerical procedure for preserving this nesting.

Adding an age-35 effect sharply narrows the NPV ranges for the same three midpoint scenarios. Panel B of Figure 3 retains the generating parameters, baseline, restrictions, valuation rule, and axis scales of Panel A. The candidate NPV ranges in Table 1 become $[6.21, 13.39]$, $[-5.90, 0.64]$, and $[-13.18, -6.14]$ thousand dollars, reductions in width of 80.4, 85.1, and 86.7 percent. The second observation matters because a single schooling gap does not determine how quickly it emerged or what follows it. Timing also matters for value: a year acquired earlier can produce earnings gains over more years but also moves forgone earnings closer to the discount base. In the timing example, the age-35 gap is only -0.043 schooling years, yet the candidates found have negative NPVs: near-convergence in later schooling need not erase the value lost during earlier delays. The candidates found for the plateau example are positive, whereas the speed example still admits

¹⁶The joint increments are $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$. This illustration satisfies the common path restrictions but is outside the two experiment grids.

both signs. Its generating speed lies close to the reference value under the one-year acquisition cap, so the stipulated effect is small. These findings do not prove sign identification: the search is finite, and one of the five timing scenarios also admits both signs after adding age 35.

The location of the additional observation matters alongside its availability. Table 2 compares selected designs within each scenario before summarizing across scenarios.¹⁷ Across the three parameter groups, adding age 20 to age 25 reduces the NPV width by a median 34.1–65.3 percent, whereas adding age 24 reduces it by 37.5–68.4 percent. Adding age 30 yields median reductions of 70.3–79.5 percent. The age-24 pair is more informative than the age-20 pair in these summaries despite the shorter distance between observations. It restricts the curve near the first observation, while later ages constrain how much of the schooling gap persists. Neither comparison supplies a general rule about nearby and distant observations: the ranking depends on the logistic family, anchor, and support. Figure B4 summarizes all observation combinations, showing both interior-scenario medians and maximum widths across all five scenarios; the individual-scenario results remain in the replication files.

Finite-window evidence can therefore leave economically consequential differences among admissible education paths, while observations at other ages can narrow the differences found. The exercise conditions on the fitted baseline, logistic family, parameter domain, and valuation calibration. It does not evaluate source-estimate sampling error or sensitivity to alternative schooling functions and earnings-response mappings. The policy applications in Section 4 add source-specific outcome definitions, timing bridges, observed mechanisms, and full welfare accounting.

4 From Education Paths to Policy Welfare

Calculating the earnings NPV of an education path is only one step toward evaluating a policy: the welfare comparison also depends on transfers, private education payments, and the net cost to the government. Building on the marginal value of public funds (MVPF) framework of Hendren and Sprung-Keyser (2020), I incorporate alternative education paths into beneficiaries’ willingness to pay and the government’s fiscal account. The broader exercise reconstructs education-related welfare calculations for 38 policy cases, whose results are brought together in Table 3 and Section 5. This section examines three cases introduced in Section 2: the STBF scholarship, the Adams Scholarship, and Tennessee Promise.¹⁸ These cases illustrate how the reconstruction procedure operates and how alternative education paths can alter welfare conclusions. Under the reference specification, the MVPF is finite and greater than one for STBF, finite and less than one for the Adams Scholarship, and classified as self-financing for Tennessee Promise. The discussion relates the candidate paths to each study’s dynamic evidence and mechanisms, distinguishing the observed education effects from the assumptions needed to project lifetime benefits and fiscal costs.

4.1 Welfare Accounting and the Education-Path Replacement

The marginal value of public funds (MVPF) measures beneficiaries’ willingness to pay for a policy change per dollar of its net cost to the government. Following Hendren and Sprung-Keyser (2020), let WTP_i denote

¹⁷Reductions are calculated within scenarios relative to age 25 alone, then summarized by the median at the 25, 50, and 75 percent interval positions. Largest widths and sign counts include both boundaries.

¹⁸The collection table normalizes both monetary components by fixed reference program costs; this section uses source dollars. MVPF is unchanged.

willingness to pay and N_i net fiscal cost for policy contrast i . Both are present values for the same population and policy unit. For a positive net cost,

$$\text{MVPF}_i = \frac{\text{WTP}_i}{N_i}, \quad N_i > 0. \quad (9)$$

Willingness to pay is the monetary value of the policy to its beneficiaries under the maintained welfare assumptions. Net fiscal cost includes the initial expenditure and subsequent fiscal effects of behavioral responses; it is not simply the program's budget. For example, higher earnings can increase private disposable income while also reducing net public cost through higher taxes and lower means-tested transfers. A policy with positive willingness to pay and nonpositive net fiscal cost is classified as self-financing and assigned an infinite MVPF.¹⁹

The education-policy accounts can be organized into earnings-related components and the remaining benefits and costs. Use the superscript ref for L1 and write

$$\text{WTP}_i^{\text{ref}} = B_i^{\text{ref}} + \text{NPV}_i^{\text{after,ref}} - C_i^{\text{priv,ref}}, \quad (10)$$

where B_i^{ref} contains the value of program transfers and other private benefits included in the source account, $\text{NPV}_i^{\text{after,ref}}$ is the disposable-income component generated by the education-related earnings projection, and $C_i^{\text{priv,ref}}$ is the associated private education payment. The corresponding public account is

$$N_i^{\text{ref}} = G_i^{\text{ref}} + C_i^{\text{pub,ref}} - \text{NPV}_i^{\text{tax,ref}}. \quad (11)$$

Here G_i^{ref} contains direct program expenditure and other net fiscal costs, $C_i^{\text{pub,ref}}$ is additional public education expenditure not already included in G_i^{ref} , and $\text{NPV}_i^{\text{tax,ref}}$ is the earnings-related fiscal return. The superscript tax includes changes in transfers when the source uses a combined tax-and-transfer rate. All components are signed policy-induced changes: a reduction in tuition payments raises willingness to pay, whereas lost tax revenue raises net cost.²⁰

I assemble the applications from education-related entries in the Policy Impacts Library and their underlying studies, focusing on calculations that project lifetime earnings and fiscal returns from education effects measured within finite windows. Of the 38 policy calculations, 37 have an original welfare calculation, denoted L1. Most use the reconstruction by Hendren and Sprung-Keyser (2020), as in Adams; STBF and Tennessee Promise use their original authors' welfare calculations (Angrist et al., 2022; Schoonover et al., 2026). The remaining application values the education channel of cannabis legalization.²¹

The 38 calculations share an outer net-income account while retaining different source mappings. Thirty-three use HSK programs, three use independent authors' forecasts, and two are explicitly limited extensions. I first replay the source monetary components, including earnings, taxes, education payments, WTP, and net cost. I then replace the education inputs, holding the source mapping fixed where it is recoverable. STBF retains its nonlinear credential–experience function. Early Head Start and Tennessee Promise use public-data

¹⁹Self-financing is a welfare convention, not the arithmetic ratio at negative cost. If both components are nonpositive, I report them without an assigned MVPF. Hendren and Sprung-Keyser also apply the marginal framework to finite policy contrasts.

²⁰Components vary by policy, and transfers must not be counted twice. Scholarship benefits need not equal expenditure dollar for dollar: Hendren and Sprung-Keyser distinguish induced students from those whose behavior is unchanged. Appendix D.1 gives accounting conventions.

²¹The 20 September 2026 library snapshot and Appendix E supply 23 source-study families. Several tax contrasts can come from one study. Inclusion requires reconstructable education and accounting inputs; the collection is not a census of education policies.

reconstructions because their full author income inputs are unavailable; their reported L1 values remain separate comparison benchmarks. No income multiplier is chosen to force these reconstructions to equal a reported total. Eleven tax cases retain the fixed finite-policy normalization in Appendix E.4.1.

Let $\mathcal{P}_i^j$ contain the relevant schooling, credential, or time-enrolled paths. The common earnings object is

$$\Delta y_i(a) = F_i(a; \mathcal{P}_i^1, z_i) - F_i(a; \mathcal{P}_i^0, z_i), \quad (12)$$

where z_i contains the retained source prices, response coefficients, and population conventions. The HSK extension uses signed completion increments with source-specific response templates. An author's nonlinear forecast instead receives the separate education inputs it originally used. In either case early losses enter this net-income effect once, without an additional opportunity-cost deduction. With ordinary age discounting, $E_i = \sum_a D_i(a) \Delta y_i(a)$; where the source discounts response phases from different bases, those phase-specific contributions are retained before aggregation (Appendix D.2).

Write T_i for the fiscal earnings component, P_i for private education payments, and C_i for public education costs. For accounts that value disposable earnings, the component replacement is

$$\text{WTP}_i^{\text{path}} = \text{WTP}_i^{\text{ref}} + (E_i^{\text{path}} - E_i^{\text{ref}}) - (T_i^{\text{path}} - T_i^{\text{ref}}) - (P_i^{\text{path}} - P_i^{\text{ref}}), \quad (13)$$

$$N_i^{\text{path}} = N_i^{\text{ref}} + (C_i^{\text{path}} - C_i^{\text{ref}}) - (T_i^{\text{path}} - T_i^{\text{ref}}). \quad (14)$$

The old component is removed before its replacement is added. In HSK cases, source unit prices and the source completion allocation also transport private and public education costs across batch timing. That additional cost-timing bridge is stated explicitly; it is not inferred from the attainment data. Independent author cases retain their specified non-income components. Early Head Start retains its source convention of gross earnings in WTP, and algorithmic placement retains source WTP while extending only the fiscal earnings channel. These exceptions are not forced into a disposable-income formula. Figure D1 displays paired WTP and net cost, so each ratio comes from the same feasible path.

The comparison curves H_i^0 and empirical restrictions come from the Census–ACS calibration library, matched to each application's geographic target and birth cohorts. STBF uses Nebraska and Adams uses Massachusetts, retaining all races, sexes, and birthplaces in the external population data. Tennessee requires an additional adjustment because its education effects are conditional on high school graduation: I match that denominator, the source's prior-year state definition, and its latest-window control means. These adjustments improve correspondence to the reported outcomes without recovering the source design's untreated potential path. Where a comparable control level is unavailable, a statewide population mean remains an external calibration rather than a measured mean for selected scholarship applicants.²²

L1 comprises the original WTP, net fiscal cost, and MVPF calculation. L2 reconstructs the education inputs retained for L1; L3 preserves L2 and adds separately reported source windows or dynamics. For accounts based on Hendren and Sprung-Keyser (2020), both reconstructions retain their outcome-conversion rules; author-reported accounts use the case-specific conversions above. Additional information need not be a directly estimated schooling-years effect: an earlier enrollment estimate can enter through the retained duration conversion, provided its placement on the schooling path is stated explicitly. The same coefficient may already enter the reference total; L3 then retains its separate timing rather than introducing a previously

²²The library covers 50 states, DC, and the nation. Multi-state targets use fixed weights; identification control states need not belong to the target population. Table E1 and Appendix D.5 document matching and support.

unused estimate. Both reconstructions hold the comparison curve, empirical support, welfare account, and L2 constraints fixed.²³

The common path-construction rule is constrained minimum distance:

$$\hat{\Theta}_{i,L} = \arg \min_{\theta_i \in \Theta_{i,L}} [\hat{\mathbf{m}}_{i,L} - \mathbf{g}_{i,L}(\theta_i)]' W_{i,L} [\hat{\mathbf{m}}_{i,L} - \mathbf{g}_{i,L}(\theta_i)], \quad L = 2, 3 \quad (15)$$

Here $\hat{\mathbf{m}}_{i,L}$ contains the fixed source point estimates, and $\mathbf{g}_{i,L}$ applies their age, cohort, and outcome operators to a candidate path. The set $\Theta_{i,L}$ imposes the anchored logistic form, calibrated support, education-level bounds, the one-year annual schooling-acquisition cap, and credential ordering where relevant. L3 retains the L2 equalities and adds source observations to the fit. $W_{i,L}$ is fixed in advance; the eight overidentified L3 applications use the identity matrix in their common within-case units, without standard-error bands. If zero residual is attainable, all exact matches minimize the same criterion. Otherwise the retained paths minimize the discrepancies subject to the unchanged L2 restrictions, and their residuals are reported. The numerical search retains distinct minimizing solutions found, not every approximately fitting path. Input uncertainty is studied separately. Table E5 identifies the eight best-fit applications, for which the search currently retains one path each; this is a conditional model selection result, not identification of the true path.

For STBF and Tennessee, the retained non-income components yield $WTP = w_0 + (1-t)E$ and $N = n_0 - tE$. At positive cost, MVPF exceeds one exactly when $E > n_0 - w_0$. The Adams account additionally transports education payments, so its welfare cannot be summarized by a fixed threshold in earnings alone. Welfare comparisons across groups also depend on social welfare weights (Hendren and Sprung-Keyser, 2020). All ranges refer to verified numerical candidates rather than certified bounds on the continuous feasible set.

4.2 STBF Scholarship

Angrist et al. (2022) study how Susan Thompson Buffett Foundation scholarships affect college enrollment, time in school, and degree completion. Awards are randomly assigned among Nebraska applicants in the 2012–2016 high-school graduating cohorts. The reconstruction uses the 2012–2014 four-year-college strata, which enter the authors' welfare calculation and have six-year follow-up. At that horizon, the archived replication estimates imply that an award offer raises time enrolled by 0.3648 years, bachelor's completion by 8.13 percentage points, and completion of any degree by 5.18 percentage points. The authors find an initial graduation delay followed by higher completion. Table D1 records the education inputs and their windows.

The authors value these effects under hypothetical public funding, following Hendren and Sprung-Keyser (2020). Their earnings model combines credentials and time enrolled with Nebraska-born ACS earnings profiles. With 3 percent discounting, projected earnings rise by \$21,150 per treated applicant. The \$32,250 scholarship transfer enters private benefits and public expenditure, while a 20 percent tax rate divides the earnings gain between recipients and the government. The resulting willingness to pay is \$49,170 and net public cost is \$28,020, giving the L1 MVPF of 1.75. Table D2 records this account.

I retain the award-offer contrast, the authors' four race–sex earnings functions, the \$32,250 transfer, and the 20 percent tax rate. The earnings model uses exclusive SC, AA, and BA shares and a quartic in potential work experience, with experience reduced by time enrolled. Time enrolled remains a separate input; it is

²³Appendix E.7 records the observation windows, conversions, and any temporal or population mapping. Institutional policy lags and conditional college-year outcomes require such mappings; they are not relabeled as individual follow-up. Splitting a single entry observation into assumed study years alone does not create L3, as in Tennessee Pell and Georgia HOPE.

not relabeled as completed schooling. The recovered source sufficient statistics reproduce the deposited workbook’s earnings NPV of \$21,129.23, \$20.77 below the printed \$21,150. I report both vintages without rescaling the income function (Appendix D.6).

L2 jointly matches four six-year inputs: time enrolled (0.3648 years), any-college attainment (0.7158 percentage points), AA+ (5.1822 points), and BA+ (8.1335 points). Each outcome has its own anchored path and fixed source age-cohort weights. The pooled path indexes transport timing to the source subgroups using their fixed reported effect ratios. This is an additional bridge between pooled controlled coefficients and subgroup controlled coefficients, whose weighted means need not coincide exactly. Credential paths are nested, and a single feasible set of source baseline shares must remain valid at every age. The ACS curves calibrate timing; they do not replace the authors’ baseline mean inputs in the income function. Figure 5 shows the four gap ranges.

The resulting earnings NPVs range from 7.76 to 61.10 thousand 2019 dollars, and the candidate MVPFs range from 1.25 to 4.05. The earnings function is held fixed across these paths; uncertainty concerns their joint continuation and the stated subgroup bridge. Four outcomes measured at the same follow-up do not themselves supply an additional observation age, so this specification remains L2. The finite search does not prove that all compatible paths have been found. Table D3 records complete joint-path witnesses, and Appendix E.9 reports alternative support and valuation assumptions.

4.3 Adams Scholarship

Cohodes and Goodman (2014) use test-score eligibility thresholds to estimate the effects of Massachusetts’ Adams Scholarship on college choice, persistence, and completion. The estimates used here concern the 2005–2006 high-school graduating classes. Eligibility shifts students toward participating public colleges. Within two years of high school, two-year enrollment rises by 0.7 percentage points and four-year enrollment changes little. At six years, bachelor’s completion falls by 2.5 percentage points, while the two-year-completion effect is approximately zero. The authors interpret similar declines in persistence and graduation as evidence favoring dropout over pure delay within follow-up.

Hendren and Sprung-Keyser (2020) convert these enrollment and completion effects into -0.018 schooling-equivalent years per eligible student, using assumed durations for each enrollment and completion margin. Their earnings projection gives an NPV of $-\$1,204.04$. Combining this loss with scholarship benefits, private education payments, public costs, and tax-and-transfer effects yields willingness to pay of $\$1,364.23$ and net cost of $\$1,896.35$, in 2004 dollars. The L1 MVPF is 0.72. This calculation concerns scholarship eligibility rather than the separate effect among students induced to switch colleges.

I retain the source conversion, signed earnings response, tax mapping, and education-cost unit schedules. A single-batch replay reproduces the reference monetary components. Replacing that batch by signed path increments changes earnings and the timing of source private and public education costs. The account is $W = w_{\text{fixed}} + E - T - P$ and $N = n_{\text{fixed}} + C - T$, with fixed noneducation components and source-derived cost schedules. It therefore does not reduce to a fixed threshold in earnings alone. Appendix D.7 states the cost-timing bridge and the retained 2004-dollar units.

The paths use the Massachusetts ACS profile for the 1987 birth cohort, anchored at age 18, under state-calibrated empirical restrictions. L2 places the converted six-year total at age 24. L3 also retains the source’s earlier entry window, assigning its 0.007-year converted contribution to age 20 (Figure 6). This uses the same enrollment and credential multipliers as L2, but preserves two timing restrictions rather than

only their final aggregate. Entry within two years does not directly measure completed schooling at age 20: that placement is an explicit temporal allocation of the conversion. Appendix D.7 gives both restrictions; Appendix E.9 reports the calibration and valuation checks.

The L2 candidates leave the welfare comparison unresolved. Adding the early converted input in L3 narrows the searched MVPFs to 0.61–0.64, with positive net cost throughout. This result depends on the source’s early entry estimate and its stated placement on the schooling path. All found L3 paths have WTP below net public cost under the retained income and education-cost mappings. Positive willingness to pay can therefore coexist with an MVPF below one because the retained private benefit need not cover public cost. The original authors’ dropout interpretation concerns the observed persistence and completion history. A candidate that later catches up describes a possibility beyond that history, not evidence against their mechanism.

4.4 Tennessee Promise

Schoonover et al. (2026) study the expansion of tuition-free community college from a Tennessee county program in 2009 to a statewide program in 2015. Their attainment analysis uses 2000–2024 ACS data for high-school graduates aged 18–24, with exposure defined by eligibility at age 17. They estimate effects for overlapping age pairs from 18–19 through 23–24, controlling for state and survey year. At ages 23–24, some college or higher rises by 3.5 percentage points, associate’s or higher by 4.0 points, and bachelor’s or higher by 2.7 points. The authors examine earlier windows and acknowledge that some participants may complete degrees after age 24.

The authors use Tennessee-specific education returns to project lifetime income within the framework of Hendren and Sprung-Keyser (2020). Private benefits include tuition relief for students who would have enrolled anyway and after-tax earnings for induced students. Scholarship and public education expenditure are offset by projected tax receipts. With a 26 percent tax rate and 3 percent discounting, they report willingness to pay of \$72,837 and net public cost of –\$10,053, giving a self-financing L1 classification. The reported tax gain of \$24,845 implies a gross earnings NPV of $E^{\text{ref}} = 24,845/0.26 = 95,557.69$ under this fiscal mapping.

The public author materials do not supply a complete age-by-credential earnings table. I therefore reconstruct Tennessee earnings from public ACS data and identify the result as a reconstruction, not an exact replication. The schedule uses 2021 dollars, two pre-completion study years for SC/AA and four for BA, zero earnings while studying, and the corresponding credential-versus-HS earnings difference after completion. Signed exclusive-credential increments aggregate these responses. The CE input serves as a proxy for SC attainment; BA+ is priced at the BA schedule without an extra graduate premium. These are additional bridges.

To connect the population attainment effects to the participant welfare unit, the main normalization combines the reported \$2,711 grant, the private-benefit remainder of \$2,124.31, and the 5.4-percentage-point entry effect. It implies a participant share of 0.249524 under the stated entry-effect bridge; this is an account-implied normalization, not observed take-up or an income-matching factor. With fixed components in the adopted 2021-dollar convention, $WTP = 2,124.31 + 0.74E$ and $N = 14,792 - 0.26E$. The source component sum differs from its printed program total by one dollar. Applying the public schedule to the nested endpoint inputs gives \$69,261.65, compared with the author’s implied \$95,557.69; the difference is retained. Appendix D.8 documents the sample, timing, and unit assumptions.

L2 matches all three latest-window effects using anchored logistic paths for the 1992–1998 birth cohorts, with fixed equal cohort weights. The Tennessee ACS reference matches the high-school-graduate denominator,

prior-year residence definition, and reported control levels. Both policy states share the age-17 anchor, and credential levels remain nested at every age (Figure 7). State-calibrated support and a restriction based on remaining completion in earlier cohorts limit the continuations; the main bachelor's plateau cannot exceed 45.4 percent. These are maintained restrictions on an external reference, not recovered untreated predictions from the source regressions.²⁴

The latest-window effects alone admit earnings NPVs from -68.71 to 203.21 thousand dollars. L3 uses all eighteen Table 2 point estimates: three credentials observed in six overlapping age windows. It preserves the three latest-window effects exactly and minimizes the equal-weight squared discrepancies for the fifteen earlier coefficients. The search retains one best-fitting joint path, with earnings NPV 113.94 thousand dollars and a self-financing MVPF under the reconstructed account. Its root mean squared residual across eighteen inputs is 0.353 percentage points, with a maximum absolute residual of 0.667 points. Table D5 reports every fitted value and residual. Figure 7 focuses on ages 17–25. The source point estimates are fixed inputs here; their uncertainty is not used to define acceptance bands.

At positive net cost, this reconstructed account has MVPF above one if $E > \$12,667.69$, while positive-benefit self-financing requires $E \geq \$56,892.31$. The comparison with the author's L1 depends on public earnings reconstruction, participant normalization, and price conventions as well as education timing. Comparisons among candidate paths hold those choices fixed. The source cautions about returns and completion after age 24 remain relevant.

Input uncertainty is examined separately from continuation at fixed education effects. Table 4 summarizes 1,000 prespecified input repetitions per application. Adams propagates linked source inputs, whereas STBF and Tennessee use researcher-constructed perturbations under stated dependence and scale choices. Tennessee's normal perturbations are not draws from the source permutation distribution. Appendix F documents these differences and unsuccessful searches; the endpoint summaries do not establish confidence coverage for the continuous path class.

The cases show how continuation and its restrictions enter distinct welfare questions: the size of the gains in STBF, the assumptions needed to locate Adams' converted schooling loss in time, and the fiscal classification of Tennessee Promise. Within each path comparison, the income and cost mappings remain fixed; the education paths entering those mappings change. Section 5 asks how frequently comparable fixed-input paths leave these judgments unresolved across the policy collection.

5 Welfare Judgments across Policies and Populations

The welfare implications of an education effect depend on its persistence, the ages at which attainment changes, and uncertainty in the education inputs. This section asks whether the observed information determines which side of MVPF one a policy lies on, whether it can finance itself, and how these conclusions change when additional comparable observations restrict the path. The collection contains 38 finite-path calculations, including 37 with L1. The tax-input conversions and account adjustments described in Section 4.1 remain identified within the full comparison. Single-parameter comparisons distinguish permanent effects on eventual attainment from temporary effects on acquisition speed or timing. Input repetitions examine a separate source of variation under each case's specified uncertainty scheme. Appendix G compares men and women within seven policies, holding terminal education effects and valuation prices fixed.

²⁴Table D4 and Appendix D.8 document donor cohorts, plateau extrapolation, level scaling, and age weights.

5.1 Continuation Values and Policy Comparisons

Figure 8 places the L2 ranges and available L3 exact-match ranges or best-fit values beside the reference MVPFs, ordered by descending L1. All 38 calculations use finite attainment paths. Among the 37 with L1, 29 admit positive-cost paths on both sides of MVPF one at fixed L2 inputs. Of these, 16 belong to the 24-account group displayed separately from the finite tax and special accounts. Taking the education coefficients as known therefore need not determine whether beneficiary welfare exceeds one dollar per dollar of net public cost.²⁵

The uncertainty can concern the signs of the monetary components, rather than only the size of their ratio. In 23 of the 37 calculations with L1, including 11 in the 24-account group, one compatible path gives negative willingness to pay and positive public cost, whereas another gives positive willingness to pay and nonpositive public cost. These paired outcomes follow from different projected earnings and tax receipts under the same education restrictions. Fixed transfers remain unchanged. The source private and public education-cost schedules are evaluated on each path, so these costs can also change. A gain that persists into high-earning ages can benefit recipients and repay the public expenditure; an initial gap that disappears or reverses can produce a different conclusion. These are possible aggregate continuations, not estimates of individual dropout or return-to-school behavior.

A large MVPF need not represent a large dollar benefit. Numerical checks find positive-cost approaches to zero net cost with a positive limiting WTP in 25 of the 37 calculations, and with a negative limiting WTP in 15. The resulting ratio diverges while both monetary components remain finite. Figure 8 therefore separates positive-cost ratio spans from self-financing states. The span does not assert that every intermediate ratio is attained, and a finite searched endpoint does not certify the boundary of the continuous feasible set.²⁶

Figure 9 applies the same lower-endpoint ordering to all 37 calculations with a reference MVPF. It first compares lower endpoints and then upper endpoints, both descending, using L3 where available and L2 otherwise. For a best-fit singleton the endpoints coincide; its rank conditions on the fitting rule as well as the path restrictions. Finite endpoints are rounded to two decimals, and equal endpoint pairs share a rank. Some tax calculations retain unbounded positive-cost ratio spans although attainment and both monetary components are bounded. The divergence comes from finite paths approaching zero net cost. Equal endpoint pairs share ranks; no path probabilities are used to break ties.²⁷ Both axes increase from small to large ranks, so a point below the diagonal has higher priority under this rule than under L1. The ordering expresses a preference over conditional endpoint summaries; it is not a social-welfare or certified worst-case ranking.

Twelve calculations with L1 admit L3 under the common rule for additional source windows or dynamics. Replacing one calculation's L2 by L3 improves its displayed priority in ten cases and leaves it unchanged in two, holding the others fixed. This is a consequence of the endpoint-ordering rule, not evidence that information raises true welfare. Additional restrictions or the best-fit criterion can exclude favorable as well as adverse continuations. For the seven best-fit L3 applications with L1, the displayed contraction includes selection by the maintained loss function. Five of these L3 reconstructions separate converted source

²⁵Several tax contrasts come from one study, so the calculation count is not a count of independent studies. The 24-account group spans 20 source-study families, with threshold pairs in 14. MVPF one weights beneficiary dollars and public resources equally; welfare priorities can also depend on distributional weights and financing alternatives (Hendren and Sprung-Keyser, 2020).

²⁶Appendix E.3 documents complete paired paths, constraint checks, and one-sided ratio sequences. A large finite ratio or a rounded zero cost is insufficient evidence of an infinite limit.

²⁷The tax calculations transport local attendance and accounting responses over a fixed finite schedule change calibrated to one point-estimate credit dollar. Appendix E.4.1 states this additional assumption.

components or dynamics under explicit time or population mappings, including Adams; their narrowing therefore depends on these mappings as well as on the additional source information. Appendix E.2 records the inputs and conversions; Figures E2–E3 display H^0 and H^1 for all 38 calculations. No probability distribution is assigned to the paths.

Table 3 reports willingness to pay and net cost alongside MVPF. The policy-type columns in the table and Figure 8 distinguish 14 direct student-aid calculations, 17 education tax-benefit calculations, four tuition or public-college-funding calculations, one college-assessment and course-placement intervention, one early-childhood intervention, and one education channel of another policy. These types describe the intervention, whereas the panels distinguish accounting specifications. Several calculations share a source study, so the category counts are not independent-study counts. Panel A contains the 24 retained comparison cases, including Tennessee’s disclosed public-data reconstruction. Panel B reports the eleven finite tax reconstructions, whose source marginal coefficients require the stated finite-policy conversion. Their WTP and net-cost extrema are finite, even when their ratios are not. Panel C distinguishes the public-data Early Head Start account, the algorithmic-placement extension, and the cannabis education channel. Monetary components in Panels A and B use a fixed reference program-cost divisor within each calculation; Panel C uses native thousands. Every MVPF combines the two components of the same path. Very large normalized amounts indicate that the allowed future response can dominate a small reference expenditure; they do not establish that such a response is likely. The main finding is the existence of conflicting welfare judgments within a fixed reconstruction, conditional on both its path restrictions and its input conversion.

5.2 Permanent and Temporary Education Effects

The same finite-window education input can support a permanent or a temporary effect, with different implications for lifetime welfare. I examine this distinction by holding two parameter families at their calibrated comparison values and solving for a change in the third that matches the retained L2 education input. A nonzero change in K produces a permanent effect by altering eventual attainment. Changing b or d with K fixed produces a temporary effect: the population acquires education faster or more slowly, but the attainment gap eventually vanishes. The observation operator, empirical support, and welfare account remain unchanged across these three exercises. This makes the continuation assumption explicit without treating the observed input as evidence that one particular parameter changed. Appendix B.5 gives the matching procedure and its relation to the main path construction.²⁸

Interpreting the observed education input as a permanent effect tends to preserve the reference welfare comparison. In Panel A of Figure 10, the finite MVPF pairs follow an upward pattern broadly aligned with the 45-degree equality line. Changing K leaves a permanent difference in eventual attainment, so the education gain or loss continues to affect projected earnings beyond the observed window. This resembles a reference calculation that uses the measured education effect to project sustained earnings gains or losses. The alignment is consistent with interpreting the reference valuations in this collection as assigning a permanent

²⁸The exercise preserves the individual components of each fitted H^0 , so a zero change reproduces that baseline exactly. K and d change additively; b changes by a common proportion. STBF permits four outcome-specific changes and Tennessee Promise permits three credential-specific changes in the active family, subject to joint restrictions. This is a diagnostic rather than a decomposition of the main candidate ranges.

effect to the short-window education input. Shared education inputs and related return calibrations contribute to this resemblance, so it does not independently establish that the education effect is permanent.²⁹

Temporary effects produce a different welfare comparison. In Panels B and C, many finite reconstructed MVPFs cluster near one across a wide range of reference values, rather than following the near-diagonal pattern in Panel A. With eventual attainment fixed, changes in b or d match the education input through faster or slower acquisition, and the attainment gap eventually fades. Figure B3 shows that these temporary effects still generate dispersed earnings NPVs, even though many corresponding MVPFs lie close to one. This pattern reflects how net earnings and education-dependent costs enter the retained welfare accounts. When earnings NPV is small relative to the retained transfers and other non-earnings components, those components largely determine willingness to pay relative to net public cost. Several accounts consequently remain near MVPF one even as their earnings NPVs differ.³⁰ Temporary effects therefore retain economic value; their influence on MVPF depends on their duration and on the scale and composition of the welfare account under the specified valuation rule. The contrast with Panel A separates two explanations of the same education input: a permanent change in eventual attainment and a temporary change in when education is acquired. A lifetime welfare calculation must distinguish them through additional evidence or an explicit continuation assumption. Greater precision within the observed window does not by itself resolve that distinction.

Table 5 gives each reference MVPF, its single-parameter reconstruction, and the required parameter change. It also records calculations without a matching solution. The matched samples differ, which limits comparisons of the plotted patterns. Holding two parameters fixed can leave no solution within the empirical support; four, eleven, and nineteen calculations have no matched plateau, speed, or timing solution, respectively. Their absence is reported rather than interpreted as a zero response. Red rings identify five plateau solutions that also pass the current additional L3 restrictions, together with one speed and one timing solution. The latter two both concern the appropriations-to-tuition calculation. More complete source evidence can therefore distinguish some otherwise admissible single-family explanations. The results do not establish that the original lifetime projections are incorrect or that their authors ignored dynamics. They show how the welfare interpretation of a retained short-window input depends on the continuation assigned to it, while the source studies' observed dynamics and mechanism evidence remain relevant to choosing among those continuations.

5.3 From Uncertain Inputs to Paths and Welfare

Uncertainty in an estimated education effect carries through to the education paths that match it and to their welfare values. This section examines that propagation, together with sensitivity to selected accounting inputs, using the specified input repetitions. Figure 11 displays point-input values in dark colors and input-repetition endpoint summaries in lighter colors. Its Type column repeats the policy categories, separately from the bracketed input-method labels. Its L1 panel reports available source 95 percent MVPF intervals. The L2 and L3 panels instead summarize the existing 1,000 prespecified repetitions per calculation, keeping the ACS calibration and specified conversion and valuation rules fixed. The source income-response and education-cost

²⁹Panel A contains 20 finite pairs, six calculations that move from a finite reference MVPF to self-financing, and seven that are self-financing in both accounts. The diagonal denotes equality on the common asinh scale; it is not an estimated regression line.

³⁰Among finite pairs, 7 of 20 temporary speed results and 5 of 14 temporary timing results lie between 0.75 and 1.25. Holding the other components fixed, $W = w_0 + (1 - t)E$ and $N = n_0 - tE$ approach the ratio w_0/n_0 as net earnings NPV E approaches zero, provided $n_0 > 0$. Path-dependent education costs can also change this limiting comparison. That ratio need not equal one. This accounting relationship does not imply general insensitivity to temporary effects: the ratio can change sharply near zero net cost.

functions stay fixed in form. All 33 HSK-based accounts propagate their paired source/accounting inputs, including earnings-return draws where available. The eleven finite tax displacements remain fixed at their point normalizations. STBF holds its recovered author-model coefficients fixed; Early Head Start and Tennessee hold their public wage schedules and cost or participation conventions fixed. Each summary runs from the 2.5th percentile of candidate minima to the 97.5th percentile of candidate maxima. The methods are intentionally named separately: propagation of archived joint source/accounting draws, constructed normal or correlated perturbations, and exercises varying economic inputs while holding education fixed.³¹

Among the 37 calculations with L1, 37 L2 summaries cross MVPF one under their specified repetition schemes. WTP summaries cross zero in 33 calculations and net-cost summaries cross zero in 34; 34 have an infinite upper MVPF endpoint under the self-financing convention or an audited positive-cost limit. Within the 24-account group, the corresponding counts are 24, 21, 21, and 21. These counts describe how the reconstructed endpoints respond to the inputs being varied. They do not measure the chance that a policy is harmful or self-financing. The monetary components remain informative when the ratio becomes unstable near zero net cost.³²

Table 4 identifies each repetition method, endpoint summary, and usable count. Across the full collection, all 38,000 scheduled reduced-information repetitions were evaluated; 37,952 yielded a finite-path set, and 37,942 yielded a defined MVPF. Failed searches and unassigned ratios remain in the record.³³ Five admitted L3 specifications lack joint repetitions for their complete input sets: Florida, Wisconsin, Kalamazoo, CUNY Pell, and Social Security student benefits. Their point-input paths remain available. Tennessee's repeated normal shocks perturb and refit the eighteen point targets; two displayed zero standard errors are assigned a 0.0005 scale based on reporting precision. This is a sensitivity convention, not a recovered standard error or a draw from the source permutation distribution.

Path uncertainty remains even if the original education effect is known exactly: different paths can match that effect and imply different welfare values. Estimation uncertainty arises because the original effect is itself estimated imprecisely. Allowing the input to vary changes the compatible paths and, in turn, the welfare range. The two sources can operate together, since each possible input may still admit several continuations. More precise estimates within the same window address estimation uncertainty; comparable observations at additional ages can help resolve path uncertainty. The repetitions reported here trace this propagation under the specified source-draw or input-sensitivity scheme.

6 Discussion

Education effects observed within a finite window can support different lifetime welfare judgments, depending on assumptions about persistence, attainment timing, and valuation. I take the source studies' causal identification as given; the additional assumptions link their estimates to education paths for the target population. Conversions must preserve policy units and observation windows, while assumed schooling durations cannot constitute new observations. The external comparison profile must also be relevant to

³¹Appendix F.1 defines the order-statistic summaries and the inputs varied. These procedures do not share a sampling law or establish confidence coverage for all compatible paths. No probability distribution is assigned to paths within a repetition.

³²The source intervals and reconstructed summaries can vary different inputs and use different dependence assumptions. Their widths are not a controlled comparison of inferential methods.

³³Social Security student benefits account for 48 failed searches. Ten HOPE/Lifetime Learning credit repetitions have monetary components but no assigned ratio. Cannabis' six-year-only repetitions are diagnostic; its primary display uses the three-window L3.

the affected population. Matching Census–ACS geography and birth cohorts improves comparability, but population averages may not describe scholarship applicants or marginal students. Historical cohorts’ unobserved early attainment is reconstructed using other cohorts’ shapes. The common anchor imposes no effect or anticipation before the specified exposure age. Fit to ACS profiles cannot establish these population and exposure assumptions.³⁴

Path and valuation assumptions further determine the comparisons. Logistic curves may omit interrupted schooling or reentry, and historical variation need not bound a policy response. The single-parameter exercises distinguish permanent from temporary explanations without identifying which applies. Earnings returns may also depend on completion age (Bárány et al., 2023). The policy reconstructions value the difference between the same income function under treated and comparison education paths. For HSK-based accounts, signed completion increments move the source negative-then-positive response template across completion ages. This assumes that the template transports across ages and adds across signed increments; it does not identify individual extra time enrolled from completed attainment. STBF retains a nonlinear credential–experience model, conditional on recovered sufficient statistics and the subgroup bridge. Early Head Start and Tennessee use disclosed public-data reconstructions where the source income tables are unavailable, so their L1 comparisons change valuation inputs as well as continuation. Source education-dependent cost schedules move with the paths. Appendix E.9 varies calibration, valuation, and search at fixed source inputs. Using L3 where available, positive-cost MVPFs cross one in 20 of 38 calculations, compared with 19 under narrower support. All seven common specifications admit negative-WTP, positive-cost paths alongside positive-WTP, nonpositive-cost paths in 15 calculations. These conditional contrasts neither validate all assumptions nor exclude unsearched paths; wider support preserves feasible paths but can change the selected minimum-distance fit. The eight overidentified L3 results report nonzero residuals and one best-fitting path each. This selection is conditional on the logistic family and fixed weights; it is not proof that the underlying educational response or its welfare value is point identified.

Interpreting the welfare ranges also requires distinguishing input uncertainty from the rule used to rank policies. Different repetition schemes and incomplete covariance information prevent a common sampling interpretation, although all repetitions hold calibration and the path family fixed. Endpoint percentiles do not establish confidence coverage (Imbens and Manski, 2004), and more repetitions can reduce simulation error without resolving these limits or assigning probabilities to paths. The lower-endpoint ordering gives weight to adverse continuations among the paths found. Its ranks depend on search coverage, calibration, and information available for L3; they are neither expected-welfare nor certified worst-case rankings. Joint reporting of WTP and net cost distinguishes large benefits from ratios driven by near-zero costs, while self-financing and paths without an assigned ratio require separate interpretation. Spending recommendations further require distributional weights, a financing comparison, and assumptions about policy expansion. To assess the underlying extrapolations, later education outcomes withheld from path construction could be used to compare logistic predictions with constant effects and flexible monotone alternatives. Observed earnings could test valuation, while comparable age windows and joint coefficient covariances would improve reconstruction and inference. Such evidence can distinguish permanent from temporary effects with different welfare implications.

³⁴Appendix D.5 records age support and donor shapes. The selected literature and policy collections also limit the generalizability of the reported frequencies.

7 Conclusion

An education effect measured within a finite window leaves two economically distinct questions open: how much of the effect persists and when the remaining attainment occurs. This paper makes those questions explicit by constructing education paths that match retained reference restrictions, fitting additional point observations under the stated criterion, and passing each path through a specified valuation rule. The simulations show how additional observations can distinguish continuations that agree at an earlier age. Single-parameter comparisons separate permanent effects on eventual attainment from temporary effects on acquisition speed or timing. Temporary effects still carry lifetime earnings consequences, even when the attainment gap eventually vanishes. In the policy applications, verified paths often support different welfare judgments even when the retained education estimates are treated as known. At fixed inputs, 29 of the 37 calculations with L1 admit positive-cost MVPFs on both sides of one. More precise estimation within the same window would not remove these particular continuation differences.

The contribution is a way to assess which welfare judgments remain dependent on unobserved education paths. Compatible paths with different classifications establish that dependence under the stated calibration and valuation rules; they do not establish either path's probability. Input repetitions answer a separate question about imprecision in the retained estimates. The distinction helps identify whether further progress requires more precise inputs, additional observations of attainment, or stronger evidence on the valuation relationship.

The results favor reporting the observed education window and the assumed continuation separately. When additional outcomes are available, their contribution depends on whether they constrain attainment at a new age for a comparable population. Reporting willingness to pay and net fiscal cost alongside MVPF also distinguishes changes in benefits from ratios driven by costs near zero. Longer follow-up, richer education histories, and observed earnings can each improve the assessment, but they answer different questions. The useful standard is whether the available evidence distinguishes the paths that matter for the welfare comparison.

References

- Abraham, K. G. and Clark, M. A. (2006). Financial aid and students' college decisions: Evidence from the District of Columbia Tuition Assistance Grant Program. *Journal of Human Resources*, 41(3):578–610.
- Acton, R. (2021). Effects of reduced community college tuition on college choices and degree completion. *Education Finance and Policy*, 16(3):388–417.
- Angrist, J. D., Autor, D. H., and Pallais, A. (2022). Marginal effects of merit aid for low-income students. *The Quarterly Journal of Economics*, 137(2):1039–1090.
- Athey, S., Chetty, R., Imbens, G. W., and Kang, H. (2026). The surrogate index: Combining short-term proxies to estimate long-term treatment effects more rapidly and precisely. *Review of Economic Studies*, 93(4):2284–2312.
- Bárány, Z. L., Buchinsky, M., and Corblet, P. (2023). Late bloomers: The aggregate implications of getting education later in life. NBER Working Paper 31874, National Bureau of Economic Research.
- Bartik, T. J., Hershbein, B., and Lachowska, M. (2021). The effects of the Kalamazoo Promise Scholarship on college enrollment and completion. *Journal of Human Resources*, 56(1):269–310.
- Bergman, P., Kopko, E., and Rodríguez, J. (2026). A seven-college experiment using algorithms to track students: Impacts and implications for equity and fairness. Author manuscript, June 2026 version.
- Bettinger, E. (2004). How financial aid affects persistence. In Hoxby, C. M., editor, *College Choices: The Economics of Where to Go, When to Go, and How to Pay for It*, pages 207–238. University of Chicago Press, Chicago.
- Bhuller, M., Mogstad, M., and Salvanes, K. G. (2017). Life-cycle earnings, education premiums, and internal rates of return. *Journal of Labor Economics*, 35(4):993–1030.
- Bound, J., Lovenheim, M. F., and Turner, S. (2012). Increasing time to baccalaureate degree in the United States. *Education Finance and Policy*, 7(4):375–424.
- Bruce, D. J. and Carruthers, C. K. (2014). Jackpot? The impact of lottery scholarships on enrollment in Tennessee. *Journal of Urban Economics*, 81:30–44.
- Bulman, G. B. and Hoxby, C. M. (2015). The returns to the federal tax credits for higher education. *Tax Policy and the Economy*, 29(1):13–88.
- Carruthers, C. K. and Welch, J. G. (2019). Not whether, but where? Pell grants and college choices. *Journal of Public Economics*, 172:1–19.
- Castleman, B. L. and Long, B. T. (2016). Looking beyond enrollment: The causal effect of need-based grants on college access, persistence, and graduation. *Journal of Labor Economics*, 34(4):1023–1073.
- Chen, J. and Ritzwoller, D. M. (2023). Semiparametric estimation of long-term treatment effects. *Journal of Econometrics*, 237(2):105545.

- Chetty, R. (2009). Sufficient statistics for welfare analysis: A bridge between structural and reduced-form methods. *Annual Review of Economics*, 1(1):451–488.
- Cocci, M. D. and Plagborg-Møller, M. (2025). Standard errors for calibrated parameters. *Review of Economic Studies*, 92(5):2952–2978.
- Cohodes, S. R. and Goodman, J. S. (2014). Merit aid, college quality, and college completion: Massachusetts’ Adams Scholarship as an in-kind subsidy. *American Economic Journal: Applied Economics*, 6(4):251–285.
- Cornwell, C., Mustard, D. B., and Sridhar, D. J. (2006). The enrollment effects of merit-based financial aid: Evidence from Georgia’s HOPE program. *Journal of Labor Economics*, 24(4):761–786.
- Deming, D. J. and Walters, C. R. (2017). The impact of price caps and spending cuts on U.S. postsecondary attainment. NBER Working Paper 23736, National Bureau of Economic Research.
- Denning, J. T. (2017). College on the cheap: Consequences of community college tuition reductions. *American Economic Journal: Economic Policy*, 9(2):155–188.
- Dynarski, S. M. (2003). Does aid matter? Measuring the effect of student aid on college attendance and completion. *American Economic Review*, 93(1):279–288.
- Fan, Y., Manzanares, C. A., Park, H., and Qi, Y. (2026). A sensitivity analysis of the surrogate index approach for estimating long-term treatment effects. Working paper, arXiv:2603.00580v1.
- García, J. L., Heckman, J. J., Leaf, D. E., and Prados, M. J. (2020). Quantifying the life-cycle benefits of an influential early-childhood program. *Journal of Political Economy*, 128(7):2502–2541.
- Goldrick-Rab, S., Kelchen, R., Harris, D. N., and Benson, J. (2016). Reducing income inequality in educational attainment: Experimental evidence on the impact of financial aid on college completion. *American Journal of Sociology*, 121(6):1762–1817.
- Haider, S. and Solon, G. (2006). Life-cycle variation in the association between current and lifetime earnings. *American Economic Review*, 96(4):1308–1320.
- Heckman, J. J., Humphries, J. E., and Veramendi, G. (2016). Dynamic treatment effects. *Journal of Econometrics*, 191(2):276–292.
- Hendren, N. and Sprung-Keyser, B. (2020). A unified welfare analysis of government policies. *The Quarterly Journal of Economics*, 135(3):1209–1318.
- Hess, A. J. (2025). *Think of the Children: The Role of Early Childhood Interventions in Long-Term Human Capital Formation*. PhD thesis, University of California, Los Angeles.
- Hoxby, C. M. and Bulman, G. B. (2016). The effects of the tax deduction for postsecondary tuition: Implications for structuring tax-based aid. *Economics of Education Review*, 51:23–60.
- Imbens, G., Kallus, N., Mao, X., and Wang, Y. (2025). Long-term causal inference under persistent confounding via data combination. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):362–388.

- Imbens, G. W. and Manski, C. F. (2004). Confidence intervals for partially identified parameters. *Econometrica*, 72(6):1845–1857.
- Kleven, H. J. (2021). Sufficient statistics revisited. *Annual Review of Economics*, 13(1):515–538.
- Li, A. (2026). Higher education or just higher? The unintended effects of recreational marijuana legalization. Author manuscript, version supplied September 2026.
- Long, B. T. (2004). The impact of federal tax credits for higher education expenses. In Hoxby, C. M., editor, *College Choices: The Economics of Where to Go, When to Go, and How to Pay for It*, pages 101–168. University of Chicago Press, Chicago.
- Manski, C. F. (2011). Choosing treatment policies under ambiguity. *Annual Review of Economics*, 3(1):25–49.
- Marx, B. M. and Turner, L. J. (2018). Borrowing trouble? Human capital investment with opt-in costs and implications for the effectiveness of grant aid. *American Economic Journal: Applied Economics*, 10(2):163–201.
- Obadović, F. (2026). Identification of long-term treatment effects via temporal links, observational, and experimental data. Working paper, revised March 2026, arXiv:2411.04380v2.
- Schoonover, P., Attridge, J., Carruthers, C. K., and Welch, J. G. (2026). A promise worth keeping? Impacts of free community college on degrees and earnings. NBER Working Paper 35226, National Bureau of Economic Research.
- Seftor, N. S. and Turner, S. E. (2002). Back to school: Federal student aid policy and adult college enrollment. *Journal of Human Resources*, 37(2):336–352.
- Turner, N. (2011). The effect of tax-based federal student aid on college enrollment. *National Tax Journal*, 64(3):839–861.
- Vohra, V. (2026). Inference for welfare metrics. *Journal of Public Economics*, 256:105603.
- Zimmerman, S. D. (2014). The returns to college admission for academically marginal students. *Journal of Labor Economics*, 32(4):711–754.

Table 1. Education valuation under one and two observed ages

Generating process	$\Delta H_c(25)$ (years)	Single-effect NPV	Generating-path NPV	Candidate NPV range	
				Age 25	Ages 25 and 35
Permanent (K)	0.265	11.20	14.08	[-3.53, 44.80]	[10.61, 14.33]
Temporary (b)	0.014	0.60	0.23	[-15.43, 42.58]	[-2.92, 0.46]
Temporary (d)	-0.402	-16.96	-6.07	[-26.41, 39.18]	[-8.74, -3.97]

Notes: NPVs are thousands of 2000 dollars per person. Each generating scenario changes only the named parameter; the other two increments are zero. The increments are (0.3160, 0, 0), (0, 0.0030, 0), and (0, 0, 6.4351), respectively. $\Delta H_c(25)$ is the implied schooling effect. The benchmark uses that effect; the generating-path value uses the full path. Permanent and temporary describe the generating path. All three parameters may vary among candidates matching the stated ages. Ranges summarize verified candidates, not confidence intervals or certified global bounds.

Table 2. How additional observation ages change the NPV range

Observed ages	Median width (\$1,000)	Largest width (\$1,000)	Reduction (percent)	Both signs found (of 5 scenarios)
<i>Panel A. Plateau (ΔK)</i>				
25	48.34	65.92	0.0	3/5
20, 25	20.78	23.32	57.0	1/5
24, 25	16.39	18.40	66.1	1/5
25, 30	7.63	8.65	84.2	0/5
25, 35	3.72	4.31	92.3	0/5
<i>Panel B. Speed (Δb)</i>				
25	58.01	72.12	0.0	4/5
20, 25	18.50	22.71	68.1	2/5
24, 25	14.57	21.10	74.9	2/5
25, 30	6.81	12.76	88.3	1/5
25, 35	3.38	7.60	94.2	1/5
<i>Panel C. Timing (Δd)</i>				
25	65.59	91.01	0.0	5/5
20, 25	49.39	49.51	33.5	4/5
24, 25	40.51	47.01	38.2	4/5
25, 30	15.76	38.76	76.0	4/5
25, 35	4.77	13.54	92.7	1/5

Notes: Each panel varies only the named generating parameter; the other two increments are zero. Candidate parameters may all vary. Median widths and percentage reductions use the three interior interval positions (25, 50, and 75 percent). Reductions are calculated within each scenario relative to age 25 alone before taking the median. Largest widths and sign counts use all five positions, including the boundaries. Both signs found requires verified NPV candidates above \$0.01 and below $-\$0.01$. A zero count does not establish that such paths are absent from the continuous feasible set. Dollar units are 2000 dollars per person; the ranges are conditional numerical candidates.

Table 3. Reference welfare values and point-input continuation ranges

Policy	Type	MVPF			WTP			Net fiscal cost		
		L1	L2	L3	L1	L2	L3	L1	L2	L3
Panel A. Primary comparisons: WTP and net cost per reference program dollar										
Tuition deduction: single, lower threshold	Tax	$+\infty^s$	$[-\infty, -8.23]^{s,\dagger}$	–	5.38	$[-104.75, 208.70]$	–	–5.10	$[-38.52, 12.72]$	–
Tennessee Promise ^r	Aid	$+\infty^s$	$[-1.49, +\infty]^s$	–	4.92	$[-3.29, 10.31]$	$[5.84, 5.84]$	–0.68	$[-2.57, 2.21]$	$[-1.00, -1.00]$
Tuition deduction: single, upper threshold	Tax	$+\infty^s$	$[-\infty, -6.40]^{s,\dagger}$	–	1.00	$[-176.23, 284.40]$	–	–1.13	$[-45.91, 27.54]$	–
Texas CC tuition	Funding	349.51	$[9.70, +\infty]^s$	–	9.31	$[5.87, 24.13]$	$[10.87, 10.87]$	0.03	$[-1.63, 0.61]$	$[-0.08, -0.08]$
Michigan CC tuition	Funding	29.46	$[-4.69, +\infty]^s$	$[13.97, +\infty]^s$	7.80	$[-33.66, 89.33]$	$[6.99, 13.44]$	0.26	$[-12.05, 7.17]$	$[-0.45, 0.50]$
DC tuition assistance	Aid	22.98	$[8.81, +\infty]^s$	–	7.62	$[6.33, 41.81]$	–	0.33	$[-2.97, 0.72]$	–
HOPE tax credit	Tax	12.58	$[-4.37, +\infty]^s$	–	5.23	$[-13.01, 47.21]$	$[17.64, 20.80]$	0.42	$[-5.10, 2.98]$	$[-1.56, -1.14]$
Florida access grant	Aid	7.42	$[2.37, 22.70]$	$[10.97, 10.97]$	7.40	$[2.65, 31.53]$	$[12.65, 12.65]$	1.00	$[1.12, 1.39]$	$[1.15, 1.15]$
Social Security student benefits	Aid	4.86	$[4.62, 10.48]$	$[5.13, 5.23]$	3.04	$[3.06, 4.33]$	$[3.22, 3.26]$	0.62	$[0.41, 0.66]$	$[0.62, 0.63]$
College appropriations: spending	Funding	4.00	$[-4.77, +\infty]^s$	$[25.23, 25.23]$	3.03	$[-13.57, 34.98]$	$[7.24, 7.24]$	0.76	$[-3.15, 2.84]$	$[0.29, 0.29]$
Georgia HOPE	Aid	4.00	$[-2.04, +\infty]^s$	–	3.12	$[-2.98, 23.68]$	–	0.78	$[-1.20, 1.46]$	–
Ohio Pell grants	Aid	2.49	$[0.18, 7.85]$	–	2.88	$[0.17, 29.85]$	–	1.16	$[0.93, 3.80]$	–
Adult Pell introduction	Aid	2.18	$[3.04, 4.75]$	–	2.71	$[3.10, 14.60]$	–	1.25	$[1.02, 3.09]$	–
Kalamazoo Promise	Aid	1.93	$[0.84, 4.52]$	$[3.61, 3.61]$	1.76	$[0.81, 4.11]$	$[3.34, 3.34]$	0.91	$[0.91, 0.96]$	$[0.92, 0.92]$
Tennessee HOPE	Aid	1.86	$[-59.36, 21.02]$	–	1.89	$[-25.67, 43.23]$	–	1.02	$[0.43, 2.06]$	–
STBF scholarship	Aid	1.75	$[1.25, 4.05]$	–	1.52	$[1.19, 2.52]$	–	0.87	$[0.62, 0.95]$	–
Wisconsin Scholars Grant	Aid	1.43	$[-1.58, 7.30]$	$[1.41, 1.41]$	1.42	$[-1.70, 6.41]$	$[1.41, 1.41]$	0.99	$[0.88, 1.08]$	$[1.00, 1.00]$
CUNY Pell grants	Aid	1.39	$[-\infty, 13.76]^\dagger$	$[-5.92, -0.26]$	1.42	$[-25.03, 38.02]$	$[-4.52, -0.25]$	1.02	$[-0.09, 2.76]$	$[0.76, 0.96]$
College appropriations: tuition	Funding	1.02	$[-7.76, +\infty]^s$	$[-1.05, -1.05]$	1.02	$[-17.25, 31.15]$	$[-1.21, -1.21]$	1.00	$[-0.94, 2.22]$	$[1.15, 1.15]$
Tennessee Pell eligibility	Aid	0.85	$[-\infty, 2.66]^\dagger$	–	0.79	$[-16.42, 25.37]$	–	0.93	$[-4.82, 9.53]$	–
Tuition deduction: joint, upper threshold	Tax	0.77	$[-3.86, +\infty]^s$	–	1.00	$[-90.02, 146.10]$	–	1.29	$[-33.93, 23.34]$	–
Adams Scholarship	Aid	0.72	$[0.52, +\infty]^{s,\dagger}$	$[0.61, 0.64]$	1.21	$[-10.01, 6.69]$	$[1.61, 2.66]$	1.68	$[-17.40, 12.91]$	$[2.53, 4.33]$
Tuition deduction: joint, lower threshold	Tax	–0.02	$[-4.29, +\infty]^s$	–	–0.03	$[-106.65, 165.16]$	–	1.38	$[-34.89, 24.86]$	–

Continued on next page

Table 3 (continued)

Policy	Type	MVPPF			WTP			Net fiscal cost		
		L1	L2	L3	L1	L2	L3	L1	L2	L3
HOPE and Lifetime Learning credits	Tax	-8.81	$[-10.75, +\infty]^s$	–	-42.82	$[-260.38, 88.45]$	–	4.86	$[-7.02, 24.23]$	–
<i>Panel B. Finite tax reconstructions: per reference program dollar</i>										
HOPE: joint, lower kink ^f	Tax	$+\infty^s$	$[-\infty, +\infty]^{s,\dagger}$	–	18.41	$[-21,373.75, 40,380.60]$	–	-3.15	$[-4,694.42, 2,529.55]$	–
AOTC: dependent single, lower kink ^f	Tax	$+\infty^s$	$[-\infty, +\infty]^{s,\dagger}$	–	23.39	$[-20,781.85, 41,722.94]$	–	-1.61	$[-4,434.02, 2,291.01]$	–
AOTC: joint, lower kink ^f	Tax	$+\infty^s$	$[-\infty, +\infty]^{s,\dagger}$	–	9.96	$[-27,403.52, 55,041.78]$	–	-1.37	$[-6,686.68, 3,453.35]$	–
HOPE: independent single, lower kink ^f	Tax	18.86	$[-\infty, +\infty]^{s,\dagger}$	–	5.91	$[-53,768.73, 101,434.66]$	–	0.31	$[-17,374.29, 9,427.43]$	–
HOPE: dependent single, upper kink ^f	Tax	11.83	$[-\infty, +\infty]^{s,\dagger}$	–	4.59	$[-17,357.02, 32,615.58]$	–	0.39	$[-3,366.47, 1,855.80]$	–
AOTC: simulated instrument ^f	Tax	10.05	$[-\infty, +\infty]^{s,\dagger}$	–	5.36	$[-82,841.87, 145,584.75]$	–	0.53	$[-16,160.55, 9,118.53]$	–
AOTC: independent single, lower kink ^f	Tax	6.75	$[-\infty, +\infty]^{s,\dagger}$	–	2.45	$[-91,289.28, 183,176.50]$	–	0.36	$[-36,378.45, 18,386.77]$	–
HOPE: joint, upper kink ^f	Tax	2.37	$[-\infty, +\infty]^{s,\dagger}$	–	8.21	$[-23,368.86, 44,169.47]$	–	3.47	$[-4,875.76, 2,696.42]$	–
AOTC: dependent single, upper kink ^f	Tax	-0.02	$[-\infty, +\infty]^{s,\dagger}$	–	-0.02	$[-21,848.03, 43,610.87]$	–	1.12	$[-4,953.33, 2,554.44]$	–
HOPE: dependent single, lower kink ^f	Tax	-1.64	$[-\infty, +\infty]^{s,\dagger}$	–	-1.91	$[-16,274.53, 30,497.00]$	–	1.16	$[-2,764.33, 1,548.51]$	–
AOTC: joint, upper kink ^f	Tax	-1.77	$[-\infty, +\infty]^{s,\dagger}$	–	-7.63	$[-28,228.40, 56,512.58]$	–	4.31	$[-7,051.93, 3,648.38]$	–
<i>Panel C. Reconstructed or extended accounts and education channel: native thousands</i>										
Algorithmic college placement ^x	Placement	$+\infty^s$	$[0.09, +\infty]^s$	–	0.22	$[0.22, 0.22]$	–	-0.09	$[-5.12, 2.38]$	–
Early Head Start ^r	Childhood	0.35	$[-1.40, +\infty]^s$	–	3.91	$[-38.43, 114.65]$	–	11.22	$[-1.10, 27.52]$	–
Cannabis legalization: education channel	Other	–	–	$[-50.08, -50.08]$	–	–	$[-2.35, -2.35]$	–	–	$[0.05, 0.05]$

Policy types: Aid, direct student financial aid; Tax, education tax benefits; Funding, tuition and public-college funding; Placement, college assessment and course placement; Childhood, early childhood services; Other, the education channel of a noneducation policy. Types describe interventions, whereas panels distinguish accounting specifications. *Notes:* L1 reproduces source values. Brackets summarize fixed-input L2/L3 endpoints, including limiting values; they are not confidence intervals or certified global bounds. WTP and net-cost endpoints need not use the same path; each positive-cost MVPF uses paired components from one complete path. Infinite range endpoints retain verified one-sided limits from finite schooling paths. Superscript s on an MVPF entry records a separate self-financing state: positive WTP and nonpositive net cost. A dagger marks additional undefined-ratio paths. Superscripts f , r , and x identify the tax conversion, public-data reconstruction, and earnings extension. Panel A divides both monetary components by one fixed reference program-cost component per case. Panel B uses the same bounded logistic family. Source attendance and monetary inputs are transported to a fixed finite schedule change calibrated to one point-estimate credit dollar, retaining the two-year schooling conversion. The conversion and its age placement are assumptions. The first-stage point normalization stays fixed across repetitions. The AOTC simulated-instrument reconstruction also corrects the direct-credit unit while retaining the reported L1. Panel C uses thousands of native dollars. Early Head Start uses unscaled public ACS income and a funded-slot cost proxy, in 2000 dollars. Its source price year is unresolved. Tennessee uses public ACS income and an account-implied participant bridge, under a disclosed 2021-dollar convention. Algorithmic placement uses the verified June 2026 source components, but its reconstruction extends fiscal earnings only, keeping source WTP fixed. Their monetary differences cannot be attributed only to continuation. Cannabis has no whole-policy L1 and uses its three-horizon L3 as the primary result; its last-horizon-only L2 remains in the appendix data. A dash denotes an unavailable reference value or an inapplicable layer; missing L3 means no eligible additional source-window restriction. NF denotes an unsuccessful search. Duration allocations alone do not qualify as L3. Appendix E.4 documents units, normalization, and account-version qualifications.

Table 4. Reported intervals and conditional input-repetition endpoint summaries

Policy	L1: reported 95%	L2: endpoints	n_2	L3: endpoints	n_3	Scope
E26 HOPE: joint, start	$[-3.51, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E28 Tuition deduction: single, start	$[-\infty, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E32 AOTC: single, start	$[-0.84, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E34 AOTC: joint, start	$[-5.96, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E37 Tennessee Promise	$[+\infty, +\infty]$	$[-1.62, +\infty]$	1000	$[+\infty, +\infty]$	1000	P
E38 Algorithmic college advising	–	$[0.08, +\infty]$	1000	–	–	P
E39 Tuition deduction: single, end	$[-\infty, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E19 Texas CC tuition	$[0.93, +\infty]$	$[-1.81, +\infty]$	1000	$[1.14, +\infty]$	1000	J
E07 Michigan CC tuition	$[-2.33, +\infty]$	$[-47.44, +\infty]$	1000	$[0.11, +\infty]$	1000	J
E02 DC tuition assistance	$[1.12, +\infty]$	$[0.96, +\infty]$	1000	–	–	X
E23 HOPE: independent single, start	$[-3.03, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E22 HOPE tax credit	$[-29.15, +\infty]$	$[-13.72, +\infty]$	1000	$[0.02, +\infty]$	1000	J
E25 HOPE: dependent single, end	$[-5.36, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E36 AOTC: simulated instrument	$[-18.36, +\infty]$	$[-\infty, +\infty]$	1000	–	–	P
E01 Florida access grant	$[0.69, +\infty]$	$[-10.17, +\infty]$	1000	NR	–	J
E35 AOTC: independent single, start	$[-1.61, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E16 Social Security student benefits	$[0.98, 52.39]$	$[0.31, +\infty]$	952*	NR	–	J
E17 College appropriations: spending	$[1.25, 20.44]$	$[-10.47, +\infty]$	1000	$[1.22, +\infty]$	1000	J
E15 Georgia HOPE	$[0.46, 22.38]$	$[-3.28, +\infty]$	1000	–	–	J
E09 Ohio Pell grants	$[0.82, 5.54]$	$[-1.72, +\infty]$	1000	–	–	J
E27 HOPE: joint, end	$[-2.22, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E20 Adult Pell introduction	$[0.71, 6.11]$	$[-2.92, 110.43]$	1000	–	–	J
E06 Kalamazoo Promise	$[0.97, 5.61]$	$[-0.55, 19.12]$	1000	NR	–	J
E10 Tennessee HOPE	$[0.92, 5.08]$	$[-\infty, +\infty]$	1000	–	–	J
E04 STBF scholarship	–	$[0.95, 4.14]$	1000	–	–	P
E05 Wisconsin Scholars Grant	$[0.96, 2.18]$	$[-2.58, +\infty]$	1000	NR	–	J
E08 CUNY Pell grants	$[-2.95, 12.88]$	$[-\infty, +\infty]$	1000	NR	–	J
E21 College appropriations: tuition	$[-1.11, 5.13]$	$[-80.12, +\infty]$	1000	$[-4.11, 29.24]$	998*	J
E11 Tennessee Pell eligibility	$[-1.59, 3.57]$	$[-\infty, +\infty]$	1000	–	–	J
E40 Tuition deduction: joint, end	$[-1.83, 57.84]$	$[-4.68, +\infty]$	1000	–	–	J
E03 Adams Scholarship	$[-0.92, 3.05]$	$[-\infty, +\infty]$	1000	$[-\infty, 7.61]$	990*	J
E18 Early Head Start	–	$[-2.21, +\infty]$	1000	–	–	P
E31 AOTC: single, end	$[-2.25, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E29 Tuition deduction: joint, start	$[-2.25, 5.52]$	$[-4.38, +\infty]$	1000	–	–	J
E24 HOPE: dependent single, start	$[-6.45, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J
E33 AOTC: joint, end	$[-17.06, +\infty]$	$[-\infty, +\infty]$	1000	–	–	J

Continued on next page

Table 4 (continued)

Policy	L1: reported 95%	L2: endpoints	n_2	L3: endpoints	n_3	Scope
E30 HOPE and Lifetime Learning	$[-\infty, +\infty]$	$[-\infty, +\infty]$	990*	–	–	J
E41 Cannabis: education channel	–	–	–	$[-58.00, -46.45]$	1000	R

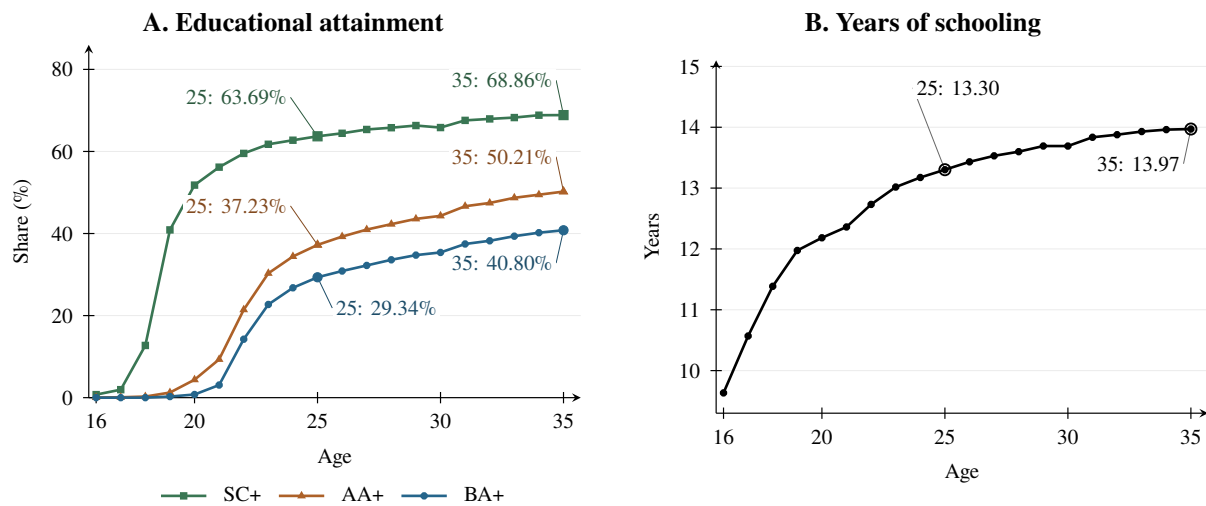
Notes: Each calculation has 1,000 prespecified L2 input repetitions. L1 reproduces available reported 95 percent MVPF intervals; a dash denotes no recovered interval. L2/L3 summarize the 2.5th percentile of candidate minima and the 97.5th percentile of candidate maxima, conditional on the specified finite-path set with a defined MVPF. n is that count out of 1,000; an asterisk marks fewer than 1,000. These are conditional endpoint summaries, not demonstrated 95 percent confidence sets for the continuous path class. No distribution over paths is imposed. Infinite endpoints retain self-financing or verified positive-cost ratio limits. L3 dashes mean not imposed; NR means no registered joint draws for the current full L3 set; prior narrower-screen results are archived; Suppr. means too few verified draws to report a summary. Only eligible reported-source timing enters L3. Input scope: J, registered source/accounting draws with their disclosed conditional input conventions; P, researcher-constructed education perturbations; X, economic inputs vary while education is fixed; R, synthetic correlated education shocks. All scopes condition on the ACS baseline and maintained calibration. Different source and reconstructed intervals can have different inputs, dependence assumptions, and coverage targets. Their widths are not a controlled comparison of methods. Appendix F gives the component summaries and explains layer availability. Cannabis covers only the education channel; the main display uses all three source horizons in L3. Its 1,000 last-horizon-only diagnostic repetitions remain in the replication files. Policy order and superscripts follow Figure 8.

Table 5. Reference and single-parameter reconstructed MVPFs

Policy	Reference	Permanent (K)		Temporary (b)		Temporary (d)	
	MVPF	ΔK	MVPF	Δb (%)	MVPF	Δd	MVPF
Florida grant	7.42	0.371	12.05	—	—	—	—
DC tuition aid	22.98	—	—	13.7	16.41	—	—
Adams	0.72	-0.0244	0.71	-1.18	0.79	0.425	0.82
STBF	1.75	Joint	1.87	—	—	—	—
Wisconsin grant	1.43	0.146	1.95	6.47	1.09	—	—
Kalamazoo	1.93	0.416	2.40	—	—	—	—
Michigan CC	29.46	0.0343	65.29	3.55	3.26	—	—
CUNY Pell	1.39	0.0197	1.83	0.686	1.14	-0.215	1.08
Ohio Pell	2.49	—	—	—	—	—	—
Tennessee HOPE	1.86	0.0575	3.36	2.28	1.19	—	—
Tennessee Pell	0.85	-0.0188	0.52	-0.808	0.95	0.246	0.98
Georgia HOPE	4.00	0.321	25.19	—	—	—	—
Social Security	4.86	—	—	—	—	—	—
Appropriations: spending	4.00	0.0822	8.11	3.92	1.79	—	—
Early Head Start	0.35	—	—	—	—	—	—
Texas CC	349.51	0.248	SF	—	—	—	—
Adult Pell	2.18	0.0815	4.36	—	—	—	—
Appropriations: tuition	1.02	8.23×10^{-4}	1.04	0.0386	1.01	-0.0154	1.01
HOPE credit	12.58	0.128	SF	6.05	3.58	—	—
HOPE ind. single: lower	18.86	1.96×10^{-4}	SF	0.00946	83.88	-0.00414	46.41
HOPE dep. single: lower	-1.64	-2.72×10^{-4}	-6.21	-0.0131	-1.89	0.00575	-2.22
HOPE dep. single: upper	11.83	3.15×10^{-4}	SF	0.0152	5.50	-0.00665	4.61
HOPE joint: lower	SF	0.00124	SF	0.0599	SF	-0.0263	SF
HOPE joint: upper	2.37	4.74×10^{-4}	22.04	0.0229	1.83	-0.0100	1.74
Tuition ded. single: lower	SF	0.0262	SF	2.02	SF	—	—
Tuition ded. joint: lower	-0.02	-0.0169	-1.09	-0.611	0.48	0.183	0.64
HOPE / Lifetime Learning	-8.81	-0.195	-10.58	—	—	—	—
AOTC single: upper	-0.02	-6.64×10^{-5}	-2.13	-0.00364	0.15	0.00174	0.01
AOTC single: lower	SF	0.00153	SF	0.0840	22.58	-0.0405	11.98
AOTC joint: upper	-1.77	-4.39×10^{-4}	-4.56	-0.0241	-1.61	0.0115	-2.03
AOTC joint: lower	SF	4.68×10^{-4}	SF	0.0257	SF	-0.0123	SF
AOTC ind. single: lower	6.75	3.17×10^{-5}	SF	0.00174	8.74	-8.31×10^{-4}	11.46
AOTC instrument	10.05	9.39×10^{-5}	SF	0.00390	3.79	-0.00136	3.02
Tennessee Promise	SF	Joint	SF	—	—	—	—
College placement	SF	0.0545	SF	5.11	SF	—	—
Tuition ded. single: upper	SF	0.000	SF	0.00	SF	0.000	SF
Tuition ded. joint: upper	0.77	0.000	0.77	0.00	0.77	0.000	0.77

Notes: Each pair varies the named parameter family; the other two remain at H^0 values. K changes eventual attainment; b and d effects eventually vanish. ΔK and Δd are in years; Δb is the proportional speed change in percent. STBF jointly changes time enrolled and three credential plateaus; Tennessee jointly changes three credential plateaus. Their outcome-specific changes are saved in the matched-solution data. † marks the same L2 solution also satisfying L3; — denotes no admissible match. SF denotes self-financing ($W > 0$, $N \leq 0$), represented by $+\infty$ in Figure 10. The table includes all 37 reference calculations; cannabis has no whole-policy reference. Appendix B.5 defines the matching procedure. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

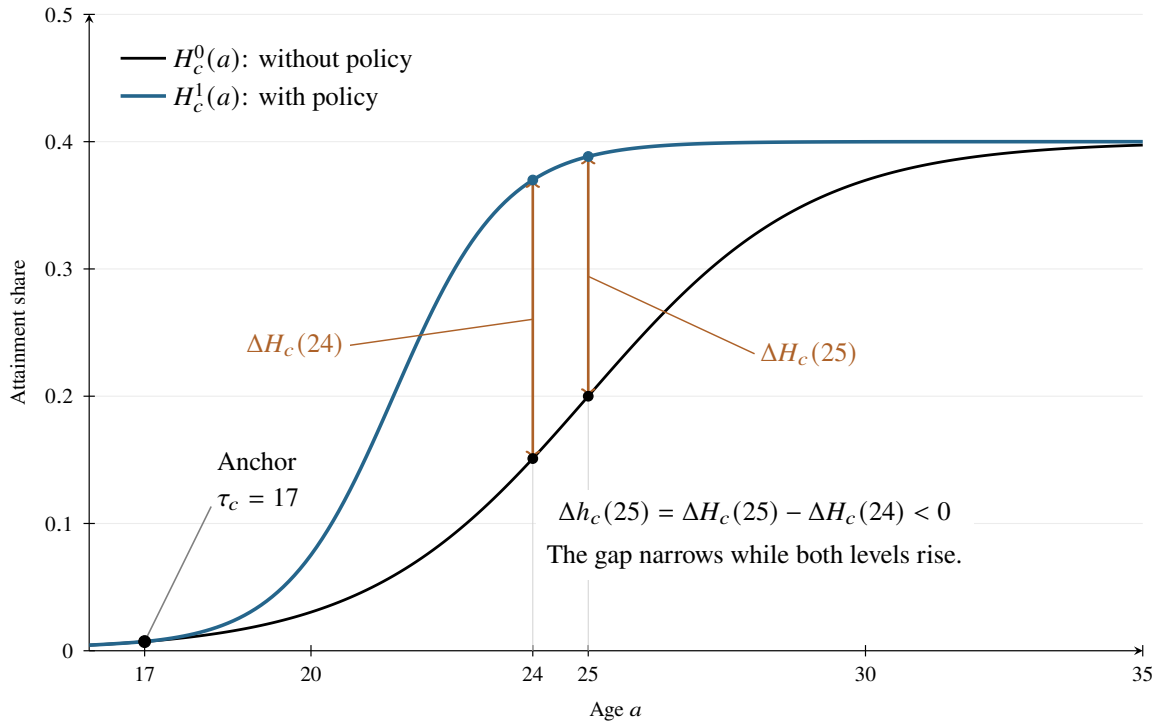
Figure 1. Education profiles for a fixed set of birth cohorts



Notes: ACS 2000–2022, birth cohorts 1985–1987, ages 16–35; all sexes, birthplaces, and racial and ethnic groups are included. Person weights apply within cohort–age cells; cohorts receive fixed one-third weights. Repeated cross sections do not fix composition within birth cohorts. SC+ denotes some college or higher, an attendance proxy; AA+ and BA+ denote cumulative degree thresholds. Schooling years use the EDUCD crosswalk. Labels report ages 25 and 35. Appendix A.5 documents these descriptive profiles.

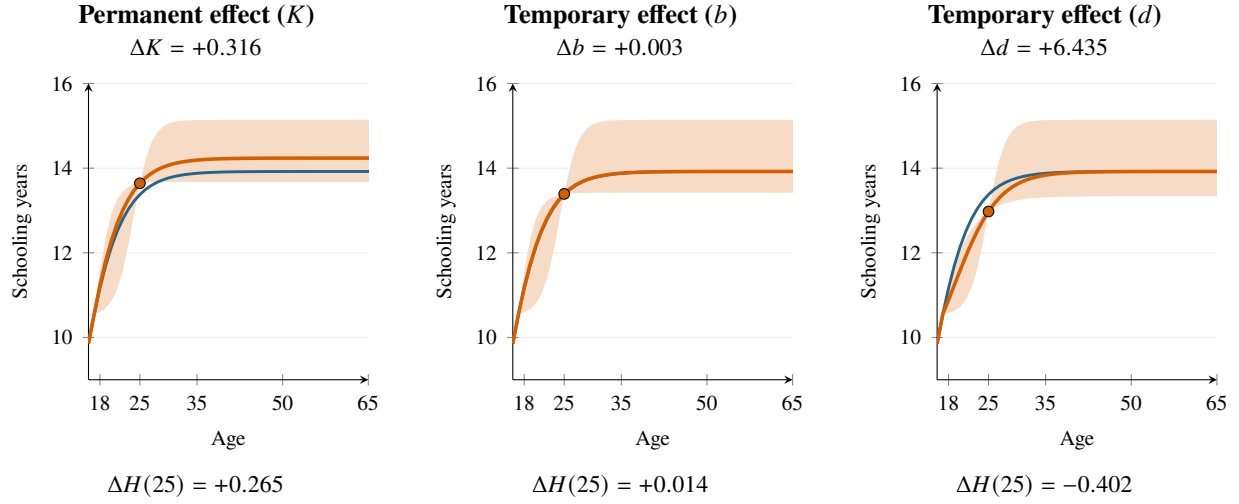
Source: Author’s calculations from ACS microdata.

Figure 2. Educational attainment, treatment effects, and the pre-policy anchor



Notes: The hypothetical reference is $H_c^0(a) = 0.4\Lambda[0.5(a - 25)]$. The treated path uses Equation (2), $(K, b, d) = (0.4, 1, 21.5)$, and the same age-17 level. Arrows mark attainment gaps at 24 and 25. Both levels rise while the gap narrows, giving $\Delta h_c(25) < 0$; both plateaus equal 0.4. These are illustrative attainment shares, not estimated effects.

Panel A. One observation: age 25



Panel B. Two observations: ages 25 and 35

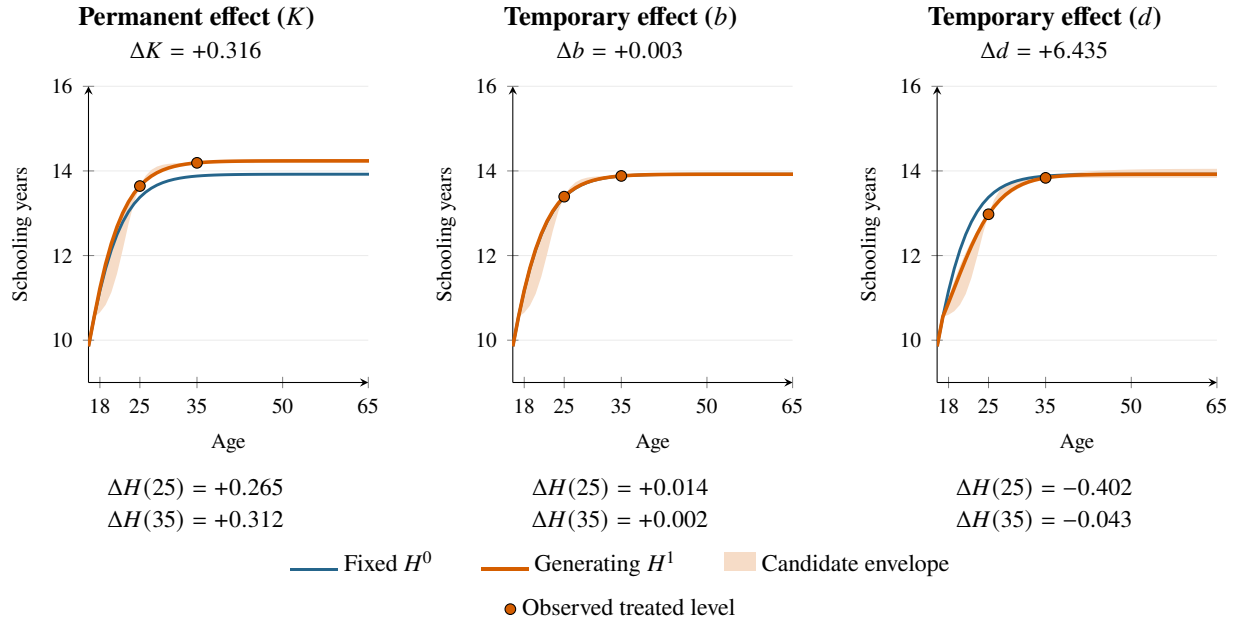
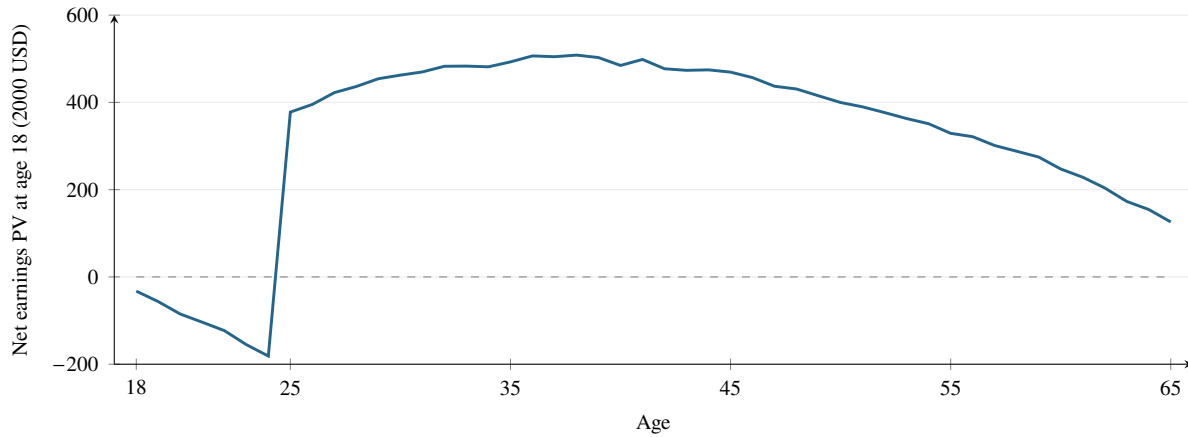


Figure 3. Compatible schooling paths with one and two observed ages

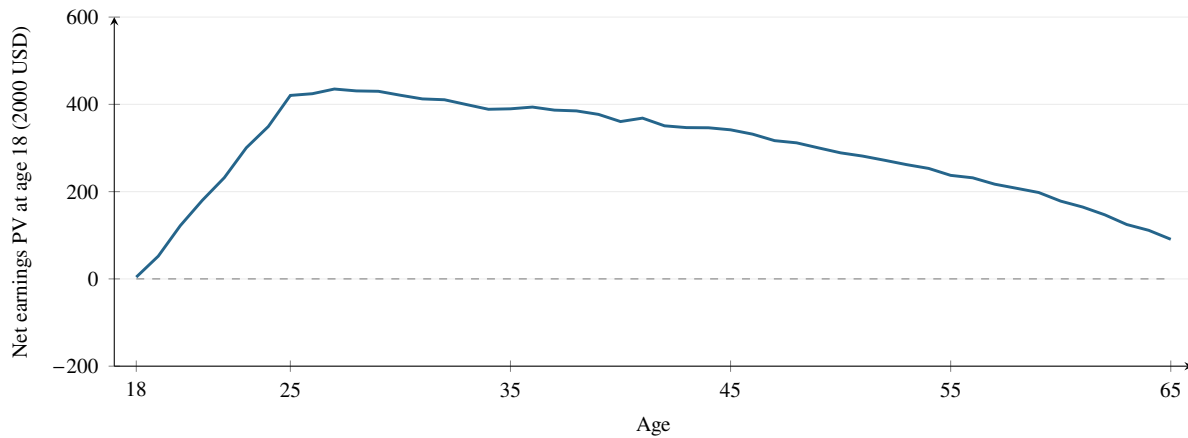
Notes: Panel A matches the age-25 effect; Panel B additionally matches the age-35 effect for the same generating process in each column. Columns vary one generating parameter at a time to the midpoint of its conditional feasible interval; the other two increments are zero. Permanent and temporary refer to the generating path, not all paths in its envelope. Blue lines show the fixed comparison path, and orange-red lines the generating path. Matching shading is the pointwise envelope of verified candidate paths, whose three parameters may all vary. Dots locate the treated levels implied by the observed effects and the fixed baseline. Every treated continuation is anchored to the comparison level at age 17. All six plots use identical age and schooling scales. Valuation results are reported in Table 1. Envelope edges need not form a single feasible path. No probability distribution is imposed on candidate paths, so the shading carries no probability interpretation. The envelopes are not confidence bands or certified global envelopes. Baseline values above age 35 are extrapolated.

A. Single completion batch at age 25

Net earnings NPV: \$15,390

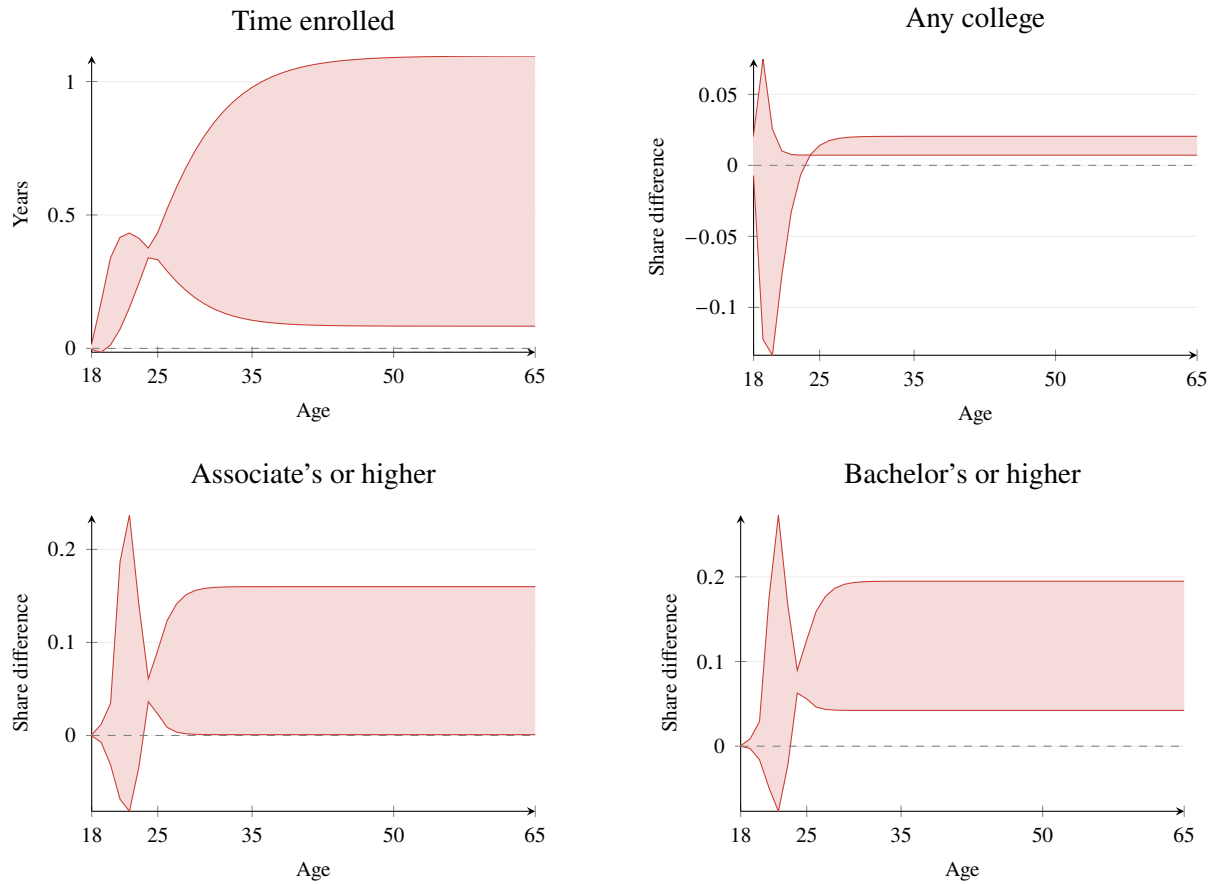
**B. Completion batches along the full path**

Net earnings NPV: \$13,986

**Figure 4.** Single-effect and full-path valuation with a permanent schooling gain

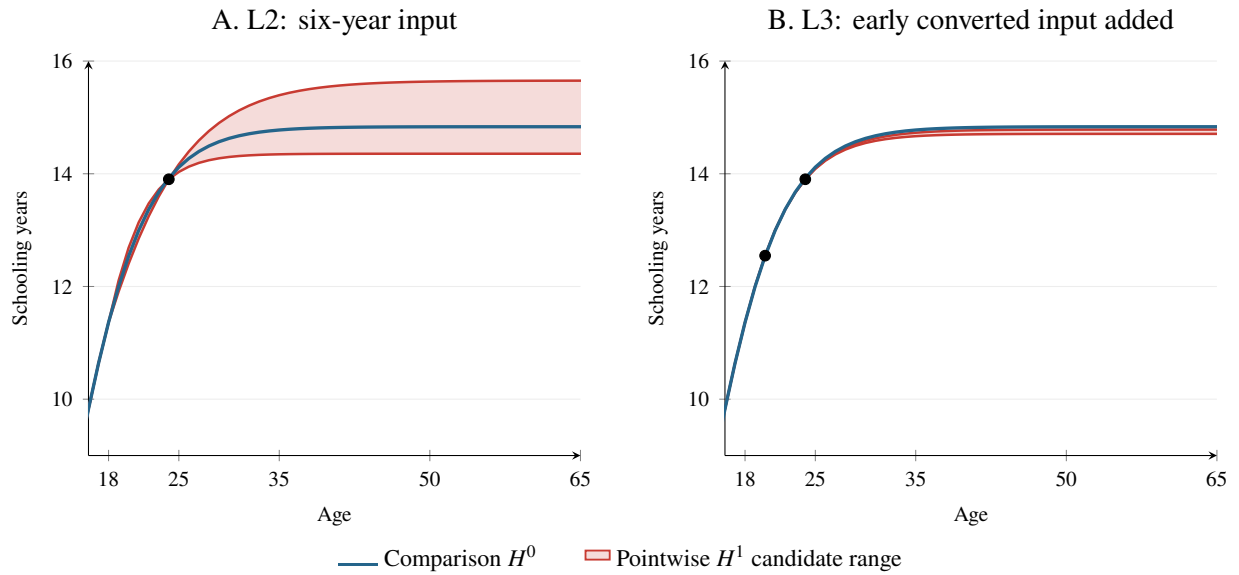
Notes: Both panels use a joint-change illustration, with $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$. Its schooling gap is 0.364 years at age 25 and 0.263 at age 65. The corresponding level paths appear in Figure B2. Each curve is a single net earnings response, including early losses and later gains. Panel A assigns the observed age-25 effect to one completion batch at 25. Panel B applies the same signed seven-year response to the signed increments of the full attainment gap. Both use income ages 18–65, the same earnings profile, and 3 percent discounting to age 18. Education paths extend through age 72 to evaluate pre-completion responses without truncating the seven-year window. No additional opportunity-cost deduction is made. This separate illustration is outside the two experiment grids.

Figure 5. STBF: four joint education-effect paths



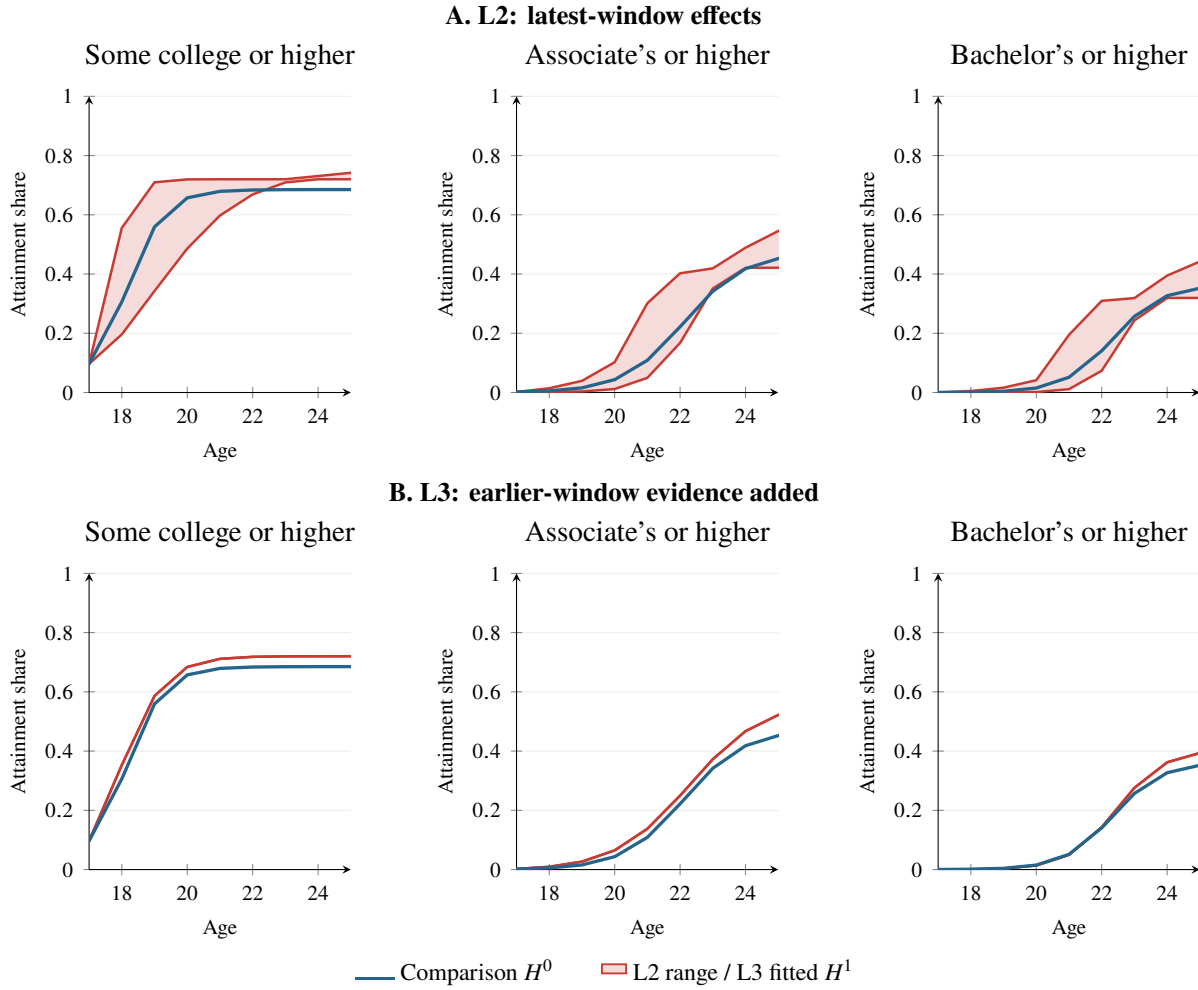
Notes: All four six-year inputs are matched jointly using the source age-cohort weights. Ranges show treatment-comparison effects, not treated levels. Credential levels are nested in every candidate, and source mean-input feasibility is checked for the full path. Time enrolled is a separate timing index, not completed schooling. Shaded boundaries can combine different joint candidates and are not confidence bands. The author nonlinear credential-experience model values each complete candidate. No additional observation age supports L3 in this specification.

Figure 6. Adams: adding the early converted education input



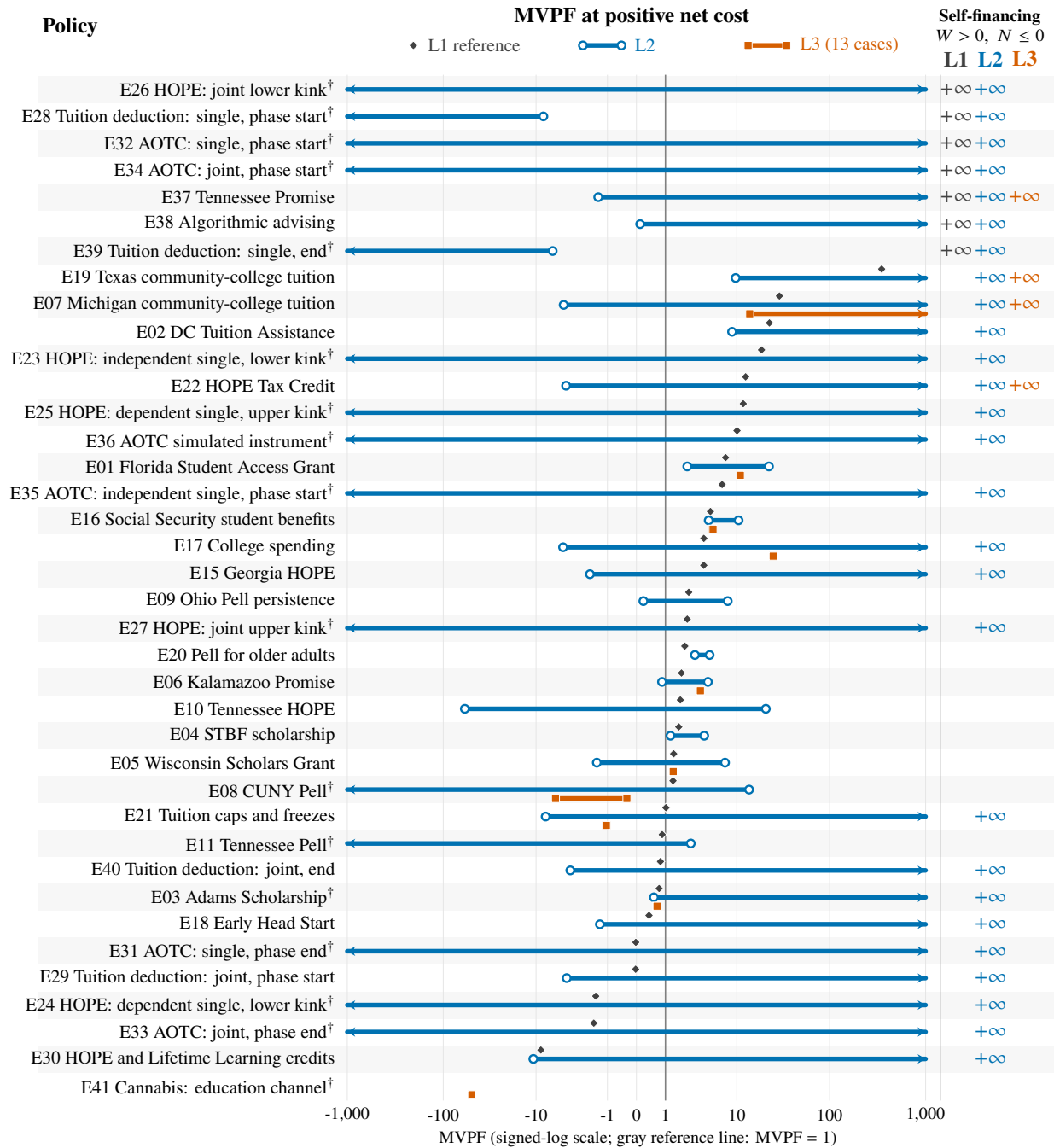
Notes: L2 matches -0.018 schooling-equivalent years at HS+6 (model age 24). Panel B additionally assigns 0.007 completed-schooling-equivalent years to HS+2 (age 20); this L3 restriction retains the early source window under an explicit temporal allocation. Dots mark converted restrictions. Blue is H^0 ; red boundaries and shading give the pointwise H^1 candidate range. Its edges may combine different paths and are not confidence limits. Scales are common. Appendix D.7 documents the calibration and allocation.

Figure 7. Tennessee Promise: L2 attainment ranges and the L3 fitted path



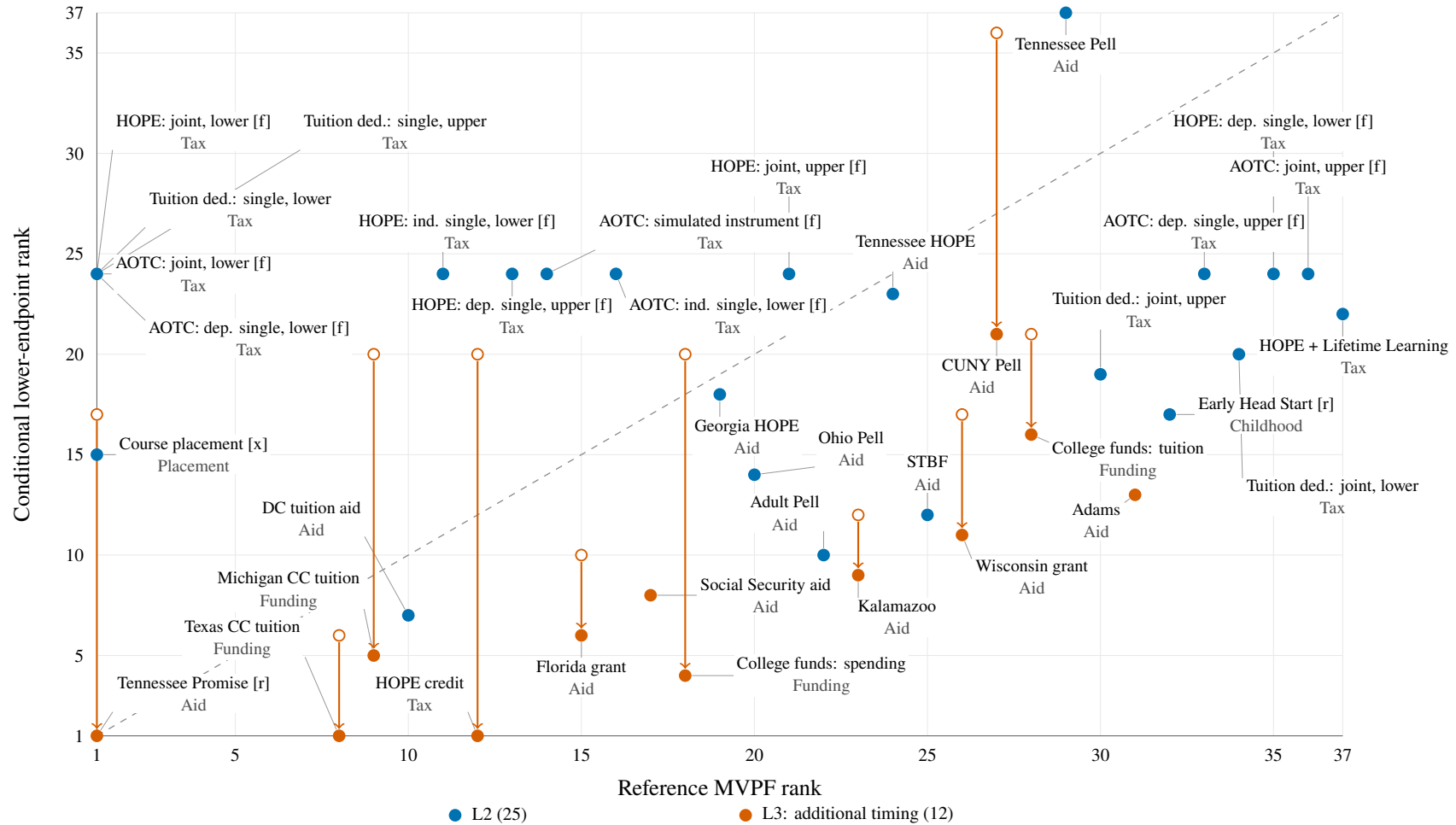
Notes: Blue is H^0 ; red boundaries and shading give the pointwise minimum and maximum H^1 among joint credential candidates. The boundaries may combine different paths across ages and credentials. L2 matches the three effects averaged over ages 23–24. L3 fits eighteen Table 2 point estimates from Schoonover et al. (2026), retaining the three latest-window effects exactly. Its coincident red boundaries represent one best-fitting joint path; Table D5 reports the nonzero residuals. Both panels use the same baseline, age-17 anchor, support, completion caps, and scales. These ranges are finite-search envelopes, not confidence bands or probability distributions. Appendix D.8 documents the restrictions; the appendix atlas shows ages through 65.

Figure 8. Education continuation ranges, ordered by the reference MVPF



Notes: Rows follow descending L1. Types: Aid, direct student aid; Tax, education tax benefits; Funding, tuition and public-college funding; Placement, assessment and course placement; Childhood, early-childhood services; Other, another policy's education channel. Gray diamonds denote L1; blue circles and red squares bound positive-cost L2 and L3 spans. Separate +∞ marks denote self-financing ($W > 0, N \leq 0$); arrows include audited positive-cost ratio limits. Gaps do not establish infeasible intermediate ratios. Daggers mark undefined-ratio paths. Missing L3 means no added restriction is admitted under the retained mapping, not no source dynamics. Superscripts f , r , and x identify tax-unit conversion, public-data reconstruction, and a fiscal-only earnings extension. Cannabis has no whole-policy L1 and shows primary L3. Eight L3 cases have coincident endpoints at one best-fitting path; the other five retain exact-match ranges. These summaries are not confidence intervals or certified bounds. Table 3 gives values. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

Figure 9. L1 and candidate-based lower-endpoint ranks for 37 policy calculations



Notes: Higher lower endpoints rank first, followed by higher upper endpoints; admitted L3 replaces L2. Finite endpoints are rounded to two decimals, and equal pairs share a competition rank. Labels give policy names and Figure 8 types; [f], [r], and [x] mark tax conversion, public-data reconstruction, and a fiscal-only earnings extension. Coincident endpoint pairs share a marker. Arrows replace only the indicated policy's L2 by L3. Fourteen calculations contain undefined-ratio paths; those paths, not the calculations, are omitted. Cannabis lacks L1. Table E2 gives all ranks. This is a candidate-based display rule, not a social-welfare or certified worst-case ordering.

• L2 match

◉ Same match also passes L3

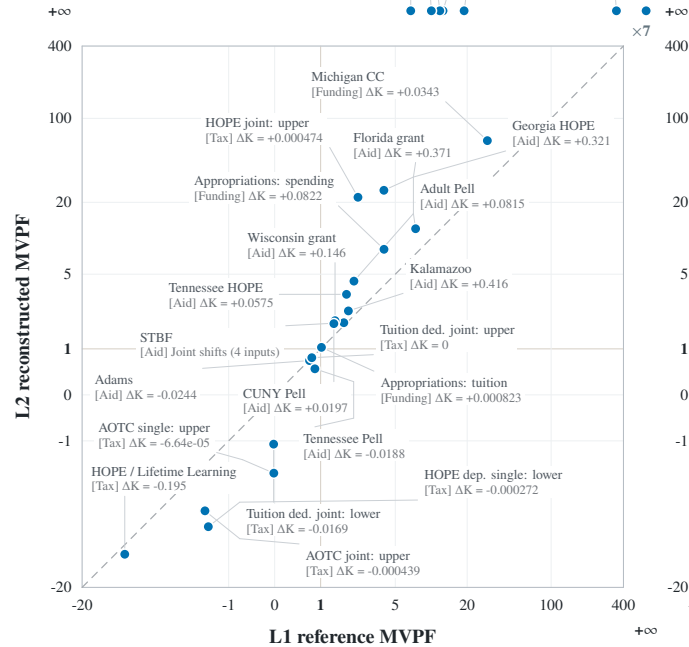
----- Equality: L2 = L1

Self-financing: outer markers; coincident cases counted

A. Permanent effect (K)

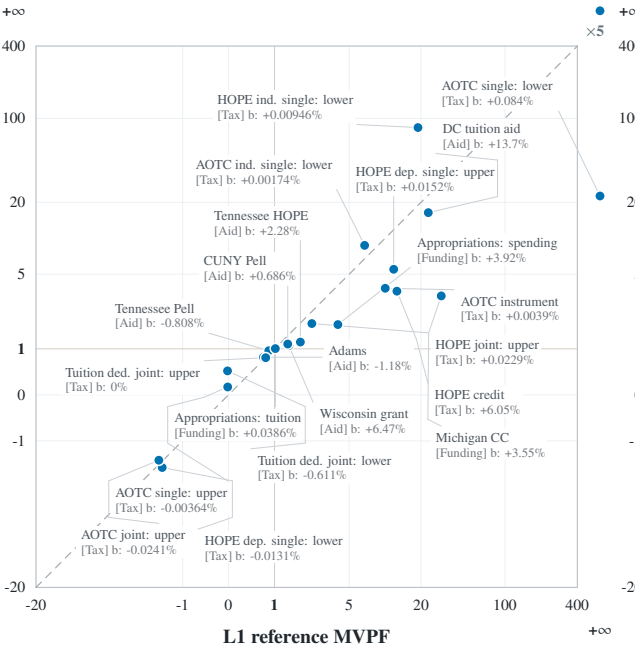
33 matched / 37; 4 no match
Both SF: Tax 5, Aid 1, Placement 1

HOPE dep. single: upper
[Tax] $\Delta K = +0.000315$
AOTC instrument
[Tax] $\Delta K = +9.39e-05$
AOTC ind. single: lower
[Tax] $\Delta K = +3.17e-05$
HOPE credit
[Tax] $\Delta K = +0.128$
HOPE ind. single: lower
[Tax] $\Delta K = +0.000196$
Texas CC
[Funding] $\Delta K = +0.248$



B. Temporary effect (b)

26 matched / 37; 11 no match
Both SF: Tax 4, Placement 1



C. Temporary effect (d)

18 matched / 37; 19 no match
Both SF: Tax 3

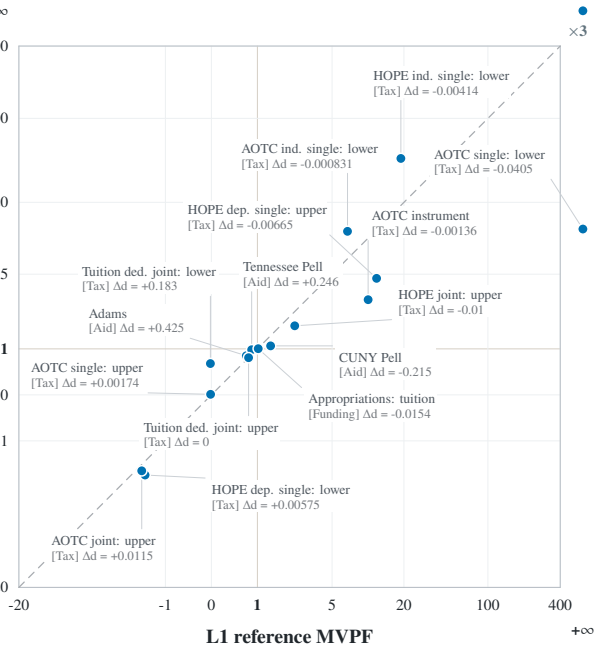
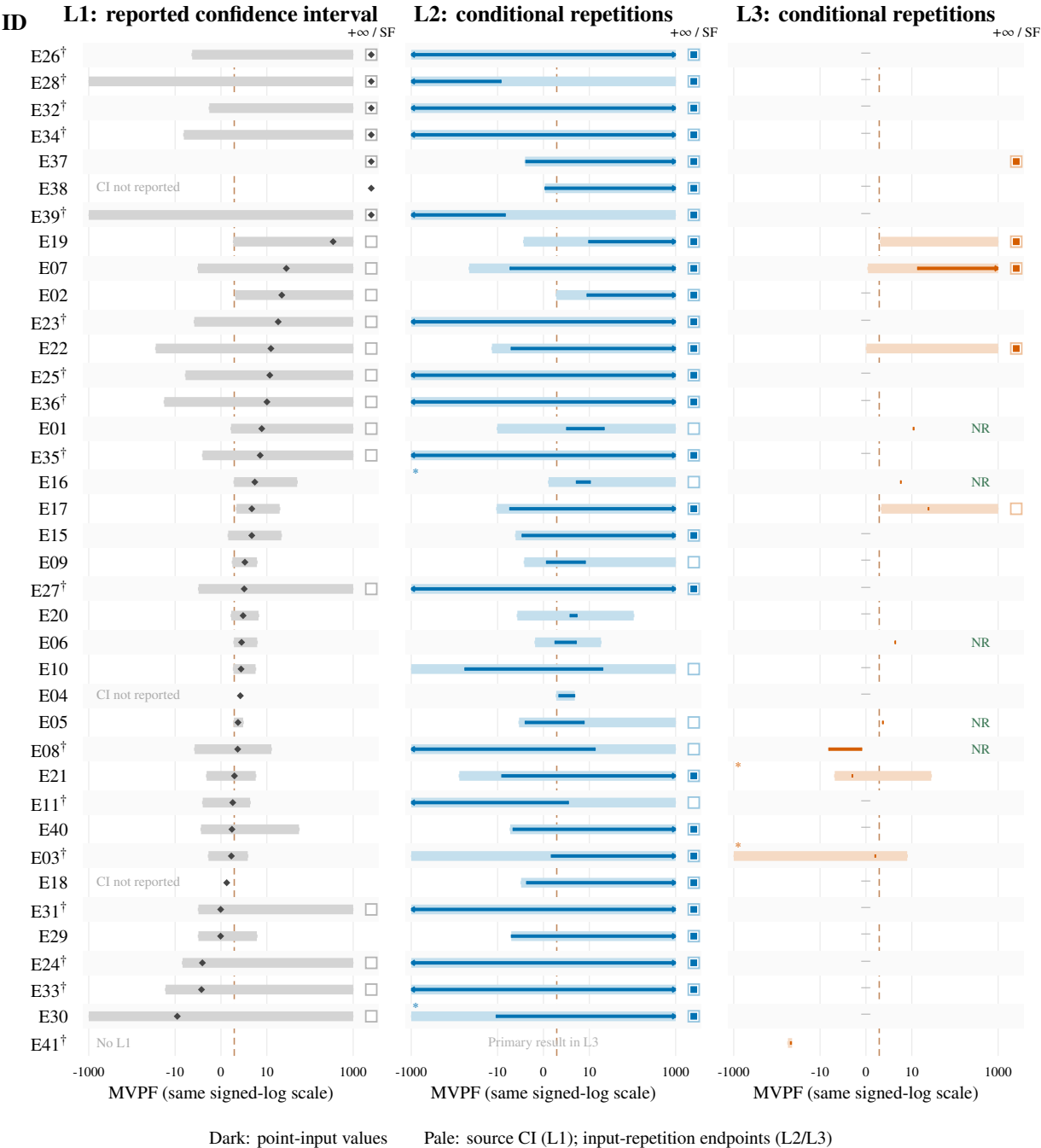


Figure 10. Reference and reconstructed MVPFs under permanent and temporary effects

Notes: Blue points match L2; red rings mark the same path also satisfying L3. Bracketed labels give the policy types defined in Figure 8. Finite axes have identical asinh scales and square dimensions. Outer markers indicate self-financing: $W > 0$, $N \leq 0$. Top markers preserve finite L1, right markers preserve finite L2, and corner counts identify coincident double-self-financing cases; their policy types appear below the panel counts. Labels give ΔK in schooling years for YS or percentage points for credentials (STBF jointly shifts time and credential inputs), proportional b changes in percent, and Δd in years. The other two parameter families stay fixed. Permanent effects change eventual attainment; temporary effects eventually vanish but can retain lifetime value. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds. Appendix B.5 gives the method and case-level details.

Figure 11. Point-input ranges, source intervals, and conditional input repetitions



Notes: Policy types follow Figure 8. Dark marks show fixed-input results; pale bands show reported L1 intervals or conditional L2/L3 endpoint summaries. Filled +∞ marks denote self-financing; open squares denote infinite summary endpoints. Arrows extend beyond the axis; the guide marks one. Daggers mark undefined ratios; asterisks mark incomplete draws; NR means no joint L2 draws. Method codes: L: registered source/accounting draws; R: constructed education perturbations; X: economic inputs with education fixed; P: synthetic correlated shocks.

A Literature Review and Descriptive Education Profiles

This appendix documents the two sources of descriptive evidence in Section 2: a review of when higher-education effects are measured and used in longer-run valuation, and ACS profiles of educational attainment across ages and birth cohorts. The literature review counts distinct papers; the policy applications elsewhere in the manuscript count policy–estimand calculations. For each reviewed paper, the accompanying records identify the source version, education outcomes, observation windows, relevant passages and exhibits, and inclusion decision. The review takes the original estimates as inputs and does not reassess each study’s causal identification.

A.1 Search sample and inclusion

The journal search covers 2000–2025 in the *American Economic Review*, *Quarterly Journal of Economics*, *Journal of Political Economy*, *Review of Economic Studies*, *Review of Economics and Statistics*, *AEJ: Applied Economics*, *AEJ: Economic Policy*, *Journal of Human Resources*, *Journal of Labor Economics*, *Journal of Public Economics*, *Economics of Education Review*, and *Education Finance and Policy*. The 674 journal-search records are the supplied, preliminarily screened higher-education records in the full-text manifest, not all articles published in those journals. Full-text inclusion and duplicate checks retain 447 for analysis. The register preserves each supplied record and its inclusion decision; the exercise is a documented review of this selected search sample, not a census of education research. Eleven additional studies underlying welfare calculations and one unpublished author manuscript on marijuana legalization bring the retained sample to 459 papers. After deduplication of the additions against the journal search, the combined register contains 686 records. The remaining 227 records include exclusions and duplicates.

Inclusion requires an original estimate relating a policy, intervention, or external exposure to a higher-education outcome observed within a finite window. The education estimate can be secondary to the paper’s main analysis. Studies reporting only test scores, aspirations, earnings, or high-school outcomes are excluded, as are purely descriptive comparisons, reviews, and editorials. The review imposes no fixed maximum follow-up length: an education outcome observed many years after exposure still provides finite-window evidence. Working and published versions of the same study count once, while distinct studies of one program can count separately. The resulting sample is selected. In particular, the additional welfare-source papers were included for their relevance to valuation, so their frequency cannot be interpreted as the prevalence of valuation in the wider literature.

A.2 Education Outcomes, Estimation Times, and Endpoints

The review distinguishes college entry (CE), associate-degree outcomes (AA), and bachelor’s-degree or higher outcomes (BA+). CE requires first entry or ever entering college; current enrollment or persistence alone does not qualify. AA preserves the source study’s distinction between an associate’s degree and at least an associate’s degree. The overall category also includes credits, completed schooling, graduate entry, and other qualifying education measures. A paper can contribute to several outcome groups, so the group counts cannot be added to obtain a total number of papers.

The timing count records distinct reported effect-estimation times or pooled windows within a comparable series. Effects reported at years 3 and 6 count as two times; effects reported separately at years 3, 4, 5, and 6 count as four. A single coefficient pooled over years 3–6 counts as one window. Alternative specifications,

treatment arms, and subgroups at the same horizon do not add times. The coding retains the time origin, reported horizons, selected series, and supporting table or figure. Independent samples or unmatched time origins are not combined into a single longitudinal sequence. The overall and outcome-specific counts can differ because they summarize different education evidence within the same paper.

Figure A2 and Table A1, Panel A, summarize these counts. All 459 papers have an overall count: 249 report one time or window and 210 report more than one. CE includes 237 eligible papers, of which 236 have a coded count; the remaining paper is excluded only from that panel's percentage denominator. AA has 40 coded papers and BA+ has 190. Among the BA+ papers, 132 report one time or window, or 69.5 percent. A single window can span several years, so these statistics describe the number of reported estimates over time rather than the length of follow-up or the duration of a policy effect.

An observation endpoint is recorded separately from the timing count and assigned to a defined clock. Attained age dates the person's education outcome. Individual follow-up measures elapsed time since an origin such as random assignment, high-school exit, or college entry. Policy or cohort event time dates an observation relative to a reform or exposed cohort and need not track the same people. These clocks are kept separate throughout. A six-year completion outcome is not assigned age 24 without evidence on the relevant entry age; academic terms are converted to years only when the source calendar supports that conversion.

Every paper receives one overall endpoint classification, reported in Table A1, Panel B. An *exact numerical endpoint* supplies a value on at least one defined clock (366 papers). An *interval or bound* restricts the endpoint without supplying a unique value (38). A *stage with unaligned timing* identifies an education stage or observation window without a common numerical endpoint (46). An *unresolved endpoint* has insufficient or conflicting information about the cutoff (9). All four categories remain in the literature sample. For example, an admission-stage effect can have one identifiable estimation window even when the paper does not report elapsed time between assignment and admission.

Figure A3 relates the number of reported times or windows to numerical endpoints for all education outcomes; Figure A4 presents the same comparison separately for CE, AA, and BA+. The overall panels contain 104 age pairs, 258 follow-up pairs, and 29 event-time pairs, representing 366 distinct papers. These counts overlap because a paper can supply more than one clock. The other 93 papers remain in the sample and in the endpoint classification, without an imputed coordinate. Bubble area is proportional to the number of papers at each coordinate. The displayed Spearman correlations are computed from the individual paper records, including ties, rather than treating each bubble as one observation. Correlations are shown only when a panel contains at least ten papers and both coordinates vary. The figures describe reporting patterns, not causal relationships or changes in treatment effects. In the overall panels, counts and endpoints can summarize different education outcomes within a paper; the outcome-specific panels narrow that comparison to the recorded education category.

A.3 Interpretation and Education-Based Lifetime Valuation

The review separates statements about finality from arguments about the adequacy of the observation window. A *finality or permanence statement* discusses whether an education difference is final or persistent, preserving the source authors' qualifications (18 papers). A *completion or window-adequacy argument only* explains why education is expected to be complete or follow-up sufficiently long, without explicitly asserting a permanent treatment effect (22 papers). The second category is assigned only when no qualifying finality statement is

recorded. The combined count of 40 therefore measures the presence of these arguments, not the number of studies establishing permanent effects.

Education-based lifetime or career valuation requires a quantitative projection from education evidence over a lifetime or working career. It includes projected earnings, fiscal effects, benefits relative to costs, and MVPF calculations; a complete welfare calculation is not required. Qualitative references to education returns, lifetime calculations unrelated to education estimates, and education costs totaled over a few years do not qualify. Thirty-one papers meet this definition. One also belongs to an interpretation category, so 70 papers satisfy at least one of these criteria and 389 have neither detected in the reviewed material (Table A2, Panel A). The latter classification does not imply that a paper contains no discussion of long-run outcomes.

Panel B of Table A2 links the 31 valuation papers to the education outcomes entering their calculations. This is a narrower question than the paper-level reporting counts above. It asks whether the valued education outcome is observed at one or several ages or individual follow-up times; it does not count earnings observations, different educational outcomes at different deadlines, or successive exposed cohorts as additional observations of one attainment path. When several outcomes enter a valuation, the classification uses the richest comparable series among those outcomes. A paper can therefore supply several inputs to a calculation while reporting only one window for each. For example, credits measured in one academic year and transfer measured in the next supply different education margins, whereas bachelor's completion reported at years 4, 5, and 6 supplies repeated observations of a degree-attainment margin. The table records available education information, not a claim that every reported time enters the authors' valuation formula.

Eighteen papers report one age, follow-up, or stage-specific window per valued education series; 12 report multiple outcome times. The remaining paper values annual institutional degree awards indexed by spending-shock time, which does not supply an individual completion deadline. Papers in the first group may still report policy-event dynamics across successive cohorts. Five of the six age-based papers supply an exact upper age, and 17 of the 20 follow-up papers supply an exact last deadline. Their endpoint spans are shown within each clock, without pooling ages and elapsed years or imputing missing cutoffs. The other follow-up records use a pooled 6–8-year window or upper bounds; the age record without an exact endpoint and the four stage-specific records remain in the table. A pooled adult-age window is not a lifetime education trajectory, and a multiple-time classification does not imply that final attainment is observed.

The accompanying 31-paper ledger records the valued outcome, reported times, time origin, endpoint qualifications, source version, and supporting pages or exhibits. It also preserves event-time information separately. These linked classifications supplement the 459-paper records rather than overwriting their broader definitions. They describe the selected valuation papers and should not be read as evidence that the original authors ignored dynamics or that their valuations are necessarily incorrect.

A.4 Source Versions and Replication Records

The review counts use recorded source versions, while the main text cites published articles where available. STBF record 628 uses the September 2020 working paper; the application uses Angrist et al. (2022), whose six-year degree results and graduation dynamics appear in Table III and Figure III. STBF applies the framework of Hendren and Sprung-Keyser (2020) but is not one of their earlier policy reconstructions. Version differences remain explicit in the evidence ledger.

For Adams, record 10 uses the May 2013 working paper and its roughly 30-percentage-point, \$300,000 illustration for marginal college switchers. The published Cohodes and Goodman (2014) use about 27

percentage points and \$270,000. Neither illustration is the 2.5-percentage-point eligibility effect or the separate policy-average MVPF in Hendren and Sprung-Keyser (2020). Tennessee record 684 uses the May 2026 version of Schoonover et al. (2026); the main BA-or-higher outcome at ages 23–24 is distinct from exact BA and the alternative ages 22–24 specification. Appendix D documents these welfare accounts, their inputs, and their authors’ discussion of dynamics and extrapolation.

Restricting the review to the journal sample leaves 130 one-window papers among 186 coded BA+ papers, or 69.9 percent, compared with 132 of 190 papers, or 69.5 percent, in the combined sample. Education-based lifetime or career valuation appears in 28 of 447 journal papers and 31 of 459 papers overall. These comparisons show how the reported counts change when the additional sources are excluded. The replication files contain the paper register, timing and interpretation decisions, endpoint classifications, evidence locations, source versions, and coordinates underlying the figures. The exhibit code aggregates those recorded judgments without recoding the papers. Bubble plots use the September 21 endpoint review, which revisited the 120 previously unplotted papers and retained the earlier reviewed coordinates for the others.

A.5 ACS sample and construction of the age profiles

Figure 1 uses ACS microdata selected within survey years 2000–2022, birth cohorts 1985–1987, and ages 16–35. Cohort is indexed by survey year minus reported integer age. The selected observations span 2001–2022, and every cohort contributes all 20 ages: 1985 over 2001–2020, 1986 over 2002–2021, and 1987 over 2003–2022. The resulting sample contains 1,978,860 respondent records from all 50 states and the District of Columbia. There are no restrictions on birthplace, race, Hispanic origin, sex, or marijuana policy status. Positive person weights and valid detailed educational attainment (EDUCD) are required; neither condition excludes a record within this selection. All 60 cohort–age cells are observed.

Person weights are applied within cohort–age cells, and the three cohorts receive equal weights at every age. Let S_{ca} be the respondents in cohort c at age a , w_i the IPUMS person weight, and H_{ik} respondent i ’s value of education measure k . For a group g containing three birth cohorts C_g , the profile is

$$\bar{H}_{gk}(a) = \frac{1}{3} \sum_{c \in C_g} \frac{\sum_{i \in S_{ca}} w_i H_{ik}}{\sum_{i \in S_{ca}} w_i}. \quad (16)$$

For Figure 1, $C_g = \{1985, 1986, 1987\}$. Fixed cohort weights prevent changes in the mixture of birth cohorts across ages. The underlying ACS observations are repeated cross sections, so the profiles need not hold immigration, survival, survey coverage, or other within-cohort composition constant. They describe the education of residents observed at each age, including people born outside the United States.

In the ACS profiles, SC+ is some-college-or-higher attainment, used as a proxy for college attendance; it is not current enrollment or a directly observed date of first entry. AA+ and BA+ denote cumulative degree thresholds. The EDUCD crosswalk assigns 12 schooling years to high-school completion, 12.5 to less than one year of college, 13 to at least one year of college without a degree, 14 to an associate’s degree, 16 to a bachelor’s degree, 18 to a master’s degree, 19 to a professional degree, and 20 to a doctorate. Lower-grade categories use the reported grade or the documented midpoint. YS is therefore an approximation based on attainment categories. The replication files provide the crosswalk, cohort–age means, respondent counts, weight sums, survey years, and source information. Using a fitted ACS profile as an untreated potential-outcome path in a policy application requires a separate calibration assumption.

A.6 Education growth after age 25 across cohorts

Figure A1 compares four nonoverlapping groups of birth cohorts: 1976–1978, 1979–1981, 1982–1984, and 1985–1987. The comparison uses ages 25–35, which are observed for all twelve cohorts within the 2000–2022 survey window. An earlier cohort cannot generally be observed at every age from 16 to 35 in these data: complete coverage of that interval would require a birth year between 1984 and 1987. The common 25–35 interval therefore permits an observed comparison of education growth after age 25 without filling in missing earlier ages. The sample contains 4,516,181 respondent records in 132 cohort–age cells and uses the same population definition, education crosswalk, person weights, and fixed one-third cohort weights as Figure 1.

Each curve shows the increase relative to its cohort group’s age-25 level. Degree and attendance changes are displayed in percentage points; schooling changes are displayed in years. This descriptive age difference is distinct from the policy effect ΔH used elsewhere in the paper. The accompanying cohort-growth data retain all levels at ages 25 and 35 and their differences.

Across the four groups, the age-25-to-35 increase in BA+ rises from 6.22 to 11.46 percentage points. The corresponding AA+ increase rises from 7.77 to 12.99 points, and the increase in approximate schooling rises from 0.447 to 0.669 years. All three increments increase successively across these groups. The college-attendance proxy follows a different pattern: its increments are 5.72, 7.28, 6.09, and 5.17 percentage points. Thus the pattern concerns later degree attainment and schooling accumulation, rather than a uniform increase across every education measure. These level differences do not measure the fraction of individual degree holders who completed after age 25, and the comparison does not isolate a causal effect of birth year or economic conditions.

B Education-Path Framework and Simulation

The Section 3 exercise asks which treated years-of-schooling paths remain compatible with stipulated age-specific effects and how their education-channel values differ under a common rule. It is a deterministic compatibility calculation. This appendix defines the observations and path restrictions, derives the matching and valuation formulas, and reports the experimental design, numerical checks, and complete results. The comparison path is the equal-cohort mean of the three fitted ACS profiles documented in Appendix C. Averaging the three cohort parameters does not make that comparison path itself a single logistic curve.

B.1 Schooling Stocks, Flows, and Observations

For a specified cumulative education outcome, $H_c^j(a)$ is its population mean at age a under policy state j . It is a share for a binary attainment outcome and is measured in schooling years for the years-of-schooling simulation. Its first difference records acquisition between adjacent ages. The effect $\Delta H_c(a)$ is the difference between the two stocks, whereas $\Delta h_c(a)$ is the difference between their acquisition flows, as defined in Equation (1). Even if both levels are nondecreasing, either effect can have either sign. For example, a treated group can acquire more schooling early and less later while retaining a positive stock difference throughout. Figure 2 illustrates these definitions with hypothetical degree-attainment shares. Both level paths increase, but the gap falls between ages 24 and 25, so $\Delta h_c(25) < 0$. These curves explain the notation and are not estimated policy effects or the ACS calibration. In the numerical schooling-year simulation, the first possible

effect age is $a_0 = 18$: the model sets $\Delta H_c(a) = 0$ before age 18 but allows $\Delta H_c(18)$ to differ from zero. The first flow is consequently $\Delta h_c(18) = \Delta H_c(18)$, relative to the zero gap at age 17.

The observation operator must retain what the source actually measures. An effect on attainment at a specified age enters as

$$\mathcal{M}_{a_*}(\Delta H_c) = \Delta H_c(a_*). \quad (17)$$

A population-weighted window average instead enters as

$$\mathcal{M}_{\mathcal{A}}(\Delta H_c) = \sum_{a \in \mathcal{A}} w_a \Delta H_c(a), \quad (18)$$

where the source's weights sum to one over $\mathcal{A}$. An effect on acquisition between two ages restricts the change in the stock gap:

$$\mathcal{M}_{a_1, a_2}^{\text{flow}}(\Delta H_c) = \Delta H_c(a_2) - \Delta H_c(a_1). \quad (19)$$

Observing two stock effects separately imposes two constraints, unlike observing only their difference. A completion rate within six years of entry is not automatically a stock measured at age 24: an attained-age mapping requires information on entry ages and the outcome definition. The experiments below use exact stock effects at one or two attained ages and require no such age bridge. They hold the target population fixed across those observations. For each compatible candidate I retain the terminal effect $\Delta H_c(65)$, the annual sum $\sum_{a=18}^{65} \Delta H_c(a)$, and the education NPV in Equation (30). The annual sum has schooling-year–age-year units and is distinct from additional completed schooling at the terminal age.

B.2 Logistic Parameters and Feasible Scenario Intervals

Write $p = (K, b, d) = (K_c + \Delta K_c, b_c + \Delta b_c, d_c + \Delta d_c)$ for the total treated-path parameters, and let $S = H_c^0(\tau_c)$. Define $L(a) = \Lambda[b(a - d)]$ and $F(a) = [L(a) - L(\tau_c)]/[1 - L(\tau_c)]$. The continuation in Equation (2) is $H_c^1(a) = S + (K - S)F(a)$ after τ_c . Holding the anchor fixed, its plateau derivative is $\partial H_c^1(a)/\partial K = F(a)$. For either shape parameter $v \in \{b, d\}$, the derivative is

$$\frac{\partial H_c^1(a)}{\partial v} = (K - S) \frac{[1 - L(\tau_c)] \partial_v L(a) - [1 - L(a)] \partial_v L(\tau_c)}{[1 - L(\tau_c)]^2}. \quad (20)$$

Here $\partial_b L(a) = (a - d)L(a)[1 - L(a)]$ and $\partial_d L(a) = -bL(a)[1 - L(a)]$. In particular, $\partial_d H_c^1(a) = -(K - S)b[1 - L(a)]F(a) < 0$ after the anchor when $K > S$ and $b > 0$. Figure B1 varies each total parameter with a common starting level. K changes the eventual plateau and scales the remaining acquisition $K - S$; b changes its speed; and d changes its timing subject to the common start. A nonzero K change thus produces a permanent education effect, whereas changes in b or d alone produce temporary effects with the same eventual attainment. These labels describe the generating path, not every candidate compatible with its observed effect. The negative derivative with respect to d means that a higher d postpones attainment while preserving the anchor and eventual plateau. Because normalization is affine in $L(a)$, the post-anchor continuation has its inflection and maximum instantaneous acquisition rate at $a = d$ when $d > \tau_c$. If $d \leq \tau_c$, that peak lies outside the continuation domain, and its acquisition rate declines after the anchor. The reference value $d = 12.564854$ is below the simulation's age-17 anchor, so it is not a post-policy age of peak acquisition. None of these parameters is an individual time to degree.

An equivalent interpretation uses the age at which the path reaches a specified attainment level. Let a_q satisfy $H_c^1(a_q) = S + q(K - S)$, where $q \in (0, 1)$ is the fraction of remaining attainment accumulated after the anchor. The normalized continuation implies

$$a_q = \tau_c + \frac{1}{b} \log \left(\frac{1 + q \exp[b(d - \tau_c)]}{1 - q} \right). \quad (21)$$

Its derivative with respect to d is $\partial a_q / \partial d = q \exp[b(d - \tau_c)] / \{1 + q \exp[b(d - \tau_c)]\}$, which lies strictly between zero and one. Holding K , b , and the anchor fixed, a higher d therefore increases the age required to reach each such level. The delay depends on q and is smaller than the change in d ; it is not a uniform horizontal shift. Here q refers to progress along the population attainment path, not the fraction of individuals who graduate. The age at which half of the remaining attainment has accumulated is $a_{1/2} = \tau_c + b^{-1} \log\{2 + \exp[b(d - \tau_c)]\}$, which generally differs from d .

The anchor implements an assumption about exposure in a fixed population. In the age-based experiment, individuals are unaffected through age 17, and a policy may first alter their measured attainment at age 18. The post-policy continuation therefore passes through the comparison level at 17, rather than attaching a curve whose value already differs there. This eliminates a mechanical level discontinuity at the transition; it does not require a zero effect at the first affected annual observation or equal derivatives at the anchor. In applications, eligibility, announcement, and exposure timing determine which age can serve as the last unaffected observation. An earlier anticipatory response would invalidate a later anchor. Equality refers to two potential outcomes for the same target population, not to equality of raw means in two selected samples. A common anchor for people first exposed at different ages requires a genuinely unaffected common age or an explicit cohort-specific construction before aggregation.

Each candidate is constrained to $0 \leq H_c^1(a) \leq 25$ schooling years at every annual age from 18 through 72. Over the same ages, its annual acquisition satisfies

$$0 \leq H_c^1(a) - H_c^1(a - 1) \leq 1. \quad (22)$$

The common-anchor construction imposes $H_c^1(a) = H_c^0(a)$ through age 17, so the first increment is the increase from that shared level along the normalized continuation. The one-year cap is a maintained bound on annual acquisition in the harmonized schooling measure, not a threshold estimated from causal policy responses. It restricts the difference between adjacent annual schooling levels, including the first post-anchor transition; it is not a bound on the instantaneous derivative or on the schooling treatment effect. The fixed comparison curve also satisfies this annual bound. Nondecreasing attainment restricts the levels and does not require a nondecreasing treatment effect. The parameter domain ensures $K > S$ and $b > 0$. Membership in this class establishes the stated mathematical restrictions, while the plausibility of its exposure and schooling assumptions remains a separate economic question.

For each coordinate $j \in \{K, b, d\}$, I fix the other two at their reference values and intersect its domain with all path restrictions. The reference triple p^0 is the mean of the fitted cohort parameters documented in Appendix C. The conditional feasible intervals are $I_K = [13.337015, 15.139521]$ for the plateau, $I_b = [0.120000, 0.401078]$ for speed, and $I_d = [8.000000, 30.000000]$ for timing. A scenario at position u in $I_j = [L_j, U_j]$ sets $p_j = L_j + u(U_j - L_j)$, leaving the other two parameters at p^0 . The midpoint scenarios use $u = 1/2$ and increments 0.3160 schooling years, 0.0030 inverse age-years, and 6.4351 age-years. The timing midpoint consequently represents later accumulation. The annual-acquisition cap binds at the upper end of the speed interval, whose midpoint lies close to the reference speed and therefore implies a

small education effect. This rule selects locations within the conditional feasible intervals without equalizing observed effects or selecting on NPV. The grid is a design for examining the maintained model, not an estimate of a typical policy response.

B.3 Exact Matching and Numerical Search

I use the constrained minimum-distance rule in equation (15). For a simulated cohort, the same criterion can be written directly as

$$\widehat{\Theta}_c(\mathbf{m}, a_0) = \arg \min_{\theta_c \in \Theta_c} [\mathbf{m} - \mathbf{g}_c(\theta_c; a_0)]' W [\mathbf{m} - \mathbf{g}_c(\theta_c; a_0)] \quad (23)$$

where $\mathbf{g}_c$ predicts the retained moments and Θ_c imposes the maintained path restrictions. W is a fixed positive-definite matrix. For the exact simulation inputs, the minimum is zero and every such matrix gives the same zero-residual set. With three path parameters and one or two post-anchor effects, the minimizer need not be unique. The simulation solves these equations directly and retains verified exact matches. The eight overidentified policy L3 cases instead report constrained best fits and their nonzero residuals. This criterion is not a claim of covariance-weighted efficient GMM. The anchor is built into the candidate path and is not another observed policy effect.

Let $S = H_c^0(\tau_c)$, let (K, b, d) denote total treated-path parameters, and define $F(a; b, d) = [\Lambda\{b(a-d)\} - \Lambda\{b(\tau_c - d)\}] / [1 - \Lambda\{b(\tau_c - d)\}]$. For one effect at $a_* > \tau_c$, the implied treated level is $y = H_c^0(a_*) + m_{a_*}$. Conditional on (b, d) , exact matching gives

$$K = S + \frac{y - S}{F(a_*; b, d)}. \quad (24)$$

I evaluate a 401×401 grid in (b, d) , recover K , and retain candidates satisfying the level, increment, and parameter restrictions. For each output metric, local constrained optimization then searches for smaller and larger values from up to two separated feasible starting candidates, using scaled coordinates, at most 100 iterations, and an objective tolerance of 10^{-11} . The observed moment remains imposed algebraically.

For two effects at $\tau_c < a_1 < a_2$, let $y_j = H_c^0(a_j) + m_{a_j}$, $z_j = y_j - S$, $R = z_2/z_1$, and $t_j = \exp[-b(a_j - \tau_c)]$. Since $F(a_j; b, d) = (1 - t_j)/(1 + qt_j)$ for $q = \exp[b(d - \tau_c)]$, define $r = R(1 - t_1)/(1 - t_2)$. Solving the two matching equations gives $q = (r - 1)/(t_1 - rt_2)$. The implied timing parameter is

$$d = \tau_c + \frac{\log q}{b}, \quad (25)$$

and the plateau is

$$K = S + z_1 \frac{1 + qt_1}{1 - t_1}. \quad (26)$$

The search is one-dimensional in b . I use 1,601 grid values, refine detected feasible-interval boundaries by root finding, and refine candidate metric extrema locally within feasible intervals. A candidate must have $q > 0$, finite parameters, and satisfy all maintained restrictions. The registered generating paths have $0 < z_1 < z_2$, so the formulas are not used for degenerate observations at the anchor or with equal treated levels. I include the stipulated generating path in each library to verify nonemptiness. This is a deterministic compatibility exercise, not an estimator-performance experiment.

A candidate is accepted only if its largest moment residual is at most 10^{-8} schooling years and every path or parameter restriction holds to tolerance 10^{-9} . If a local optimizer stops just outside a boundary, I move

its proposal a small fraction toward its verified starting point, recover K from the exact matching equation, and check the conditions again; the feasibility tolerance is not relaxed. Independently feasible outputs are retained even when the optimizer does not report convergence, because these outputs supply candidate paths rather than certified extrema. Libraries are deduplicated using parameter values rounded to 12 decimal places. Every verified two-age candidate is also supplied to both component single-age libraries, after which their metric endpoints and pointwise path envelopes are recomputed. This preserves numerical nesting without treating the finite search as a proof of completeness.

B.4 A Unified Net-Earnings Response

The simulation uses annual comparison earnings $Y_0(a)$, income ages 18–65, and discount factors $D(a) = 1.03^{-(a-18)}$. Amounts are in 2000 dollars. With $m = \Delta H(25)$ and the retained HSK signed return schedule, the single-effect benchmark is

$$E^{\text{point}} = m \sum_{a=18}^{65} D(a) Y_0(a) \left[r^- \mathbf{1}\{18 \leq a < 25\} + r^+ \mathbf{1}\{a \geq 25\} \right]. \quad (27)$$

The full-precision coefficients are $r^- = -0.0667064939863901$ and $r^+ = 815/7241 = 0.1125535147$. They retain the benchmark conversion of early and later earnings effects associated with the education estimate (Zimmerman, 2014; Hendren and Sprung-Keyser, 2020). This single-batch representation is an equivalent accounting construction, not a claim that the source observes a universal graduation age.

For an arbitrary attainment gap, define $q(\tau) = \Delta H(\tau) - \Delta H(\tau - 1)$ and retain the common pre-policy value $\Delta H(17) = 0$. The linear extension is

$$R(a, \tau) = r^- \mathbf{1}\{\tau - 7 \leq a < \tau\} + r^+ \mathbf{1}\{a \geq \tau\}, \quad (28)$$

$$\Delta y(a) = Y_0(a) \sum_{\tau=18}^{72} R(a, \tau) q(\tau), \quad 18 \leq a \leq 65, \quad (29)$$

$$E^{\text{path}} = \sum_{a=18}^{65} D(a) \Delta y(a). \quad (30)$$

The policy-onset floor truncates pre-completion responses before age 18. Acquisition after age 65 can affect earnings before 65 through the seven-year window, which is why the education support extends to 72. No earnings after 65 are valued. The annual acquisition cap is imposed on the full modeled education path; the reported terminal gap remains $\Delta H(65)$.

Summing the increments gives the equivalent expression

$$\Delta y(a) = Y_0(a) \left[r^+ \Delta H(a) + r^- \{\Delta H(a + 7) - \Delta H(a)\} \right]. \quad (31)$$

The program checks present values both through the increment sum and through this stock expression. A step of size m at 25 recovers Equation (27) exactly. For two steps at 23 and 25, the response is the sum of their respective shifted templates. Signed increments also admit delayed acquisition and catch-up: a negative increment is a difference between population acquisition flows, not a negative graduation probability. A seven-year negative-response window per schooling-equivalent increment is an extrapolation template, not seven additional years physically spent in school.

The continuous-age analogue is the signed Stieltjes integral

$$\Delta y(a) = Y_0(a) \int R(a, \tau) d\Delta H(\tau). \quad (32)$$

When ΔH is differentiable, $d\Delta H(\tau) = \Delta H'(\tau)d\tau$; when it has jumps, the same expression includes the corresponding discrete batches. The calculations use annual sums, not numerical estimates of a continuous-time structural model. The finite path family does not identify the kernel from education observations. Its use assumes linear transport across acquisition ages and maintains the earnings profile, response coefficients, and policy-onset convention.

Both early losses and later gains enter one net-earnings curve. No additional contemporaneous schooling-flow charge or separately calibrated cost-share parameter enters the current specification. In particular, the method does not impose equality between completed-schooling increments and extra enrollment time. Tuition, public education resources, transfers, and fiscal responses remain separate welfare-account items, because they are not all contained in earnings.

Figure 4 evaluates the joint-change illustration $(\Delta K, \Delta b, \Delta d) = (0.2625, 0.0300, -0.4500)$ using the same template for the single-batch benchmark and the full path. Its maximum annual schooling increment is 0.766 years. The two NPVs are \$15,390 and \$13,986. The curves display discounted net earnings at each age, so their annual sums give those NPVs directly. Their positive and negative portions need not identify separate labor-supply or wage-return mechanisms.

The policy collection shares this outer net-income accounting, while retaining source-specific inner mappings. HSK cases retain their source response phases and monetary conventions; STBF retains its nonlinear credential–experience model; Early Head Start and Tennessee use explicitly identified public-data reconstructions when the author inputs cannot be recovered. Appendix D documents these distinctions and source-reproduction checks.

B.5 Single-Parameter Explanations of Policy Education Inputs

The policy exercise asks whether an observed education input can be matched by a permanent effect through K alone or a temporary effect through b or d alone. It differs from the deterministic experiments in Section 3: the policy input is held fixed, and the active parameter is solved for rather than selected to generate a new input. Each calculation retains its geographic and cohort calibration, observation operator, policy unit, and valuation account. A weighted stock input and a difference between two schooling stocks are both admissible inputs to this exercise. Neither is replaced by an effect at a convenient single age. The eleven finite tax calculations retain their policy-dose conversion and acquisition-window mapping.

I preserve every fitted component of the comparison path. Let j index those components within policy i , with fixed weights ω_{ij} and baseline parameters $z_{ij}^0 = (K_{ij}^0, \log b_{ij}^0, d_{ij}^0)$. For parameter family $q \in \{K, b, d\}$, the experiment uses

$$z_{ij}^{1,q}(t) = z_{ij}^0 + te_q, \quad (33)$$

where e_q selects the corresponding coordinate. A plateau or timing experiment adds the same t to every component's K or d . A speed experiment multiplies each component's b by $\exp(t)$, so its labels report $100[\exp(t) - 1]$ percent. The other two parameter families, component weights, and pre-policy anchors remain fixed. Applying the anchored continuation component by component and then averaging reproduces H_i^0 exactly at $t = 0$. This construction preserves baseline heterogeneity. A nonzero K shift changes the

limiting attainment level and is therefore permanent. Changing b or d with each component's K fixed leaves the limiting aggregate gap at zero and is therefore temporary, although the gap can persist beyond the valuation horizon. It is a separate diagnostic and need not be a coordinate slice of the main search's common-parameter specification.

Let $\Delta H_i^q(\cdot; t)$ denote the resulting aggregate education effect. The matching calculation retains values in

$$\mathcal{T}_{iq} = \left\{ t : H_i^{1,q}(\cdot; t) \in C_i, \quad \mathcal{M}_i\{\Delta H_i^q(\cdot; t)\} = \mathbf{m}_i^{L2} \right\}. \quad (34)$$

Here C_i includes the level and annual-acquisition restrictions, parameter bounds, timing support, and calibrated covariance ellipsoid. The ellipsoid is applied to the weighted parameter center in $(K, \log b, d)$ coordinates. Holding two coordinates fixed can substantially shorten the remaining interval relative to that coordinate's marginal bounds. The numerical routine scans this conditional interval, brackets sign changes, and checks local minima of the absolute matching residual for tangencies. Reported solutions have input residuals below 10^{-8} in the input's units. No input standard errors are added to obtain a match. Missing solutions remain missing rather than being assigned zero or an unrestricted parameter value.

STBF has four jointly matched inputs: time enrolled, any college, AA+, and BA+. Its active family permits one shift for each outcome, subject to credential nesting and feasibility of the recovered subgroup mean inputs. These shifts have different units and are not summed. Only the plateau family yields a verified joint match. Tennessee Promise has three education inputs. Its family experiment permits a separate change in the active parameter for SC+, AA+, and BA+, holding the other two parameters of each credential path fixed. The three latest-window effects must be matched jointly, and the paths must satisfy credential ordering and the late-completion restrictions. Thus, its plateau point represents three credential-specific plateau changes, reported in percentage points, rather than one schooling-year shift. Only this family has a jointly admissible solution. Red rings in the policy figures indicate that an already matched L2 solution also satisfies the admitted additional L3 restrictions. They do not represent a second optimization, an additional time-allocation assumption, or a confidence region.

Figure B3 compares the reference earnings NPV with the earnings NPV of each matching path. Each saved path value is already expressed in its current account units. A paired dollar comparison is made only when the monetary and discount-base units can be reconciled with L1. Both coordinates are then divided by the same positive reference program expenditure. This normalization permits a comparison within each policy without mixing its currency years or policy doses. The reference earnings component is recovered from the original account; it is not inferred from the MVPF ratio. Algorithmic placement has no source lifetime-earnings projection and is excluded from this comparison, rather than assigned a reference NPV of zero. The source monetary years for Early Head Start and Tennessee cannot be established sufficiently for a matched dollar comparison, so their reconstructed amounts are not plotted against L1 dollars. Their dimensionless MVPF comparisons remain available. Cannabis has no whole-policy L1. The paired samples contain 31, 25, and 18 calculations for K , b , and d , respectively. Table B3 reports the normalized values and parameter changes for all 37 reference calculations, including unpaired cases. A self-financing welfare account can still have a finite earnings NPV and remains in these plots.

The income comparisons retain each application's valuation rule, including its signed early-income response or author income function. They therefore differ from the common synthetic-cohort calibration used in the Section 3 simulations. Differences from a source earnings projection may reflect both the education continuation and the valuation mapping. For example, the Adams reference earnings loss is about \$1,204 in the account's 2004-dollar units. Matching its converted -0.018 schooling-year input with plateau, speed, or

timing changes gives losses of about \$1,691, \$416, and \$355. The latter two temporary-effect paths converge to the comparison plateau, reducing the duration of the earnings loss; the permanent-effect path retains a lower plateau. This comparison does not establish which path occurred or overturn the source's evidence of dropout during follow-up.

The matched samples also differ across parameter families. In the separate subset of 14 calculations with one YS level input, the plateau, speed, and timing exercises yield 11, 7, and 3 matches. This subset allows cohort-age weights but excludes acquisition contrasts and joint credentials. For Florida, the required input is 0.1819 schooling years, whereas the saved speed and timing scans reach at most 0.0914 and 0.0089 years. The absent points therefore reflect the input and the conditional support. They do not show that timing responses are impossible outside this specification. The full parameter sweeps, matching results, exclusions, earnings-account bridges, and policy classifications are retained in the replication files.

B.6 Two Experiments and Summary Measures

The first experiment changes the generating process while fixing the observed age at 25. For each parameter j , I use $u = 0, 0.01, \dots, 1$ in the scenario rule from Appendix B.2 and additionally retain its reference value, yielding 102 registered scenarios per parameter and 306 rows. The two parameters not varied in a generating scenario remain at their reference values. I then discard that knowledge of the generating mechanism and search the common three-parameter class for paths matching the scenario's age-25 effect. The three reference rows describe the same generating path and are retained within each mechanism's reporting grid; they are not three independent pieces of evidence. The complete finite grids remain in the replication results; they do not exhaust the continuum of real-valued parameters.

The second experiment holds a generating process fixed and changes the observed ages. It uses five interval positions, $u \in \{0, 0.25, 0.5, 0.75, 1\}$, for each parameter, giving the 15 scenarios in Table B1. For each scenario the design set contains all 18 single ages from 18 through 35 and all $\binom{18}{2} = 153$ distinct age pairs, producing $15 \times 171 = 2,565$ scenario–design comparisons. The age-25 results for these 15 scenarios are shared with the first experiment, leaving 2,856 distinct scenario–design identifiers across the two output files. The three midpoint scenarios in Figure 3 are unchanged between Panels A and B. A fixed generating process implies age-specific effects that generally differ across ages; it does not mean that the same numerical treatment effect is imposed at every age. Every verified pair candidate is also inserted into both component single-age libraries. This ensures that their numerical summaries respect the nesting implied by the observation restrictions.

For a scenario s and observation design D , let $\widehat{\mathcal{P}}_s(D)$ denote its finite library of verified candidates. For a candidate treated path h , write $\text{NPV}_c^{\text{path}}(h - H_c^0)$ for the value of its associated education-effect path under Equation (30), holding a_0 , A , and ρ fixed. Its smallest NPV is $\widehat{\text{NPV}}_s^-(D) = \min_{h \in \widehat{\mathcal{P}}_s(D)} \text{NPV}_c^{\text{path}}(h - H_c^0)$, and its largest is $\widehat{\text{NPV}}_s^+(D) = \max_{h \in \widehat{\mathcal{P}}_s(D)} \text{NPV}_c^{\text{path}}(h - H_c^0)$. The reported width is

$$W_s(D) = \widehat{\text{NPV}}_s^+(D) - \widehat{\text{NPV}}_s^-(D). \quad (35)$$

For comparisons with observing age 25 alone, the proportional width reduction is $1 - W_s(D)/W_s(\{25\})$, calculated within each scenario before summarizing. A zero benchmark width would make that proportional measure undefined rather than justify recording a zero reduction. Median widths and median proportional reductions use the three interior positions, $u = 0.25, 0.5, 0.75$. Largest widths and sign counts use all five positions, keeping the two boundary cases visible. A pair with a positive candidate above \$0.01 and a negative

candidate below $-\$0.01$ is recorded as admitting both NPV signs. The cents-level threshold avoids treating numerical zero as a sign reversal. These are equal-weight summaries of the experimental scenarios, not estimates of the frequency of mechanisms among real policies.

B.7 Numerical Checks and Interpretation

All 306 first-experiment rows and 2,565 second-experiment rows were completed. The largest observed-moment residual among retained candidates was 3.55×10^{-15} schooling years, and the most negative recorded constraint margin was -9.99×10^{-10} , within the declared numerical tolerance. The audit checked 2,295 pair designs for nesting in their component single-age libraries; the largest recorded containment discrepancy was 1.58×10^{-10} in the numerical metric comparisons. An independent calculation from the stored endpoint parameters recomputed 17,136 metric-endpoint records using the normalized logistic difference directly. Its largest NPV difference was 2.51×10^{-10} dollars, and its largest terminal-schooling and cumulative-gap differences were 1.78×10^{-14} schooling years and 1.81×10^{-13} schooling-year-age units, respectively. Some endpoint records correspond to the same path because multiple metrics or designs can select it.

The local optimizer made 6,708 recorded constrained-refinement calls: 4,586 converged and passed feasibility checks, 1,753 did not report convergence but supplied independently verified feasible candidates, and 369 outputs were rejected. Boundary repair toward a verified seed, as described in Appendix B.3, was used for 1,102 accepted outputs without relaxing the tolerances. Feasible outputs contribute witnesses and candidate extrema regardless of their stopping flag; they are not certified optima. A pilot comparison of the 201×201 and 401×401 single-age grids, together with 801 and 1,601 pair-grid values, changed the checked NPV endpoints by at most about $\$0.098$. That pilot supports local numerical stability in the designs checked, not completeness of the continuous search. No sampling bootstrap or Monte Carlo coverage calculation is part of these two experiments.

The numerical checks establish that saved candidates meet the maintained restrictions, not that those restrictions hold for real policies. ACS fit is informative within observed ages, while extrapolation above age 35, population transport, the logistic family, empirical support, and the flow-to-cost mapping remain assumptions. Finding opposite-sign values establishes conditional ambiguity; finding only one sign does not certify sign identification. Finite-bank envelopes can switch candidate identity across ages, so their edges need not be feasible paths. No probabilities or confidence levels are attached to them. Alternative families, calibrations, and noisy observations require separate analyses.

B.8 Results across the Complete Experimental Grids

Table B2 summarizes all 102 age-25 scenarios per parameter. The complete 306-row file retains the generating parameter, observed effect, terminal schooling and cumulative-gap ranges, NPV endpoints, and generating-path NPV. Endpoints for different metrics can belong to different paths. Median NPV widths are $\$36,725$, $\$43,977$, and $\$52,775$ for the plateau, speed, and timing scans; 53, 83, and 102 scenarios, respectively, admit both signs. These counts summarize the chosen grid and assign no probabilities to policy mechanisms.

Table 2 reports selected observation designs, and Figure B4 covers every single age and age pair. Median widths use the three interior generating scenarios; maximum widths use all five, preserving the most permissive boundary or interior result. Diagonal cells represent one age and upper-triangle cells two distinct ages. All

panels share one dollar-width scale. The five scenarios' individual heatmaps and all 2,565 scenario–design rows remain in the replication material, allowing the aggregate patterns to be checked against every generating path.

The two panels of Figure 3 use the same midpoint scenarios and common axes, so their visual difference comes from the additional age-35 constraint and the accompanying candidate search. For these three scenarios the NPV-width reductions are 80.4–86.7 percent. The candidates found for the plateau example have positive NPVs, those for the timing example have negative NPVs, and those for the speed example still span both signs. The speed midpoint is close to the reference speed within its cap-restricted feasible interval. These midpoint results do not describe every registered scenario: the additional age-35 observation admits both signs in one of the five speed scenarios and one of the five timing scenarios. Other age pairs need not produce the same reduction. Within the registered interior scenarios, adding age 24 to age 25 gives larger median proportional reductions than adding age 20 in each parameter group, despite the smaller age separation. The result concerns where observations constrain this restricted curve family, and should not be generalized to an optimal sampling design under noisy estimates without a separate analysis.

C Comparison Schooling Paths and Empirical Support

C.1 Construction of the Comparison Curve

For the schooling-year exercise in Section 3, I hold fixed a comparison path formed from three ACS birth-cohort fits. Let $\hat{H}_{0q}(a)$ be the fitted mean completed schooling profile for birth cohort q . The reference path is

$$H_c^0(a) = \frac{1}{3} \sum_{q=1985}^{1987} \hat{H}_{0q}(a). \quad (36)$$

The subscript c in the simulation therefore refers to this synthetic target group, not to one of the three birth cohorts. The ACS figure in Figure 1 shows descriptive attainment at observed ages; Equation (36) instead averages fitted cohort curves and is used as a fixed input to the path search. Neither object is an estimated policy counterfactual for a specific case in Section 4.

The 60 person-weighted cohort–age means and their population weights come from the sample and education crosswalk documented in Appendix A.5. All three cohorts contribute every age from 16 through 35. The following fit uses these cell means, with no additional demographic restriction.

Let $\overline{Y}_{Sqa}$ denote the person-weighted mean in cell (q, a) and ω_{qa} its population weight normalized to sum to one over the 20 ages within cohort q . I estimate each cohort's parameters separately by nonlinear weighted least squares:

$$(\hat{K}_q, \hat{b}_q, \hat{d}_q) = \arg \min_{K, b, d} \sum_{a=16}^{35} \omega_{qa} \left[\overline{Y}_{Sqa} - K \Lambda\{b(a - d)\} \right]^2. \quad (37)$$

The fitting bounds are $K \in [0.01, 25]$, $b \in [0.001, 5]$, and $d \in [6, 75]$. Five starting values lead to the same retained objective in each cohort; each fit allows at most 3,000 function evaluations with numerical tolerances of 10^{-13} . These fitting bounds differ from the narrower treated-path search domain in Section 3. Table C1 reports the cohort sizes, parameter estimates, and weighted root mean squared errors. I then average fitted curves with fixed one-third weights, rather than allowing cohort shares to change with age.

C.2 Observed Ages and Extrapolation

Each selected cohort contributes all ages 16–35, so age 25 and age 35 are both within every cohort’s observed support. Figure C1 compares the mean fitted curve with the mean of the observed cohort–age schooling measures over that common range. Cohort-specific weighted root mean squared errors are 0.112–0.115 schooling years, and the largest absolute cell residual is 0.327 years. This is an in-sample descriptive fit diagnostic. Every comparison-path value above age 35 is extrapolated. The age-17 fitted value is the shared level at which candidate continuations are anchored, under the assumption that the hypothetical policy first affects measured attainment at $a_0 = 18$. Fixing cohort weights removes changes in the included birth-cohort shares; it does not turn repeated cross sections into a panel or establish the profile as the untreated path for a policy population. The simulation conditions on the fitted curve rather than propagating its sampling or extrapolation uncertainty.

C.3 Parameter Reference and Restrictions

The reference triple used to express $\theta_c = (\Delta K_c, \Delta b_c, \Delta d_c)$ is the arithmetic mean of the three fitted cohort parameter triples. Its values are approximately (13.922226, 0.257500, 12.564854). Its logistic curve is not generally equal to the mean of three logistic curves in Equation (36); their largest difference over ages 16–65 is 0.000412 schooling years. After anchoring the reference-parameter continuation at the common age-17 level, its largest discrepancy from H_c^0 over ages 18–65 is 0.000226 schooling years. Thus, the zero-increment reference exactly shares the anchor but does not impose an exactly zero effect at later ages.

The plateau restriction $K \in [13.3370, 15.1395]$ is the weighted first-to-99th percentile interval of a historical bank of 716 state-by-birth-cohort logistic fits. The bank covers the 50 states and the District of Columbia and birth cohorts 1980–1994, using ACS observations from 2000–2023. Its source construction retains U.S.-born non-Hispanic White, non-Hispanic Black, and Hispanic respondents, valid education records, positive person weights, and recorded male or female sex. State–cohort profiles are fitted by population-weighted nonlinear least squares. Retained fits have at least 14 age cells spanning at least 13 years, begin by age 20, reach at least age 29, converge, and lie inside the fitting bounds.³⁵ The interval restricts candidate eventual schooling to a broad central range of historical fitted profiles. Its transfer to the current full-population, 1985–1987 baseline is a maintained calibration assumption: the populations and survey windows differ, and the interval is neither a confidence interval for that baseline nor validation of policy-induced endpoints.

The speed and timing domains, $b \in [0.12, 0.65]$ and $d \in [8, 30]$, and the one-year annual acquisition cap are experimental design restrictions rather than estimated quantiles from the current cohort sample. The simulation does not impose the historical bank’s timing-index bounds or joint parameter ellipsoid. Appendix B.6 describes the parameter sweeps within these restrictions. The reported widths are conditional on this domain, the baseline, and the valuation calibration; sensitivity to a population-matched support bank, alternative bounds, or other path families has not been evaluated.

³⁵Plateau weights sum person weights over each profile; quantiles interpolate cumulative normalized weights. The resulting limits are 13.337015059499 and 15.139521154274. Observations span ages 16–43 collectively, so plateaus remain fitted asymptotes rather than observed lifetime attainment.

D Policy Welfare Accounting and Education-Path Replacement

The welfare reconstruction replaces specified education-related components of a source account while retaining its policy contrast and the remaining accounting assumptions. This appendix defines the units and component boundaries, maps an education path into private income and fiscal returns, and states the extra information needed to let other education costs vary. The formulas describe the accounting procedure; they do not establish that a candidate path is the true policy response or that a finite search has recovered all compatible paths.

D.1 Policy Units and the Reference Ledger

Index the policy contrast by i . L1 reports WTP_i^{ref} , N_i^{ref} , and their associated MVPF, using the components in Equations (10) and (11). Each entry uses the same target population, treatment contrast, currency year, and discount base. Values may be stated per eligible person, per recipient, or per specified change in aid, but all components within an account must use the same unit. Multiplying every component by the same positive scale leaves the MVPF unchanged; mixing components from different populations or treatment margins does not. An intention-to-treat effect and an effect for students induced to switch institutions therefore require a documented conversion before they can enter one account. Monetary values from different cases also require conversion to common dollar units before their levels, rather than their dimensionless ratios, are compared.

The terms B_i^{ref} and G_i^{ref} collect the private and public components outside the replaced earnings projection and the separately recorded education payments. They are not necessarily pure cash transfers and statutory program costs. Depending on the source, they may include other measured private benefits, effects on public spending, or fiscal receipts. The private payment $C_i^{\text{priv,ref}}$ and public cost $C_i^{\text{pub,ref}}$ must follow the source's tuition and subsidy convention. For example, if the benefit of a grant is recorded separately in B_i^{ref} , the same grant cannot also be deducted from the private tuition charge to create a second benefit. Public education costs must likewise exclude payments already included in G_i^{ref} . Where the source nets private tuition receipts from institutional expenditure, that convention is retained. The original component sum, not a generic cost-per-degree formula, determines the reference values.

Interpreting income changes as willingness to pay retains the welfare assumptions of the source calculation. In their college-admission example, Hendren and Sprung-Keyser (2020) treat the earnings gain as a return to human capital rather than compensation for greater effort, and value the income change net of taxes and education expenses. A benefit requiring additional work effort, an unmeasured consumption value of college, or an effect on other people's earnings would require a corresponding welfare adjustment. Scholarship transfers also depend on whether the recipient would have attended without the award. These assumptions are not consequences of the logistic path specification. For the three main applications, the reference ledger must follow the Adams calculation in Hendren and Sprung-Keyser (2020) and the STBF and Tennessee Promise calculations in Angrist et al. (2022) and Schoonover et al. (2026), respectively.

The accounting check is to recover the source's willingness to pay and net cost by summing its recorded components before substituting any path. Feeding the original replaced components back into Equations (13) and (14) must then return those same totals. This is an arithmetic identity check, not evidence that the original continuation belongs to the candidate logistic family. If a source does not separately report a component needed for the subtraction, recovering it requires a documented calculation or an explicit additional calibration.

A missing component cannot be silently set to zero. The case records report original education inputs and timing in Table D1, followed by the reference and replacement components in Table D2.

D.2 From Attainment to Disposable Income and Fiscal Returns

The observation operator in Equation (3) determines what a candidate education path must reproduce. A stock effect at one age, an average over an age window, and an effect on schooling acquired during an interval constrain different features of the path. Converting credits or credentials into schooling years is a separate step and must preserve the source’s treatment margin. In particular, a change in the probability of earning a bachelor’s degree is not automatically four additional years per person: the appropriate conversion depends on schooling already counted through enrollment, partial completion, and other credentials. The anchor must precede policy exposure or anticipation for the target population. Neither age 17 nor the simulated 1985–1987 comparison group is imposed on every application merely because it is used in Section 3.

Let $e_i^s(a)$ be the annual pretax earnings effect in account $s \in \{\text{ref}, \text{path}\}$, and let $\mathcal{A}_i$ be the set of ages included in the replaced projection. The discount factor is $D_i(a) = D_{\rho_i}(a; a_{d,i})$, with source-specific rate ρ_i and discount-base age $a_{d,i}$. That base need not equal the first age at which attainment can change. All annual earnings effects are expressed in the same currency year, and their gross present value is

$$\text{NPV}_i^{\text{earn},s} = \sum_{a \in \mathcal{A}_i} D_i(a) e_i^s(a). \quad (38)$$

For HSK schooling-year inputs, the path extension uses

$$e_i^{\text{path}}(a) = Y_{0i}(a) \sum_{\tau} R_i(a, \tau) [\Delta H_i(\tau) - \Delta H_i(\tau - 1)]. \quad (39)$$

The source programs determine each phase’s effective return, eligible-person conversion, age range, monetary year, discount convention, and tax rate. A single batch at the source reference completion age reproduces earnings PV, fiscal PV, private and public education costs, WTP, and net cost separately. All 33 point replays and the 33,000 registered source-input replays pass these component checks. Source code and reported L1 vintages remain distinct: for example, the Tennessee Pell program yields 0.8437 while the retained library L1 is 0.85.

The common template has seven negative-response years followed by a positive response, but source exceptions are retained. Ohio Pell has six early response years and different early and later discount bases. The independent-single tax cases retain their later source income clock, and adult Pell retains its adult entry clock and 1973-dollar account. Kalamazoo’s released program overlaps a positive response beginning at 18 with a negative response at 18–24 and discounts the positive block from age 11. This code convention is reproduced and transported with its own phase timing; it is not silently replaced by the illustrative post-25 template. Georgia HOPE retains its sector mixture, including its zero community-college early component.

When source phases b use different discount bases, the implemented present value is $E_i = \sum_a \sum_b D_{ib}(a) e_{ib}(a)$, and fiscal value is computed with the same phase’s retained tax rule. The nominal net-income curve is $\Delta y_i(a) = \sum_b e_{ib}(a)$; the unified discounted curve is $\Delta y_i^{PV}(a) = \sum_b D_{ib}(a) e_{ib}(a)$. Its age sum equals the implemented NPV. A single common $D_i(a)$ is used only when it reproduces the source phase conventions. The source-block export records the distinctions, avoiding an apparent source match created solely by a final scaling factor.

Independent forecasts instead use the original function's input structure when recoverable. STBF's F_i is nonlinear in credential shares and potential experience. Early Head Start uses the public BA–HS schedule described below; Tennessee uses signed exclusive-credential batches. No case receives an extra schooling-flow opportunity-cost charge on top of its net-income response. Negative portions of that response are earnings losses, not separately identified pure costs of time in school.

With a retained linear tax-and-transfer schedule, let $\vartheta_i(a)$ denote the fraction of an earnings change that accrues to the government through taxes and reduced transfers. The disposable-income and fiscal components are

$$\text{NPV}_i^{\text{after},s} = \sum_{a \in \mathcal{A}_i} D_i(a) [1 - \vartheta_i(a)] e_i^s(a) \quad (40)$$

and

$$\text{NPV}_i^{\text{tax},s} = \sum_{a \in \mathcal{A}_i} D_i(a) \vartheta_i(a) e_i^s(a). \quad (41)$$

The two components sum to Equation (38). Negative earnings effects reduce private income and, under a positive rate, reduce fiscal revenue. The same schedule is used in the reference and path calculations when the comparison is intended to hold taxation fixed. If the source uses different short- and long-run rates, the age-specific rates remain in the sum; a single average rate generally cannot reproduce both components for every candidate path. Nonlinear tax rules or separate transfer responses require their source-specific mapping rather than these linear formulas. In particular, applying a nonlinear tax schedule to mean earnings need not recover mean taxes without distributional information.

Holding the education-payment components fixed, when $\vartheta_i(a) = \bar{\vartheta}_i$ is constant, the income bookkeeping is especially transparent. Writing $\Delta E_i = \text{NPV}_i^{\text{earn,path}} - \text{NPV}_i^{\text{earn,ref}}$, the income replacement changes willingness to pay by $(1 - \bar{\vartheta}_i)\Delta E_i$ and net fiscal cost by $-\bar{\vartheta}_i\Delta E_i$. The private gain and public revenue gain partition the same pretax earnings change. Where education payments vary across paths, their changes also enter WTP and net cost. This conditional special case explains the signs in Equations (13) and (14); it does not impose one tax rate across policies. If the original income forecast and the candidate forecast use different return or opportunity-cost rules, their difference incorporates that accounting change as well as the education continuation. Separating the two effects requires evaluating a specified reference path under the same candidate valuation rule, when such a reference path can be recovered from the source.

D.3 Education Payments, Resource Costs, and the Scope of Replacement

Source education-cost functions are preserved through an explicit timing bridge where they are recoverable. For each HSK account, the export separates fixed noneducation components from the source private-payment and public-education amounts. The source schooling quantity and completion allocation identify their per-unit schedules. Applying those schedules to signed completion batches reproduces the original education costs at the source reference batch, while allowing payments to vary with path timing and quantity. This transports the source institution mix and marginal unit costs; it does not infer actual enrollment, sector switching, or aid eligibility from attainment alone. Where a source aggregate combines enrollment and credential margins, holding that composition fixed per schooling equivalent is an additional restriction.

The three independent cases retain their stated non-income amounts while replacing income. This isolates a conditional income-channel comparison; it does not let every alternative path change grant duration, tuition, or public instruction resources. Algorithmic placement retains original WTP and adds only fiscal earnings.

Cannabis prices its education-resource flow separately from its signed income response. These scopes are recorded by case rather than described as one common fixed residual for all 38.

With fixed conversion rules, prices, and valuation coefficients, the account can also be collected into a ratio of affine functions of path summaries:

$$\text{MVPF}_i(\Delta H_i) = \frac{\alpha_i + \mathbf{w}_i^\top \mathbf{z}_i(\Delta H_i)}{\beta_i + \mathbf{n}_i^\top \mathbf{z}_i(\Delta H_i)}, \quad N_i(\Delta H_i) > 0. \quad (42)$$

The constants α_i and β_i collect retained components; $\mathbf{z}_i$ contains the specified discounted earnings and, where applicable, education-cost summaries. The coefficients $\mathbf{w}_i$ and $\mathbf{n}_i$ map those summaries into willingness to pay and net cost. For the main replacement alone, take $\mathbf{z}_i = (\text{NPV}_i^{\text{after,path}}, \text{NPV}_i^{\text{tax,path}})^\top$, $\mathbf{w}_i = (1, 0)^\top$, and $\mathbf{n}_i = (0, -1)^\top$, with $\alpha_i = \text{WTP}_i^{\text{ref}} - \text{NPV}_i^{\text{after,ref}}$ and $\beta_i = N_i^{\text{ref}} + \text{NPV}_i^{\text{tax,ref}}$. This is a representation of the retained accounting, not a claim that welfare is linear in the logistic parameters or that every education-policy valuation uses fixed coefficients. Returns that depend on education expansion, endogenous prices, or additional behavioral channels would require further modeling.

D.4 Comparing Paths Under a Common Welfare Account

L1 is the original welfare calculation. L2 imposes the education moments selected from that calculation and searches the maintained continuation class. L3 adds comparable source information while retaining the L2 moments and all other conditions. Write $\widehat{\mathcal{D}}_{i\ell}$ for the finite set of verified education-effect paths found for policy i at information layer $\ell \in \{2, 3\}$. The joint welfare values are

$$\widehat{\mathcal{V}}_{i\ell} = \left\{ \left(\text{WTP}_i^{\text{path}}(\Delta H_i), N_i^{\text{path}}(\Delta H_i) \right) : \Delta H_i \in \widehat{\mathcal{D}}_{i\ell} \right\}. \quad (43)$$

The numerator and denominator of each MVPF come from the same member of this set. Pairing the largest willingness to pay with the smallest net cost from different paths can create a ratio that no candidate attains. Likewise, the reference MVPF falling inside a candidate range does not establish that the reference continuation is a member of the path family. A later observation refines the admissible set only when it concerns the same population and outcome under a compatible time mapping. Changing the sample, functional form, or cost calibration defines an alternative specification rather than a nested information addition.

For positive net costs, $\text{MVPF}_i > 1$ if and only if $\text{WTP}_i - N_i > 0$. Finding two compatible, positive-cost paths on opposite sides of this threshold establishes ambiguity conditional on the maintained restrictions and account. Finding candidates on only one side does not establish that all compatible paths lie there. The self-financing region requires positive willingness to pay and nonpositive net cost and is reported separately. Ratios can also become arbitrarily large in magnitude along a sequence of positive-cost paths whose costs approach zero; an education effect crossing zero is not sufficient to establish such a sequence. A large finite ratio or a rounded zero denominator is therefore not a proof of an infinite bound. The analysis assigns no probabilities to candidate paths, and any sampling analysis of the source estimates remains distinct from this continuation exercise.

The threshold at one compares a dollar of measured private willingness to pay with a dollar of net public cost. It is not a universal policy-adoption rule. As Hendren and Sprung-Keyser (2020) show, a budget-neutral comparison of policies affecting different people also depends on the social weights attached to their beneficiaries and on the financing alternative. The cross-policy comparisons in Section 5 retain that distinction. Within a policy, the accounting exercise asks the narrower question of whether the same education evidence supports different threshold or fiscal classifications under a fixed valuation rule.

D.5 Geographic References, Source Control Levels, and Empirical Support

The geographic library uses the unrestricted 2000–2022 IPUMS extract: no race, sex, or birthplace restriction is imposed. It contains the 50 states, the District of Columbia, and a national aggregate. Person weights form outcome means within state–birth-cohort–age cells; birth cohort is survey year minus integer age. The 2000 component is Census sample 200001, followed by annual ACS samples in 2001–2022. The sample inventory is retained with the replication files. The full library supplies years of schooling and some-college-or-higher (SC+), associate’s-or-higher, bachelor’s-or-higher, and postgraduate outcomes. Case-specific geographic and cohort weights are fixed across ages, so changing the observed age does not change those chosen weights. This does not turn repeated cross sections into individual histories or eliminate migration and other within-cell composition changes. The library is an external description of education profiles, not an additional source of policy identification.

For each geographic cell and outcome, I fit a logistic profile to person-weighted age means, using observed ages 16–35 when the coverage supports estimation. State shapes are regularized toward their national counterparts. A cohort with insufficient young-adult support instead borrows a national shape and uses its available observations to calibrate the level. The data therefore involve both forward extrapolation for young cohorts and backward reconstruction for historical cohorts. The 1947 cohort used for Adult Pell is calibrated at ages 53–65, and the 1963 cohort used for Social Security student benefits at ages 37–59. Both borrow the 1978 national shape, itself fitted at ages 22–35. Their early-life paths are not observations of those cohorts at ages 16–35. Table E1 reports actual support and fit modes; the accompanying registry lists every state–cohort–outcome cell, donor, and level-calibration range. These records also distinguish Tennessee’s subsequent source-population and control-level adjustments. The library contains 10,920 fits across 52 geographic units, 35 birth cohorts, and six outcome definitions.

A published control mean is used only when its population, outcome, and clock match the effect under study. In Tennessee, the effects concern high school graduates and residence one year before interview. I therefore construct the corresponding Tennessee profiles using migration information and rescale each outcome to its reported age-23–24 control mean. The common cohort weights remain fixed. This operation supplies an external level and shape; it does not recover the regression’s conditional untreated prediction, and conditioning on high school graduation is itself part of the source population definition. Adams uses a full-population schooling profile because the source degree means do not identify total completed schooling. STBF uses separate external time and credential timing indexes, while its income function retains recovered source mean inputs. Its 3.93-year enrollment mean is not converted to total schooling by adding twelve. The distinction between source-matched levels and external population proxies is explicit in the case registry.

The empirical support is constructed before searching policy paths. Let $x = (K, \log b, d)^\top$. For each outcome I retain fitted profiles for the 1983–1987 donor cohorts in the 51 state-level regions. A raw logistic profile with its own level at the anchor can be written exactly in the anchored normalized form, using the same parameter triple; a separate post-anchor refit is therefore unnecessary. Such refits are saved as diagnostics because many schooling fits have weakly determined timing parameters at the numerical boundary. The main covariance averages the relevant state’s within-donor covariance and the pooled-state covariance with equal weights. National targets use the pooled covariance; multi-state targets average the state covariances with the fixed target weights. The support center is the weighted parameter location of the target cohort’s fitted reference, and the restriction is

$$(x - \bar{x}_i)^\top \Sigma_i^{-1} (x - \bar{x}_i) \leq 11.34486673. \quad (44)$$

The cutoff retains the earlier 99-percent chi-square scale only as an empirical calibration convention. It is not a confidence region for a treatment response and assigns no probability to candidate paths. Centering historical dispersion on a different population or cohort is a maintained transport restriction. The wider sensitivity doubles Σ_i and retains every previously verified main candidate.

Timing restrictions refer to fractions of the increase after the anchor, rather than to the unanchored curve's half-plateau age. For anchor τ , the age at fraction q of the remaining logistic increase is

$$a_q(b, d; \tau) = \tau + \frac{1}{b} \log \left\{ \frac{1 + q \exp[b(d - \tau)]}{1 - q} \right\}. \quad (45)$$

For $q = 0.1$ and 0.9 , pooled donor deviations between the 0.5th and 99.5th percentiles are transported around the corresponding target reference. The wider sensitivity multiplies these deviations by $\sqrt{2}$. Schooling paths remain nondecreasing and acquire at most one year per annual step after the anchor; probability paths stay between zero and one. These restrictions describe the admitted economic continuations. They are not validated causal restrictions merely because they are informed by ACS profiles.

For L2, a deterministic Sobol shape grid and a 97×97 regular grid cover $(\log b, d)$. Conditional on a shape pair, the retained moment determines K algebraically. Five exact-match L3 cases additionally solve a second moment equation by bracketing and root refinement. The other eight L3 cases minimize the fixed equal-weight point discrepancies in equation (15), preserving L2 exactly. The main scalar fits use up to 64 diverse feasible starts and Tennessee uses 42 joint starts, supplemented by separate credential fits; constrained sequential quadratic programming is followed by an interior least-squares polish only when it passes every original restriction. Distinct minimizing basins are retained; repeated optimizer outputs near the same minimum are not interpreted as a welfare interval. All eight point searches currently retain one best-fitting path with nonzero additional-input residuals. Table E5 summarizes fit quality and Table E6 reports the schooling-input residuals. Tennessee's complete residuals appear in Table D5. Annual paths, the L2 equalities, credential ordering, and welfare accounts are independently reconstructed. These checks establish the validity of the saved solutions, not certified global optimality or uniqueness.

D.6 STBF: The Author Credential–Experience Income Function

The source is the deposited main and appendix workbooks and their accompanying earnings programs for the 2012–2014 four-year strata (Angrist et al., 2022). The four pooled controlled effects at six years are $m_T = 0.3647777438$, $m_C = 0.00715823844$, $m_A = 0.05182238668$, and $m_B = 0.08133502305$. Their marginal SEs are 0.04131259769, 0.002961894264, 0.015011851676, and 0.015532993712. Time enrolled is measured in years, and the other inputs are cumulative credential shares. Each outcome has a separate anchored path, with the source's fixed cohort and high-school-age weights:

$$\mathcal{M}_o\{\Delta H_o\} = m_o, \quad o \in \{T, C, A, B\}. \quad (46)$$

The three credential levels satisfy $H_C^s \geq H_A^s \geq H_B^s$ at every age. The time-enrolled effect uses the external schooling-shaped family as a timing index, not as an observed level of enrollment or completed schooling. All four same-horizon estimates enter L2; they do not create a second follow-up date or an L3 specification.

For each of four race–sex groups g , the source Poisson mean-input forecast can be written

$$F_g(a; \Delta p_g, \Delta T_g) = \exp[\alpha_g + (B_g + \beta'_g \Delta p_g(a)) \mathbf{1}\{x_g(a) > 0\} + f_g(x_g(a))], \quad (47)$$

$$x_g(a) = \max\{a - 18 - \bar{T}_g - \Delta T_g(a), 0\}. \quad (48)$$

Here f_g is the retained source quartic with its original scaling, $\Delta p_g = (\Delta C_g - \Delta A_g, \Delta A_g - \Delta B_g, \Delta B_g)$ contains exclusive SC, AA, and BA changes, and $B_g = \beta'_g \bar{p}_g$. The forecast evaluates mean inputs; it is not an estimate of the expectation of this nonlinear function over an unobserved individual distribution. Group-specific coefficients, 3 percent discounting to age 18, income ages 18–65, and 2019-dollar conventions are retained. No HSK return coefficient is substituted into this function.

The source's 3 and 5 percent income PVs identify the sufficient statistics $(\bar{T}_g, B_g)$ conditional on its coefficients and education effects. In source group order, the recovered pairs are (1.90624931, 0.28136372), (2.01871376, 0.32275122), (1.56582623, 0.23536598), and (1.62036888, 0.21935042). Individual baseline credential means remain underidentified. Their feasible affine sets are retained, and each candidate must permit one fixed baseline share vector per group that remains valid at all ages. No arbitrary feasibility witness is presented as an author mean. Conditional on the source control total of 1,924 and the exported aggregate PVs, integer weights are proportional to (522, 798, 242, 362); this recovery assumption is recorded separately from directly supplied coefficients.

Pooled and subgroup controlled effects are not exactly aggregation-consistent. The main bridge transports each pooled timing index to group g according to $\Delta H_{go}(a) = m_{go} \Delta H_o(a) / m_o$, holding the source effect ratios fixed. Thus the pooled paths are timing indexes, not claims to equal the weighted subgroup means at every age. Reconstructing the original fixed effects recovers the source group PVs and the workbook aggregate of \$21,129.2291. The published text instead reports \$21,150; both are retained, and no multiplicative correction is made. Current path income is $\sum_g w_g [F_g(a; \Delta p_g, \Delta T_g) - F_g(a; 0, 0)]$. WTP is $32,250 + 0.8E$ and net cost is $32,250 - 0.2E$.

The 1,000 conditional repetitions multiply all four marginal SEs by the same registered normal draw. This imposes perfect dependence rather than recovering an author joint bootstrap. Group coefficients, sufficient statistics, effect ratios, and external ACS calibration remain fixed. The complete eight-dimensional joint path search is repeated for each input draw.

D.7 Adams: Education Conversion and Additional Source Timing

The Adams reconstruction retains the eligibility RD contrast for the 2005–2006 graduating classes and the source program used by Hendren and Sprung-Keyser (2020). The source observes college entry immediately after high school and within two years, persistence in the fourth year, and graduation within four and six years. The retained conversion assigns one year to additional two-year enrollment, two years to additional four-year enrollment, half a year to two-year completion, and one further year to bachelor's completion. The corresponding multipliers are (1, 2, 0.5, 1), giving $0.007 + 2(0) + 0.5(0) - 0.025 = -0.018$ schooling-equivalent years per eligible student. The baseline is the full-population Massachusetts profile for birth cohort 1987, with age 18 as the last unaffected observation. L2 imposes

$$\Delta H_{\text{Adams}}(24) = -0.018. \quad (49)$$

L3 preserves this endpoint and places the separately reported early entry contribution at age 20:

$$\Delta H_{\text{Adams}}(20) = 0.007. \quad (50)$$

These are temporal allocations of the source schooling conversion, not directly estimated annual YS effects. The source's local population near the eligibility threshold is not interchangeable with all Massachusetts

residents; that external-baseline approximation is retained explicitly. Its control bachelor's mean of 0.631 provides degree context, but does not identify a comparable total-schooling control curve.

The root search gives 21 L3 paths, all contained in the 5,851 L2 candidates. The source supplies two distinct windows, and the retained conversion assigns schooling-equivalent values to their inputs. L3 additionally maintains that the converted entry contribution is acquired by the early modeled endpoint. It does not claim that the source directly estimates annual completed schooling. Michigan's early-entry and later-credential reconstruction follows the same rule. By contrast, a second time point obtained solely by splitting one entry estimate using an assumed duration is excluded. The original authors' evidence favors dropout over delay within follow-up; later recovery among candidate paths remains an unobserved continuation, not a rejection of that evidence.

The source program yields earnings NPV $-\$1,204.04$, WTP $\$1,364.23$, and net cost $\$1,896.35$ in 2004 dollars. The new single-batch replay reproduces these components individually. It retains the original phase-specific earnings and fiscal rules and transports the source private and public education-cost schedules with the signed schooling-equivalent batches. The fixed noneducation parts are

$$w_{\text{fixed}} = W^{\text{ref}} - E^{\text{ref}} + T^{\text{ref}} + P^{\text{ref}}, \quad (51)$$

$$n_{\text{fixed}} = N^{\text{ref}} + T^{\text{ref}} - C^{\text{ref}}. \quad (52)$$

For a candidate path, $W = w_{\text{fixed}} + E - T - P$ and $N = n_{\text{fixed}} + C - T$. Private payments and public education costs can therefore vary with the candidate under the source cost-timing bridge. There is no fixed threshold in E alone. The source institution mix per schooling equivalent remains a maintained transport assumption when the total converted education effect is negative. All kernels are already in 2004 dollars; no second 188.9/172.2 conversion is applied.

L3 earnings NPVs range from -6.98 to -3.12 thousand dollars, WTP from 1.82 to 3.00, and net public cost from 2.85 to 4.89 thousand dollars. The candidate MVPFs lie between 0.61 and 0.64. These values use the same signed completion-response and education-cost mappings in L2 and L3. The source's within-follow-up dropout evidence is retained; a later catch-up path is a conditional extrapolation beyond that evidence.

D.8 Tennessee Promise: Matched Credential Paths and Earlier Windows

The Tennessee source uses 2000–2024 ACS data and overlapping age groups among high school graduates. Exposure is the share of a cohort eligible at age 17, with geographic assignment based on state of residence one year earlier. The regressions include state and survey-year fixed effects. The reconstruction uses fixed equal weights across the 1992–1998 birth cohorts, which reach age 17 during the county-to-statewide expansion in 2009–2015. These are the model cohorts, not a claim that the source regressions use only those birth years. Each outcome's Tennessee age shape is fitted to the corresponding external population and rescaled so that its age-23–24 mean equals the source control mean: 0.685 for some college or higher, 0.380 for associate's or higher, and 0.292 for bachelor's or higher. The treated curves share their outcome-specific age-17 reference levels. This reproduces selected population levels, not a fitted untreated DID trajectory; the control mean may itself combine treatment states, covariates, and cohorts differently from the regression contrast.

For $j \in \{C, A, B\}$, the retained source effects are $m_j \in \{0.035, 0.040, 0.027\}$. I impose

$$\frac{\Delta H_j(23) + \Delta H_j(24)}{2} = m_j. \quad (53)$$

Equal age weights approximate the unavailable regression operator. At every annual age, both reference and treated levels satisfy $H_B^s(a) \leq H_A^s(a) \leq H_C^s(a)$. Each credential has its own anchored parameter triple and state-centered support. The constituent shape pools are combined with a deterministic Sobol design, supplemented with combinations near earnings extremes and earlier-window fits. The main bank contains 92,955 complete joint candidates. Each reported welfare pair uses one such joint candidate; separate credential extrema are never combined as if they were one feasible path.

The additional late-completion restriction is relative to those who have not yet attained the outcome. Let $\bar{H}_{jc}$ be a donor cohort's mean level at ages 23–24. Define the historical completion fraction

$$R_j = \max_{c \in \{1983, \dots, 1987\}} \frac{K_{jc} - \bar{H}_{jc}}{1 - \bar{H}_{jc}}. \quad (54)$$

If necessary, the main allowance also includes the fitted target reference's own fraction. The candidate restriction is

$$K_j^1 \leq \bar{H}_j^1 + (1 - \bar{H}_j^1)R_j. \quad (55)$$

The donor profiles use the same Tennessee high-school-graduate denominator. This rule avoids treating a statewide attainment share as a universal ceiling for a selected population. It still assumes that remaining completion among the treated can be bounded by the chosen donor experience, whose plateau is itself extrapolated. The main bachelor's ceiling is 0.4537. Table D4 gives the other quantities; Appendix E.9 separately varies this allowance and the parameter support, so the two restrictions are not changed together.

L3 compares predicted adjacent-age averages with all eighteen point estimates in Table 2, retaining the three latest-window effects exactly. The fifteen earlier coefficients enter equal-weight least squares in probability units. The source reports clustered SEs and uses policy permutations for inference; neither the SEs nor the graphical permutation intervals form acceptance bands in this fixed-input exercise. The best-fitting joint path has eighteen-input RMSE 0.003525 and maximum absolute residual 0.006673 in probability units. Table D5 records all targets, predictions, and residuals. The same ACS reference, cohort weights, logistic family, support, ordering restrictions, late-completion caps, and income account apply in L2 and L3. The retained L3 path is also supplied to the L2 bank.

The source welfare account applies Tennessee-specific education returns to project earnings and taxes. It assumes 2.3 years of Promise participation and five years to a bachelor's degree. Direct scholarship spending averages \$2,760 before discounting; the source reports total public cost before tax offsets of \$14,793. Tuition relief for students who would have enrolled without the program and projected after-tax earnings for induced students enter willingness to pay. The reported totals are \$72,837 for willingness to pay, −\$10,053 for net public cost, and \$24,845 for additional taxes. The authors qualify the use of average education returns for induced students and the public-expenditure assumptions. The reconstruction preserves the monetary units and policy unit of this account.

The author age-by-credential income table could not be recovered. Public ACS 2011–2021 data supply reconstructed mean wages for employed, out-of-school Tennessee residents under the documented prior-residence sample, in 2021 dollars. They are not author-supplied income inputs. Write $Y_j(a)$ for earnings in exclusive category $j \in \{SC, AA, BA\}$ and $Y_{HS}(a)$ for high-school earnings. The completion kernel is $-Y_{HS}(a)$ during d_j study years before completion and $Y_j(a) - Y_{HS}(a)$ from completion onward, with $(d_{SC}, d_{AA}, d_{BA}) = (2, 2, 4)$. It is zero outside these periods and before policy onset. This approximates zero earnings while studying; the source's 2.3-year Promise duration and five-year BA assumptions are not claimed to have been recovered as annual income paths.

Let $q_j(\tau)$ be signed increments of the exclusive effects $(\Delta C - \Delta A, \Delta A - \Delta B, \Delta B)$. The reconstructed account is

$$E = \frac{1}{p} \sum_{a=18}^{64} 1.03^{-(a-18)} \sum_j \sum_{\tau} K_j(a, \tau) q_j(\tau). \quad (56)$$

The CE path proxies SC acquisition; BA+ is priced at BA earnings without a graduate premium. Education paths extend to 68 for the four-year pre-completion window. Negative exclusive increments are retained as differences between two credential distributions.

The unit bridge is explicit. With grant $g = 2,711$, private remainder $w_0 = 2,124.307692$, and entry effect 0.054 , set $u = w_0/g$ and $p = 0.054/(1 - u) = 0.2495243215$. Under the maintained interpretation, u is the uninduced participant share and p maps population effects into participant units. This is account-implied, not observed take-up. It does not select p to match earnings PV. The earnings scale is one. The source-rounded exclusive endpoint exercise gives \$59,881.13; the nested endpoint vector used in the path model gives \$69,261.65. Both differ from the reported implied \$95,557.69, and neither is rescaled to that total. Fixed source non-income amounts are assigned the 2021-dollar convention to match reconstructed wages; their exact original price vintage is unavailable. Consequently the L1–L2 comparison includes price and unit assumptions in addition to continuation.

The private remainder is obtained by removing the reference after-tax earnings contribution from total willingness to pay:

$$w_0 = 72,837 - 0.74 \left(\frac{24,845}{0.26} \right) \simeq 2,124.31. \quad (57)$$

Thus, \$2,124.31 is the private transfer and tuition component left after the earnings contribution is removed from the reported total. It is recovered from the account, not an additional estimate of a policy effect. Adding the projected tax receipts back to net public cost gives

$$n_0 = -10,053 + 24,845 = 14,792. \quad (58)$$

This is the reconstructed pretax fiscal cost, containing scholarships and education spending. It differs by one dollar from the source's displayed \$14,793 total because the reported components do not sum exactly at their published precision. I retain the reported net-cost and tax totals for an internally consistent reconstruction.

The candidate account is therefore $WTP = 2,124.31 + 0.74E$ and $N = 14,792 - 0.26E$, using unrounded remainders in the calculations. Equivalently, it starts from the reported willingness to pay and net cost and replaces the private and fiscal shares of E^{ref} with those of E . Returning E^{ref} recovers the source totals. Fiscal break-even occurs at $E = n_0/0.26 = 56,892.31$; with positive willingness to pay, net cost at or below zero yields the self-financing classification. Grant duration, tuition, and public education resources remain in the fixed remainders. Their invariance across paths is a restriction on this comparison, not a prediction about how enrollment costs respond to each education continuation.

Among the L2 candidates, willingness to pay ranges from -48.72 to 152.50 thousand dollars and net cost from -38.04 to 32.66 thousand dollars. The L3 best fit gives willingness to pay of 86.44 thousand dollars and net cost of -14.83 thousand dollars. Its positive benefit and negative net cost imply self-financing. Figure 7 displays L2 attainment ranges and the L3 fitted path alongside H^0 . Appendix E.9 refits this point criterion under alternative calibration and valuation assumptions. Pixel-width sensitivity is inapplicable to the point-estimate criterion.

D.9 Source-Input Repetitions

Table 4 reports 1,000 prespecified repetitions per application, holding calibration and path restrictions fixed. STBF uses constructed normal perturbations; Adams uses linked education and accounting draws. Tennessee perturbs the eighteen point coefficients using displayed clustered SEs and refits the same constrained minimum-distance criterion in each repetition. This does not recover the source permutation distribution. Verified and unsuccessful draw counts are reported in Appendix F; no unsuccessful input draw is replaced. Appendix F defines the quantiles and explains the statistical interpretation, including Tennessee’s SE convention.

E Full Policy Collection and Case Registry

This appendix documents the 38 policy calculations in Section 5: their source inputs, calibration, welfare comparisons, and ordering rule. Separate profiles describe the 35 cases outside the main applications. Appendix F treats input uncertainty; the replication materials supply full-precision observation operators and complete paths.

E.1 Cases, Source Families, and Calibration

Table E1 lists all 38 calculations. The collection retains 34 earlier calculations and adds algorithmic college placement, two further tuition-deduction contrasts, and the cannabis education account. Several calculations can come from one paper. The source-account cases represent 22 research families, with one further family for the author cannabis study. The grouping combines related Bulman–Hoxby versions and tax-schedule calculations, the spending and tuition calculations associated with Deming–Walters, and the Kalamazoo Promise research versions. The source-authors string is retained separately. A family-level witness count records whether at least one member supplies the specified paths; it is not an estimate of a policy-population frequency.

The geographic registry selects the source’s treated population rather than combining it mechanically with control states. Birth cohorts and mixture weights remain fixed across candidates. Geographic matching improves comparability, but an ACS mean is not an identified untreated trajectory for a source study’s selected population. The source-control status records that distinction, including Tennessee’s high-school-graduate denominator and control-level calibration. Education conversion, age mapping, the anchored logistic continuation, empirical support, and transport of welfare prices remain maintained assumptions. Reproduction of a source coefficient or account is distinct from validation of those assumptions.

The accompanying replication materials record each moment’s outcome, reconstruction, target, available standard error, and nonzero age-cohort weights, together with the comparison profiles, anchors, valuation coefficients, and sources. A converted coefficient is not relabeled as observed completed schooling. In particular, current enrollment and accumulated education are different outcomes; the source-specific bridge remains part of each reconstruction. Full-precision inputs permit the matching equation to be reproduced without inferring an age from a figure or replacing a window by its midpoint.

E.2 Education Inputs and Observation Windows

These profiles document the 35 calculations not developed in Section 4. Each first identifies the source outcome, observation window, and numerical effect; then states the schooling conversion and restriction

used in L2; and finally lists the additional inputs in L3 or explains why that layer is absent. The three main applications are documented in Appendices D.6, D.7, and D.8. Unless stated otherwise, the reference conversion is that of Hendren and Sprung-Keyser (2020) and the associated Policy Impacts calculation. Here $\Delta H(a)$ denotes the schooling-years effect for the case's target population, except in Early Head Start, where it denotes the BA+ attainment effect. Probability coefficients are reported in probability units unless explicitly expressed in percentage points. Credits divided by 24, enrollment multiplied by an assumed duration, and credentials assigned additional schooling years are conversions of source estimates, not directly observed years-of-schooling effects. Table E1 records the matched comparison populations. I distinguish the source's observation window from the ages assigned in the reconstruction throughout.

For the point-input results, L2 retains the stated converted input as an equality. An admitted L3 preserves that equality and adds comparable source point estimates. Adams, Michigan, CUNY, Social Security student benefits, and HOPE tax credit admit exact matches. The other eight L3 applications use constrained equal-weight minimum distance and report residuals when the additional points cannot all be matched. Standard errors do not define bands or fitting weights in the main exercise. The accompanying registry retains standard errors for the separate uncertainty analysis, together with full age-cohort weights and observation operators. A case with no L3 can still contain several source coefficients: adding information requires separately observed comparable timing, rather than splitting one observation by an assumed duration.

Florida Student Access Grant

Castleman and Long (2016) estimate the effect of eligibility for the Florida Student Access Grant near the financial-aid cutoff. Their credit outcomes cover the first semester and the first four years after high school. Table 4 reports effects of 0.355 credits in the first semester and cumulative effects of 1.098, 2.666, 3.847, and 4.366 credits after years one through four. L2 uses only the fourth-year total, imposing $\Delta H(22) = 4.366/24 = 0.181917$ schooling years under a representative entry age of 18. L3 retains this endpoint and adds the first-semester and first-, second-, and third-year inputs. Their converted values are 0.014792, 0.045750, 0.111083, and 0.160292 schooling years at assigned ages 18.5, 19, 20, and 21, respectively, as fixed point targets in the constrained fit. Thus, L3 restricts when the credit gain accumulates before the same fourth-year total. The source also reports subsequent credential outcomes; these are not imposed as additional credit observations without a further credential-to-schooling conversion.

District of Columbia Tuition Assistance Grant

Abraham and Clark (2006) study the District of Columbia Tuition Assistance Grant's effect on college enrollment. The retained outcome is first-time freshman enrollment among students who completed high school in the preceding 12 months. The reference calculation uses an enrollment increase of 0.089, or 8.9 percentage points, and assumes two additional schooling years per induced entrant. L2 therefore imposes $\Delta H(20) = 2(0.089) = 0.178$ schooling years. Age 20 is the assigned endpoint of that duration conversion, not a directly observed schooling outcome for all source students. There is no L3: placing 0.089 schooling years at age 19 would divide the same assumed two-year total between years, rather than add a separately estimated follow-up.³⁶

³⁶The reference uses published Table 7A's 0.089 coefficient. The recovered 2003 working paper is an earlier version and does not independently verify that coefficient or its SE.

Wisconsin Scholars Grant

Goldrick-Rab et al. (2016) estimate the effects of the randomized Wisconsin Scholars Grant on semester credits, cumulative credits, and completion for three entering cohorts. The retained cumulative-credit estimates are 0.9 credits after three years for cohort 1 and 2.1 credits in the pooled observation of cohort 2 after three years and cohort 3 after two years. L2 combines these inputs with weights 0.314570 and 0.685430 and divides by 24, yielding 0.071772 schooling years. It imposes one weighted age-cohort restriction, rather than treating that pooled result as an observation at a single age.³⁷ L3 preserves this restriction and separately fits the 0.9- and 2.1-credit cumulative point inputs. It also uses Table 5's semester estimates to form annual credit effects: 0.5, 0.2, and 0.1 credits in years one, two, and three for cohort 1; 0.4 and 1.0 credits in years one and two for pooled cohorts 2–3; and 0.5 credits in year three for cohort 2 alone. Each is divided by 24 and restricts the change in ΔH across the corresponding year, with entry assigned to age 18. The third year of cohort 3 is unobserved and is not imputed. These reported values enter the point criterion without reconciliation. Cohort 1's six semester effects sum to 0.8 credits whereas the reported cumulative effect is 0.9. The cumulative and semester control means also differ, and the later cohorts have different follow-up windows. The published table does not establish an exact adding-up identity for these regression estimates. I therefore preserve the source precision and population mappings; the reported residuals can reflect input aggregation and model approximation, not only logistic curvature.

Kalamazoo Promise

Bartik et al. (2021) estimate Kalamazoo Promise eligibility effects on enrollment, attempted credits, and credentials. The credit estimates used here are 3.236, 4.309, and 6.562 additional attempted credits after two, three, and four years, respectively. The reference calculation retains the fourth-year estimate rounded to 6.56 and converts it at 24 credits per schooling year. L2 imposes $\Delta H(22) = 6.56/24 = 0.273333$. L3 keeps this exact fourth-year input and adds targets of $3.236/24 = 0.134833$ at age 20 and $4.309/24 = 0.179542$ at age 21, as fixed point targets. These ages assume entry at 18. The source also observes enrollment within six and 12 months and later credentials, including four- and six-year completion. Those outcomes are not interchangeable with attempted-credit totals under the retained conversion. Because the cohorts with sufficient follow-up differ across the credit horizons, interpreting them as restrictions on a common response also requires population comparability.

Michigan community-college tuition

Acton (2021) estimates the effect of a \$1,000 reduction in local community-college tuition for Michigan high-school graduates. For the 2009–2011 graduation cohorts, enrollment within one year changes by 0.036 at local community colleges, -0.021 at other community colleges, -0.004 at vocational colleges, and -0.003 at four-year colleges. Completion outcomes observed in 2017 change by -0.002 for certificates, 0.003 for associate's degrees, and 0.011 for bachelor's degrees. The reference conversion weights these seven probability effects by (1, 1, 1, 2, 0.5, 1, 2) schooling years, respectively, giving 0.029 years. L2 imposes

³⁷L2 uses $0.314570\Delta H_{1990}(21) + 0.342715\Delta H_{1991}(21) + 0.342715\Delta H_{1992}(20)$, with unrounded weights in computation. The 475 and 1,035 group counts motivate the retained group weights; the pooled cohorts are split equally as a calibration assumption, not as recovered regression weights. Entry at 18 and common response parameters are also assumptions. The third cohort's unobserved third year has zero observation weight.

$\Delta H(25) = 0.029$; age 25 represents the pooled 2017 completion observation, which occurs six to eight years after high school across the source cohorts. L3 retains the early entry window separately, imposing $\Delta H(19) - \Delta H(18) = 0.005$ in addition to the L2 equality. The 0.005 is $0.036 - 0.021 - 0.004 + 2(-0.003)$ under the same reference conversion. Assigning this duration-weighted contribution to the first modeled year is a temporal allocation, not an observed completed-schooling increment. Unlike a split of one isolated entry observation, it distinguishes the source's entry and later credential windows. The alternative semester and credit outcomes are not added to a composite that already uses enrollment and completion.

CUNY Pell grants

Marx and Turner (2018) estimate the effects of an additional \$1,000 of Pell Grant aid on first-year CUNY students entering in 2007–2011. Table 8 reports 0.212 additional academic credits earned in the current year and 0.223 cumulative credits three years after entry. L2 uses the latter, imposing $\Delta H(21) = 0.223/24 = 0.009292$ schooling years under an entry-age mapping of 18. L3 retains that equality and adds $\Delta H(19) = 0.212/24 = 0.008833$ exactly in the point-input exercise. The difference between the two inputs therefore concerns the timing of earned credits under the same conversion. The first- and third-year estimates use 38,100 and 32,271 observations, so their interpretation as points on one response path requires comparability of the available samples. The table's separate returning-student estimate, 0.492 current-year credits, concerns another risk set and is not a third observation of the first-year entrants.

Ohio Pell grants

Bettinger (2004) studies withdrawal between the 1999–2000 and 2000–2001 academic years in Ohio. The reference input is the Table 5.3, column 4 estimate for students initially enrolled at four-year colleges: a \$1,000 increase in imputed Pell aid reduces the withdrawal probability by 0.064, or 6.4 percentage points. The reference conversion reverses this sign and assigns one additional schooling year to persistence, yielding 0.064 expected schooling years. L2 imposes $\Delta H(20) = 0.064$, with age 20 assigned to the persistence endpoint. There is no L3. Other columns change the estimation sample or specification for the same transition; they do not observe later attainment for this target group. In particular, the retained estimate is the four-year-student contrast in Table 5.3, not an additional follow-up from Table 5.5.

Tennessee HOPE scholarship

Bruce and Carruthers (2014) study Tennessee HOPE eligibility and post-high-school college choice for 2006–2009 graduates. In the retained low-GPA eligibility comparison, Table 3 reports a -0.022 effect on two-year college enrollment, a reduction of 2.2 percentage points. The reference calculation uses a one-year schooling-duration contrast for the sector response, reversing the sign to obtain 0.022 schooling years. L2 imposes $\Delta H(20) = 0.022$ at the assigned endpoint. This is the reference conversion of the eligibility estimate; it is not rescaled into an effect per scholarship recipient. There is no L3 because the sector choices and robustness estimates do not supply later completed-schooling observations for the same contrast. The records end in 2009, so follow-up differs across graduation cohorts. The program's requirement to enroll within 16 months of high-school graduation is an eligibility rule, not a common observed follow-up duration.

Tennessee Pell eligibility

Carruthers and Welch (2019) estimate initial-fall college-entry responses at the Pell eligibility cutoff for 2006–2009 Tennessee high-school graduates. The reference calculation retains probability coefficients of -0.009200 , 0.002450 , and 0.000514 for Tennessee public, out-of-state public, and private four-year institutions, respectively. The public two-year coefficients are 0.002260 in Tennessee and 0.001570 out of state. The private for-profit enrollment coefficient is 0.001360 ; the reference conversion groups this margin with two-year entry. The resulting four-year and two-year-equivalent sums are $P_4 = -0.006236$ and $P_2 = 0.005190$. Applying the reference durations of two schooling years for four-year entry and one for two-year entry gives $2P_4 + P_2 = -0.007282$. L2 matches this total at the assigned second post-high-school year, averaging over graduation ages 17, 18, and 19 rather than placing everyone at one attained age.³⁸ There is no L3: $P_4 + P_2 = -0.001046$ is a sum of the same initial-entry coefficients, not a separately observed first-year schooling total. Imposing it at an earlier age would assume how the converted duration is allocated over time.

Georgia HOPE scholarship

Cornwell et al. (2006) estimate Georgia HOPE effects on first-time freshman enrollment using a state-year panel from 1988–1997. The reference inputs include log-enrollment effects of 0.086 for public four-year colleges, 0.132 for private four-year colleges, and -0.045 for public two-year colleges, together with a reduction of 560 in recent high-school graduates leaving the state for college.³⁹ These are enrollment flows across successive cohorts, not follow-up outcomes for one entering class. The reference conversion transforms each log coefficient using $\exp(\beta) - 1$, applies pre-policy sector counts and its recipient denominator, and accounts for reduced out-of-state attendance. This gives net enrollment responses of 0.055988 in four-year colleges and -0.014674 in two-year colleges. With durations of two and one years, the schooling input is $2(0.055988) - 0.014674 = 0.097302$. L2 imposes that total at age 20, equally averaging the calibrated 1975–1979 birth cohorts. There is no L3: assigning the sector sum of 0.041314 to age 19 would allocate the assumed duration, rather than incorporate a separately reported attainment window.

Social Security student benefits

Dynarski (2003) estimates how Social Security student-benefit eligibility affects college attendance and completed schooling. Table 3 reports 0.564 additional completed schooling years by age 23. By age 28, the lower-bound treatment of attrition gives 0.679 years; excluding attriters gives 0.754 , and assigning their last observed schooling gives 0.727 . L2 retains the lower-bound estimate used for L1, imposing $\Delta H(28) = 0.679$. L3 adds $\Delta H(23) = 0.564$ exactly while retaining the same age-28 convention. These are two actual observation ages, so no enrollment-duration conversion or assumed division across years is required. The 0.754 and 0.727 estimates are alternative treatments of missing outcomes at age 28, not additional follow-up periods, and are not imposed alongside the selected lower-bound input.

³⁸Graduation-age weights are 0.065207 , 0.575880 , and 0.358913 ; each graduation cohort receives one-quarter weight. The second-year restriction therefore uses ages 19–21. These are calibration weights, not additional source effects.

³⁹These values follow the Hendren and Sprung-Keyser replication. The accessible 2004 working paper differs from the published specification, including the private four-year coefficient; it does not independently verify every retained value.

College appropriations: spending

Deming and Walters (2017) estimate how public-college resources affect institutional enrollment using institution-year data from 1990–2013. The reference input comprises enrollment elasticities of 0.304, 0.796, 0.845, and 0.830 in the spending-change year and the next three years. Following the reference calculation, I sum these coefficients, divide by the 2.02 persistence adjustment, and scale by \$1,000 relative to average spending of \$29,856.23. The resulting schooling-equivalent input is 0.046013 years. L2 imposes $\Delta H(22) - \Delta H(18) = 0.046013$. Ages 18–22 place the aggregate enrollment response on the modeled schooling path; the source does not follow an individual birth cohort through those ages. L3 retains the four reported lag responses separately. Applying the same conversion to each gives annual schooling-equivalent targets of 0.005041, 0.013199, 0.014011, and 0.013762 at modeled ages 19–22, as fixed point targets, while the L2 total stays exact. This transports the institutional policy-time sequence to the representative population's acquisition years; it does not recover an individually observed cohort path. The reference calculation already uses all four coefficients in its sum, so L3 adds their separate dynamic restrictions rather than previously omitted coefficients.

Early Head Start

Hess (2025) estimates the effects of county access to Early Head Start on subsequent educational attainment. Table 1.2 distinguishes partial from full exposure and population intent-to-treat from participant effects. For full exposure, the retained bachelor's-completion estimate is 0.168 for participants, or 16.8 percentage points; the corresponding intent-to-treat estimate is 0.012. L2 models the BA+ outcome directly and imposes a weighted age-22 gap of 0.168 over the calibrated 1989–1999 birth cohorts. It does not multiply the coefficient by schooling years. Age 22 is the reconstruction's representative completion endpoint: the source pools adult attainment from linked Census–ACS records and limits its exposure cohorts so that college graduation can be observed from that age. There is no L3 because no additional comparable age-specific BA+ estimate is retained for the same exposure group. Partial-exposure estimates, sex-specific effects, and the pooled some-college outcome change the estimand rather than adding another BA+ observation age.

Texas community-college tuition

Denning (2017) estimates the effects of community-college district access and its associated tuition reduction for Texas high-school graduation cohorts from 1994–2005. The enrollment inputs cover the high-school graduation year and each of the next six years. The community-college probability effects, in time order, are 0.032, 0.036, 0.027, 0.024, 0.0095, 0.0083, and 0.014. The university effects at the same times are –0.00045, 0.010, 0.0082, 0.0089, 0.017, 0.0068, and 0.0082. Online Appendix Table A4 reports bachelor's-completion effects of 0.0033 after six years and 0.0078 after eight years. The reference schooling conversion sums the seven annual enrollment responses across sectors, obtaining 0.209450 years, and adds $2(0.0078 - 0.0033) = 0.009000$ years for the later completion increment. L2 therefore imposes $\Delta H(26) = 0.218450$, using a graduation-age mapping of 18. L3 preserves that equality and fits the earlier cumulative point sums: 0.031550, 0.077550, 0.112750, 0.145650, 0.172150, 0.187250, and 0.209450 years at assigned ages 18–24, respectively. The eight-year completion input supplies the age-26 endpoint; no year-seven enrollment effect is imputed. L3 thus retains separately reported timing under the reference enrollment-to-schooling conversion.

Adult Pell Grant introduction

Seftor and Turner (2002) estimate the effect of introducing federal aid for adult students on current college enrollment among people ages 22–35. The retained probability effects are 0.015 for men and 0.014 for women. The reference calculation weights them by sample sizes of 47,407 and 62,405, giving a pooled enrollment response of 0.0144317, and assumes two additional schooling years per induced enrollment. L2 imposes $\Delta H(30) - \Delta H(28) = 2(0.0144317) = 0.0288634$. Ages 28 and 30 are representative placements of this assumed duration; the source does not observe the same adults' cumulative schooling at these two ages. There is no L3. Estimates for different age groups or policy periods do not supply a later observation of the same target population, and dividing the two-year conversion into separate annual effects would add an assumption rather than an observed window.

College appropriations: tuition

Deming and Walters (2017) also estimate institution-level enrollment responses to tuition changes over 1990–2013. The four retained log-enrollment/log-tuition coefficients are -0.016 , 0.066 , 0.031 , and -0.073 in the current year and the next three years. The reference transformation divides their sum, 0.008 , by 2.02 and multiplies by $1,000/8,599.86$, yielding 0.000460519 schooling years. L2 imposes $\Delta H(22) - \Delta H(18) = 0.000460519$, preserving the reference calculation's policy-direction convention.⁴⁰ L3 adds the separate converted lag targets -0.000921 , 0.003799 , 0.001785 , and -0.004202 for modeled annual acquisition at ages 19–22, as fixed point targets, while retaining the exact L2 sum. As in the spending calculation, this is conditional on transporting institution-level policy time to the representative age path. The coefficients already enter the reference total; the added restriction preserves their sequence.

HOPE tax credit

Turner (2011) estimates enrollment responses to tax-based student aid among people ages 18–19. The reference inputs are a 0.235-percentage-point increase in first-year full-time enrollment per \$100 of aid and a 0.210-percentage-point increase in second-year persistence conditional on completing the first year.⁴¹ Using the reference subsidy of \$1,104 and 1.5 additional years for the second margin gives $(1,104/100)[0.00235 + 1.5(0.00210)] = 0.060720$ schooling years. L2 imposes $\Delta H(22) - \Delta H(18) = 0.060720$. The four-year age interval is the retained placement of this converted total, not four observed years of individual attainment. L3 preserves the first-year input separately as $\Delta H(20) - \Delta H(18) = (1,104/100)(0.00235) = 0.025944$, while retaining the L2 total. The remaining 0.034776 follows the same 1.5-year conversion of the second margin. The first component's placement in the earlier age block and the reference combination of conditional risk sets are maintained mappings, not separately estimated unconditional cohort effects. The source's overall enrollment effect, 0.320 percentage points per \$100, and its both-years indicator, 0.372 percentage points, overlap the retained margins and are not extra follow-up observations.

The eleven HOPE/AOTC contrasts use the finite policy-unit conversion in Appendix E.4.1. For the ten kink estimates, a fixed schedule displacement calibrated to one point-estimate credit dollar gives an attendance

⁴⁰The positive input preserves the released Hendren and Sprung-Keyser tuition-cut coding. Applying a literal negative tuition change would reverse its sign; that is an alternative specification, not additional timing information.

⁴¹The conditional second-year input follows the Hendren and Sprung-Keyser replication. The accessible 2010 working paper reports a different, unconditional second-year specification; it is not an independent verification of this input.

response of β_i/φ_i ; the two-year conversion gives $2\beta_i/\varphi_i$ schooling years. The assumed acquisition years are 19–20 for the age-19 cell and 20–21 for the age-20 cell. Only current attendance is observed: neither the second year nor contemporaneous sector outcomes supplies L3. The case profiles below retain each coefficient, first stage, and converted restriction. All use the common finite anchored path support and corresponding valuation schedules. The simulated-instrument coefficient is already per credit dollar and is treated separately.

HOPE credit: independent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 3×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1156229. The retained two-year duration converts it to 0.000051893 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1987–1988 ACS cohorts.

HOPE credit: dependent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is -5×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1386706. The retained two-year duration converts it to -0.000072113 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1987–1988 ACS cohorts.

HOPE credit: dependent single filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 6×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1441341. The retained two-year duration converts it to 0.000083256 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1987–1988 ACS cohorts.

HOPE credit: joint filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 12×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.0731243. The retained two-year duration converts it to 0.000328208 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1987–1988 ACS cohorts.

HOPE credit: joint filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2007. The signed input is 4×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.0638205. The retained two-year duration converts it to 0.000125352 schooling-equivalent years. L2

imposes this converted total through the joint cohort–age operator in Equation (60), using national 1987–1988 ACS cohorts.

Tuition deduction: single filers, phase-start

Hoxby and Bulman (2016) study college attendance around federal tuition-deduction thresholds. For single filers at the deduction’s start, the reference calculation retains a current-attendance effect of 0.005, or 0.5 percentage points, from its Table 5 input. The assumed two-year duration gives 0.010 schooling years, and L2 imposes $\Delta H(23) - \Delta H(21) = 0.010$. The source also reports Table 7 effects of (0.020, 0.014, 0.008, –0.011, 0.007) in years zero through four relative to expected on-time high-school graduation. These are actual additional reported windows, but they are not admitted as nested L3 in the current calculation. The Table 7 year-one coefficient differs from the retained Table 5 input, and the common sample/specification and age mapping have not been reconciled. The full five-value sequence is preserved as a separate timing exercise; it does not silently replace the L2 input or its observation operator.

Tuition deduction: joint filers, phase-start

Hoxby and Bulman (2016) report a current-attendance effect of –0.003, or –0.3 percentage points, for joint filers at the tuition-deduction start in the input retained by the reference calculation. Applying its two-year duration gives –0.006 schooling years. L2 imposes $\Delta H(20) - \Delta H(18) = -0.006$. Table 7 separately reports attendance effects of (0.016, –0.003, 0.010, 0, –0.001) in years zero through four relative to expected on-time high-school graduation. Although the year-one coefficient agrees with the retained input in this contrast, the dynamic sequence is an attendance measure, whereas L2 constrains a two-year schooling increment. A compatible mapping that adds these observations while preserving the original restriction is not established. There is therefore no admitted L3; all five reported values remain in the separate timing exercise. The source clock is expected graduation time, not years since a policy event.

HOPE and Lifetime Learning tax credits

Long (2004) studies the introduction of the HOPE and Lifetime Learning tax credits using repeated cross sections. The reference input comes from current college enrollment among people ages 18–24 in Table 3.11: an odds ratio of 0.9342 and a baseline enrollment probability of 0.463. The retained reference approximation is $(0.9342 - 1) \times 0.463 = -0.0304654$ for the enrollment change. Multiplication by the assumed two schooling years gives –0.0609308, so L2 imposes $\Delta H(23) - \Delta H(21) = -0.0609308$. The odds-ratio conversion is the reference approximation, rather than an exact conversion of a logistic odds ratio into a probability difference. Ages 21–23 place the converted total on the model path; they do not describe a two-wave source observation. There is no L3 because comparisons across calendar years or age groups do not supply repeated attainment observations for the same cohort.

AOTC: dependent single filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is -2×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.2164388. The retained two-year duration converts it to –0.000018481 schooling-equivalent years. L2

imposes this converted total through the joint cohort–age operator in Equation (60), using national 1991–1992 ACS cohorts.

AOTC: dependent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is 43×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.2019312. The retained two-year duration converts it to 0.000425888 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1991–1992 ACS cohorts.

AOTC: joint filers, phase-out end

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is -7×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1146557. The retained two-year duration converts it to -0.000122105 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1991–1992 ACS cohorts.

AOTC: joint filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is 7×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.1074026. The retained two-year duration converts it to 0.000130351 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1991–1992 ACS cohorts.

AOTC: independent single filers, phase-out start

Bulman and Hoxby (2015), Table 14, estimate current-attendance effects for ages 19–20 in 2011. The signed input is 1×10^{-6} probability units per income-margin dollar; the matching credit first stage in Table 13 is 0.2268326. The retained two-year duration converts it to 0.000008817 schooling-equivalent years. L2 imposes this converted total through the joint cohort–age operator in Equation (60), using national 1991–1992 ACS cohorts.

AOTC: simulated instrument

Bulman and Hoxby (2015), Table 16, report attendance responses of 1.4×10^{-5} per actual credit dollar at age 19 and -1.6×10^{-5} at age 22 over 2006–2011. L2 retains the age-19 estimate. Under the two-year conversion, the fixed one-dollar input is 0.000028 schooling-equivalent years, constraining the average $\Delta H_c(20) - \Delta H_c(18)$ across the 1987–1992 cohorts. This maps two assumed acquisition years onto finite logistic levels; it does not treat the second year as observed. The direct-credit component is one dollar because the source coefficient is already second stage. Appendix E.4.1 documents the correction from approximately 0.38, retained in every repetition. The age-22 result concerns different cohorts and does

not form a longitudinal L3 for the age-19 population. L1 remains the reported welfare calculation, so its comparison with L2 includes the unit correction.

Algorithmic college placement

Bergman et al. (2026) randomize algorithmic rather than test-only course placement at seven New York community colleges. In the June 2026 version, Table 3 reports a pooled effect of 0.562 additional college credits earned over the available follow-up. The entering cohorts have different follow-up lengths through 2018–2019; this is not a common two-year endpoint. The reconstruction divides the pooled estimate by 24 and imposes $\Delta H(23) - \Delta H(21) = 0.562/24 = 0.023417$ in L2. Ages 21–23 are a representative placement of the converted increment. The source also reports 1.717 additional earned credits for the more mature fall-2016 entering cohort, observed for roughly two and a half years. This is not admitted as L3 because it changes the cohort relative to the pooled L2 estimate, rather than adding a later observation of the same target. First-term remedial or college-level course outcomes likewise cannot be imposed as total earned-credit stocks without another conversion. Thus, the main calculation has L2 only.

Tuition deduction: single filers, phase-end

Hoxby and Bulman (2016) provide the tuition-deduction phase-out-end contrast for single filers. The current-attendance coefficient retained in the reference Table 5 input is zero. With the same two-year duration conversion, L2 imposes $\Delta H(20) - \Delta H(18) = 0$; this does not impose a zero effect at every subsequent age. Table 7 reports attendance effects of (0.008, 0.015, 0.008, 0.009, –0.002) in years zero through four relative to expected on-time high-school graduation. These values are preserved in the separate dynamic exercise. There is no admitted L3 because the year-one coefficient of 0.015 differs from the retained zero input, and the correspondence of the source specifications and observation operators remains unresolved. The source’s additional windows are therefore documented rather than described as absent, but they are not imposed as nested restrictions on the existing L2 path.

Tuition deduction: joint filers, phase-end

Hoxby and Bulman (2016) report the tuition-deduction phase-out-end contrast for joint filers. The retained current-attendance input is zero, which the reference two-year conversion also maps to zero schooling years. L2 imposes $\Delta H(20) - \Delta H(18) = 0$. The source’s Table 7 sequence is (0.010, 0, 0.015, 0.005, 0.002) in years zero through four relative to expected on-time high-school graduation. Its year-one coefficient agrees with the retained zero, but an attendance sequence is not the same object as L2’s converted two-year schooling increment. A compatible nested mapping has not been established. There is consequently no admitted L3; all five source values remain in the separate timing exercise. The zero L2 increment leaves later continuation unrestricted except by the maintained path and empirical-support conditions.

Cannabis legalization: education channel

Li (2026) estimates recreational-cannabis legalization effects on bachelor’s completion among first-time, full-time college entrants, using entering cohorts from 2006–2017 and exposure after college entry. The reported effects are –0.016, –0.011, and –0.018 on completion within four, five, and six years, respectively.

The education-channel reconstruction assumes two additional schooling years per bachelor's completion, giving targets of -0.032 , -0.022 , and -0.036 schooling years. With entry assigned to age 18, these correspond to ages 22, 23, and 24. The primary result is L3: it retains the six-year target exactly and adds the four- and five-year targets as fixed point targets in the constrained minimum-distance criterion, equally averaging the calibrated 1988–1999 birth cohorts. The six-year-only restriction is stored as an L2 diagnostic and is not presented as a separate main result. The source's three-year outcome for two-year institutions concerns a different entrant population and is not added to the bachelor's-completion path. Thus, the primary reconstruction uses all three comparable BA windows without combining unlike credential samples.

E.3 Welfare Witnesses and Information Layers

For finite-policy calculations, let $\mathcal{B}_{i\ell}$ be the saved finite bank for calculation i and layer ℓ . Crossing MVPF one at positive cost requires a path h^- with $N_i(h^-) > 0$ and $W_i(h^-) < N_i(h^-)$ and another path h^+ with $N_i(h^+) > 0$ and $W_i(h^+) > N_i(h^+)$. Both satisfy the same retained information and accounting restrictions. A harmful-costly versus self-financing comparison instead pairs $W_i(h^-) < 0$, $N_i(h^-) > 0$ with $W_i(h^+) > 0$, $N_i(h^+) \leq 0$. The replication materials identify each complete path and its paired welfare values; they do not combine marginal extrema. Table 3 supplies the separate component extrema; they are descriptive summaries and cannot be combined into a ratio.

All main L3 applications take the selected source estimates as fixed point inputs and retain the L2 equalities. Adams, Michigan, CUNY, Social Security student benefits, and HOPE tax credit admit verified exact matches. Florida, Wisconsin, Kalamazoo, public-college spending, tuition caps, Texas tuition, Tennessee Promise, and the cannabis education channel use the same minimum-distance rule when the added point values cannot all be matched exactly by the retained family. Each of these eight searches currently returns one best-fitting path with nonzero residuals. The original source estimates are not changed to make the family fit. Tables E5, E6, and D5 disclose the discrepancies. Their zero-width fitted welfare summaries describe conditional selections, not absence of statistical uncertainty or certified point identification. Source standard errors enter only the separate input-uncertainty exercises.

A main L3 restriction must preserve a time window actually reported by the source while retaining the education-conversion rule used in L2. Wisconsin qualifies because Table 5 reports semester credit effects and cumulative totals. Its annual flows sum adjacent observed semesters under the retained credits-to-schooling conversion. The later-cohort third-year restriction uses cohort 2 alone because cohort 3 is not observed that long; no missing year is imputed. The entry-age mapping, cohort weights, and common response parameters are already maintained in L2. For the separate uncertainty sensitivity, annual-flow SE upper bounds sum marginal semester SEs where covariance is unavailable; these SEs do not weight or relax the main point fit. The circle marker therefore identifies added source timing, not a reconstruction free of calibration assumptions.

L3 may preserve converted source dynamics rather than directly measured annual schooling. The two college-appropriation calculations use four reported policy-lag responses, retaining the reference persistence and enrollment conversion and assigning the resulting sequence to modeled acquisition years. This requires transport from institutions to a representative population. Adams, Michigan, and HOPE tax credit likewise retain explicit timing or risk-set mappings. These assumptions are recorded alongside the source information; the L3 label does not establish their validity. Tennessee Pell and Georgia HOPE remain L2 only because their proposed extra years merely split a single sector-entry observation using assumed durations. They add no separately observed window or dynamic response.

Every retained main L3 row, including each best-fit selection, is included in L2, and extrema are recomputed from the saved rows. Component-width ratios therefore cannot exceed one apart from numerical tolerance. Nesting of recorded banks does not establish that either exhausts the continuous feasible set. Failure to verify a candidate remains a search outcome. A span between finite endpoints does not assert that every intermediate value has been verified. No distribution over candidates or over the continuous continuation class is specified.

E.4 Monetary Units and Reference Components

Table 3 uses a positive reference program-cost component as a display denominator wherever its provenance is established. The same denominator divides willingness to pay and net cost in L1 and both reconstructions of a calculation. It is not the path-dependent net cost, and this scaling leaves each MVPF unchanged. The component follows the native account, such as a scholarship's discounted transfer or a tax credit's direct revenue cost; it need not represent an identical policy dose across applications. In tax-schedule calculations, the original policy unit can be a small change in a credit or deduction. A large normalized component can therefore combine a broad continuation range with a small reference expenditure. It should not be read as a large absolute dollar amount for an average program participant. The replication materials document the normalization, source, and monetary base for all 38 calculations. The underlying point-input ranges and their units are preserved separately from the rounded table.

The reference component pair is taken from the same source account used to identify L1, with unrounded ledger values retained when they match that record. Reported reference ratios are not silently replaced by ratios recomputed from rounded components. For Early Head Start, the library account gives $E^{\text{ref}} = 21,944$, $W^{\text{ref}} = 3,906$, and $N^{\text{ref}} = 11,216$. Its fiscal sum is $24,257 + 1,540 - 10,478 - 4,103 = 11,216$: program cost plus the community-college cost change, less university savings and additional taxes. The dissertation reports net cost of about 12,755. Its displayed calculation appears to add the 1,540 community-college term again after including it in the net education-cost change. I use the library ledger consistently and retain the dissertation amount as an alternative account, rather than mix their components. The source price year remains unresolved.

Early Head Start is a public-data reconstruction rather than a source-total alignment. The 2017–2019 national ACS provides age-specific BA-versus-HS/GED wages, including zero wages, with person weights and documented dollar adjustments. The reconstruction uses 2000 dollars and values the retained 0.168 ATT degree input through its age-specific path. Source age placement at 22 is a modeling convention, not an exact age identified by the reported adult outcome. No early signed coefficient is invented for this childhood intervention and no HSK schooling-flow charge is added.

The program-cost proxy uses the available FY2001–2008 statutory EHS allocation per funded slot, valued for three years; it is \$28,158.29 in discounted 2000 dollars. It is an average funded-slot proxy rather than the experimental sample's measured marginal cost. The retained public-source college-price reconstruction gives a fixed private education payment of \$11,463.05 and a public college-cost change of $-\$5,680.05$. Thus $W = E - 11,463.05$ follows the source convention of gross earnings in WTP, and $N = 22,478.24 - 0.187E$. Using gross earnings in WTP while also valuing fiscal revenue is an inherited convention, not a universal private-disposable-income rule. The public college prices require an explicit 2021 publication-price convention before conversion to 2000 dollars because their exact source vintage is unavailable. A constant 0.168 degree gap gives \$78,683.28 and an age-22 step gives \$77,910.64; both differ

from the reported \$21,944. No multiplier is used to conceal that difference. The alternative program-cost exercise uses the FY2001 slot proxy. Input repetitions condition on the fixed public wage and cost schedules.

Algorithmic college placement retains the June 2026 source components, $W = 220$ and $N = -90$. The reconstruction retains source WTP of 220 and extends only the fiscal account using the signed education-related income mapping. It does not add another private earnings benefit. It remains an explicitly marked extension in the full ordering, rather than evidence that continuation alone overturns an otherwise identical account. Cannabis is separate because it has no whole-policy reference and values only the education channel.

E.4.1 Marginal Tax Inputs in a Finite Policy Experiment

The eleven HOPE/AOTC reconstructions use the same finite, anchored logistic attainment levels as the other applications. Their source coefficients require an additional policy-unit conversion. Table 14 of Bulman and Hoxby (2015) reports changes in attendance–income slopes at tax-schedule kinks for ages 19–20, in 2007 for HOPE and 2011 for AOTC. Table 13 supplies the corresponding credit–income first stages. Let β_i be the attendance coefficient and $\varphi_i > 0$ the credit first stage, both signed in the reference policy direction. I fix a schedule displacement $q_i = 1/\varphi_i$, calibrated to raise the credit by one dollar at the point estimates. I assume that the local attendance and accounting responses transport linearly over this displacement. This specifies a finite counterfactual; the source derivative alone does not identify its exact effect. Keeping the reference two-year enrollment-to-schooling conversion gives the input

$$m_i^{YS} = 2\beta_i q_i = \frac{2\beta_i}{\varphi_i}. \quad (59)$$

For the Table 16 simulated-instrument estimate, the attendance coefficient is already per actual credit dollar, so $q_i = 1$ without another first-stage division. These conversions define schooling-equivalent inputs, not directly observed completed-schooling effects.

The source ages determine the cohort calibration; assumed duration is stated separately. Nationwide ACS profiles use the 1987–1988 birth cohorts for HOPE, 1991–1992 for the AOTC kinks, and 1987–1992 for the simulated-instrument input observed at age 19 during 2006–2011. Equal cohort-cell weights are maintained because the source regression weights have not been recovered. In the annual acquisition convention, current enrollment at age a contributes to schooling in years a and $a + 1$ under the two-year duration assumption. The converted total therefore restricts $\Delta H_c(a + 1) - \Delta H_c(a - 1)$. For HOPE, the joint restriction is

$$\frac{1}{2} [\Delta H_{i,1987}(21) - \Delta H_{i,1987}(19)] + \frac{1}{2} [\Delta H_{i,1988}(20) - \Delta H_{i,1988}(18)] = m_i^{YS}. \quad (60)$$

The AOTC kink operator replaces these cohorts by 1991 and 1992. The simulated-instrument operator averages $\Delta H_c(20) - \Delta H_c(18)$ across its six age-19 cohorts. The second acquisition year is a duration allocation, not another source observation, and does not create L3. All tax paths assume no attainment effect through age 17, allowing first effects at age 18. This exposure restriction is maintained rather than estimated. Earlier placements at ages 21–23 or 25–27 are not treated as source observation windows. Independent-filer earnings schedules beginning at 25 remain valuation assumptions. The age-22 simulated-instrument result refers to a different cohort composition and remains outside the main L3 set.

The restored paths retain the current full-population ACS calibration, the transported joint parameter support, completion-timing restrictions, nondecreasing levels, and the one-schooling-year annual acquisition cap. A common finite treated parameter triple is used across each application’s cohort components, with their

separate baseline anchor levels. Thus both H^0 and H^1 are finite attainment paths throughout the comparison. For example, the HOPE independent-single contrast has $\beta_i = 3 \times 10^{-6}$ and $\varphi_i = 0.1156229$. Its fixed schedule displacement is about 8.649 income-margin units, and its two-year input is approximately 0.000051893 schooling years. This is a local-linear calibration of a one-dollar credit change, not the effect of adopting the whole tax program. Matching an ACS support restriction does not establish the plausibility of every policy response inside that support.

The monetary conversion follows the same fixed displacement. For the ten kink contrasts, the non-earnings components become $w_{0i}^F = q_i w_{0i}^{\text{source}}$ and $n_{0i}^F = q_i n_{0i}^{\text{source}}$. The per-schooling-year earnings and fiscal schedules retain their units and values. The account evaluates the finite cohort-average gap, $\overline{\Delta H}_i(a)$, as

$$W_i^F(h) = w_{0i}^F + \sum_a w_i(a) \overline{\Delta H}_i(a). \quad (61)$$

The corresponding net cost is $N_i^F(h) = n_{0i}^F + \sum_a n_i(a) \overline{\Delta H}_i(a)$. The reported L1 ratio and interval are unchanged. Its WTP, net cost, and program-cost display divisor receive the same normalization, so their normalized values are unchanged as well. For the simulated instrument, I retain the unit correction that sets the direct credit payment to one dollar rather than the first-stage amount of approximately 0.38. At each input repetition, the correction adds $1 - \text{tax_value}$ to net cost and $(1 - \text{tax_value}) \min(1 - \beta_i, 1)$ to the non-earnings WTP component. Its L1–L2 comparison therefore includes this accounting correction as well as continuation.

All 1,000 archived joint input IDs are retained. The point normalization q_i stays fixed across repetitions; education, tax-credit, cost, and earnings inputs still vary together according to their registered scheme. In particular, five negative first-stage draws are retained, rather than discarded or divided into the education coefficient. Their credit changes are not constrained to equal one dollar away from the point estimates. The simulated-instrument repetitions remain a constructed reference-input exercise, not draws from the source's sampling distribution. Converting monetary units from dollars to cents leaves the same physical policy experiment, path constraints, and MVPF unchanged. Changing the size of the finite policy experiment is instead a substantive sensitivity, because the ACS support restricts levels rather than responses per dollar.

A limited dose sensitivity repeats the finite-path screening for point credit changes of 0.50, 1, and 2, scaling the education input and non-earnings account components together. At all three doses, each tax calculation has verified negative-WTP, positive-cost paths and positive-WTP, nonpositive-cost paths. This checks the existence of these conflicting classifications on the tested grid; it does not establish invariant endpoints or welfare bounds across policy sizes. Currency-unit checks separately preserve the physical policy experiment and recover the same schooling restriction and MVPF.

The replication materials document the source coefficients, first stages, joint age-cohort operators, equation residuals, and complete finite paths. WTP and net-cost extrema are finite. Some MVPF endpoints nevertheless remain infinite because verified finite paths approach zero net cost from above with nonzero WTP. Those ratio limits are distinguished from self-financing states. Equal endpoint pairs retain tied ranks; no additional responsiveness cap or path probability is introduced to separate them.

E.5 Candidate-Based Lower-Endpoint Ordering

Figure 9 ranks all 37 calculations with a reference MVPF. Twelve use L3: Florida, Adams, Wisconsin, Kalamazoo, Michigan, CUNY Pell, Social Security student benefits, both college-appropriation calculations, HOPE tax credit, Texas CC tuition, and Tennessee Promise. The other twenty-five use L2, including the

eleven finite tax reconstructions just defined. Early Head Start uses the public-data reconstructed account, and algorithmic placement retains its marked earnings-channel extension. Cannabis has no whole-policy reference. Table E2 reports every rank and source window; the restricted 24-case ordering remains in the data.

For each selected finite-path specification, let $\underline{M}_i$ and $\overline{M}_i$ be the lower and upper endpoints of defined MVPFs, including audited one-sided limits and the self-financing convention. Let r_2 round finite numbers to two decimals and leave infinities unchanged. The display orders policies lexicographically in ascending order of

$$(-r_2(\underline{M}_i), -r_2(\overline{M}_i)). \quad (62)$$

Equal endpoint pairs receive the same competition rank, one plus the number of strictly preferred calculations. Reference ranks compare the recorded L1 values, preserving ties. Repeating the full ordering with three- or four-decimal endpoint rounding leaves all ranks unchanged in the current results. This numerical stability check does not establish global extrema: a newly discovered adverse path could change an ordering based on candidate minima.

An open marker replaces only that policy's selected L3 set by L2, holding other selected sets fixed. Eleven of the twelve added-information cases improve under the rule and one remains unchanged. The arrows describe the rule's response to restrictions, not an increase in the policy's true welfare. Fourteen ranked calculations admit paths with nonpositive WTP and nonpositive net cost. Those unassigned-ratio paths, rather than the calculations containing them, are omitted from the ordering; their monetary components remain in the collection table. The rule assigns neither artificial finite values to infinity nor probabilities to paths, and it is not a social-welfare or certified maximin ranking.

E.6 Reported Timing and Its Use in Reconstruction

Table E3 distinguishes source timing coverage from imposed path restrictions. It counts distinct age or individual-follow-up windows for the relevant contrast and outcome family. An overlapping two-year age interval counts once; additional credentials, estimators, subgroups, and tax contrasts at that window do not create more horizons. Institutional policy lags and college-grade risk sets retain their own clocks. The count measures neither precision nor independent observations.

Among the 37 calculations with L1, 17 have one verified window, five have two, and 11 have at least three. Four use other clocks: Georgia's state-year freshman flows, the two Deming–Walters institution-level calculations, and Turner's college-grade and pooled enrollment outcomes. The published Tennessee HOPE paper now verifies one initial-college-choice window. Its 16-month eligibility deadline is not a common observed follow-up, and its alternative sectors and specifications do not constitute additional attained-education times. The unranked cannabis application reports completion at three horizons. All 38 calculations therefore receive a coverage record, and all 37 calculations with L1 enter the expanded ordering with their specification groups identified.

Several multiple-window studies still require a population, outcome, or specification bridge before their results constrain one path. Absence of L3 does not imply that a source lacks dynamic evidence. Nor does L3 imply that the reference valuation ignored an estimate: Texas can use separately reported timing while retaining coefficients already used in its aggregate conversion. Adams also enters L3 through its separately reported early entry input, conditional on the temporal allocation in Appendix D.7. The registry separates reference inputs, source windows, imposed restrictions, and their empirical or assumed origins, with source

locations and hashes. It does not treat the number of windows as a causal determinant of ranking changes. Several policy contrasts also share a study family.

E.7 Source Comparability and Numerical Verification

Additional coefficients constrain the same path only if their population, policy contrast, outcome, and clock can be reconciled with L2. The case profiles identify unresolved specification differences for tuition deductions, a changed observation operator for the AOTC simulated instrument, and different populations for the pooled and earliest-cohort algorithmic-placement estimates. Those comparisons remain separate specifications. DC tuition assistance retains an input traceable through L1, but its final publication coefficient and missing SE are not independently recovered from the available earlier paper. These source limitations remain distinct from numerical feasibility.

The main L3 fits retain Florida’s first-semester and annual credit totals and Wisconsin’s reported semester credit flows. Their full input sets lack registered joint draws, so no full-L3 repetition summary is imputed. Appendix F.4 reports a separate marginal-input sensitivity, which is not an uncertainty interval for the new minimum-distance selector. Tennessee uses eighteen table point estimates; the prior graphical interval screen is archived. There are thirteen admitted L3 specifications: five exact-match banks and eight best-fit selections.

For finite-policy calculations, all saved point candidates are independently evaluated at annual ages through 72, or through 68 for Tennessee, while income is valued only over each registered earnings horizon, checking their anchors, nondecreasing attainment, credential nesting where required, encoded moments, empirical support, and paired welfare values. Every verified L3 candidate also enters L2. Searches supplement shape grids with multiple-start component optimization and equations for zero WTP, WTP equal to cost, zero net cost, and selected ratios. At a zero-cost root with nonzero WTP, decreasing positive-cost targets check the one-sided limit. These checks prevent a sparse grid from obscuring narrow positive-cost branches, but do not certify global finite endpoints or connectedness of the feasible set. The same checks apply to registered input repetitions. An unsuccessful search remains distinct from evidence of infeasibility.

The numerical checks use each calculation’s current calibration and observation operator. In particular, Wisconsin’s point paths and all 1,000 L2 repetitions use the joint cohort–follow-up operator in Appendix E.2. Its input sequence, external baseline, and welfare account are held fixed. The separate replication guide identifies the current records and archived specifications.

E.8 The Complete Path Display

Figure E2 displays comparison and treated schooling levels for the 35 single-schooling-equivalent calculations. Figure E3 covers STBF’s four joint effect paths, Early Head Start’s BA+ path, and Tennessee Promise’s three joint credential paths. Together they cover all 38 finite-policy calculations, including the three main applications. Black lines show H^0 . Blue and red boundaries and shading show the pointwise minimum and maximum H^1 in the saved L2 and admitted L3 sets. These ranges describe attainment without selecting paths by their welfare values. Tennessee’s L2 ranges and L3 fitted path satisfy credential nesting; the cannabis education account displays only its primary L3 best fit. For the eight selected fits, coincident upper and lower traces represent one retained solution.

The envelope edges can switch candidate identity across ages and credentials and need not themselves form feasible paths. They summarize the current finite search, not a probability distribution, sampling interval,

or certified boundary of all possible continuations. Complete candidate paths and their paired welfare values remain in the replication records. All schooling panels share one age and schooling scale, and all credential panels share one age and attainment-share scale. The source profiles in Appendix E.2 and the expanded operators distinguish observed follow-ups from their assigned ages. Neither a later crossing nor a widening modeled gap establishes an individual behavioral mechanism or shows that the source researchers overlooked dynamics.

E.9 Point-Input Sensitivity across All Applications

Figure E4 and Table E4 compare all 38 calculations under alternative calibration, valuation, and numerical-search choices. The selected information set is L3 for thirteen calculations, including the cannabis education channel, and L2 for the other twenty-five. Both layers remain in the accompanying data where available. These counts differ from the L2-only comparison of the 37 cases with L1.

E.9.1 Changes to Calibration, Valuation, and Search

Narrower and wider support multiply the empirical covariance by 0.5 and 2 and transported timing deviations by their square roots. Source moments, anchors, natural level bounds, and the one-year annual schooling-acquisition cap are retained. The geographic alternative uses national profiles and support for the same cohorts; cases already nationally calibrated repeat that baseline. Tennessee retains its high-school-graduate denominator and source-control matching, while its late-completion allowance is varied separately. These exercises assess sensitivity to population transport; they do not estimate confidence levels.

The income alternatives multiply each path's net income and associated tax effects by 0.75 or 1.25. Fixed transfers and education-resource schedules retain their original values. Negative and positive portions of income are scaled together, without adding or subtracting another opportunity cost. For STBF, the multiplier applies after evaluating the nonlinear author income function. For Early Head Start and Tennessee, it applies to their disclosed public-data reconstruction.

Every alternative re-evaluates the relevant path rule. L2 and exact-match L3 banks retain monetary-extremum searches; the eight overidentified L3 applications instead refit the fixed-point minimum-distance criterion before valuation. Their denser check uses more optimizer starts and the denser feasible seed bank. A wider support can change a selected minimum even though it contains the old feasible paths; the minimizer set itself need not expand. The denser scalar L2 grid uses 2^{16} shape pairs, with supplemented joint pools for the two joint models. All common specifications have verified results. Annual monetary accounts are independently checked. These finite searches do not certify global extrema or uniqueness and are separate from the 1,000 input repetitions.

E.9.2 Which Welfare Comparisons Survive?

Positive-cost MVPFs cross one in 20 of the 38 main selected sets. The counts are 19 under narrower support, 21 under wider support, 19 with national calibration, 20 at the 75 percent income scale, and 19 at the 125 percent scale. The denser search also gives 20. Main and denser searches both find self-financing paths in 24 selected sets. Eight main L3 entries are best-fit singletons, so their apparent contraction reflects a maintained fitting selection as well as added observations. These counts concern saved numerical results and cannot establish that no other classification is feasible.

Eighteen main sets contain both a negative-WTP, positive-cost path and a positive-WTP, nonpositive-cost path. Fifteen calculations retain both types in every common specification. The corresponding count is 16 under narrower support, 19 under wider support, 18 under national calibration, 15 at the lower income scale, and 19 at the higher scale. Several tax cases share a source study, so these frequencies count calculations rather than independent replications. Functional form, population transport, fitting selection, and source identification remain maintained.

E.9.3 Source-Specific Checks

STBF replaces the unrounded time-enrolled input by the printed 0.360 years, keeping all three credential inputs. A separate diagnostic sets the experience-delay channel to zero within the author income function while retaining the joint education restrictions. It does not remove a separately estimated cost. All its found positive-cost MVPFs exceed one across the common and source-specific checks. The main range is approximately [1.25, 4.05].

Adams retains the explicit early-window allocation in its admitted L3. Its current main positive-cost range is approximately [0.614, 0.636], with positive WTP and positive net cost. Setting the negative portion of the completion-response template to zero changes income valuation without supplying an additional education observation. Private and public education-cost functions continue to move with the candidate path.

Tennessee refits the eighteen point targets under alternative source-control-level matching, national calibration, and late-completion allowances, as well as the common support and income changes. Every retained L3 fit in these checks is self-financing under its corresponding public-data account. Graphical interval-width variations are retired because no interval enters the new main criterion. No income multiplier is chosen to force agreement with the published total; the separately documented participant-unit bridge remains fixed within each comparison.

The eleven finite tax experiments additionally use point credit changes of 0.50 and 2, scaling education inputs and fixed monetary components together. Early Head Start substitutes the FY2001 funded-slot program-cost proxy for the FY2001–2008 average, with the same public wage schedule. Cannabis varies education-cost sectors, state weights, and the negative portion of the income response. Its main tax base is net income, and every specification concerns education alone. Monetary components from different paths are never combined to form a ratio. One-at-a-time changes do not establish robustness to all simultaneous departures.

F Conditional Input Repetitions and Additional Information

The uncertainty exercise separates variation across education paths from variation under the specified source-input and accounting perturbations. Each of the 38 policy calculations uses 1,000 prespecified L2 input repetitions and the same observation operators and response restrictions as its point-input calculation. L3 refits the same exact-match or minimum-distance rule where complete registered input perturbations are available; five cases retain point-input L3 only for their current full constraint sets. The geographic comparison curves, empirical support, education conversions, and specified valuation relationships remain fixed. Table 4 places the resulting MVPF endpoint summaries alongside available source estimates and reported 95 percent intervals. Table F1 reports willingness to pay and net cost separately, including cases in

which the ratio is poorly behaved near zero net cost. These are within-case comparisons: monetary years, policy units, populations, and the uncertainty included in the source accounts can differ across cases.

F.1 Conditional Endpoint Summaries

For repetition r , let $\widehat{D}_{i\ell}^{(r)}$ be the finite verified path bank for case i and information layer ℓ . For a path bank, the numerical search includes targeted zero-cost checks. For a retained best-fit singleton, both within-draw endpoints equal its fitted welfare value; no old interval-screen endpoints are carried over. Its lower and upper summaries, including numerically audited one-sided ratio limits, are denoted by $\underline{M}_{i\ell}^{(r)}$ and $\overline{M}_{i\ell}^{(r)}$. I report the order-statistic summary

$$\left[q_{.025} \left(\{ \underline{M}_{i\ell}^{(r)} \}_{r \in \mathcal{R}_{i\ell}} \right), q_{.975} \left(\{ \overline{M}_{i\ell}^{(r)} \}_{r \in \mathcal{R}_{i\ell}} \right) \right], \quad (63)$$

where $\mathcal{R}_{i\ell}$ contains repetitions with a verified finite-path set and a defined ratio. The valid count is always reported relative to the original 1,000 repetitions. For WTP and net-cost endpoints, the same quantile operation uses all repetitions with verified monetary components, including repetitions with no defined MVPF. Thus component and ratio summaries can have different usable counts. For the eleven tax calculations, the policy normalization in Appendix E.4.1 stays fixed across repetitions. Infinite entries from the positive-benefit, nonpositive-cost convention and from numerically audited one-sided positive-cost limits remain in the order statistics. When both components are nonpositive, the components remain reportable but that path is not assigned a self-financing MVPF. Component extrema are computed separately and are never combined to manufacture a ratio.⁴²

Equation (63) describes the conditional input exercise; it is not a demonstrated 95 percent confidence region for the entire continuous continuation set. Finite search can miss extrema, selected input draws can lack a verified candidate, and external calibration uncertainty is not resampled. Moreover, the input construction differs according to what the source releases. Registered joint source and accounting draws are retained where available. STBF, Early Head Start, algorithmic college placement, the simulated-instrument education input, and Tennessee Promise instead use researcher-constructed perturbations of reported education effects. DC tuition assistance varies economic inputs while its education effect remains fixed because a corresponding source SE is unavailable. The current HSK reconstruction replays 1,000 source-account repetitions for each of its 33 cases, retaining paired earnings-return inputs, including Georgia HOPE and adult Pell. The source education perturbation itself can still be conditional or researcher constructed. The case registry distinguishes these methods rather than labeling every row an original-study bootstrap.⁴³

The source comparison is likewise qualified by what can be recovered. Reported MVPF intervals are available for 34 of the 38 calculations. The main comparison uses the frozen Policy Impacts Library snapshot of 20 September 2026; for Tennessee Promise, Schoonover et al. (2026), Section 5.2.2, explicitly report both endpoints as infinite. No comparable reported interval was recovered for STBF, Early Head Start, or algorithmic college placement, and the author cannabis study does not supply an original MVPF calculation. These entries are left blank. The comparison preserves the source's reported method and account rather than relabel a conditional reconstruction as its published interval. Differences in width can reflect continuation, source inputs, dependence assumptions, valuation calibration, and coverage targets simultaneously.

⁴²Quantiles use the inverse empirical distribution without interpolating finite and infinite values. With 1,000 complete repetitions, endpoints use the 25th minimum and 975th maximum order statistics. Each ratio still pairs components from one path and draw.

⁴³Registered draw dependence is preserved; failures are not replaced. Newly verified L3 paths also enter the corresponding L2 sets. Search failures remain recorded without a claim of infeasibility. No distribution over paths is imposed.

F.2 Additional Information and Unsuccessful Searches

Additional timing and repeated inputs require separate availability checks. Table E1 identifies thirteen admitted L3 specifications and 25 calculations without one. All thirteen have verified point-input candidates; Table 4 records which also have joint input repetitions. Appendix E.7 explains the population, outcome, and timing requirements. Thus an absent L3 interval can mean that no additional restriction is admitted or that repeated inputs are unavailable, rather than that the source reports no dynamics.

Missing additional information and failure to find a compatible path have different implications. All 38,000 prescribed L2 repetitions complete; 37,952 yield finite-path sets. The 48 unsuccessful searches occur in the Social Security calculation and remain recorded without replacement or a claim of infeasibility. Ten HOPE/Lifetime Learning credit draws yield monetary components but no defined MVPF. Florida, Wisconsin, Kalamazoo, CUNY, and Social Security lack joint input mappings for their complete L3 constraints, so their expanded L3 sets have no main repetition interval. Appendix F.4 instead uses their available marginal uncertainty. Adams, Michigan, both appropriations calculations, and HOPE tax credit propagate 1,000 linked inputs through the same conversion and time or population mappings as their point reconstructions. These mappings remain fixed assumptions.

Tennessee has 1,000 verified L3 repetitions. Each perturbs the eighteen Table 2 point coefficients and refits equal-weight minimum distance, keeping the three perturbed latest-window inputs exact. The perturbation scales use displayed clustered SEs, with a 0.0005 floor for the two bachelor's coefficients whose SEs print as zero. The floor is an imposed reporting-precision convention, not a recovered standard error. The source's warning that clustered SEs may understate uncertainty therefore limits these repetitions. They do not recover the source permutation distribution or establish coverage for the best-fit selector. Public-college spending, Texas tuition, tuition caps, and cannabis likewise refit their point criteria for all 1,000 registered input draws each. Tennessee Pell and Georgia HOPE remain L2 only: splitting one entry observation by assumed duration supplies no additional source window.

F.3 Cannabis: An Education-Channel Welfare Account

The cannabis calculation values the education channel of the registered policy contrast without claiming to value legalization as a whole. Its retained bachelor's-completion effects at four, five, and six years after the source's reference event are -0.016 , -0.011 , and -0.018 . A fixed two-schooling-years-per-degree bridge converts these to -0.032 , -0.022 , and -0.036 schooling-equivalent years. The registered age mapping places them at model ages 22, 23, and 24, with age 18 as the last unaffected observation; it is a maintained mapping rather than individually observed exposure ages. L2 uses the last moment. L3 retains it exactly and minimizes the squared discrepancies from the three fixed point targets, without SE bands. The baseline uses the 14 adopting states and the 1988–1999 birth cohorts, with equal cohort weights and fixed state weights from the age-18 population. These state weights are external calibration weights, not recovered weights from the source's causal estimator. Appendix E.9 includes an equal-state point-calibration sensitivity.

Let $E_i = \sum_a D_i(a)\Delta y_i(a)$ be net education-related earnings, including the signed pre-completion and post-completion response, and let F_i be the discounted schooling-equivalent acquisition flow priced by the education-resource schedule. Under the fixed effective tax rate, $T_i = t_i E_i$. The education account is

$$W_i^{\text{edu}} = E_i - T_i - p_i F_i, \quad (64)$$

$$N_i^{\text{edu}} = g_i F_i - T_i, \quad (65)$$

where p_i is private payment and g_i is public education-resource cost net of that payment. The main schedule uses $t_i = 0.189$, $p_i = 5,997.02$, and $g_i = 8,557.30$, in 2000 dollars. No separate forgone-work cost is deducted. Private payments and public resource costs use the 2015 public four-year-college inputs in the HSK replication, aggregated by full-time-equivalent enrollment within states, deflated, and combined with the fixed geographic weights. Pricing a schooling-equivalent increment as one full-time-equivalent education year and treating average expenditure as marginal cost remain calibration assumptions. Attainment effects do not identify enrollment sector or actual time out of employment.

The paired components remain central. The main L3 search retains one best-fitting path, giving earnings NPV $-\$3,421.32$, WTP $-\$2,354.48$, and positive net education cost $\$47.02$. Its MVPF is -50.08 . These are paired values from that fitted path, not endpoints inherited from the previous interval screen. Alternative sector, state-weight, support, and income-response assumptions are refit separately in Appendix E.9. The complementary resource comparison remains $W_i^{\text{edu}} - N_i^{\text{edu}} = E_i - (p_i + g_i)F_i$, since the tax transfer cancels. A fitted singleton is conditional on the path family and target criterion, not evidence that the true education-channel value is known.

Other legalization components remain outside this account. For example, adding a specified direct noneducation cost C_i would give net cost $C_i + N_i^{\text{edu}}$, so, for a positive-benefit path, the largest such cost consistent with fiscal self-financing is $-N_i^{\text{edu}}$ and the net-benefit break-even cost is $W_i^{\text{edu}} - N_i^{\text{edu}}$. These are conditional accounting thresholds, not estimates of the actual policy budget. Cannabis excise revenue, health, crime, enforcement, and consumption value would require separate evidence before a total-policy MVPF could be reported. In particular, a fiscal surplus in the education account does not establish that legalization itself is self-financing.

The cannabis repetition results also have a narrower interpretation than a new microdata bootstrap. They extend the registered calibration using 15 equal-weight synthetic Rademacher clusters and an imposed cross-moment correlation parameter of 0.95, scaled by the retained marginal SEs. These are researcher input perturbations, not recovered cluster influence functions or a re-estimation of the earlier CBASE model. The resulting endpoint summaries in Tables 4 and F1 condition on that perturbation rule, the degree-to-schooling bridge, and the fixed education-cost schedule. Keeping this limitation explicit permits an education-related valuation while leaving unsupported noneducation components outside the calculation.

F.4 A Separate Sensitivity without Joint Input Draws

For Florida, Wisconsin, Kalamazoo, CUNY, and Social Security student benefits, Table F2 examines a simultaneous region formed from the reported marginal standard errors. For J encoded source moments, the normal Bonferroni critical value is $\Phi^{-1}(1 - 0.05/(2J))$. Each moment may vary within its estimate plus or minus that value times its SE. Wisconsin's annual credit sums and aggregate input use conservative SE upper bounds obtained by summing the relevant marginal SEs, since their covariance is unavailable. Florida includes the recovered first-semester coefficient. Under the marginal normal approximations, the union bound gives an approximate 95 percent simultaneous region for the encoded input means without requiring their covariance matrix. This approximation remains conditional on the source's population and attrition conventions; it does not resolve missing outcomes or transport between different samples. All encoded inputs, including the L2 endpoint, vary inside the region, while ACS calibration and accounting prices remain fixed.

The table reports the extrema among paths found within this input region. This exercise projects moment-compatible paths within a varying input region; it does not refit the minimum-distance selector at

every input vector and is therefore not a confidence or outer region for that selector. A finite search also need not recover the complete projection. The reported spans are therefore sensitivity results, not certified welfare confidence limits. They are kept outside the main input-repetition panels and the point-input ranking. This distinction permits use of the available marginal uncertainty without fabricating a joint draw distribution or assigning probabilities to continuation paths.

G Policy-Specific Male and Female Education Paths

This appendix uses published subgroup treatment effects to construct male and female continuations within seven policies. The purpose is to distinguish differences in the size of an education effect from differences in when education is acquired. No restriction presumes that women graduate later or face larger timing losses. These degree-channel exercises preserve the original subgroup outcomes; they do not replace the pooled-policy education inputs or welfare accounts used elsewhere in the paper.

G.1 Source Effects and the Reference Populations

Table G1 records the retained point inputs and the resulting standardized education valuations. The college-aid exercises use reported degree outcomes. For Adams, Table 7, Panel D, column 5 of Cohodes and Goodman (2014) reports effects on graduation from any four-year college within four years of high school. The sample contains the 2005–2008 high-school classes, rather than the 2005–2006 classes used for the pooled six-year result. The source test does not reject equality of the male and female coefficients. For STBF, Online Appendix Figure A4 of Angrist et al. (2022) reports six-year BA effects for the four-year applicant strata, with both estimates rounded to 0.08. The reported year-six gender contrast is 0.001, with a standard error of 0.005. Equal displayed coefficients therefore do not establish identical underlying estimates. The figure also reports earlier degree outcomes, but its samples differ by year. I retain the six-year inputs and do not combine the earlier values into a sex-specific L3 without an additional sample-transport assumption.

For Wisconsin, Table 7 of Goldrick-Rab et al. (2016) reports the first cohort’s administrative-sample four-year BA effect as 4.7 percentage points, with a treatment-by-male interaction of 0.0 points at the printed precision. Both subgroup inputs are therefore 0.047 in this conditional calculation. The gender level coefficient is not a treatment effect. This specification differs from the table’s smaller financial-aid sample and from the pooled credit-based account used in the collection. The table does not supply the covariance needed to recover a male subgroup standard error. For Michigan community-college tuition, Table 7, Panel B, column 7 of Acton (2021) reports the effects of in-district residence on BA completion. These are not per-dollar tuition slopes. The degree sample contains the 2009–2011 high-school classes and excludes degrees obtained in high school; its outcomes are observed through 2017. The source test does not reject equality of the male and female BA effects. The enrollment columns use additional cohorts and do not supply an extra BA observation for the same population.

Early Head Start retains the full-exposure college-graduation ATT in Table 1.2 of Hess (2025): 0.130 for men and 0.212 for women, with source standard errors of 0.040 and 0.050. The corresponding ITT estimates are 0.009 and 0.015. The ATT inputs cannot be described as effects on the entire population exposed to county program availability. The source pools adult attainment; the reconstruction assigns age 22 as the completion endpoint and uses the main case’s BA-channel calibration. I retain its national 1989–1999 cohort mixture and full-population denominator, rather than imposing the high-school-graduate restriction used

for the college-aid references. Transporting a national age profile to the program’s participants remains an external calibration assumption.

For Tennessee, Schoonover et al. (2026) report BA-or-higher effects of 0.012 for men and 0.041 for women at ages 23–24. The female estimate is statistically significant at the 10 percent level; the male estimate is not statistically significant. This contrast does not itself test equality between the two effects. Figure 4F supplies estimates and marginal permutation intervals for the earlier overlapping two-year age groups, 18–19 through 22–23. The reference population follows the main case’s high-school-graduate denominator and prior-year Tennessee residence. The fixed 1992–1998 cohort mixture is retained, separately by sex. No pooled control mean is relabeled as a sex-specific control mean.

For Kalamazoo, Table 8 of Bartik et al. (2021) reports within-district IV effects on obtaining a BA/BS within six years after high-school graduation: 0.041 for men and 0.132 for women, with heteroskedasticity-robust standard errors of 0.070 and 0.077. The source test does not reject equality of the sex-specific BA/BS effects. The corresponding pre-Promise eligible-group means are 0.295 and 0.304. This is not the alternative between-district estimand in the source’s Table 9. It is also not the attempted-credit input used in the pooled Kalamazoo Promise account. The subgroup exercise retains the reported degree outcome directly.

All calibrations use the read-only ACS 2000–2022 extract, valid person weights and education codes, and all race and birthplace categories. Means are calculated within sex–cohort–age cells, retaining fixed cohort weights that are identical across sexes within a policy. The college-aid references use high-school graduates; Early Head Start uses the full population. Geography is current residence except for Tennessee’s known prior-year residence. Massachusetts birth cohorts 1987–1990 approximate the Adams high-school classes under graduation at 18, mapping four-year completion to age 22. Wisconsin’s 2008 entering cohort is approximated by birth cohort 1990 and entry at 18, also mapping its four-year result to age 22. These state profiles are proxies for the respective scholarship populations, not measured control paths at the RD cutoff or among grant-eligible college entrants.

STBF retains the main case’s fixed entry-age mixture for the 2012–2014 cohorts and applies the six-year degree moment at the corresponding ages 23, 24, and 25. The same age and cohort weights apply to both sexes; subgroup-specific entry-age distributions were not recovered. Michigan’s 2009–2011 high-school classes are approximated by birth cohorts 1991–1993. Completion through 2017 then enters a fixed equal-weight mixture at ages 26, 25, and 24, rather than a single midpoint. Kalamazoo’s 2006–2008 post-policy graduating cohorts are approximated by birth cohorts 1988–1990 under graduation at 18. Six-year follow-up is mapped to age 24, and the state’s sex-specific age shape is rescaled to the source subgroup mean at that age. This transports a pre-policy eligible-group level and is not an identified untreated path for the marginal IV population. Source age distributions would permit less restrictive mappings.

Logistic reference curves are fitted separately by cohort and sex using available means through age 35 and the national-shape shrinkage rule. Recent cohorts lacking later ages borrow a same-sex shape and fit their own level. The Adams, STBF, Wisconsin, Michigan, and Early Head Start calibrations use 96 successful reference and donor fits, with a maximum weighted root-mean-square error of approximately 7.5 percentage points. Figures G1, G2, and G3 display all seven policies. The common-reference sensitivity below removes differences between the transported male and female profiles.

G.2 Continuation Restrictions and Earlier Source Information

For subgroup $s \in \{m, f\}$, the treated continuation follows the anchored Logistic specification in Section 3. Each component equals its reference curve through the retained degree-channel anchor, is nondecreasing, and lies in $[0, 1]$. The college-aid anchors use the calibrated entry ages. Early Head Start retains the main case's approximation of zero BA attainment through age 19; that is a degree-outcome approximation, not a claim that childhood treatment has no earlier effects. The subgroup moment restriction is

$$\mathcal{M}_i\{H_s^1 - H_s^0\} = \widehat{\delta}_{is}, \quad (66)$$

where the operator retains each policy's component ages and weights. Tennessee averages ages 23 and 24, while the other operators use the mappings described above. The subgroup coefficients are treated as fixed point inputs, including their published rounding. Sampling uncertainty is not eliminated by this conditioning. The pooled-policy repetitions do not become subgroup draws or confidence intervals.

The geographic joint-support covariance is shared across sexes and transported to each reference center, using the same state/pooled construction as the main reconstruction. A common residual-attainment cap limits the fraction of those without a BA at the observed window who can acquire one later. The cap takes the largest residual share among the two subgroup reference curves and the corresponding 1983–1987 sex-specific donor curves. Donor attainment is evaluated using the source operator's age weights. The cap is fixed before the candidate search and is identical for men and women within a policy. This avoids imposing a systematically more permissive continuation on either sex; it remains an empirical transport restriction, not a proven policy-response bound.

Tennessee's earlier subgroup intervals are recovered from the embedded image of Figure 4F. I retain the image coordinates and scale, round the recovered limits outward to 0.001, and add a 0.002 outward allowance for graphical resolution. The extraction includes marker pixels if they extend beyond a very short interval, so this screen is deliberately conservative relative to the digitized plot. Earlier model moments must fall inside all five such intervals. They are not imposed as exact point estimates, converted to normal standard errors, or interpreted jointly as a 95 percent confidence region. The author's textual last-window estimates remain the exact inputs. Both endpoint-only and screened results are saved. No other policy in this subgroup exercise receives a second degree moment by distributing one coefficient across assumed years or combining different outcomes or samples. This auxiliary subgroup interval sensitivity is separate from the pooled application, whose current L3 fits fixed Table 2 point estimates. The subgroup allowance concerns graphical uncertainty and does not establish joint coverage or supply the pooled L3 rule. These degree-channel exercises do not alter the pooled L3 classifications.

G.3 Separating Timing from the Terminal Effect

A larger early effect for women does not by itself show a greater delay loss. To separate timing from the eventual gain, I additionally impose the same terminal BA effect and valuation prices on the two sexes. Earlier attainment changes the timing of both negative and positive portions of the income response. I apply the Section 3 signed completion template with a fixed two-schooling-years-per-degree conversion for BA relative to some college. This is an explicit common-price standardization, not an estimate of observed schooling or the full credential mix for either sex:

$$V_{is}^{\text{BA}} = 2 \sum_{a=18}^{65} D_{.03}(a; 18) Y_0(a) \left[r^+ \Delta H_{is}^{\text{BA}}(a) + r^- \{\Delta H_{is}^{\text{BA}}(a+7) - \Delta H_{is}^{\text{BA}}(a)\} \right]. \quad (67)$$

The coefficients are $r^+ = 815/7241$ and $r^- = -0.066706494$. Paths extend through age 72 so that the seven pre-completion response years are included for earnings through age 65. The terminal comparison below still fixes the age-65 effect. No additional opportunity cost is subtracted. Prices are common across sexes and policies, in 2000 dollars discounted to age 18. This standardized channel excludes sex-specific wage returns, other credentials, transfers, and fiscal costs; it does not estimate a full sex-specific MVPF. Direct annual net-income summation agrees with the stock-kernel calculation.

The unrestricted point-input spans in Table G1 mix timing and eventual attainment. For the timing comparison, each subgroup must additionally satisfy

$$\Delta H_{im}^{BA}(65) = \Delta H_{if}^{BA}(65) = \eta, \quad (68)$$

for a fixed terminal-effect grid. Adams uses $\eta \in \{-0.15, -0.10, -0.05, 0\}$, allowing the same terminal reduction or zero effect after its negative four-year inputs. The other policies use $\eta \in \{0.02, 0.05, 0.10, 0.15\}$. These grid points are comparisons, not quantiles of a path distribution. Table G2 reports every value and the number of verified paths for both sexes. The common-reference sensitivity uses the equally weighted mean of the two subgroup reference curves within each component, retaining the source effects, age weights, support construction, residual cap and terminal grid.

Female-minus-male NPV endpoints compare independent complete paths at the same η , common prices, and common exposure-age weights within the policy. Each endpoint retains a pair of parameterized path witnesses. A negative difference supplies a witness of lower female education NPV at that terminal effect. It would not establish that such a path occurred, that it is more likely, or that pregnancy or marriage caused it. Figure G4 summarizes the smallest and largest paired differences across each policy's tested grid. Its endpoints can come from different terminal effects, but each male-female pair shares one terminal effect. The figure does not rank policies or assign probabilities to the continuations.

Six policies have a verified equal-terminal comparison under separate reference curves at one or more grid values. Adams, STBF, Wisconsin, and Michigan community-college tuition each permit both signs of the NPV difference at an individual fixed terminal value. Early Head Start and Tennessee have positive differences in the saved comparisons. For Tennessee, when both terminal effects are five percentage points, the saved paths give women a standardized education NPV approximately \$46–\$277 higher than men's, in 2000 dollars. The common-reference sensitivity has the same direction. The positive comparisons describe the paths found and do not establish that every compatible continuation favors women. Wisconsin admits both sexes at a two-point terminal gain, while no female path is found at five points. The separate-reference Kalamazoo banks do not yield a common terminal value on the four-point grid. Under the common reference, a ten-point terminal gain is supported for both sexes, with higher female NPV in the saved comparisons. STBF and Wisconsin, which have identical rounded point inputs, permit both NPV orderings under the common reference. Source precision, population transport, finite search, age mapping, and common-price standardization limit conclusions about subgroup welfare.

These findings motivate examining subgroup dynamics without imposing a female-delay interpretation. Marriage, childbearing, work, and college access may influence education timing, but the reconstructed paths do not identify those mechanisms. Tennessee's authors themselves discuss delayed transitions in schooling, work, and family formation. Their evidence does not establish that marriage or childbirth caused women to delay graduation. Distinguishing that mechanism requires additional subgroup or longitudinal evidence.

Table A1. Literature coverage, estimation times, and observation endpoints

<i>A. Reported effect-estimation times or windows</i>						
Outcome	Full sample				Journal sample	
	Eligible	Coded	One	One (%)	One/coded	One (%)
All education outcomes	459	459	249	54.2	244/447	54.6
College entry (CE)	237	236	193	81.8	189/231	81.8
Associate's degree (AA)	40	40	32	80.0	32/39	82.1
Bachelor's degree or higher (BA+)	190	190	132	69.5	130/186	69.9

<i>B. Availability of a numerical observation endpoint</i>		
Endpoint classification	Papers	Percent
Exact numerical endpoint	366	79.7
Interval or one-sided bound	38	8.3
Stage identified; clocks not aligned	46	10.0
Endpoint missing or conflicting evidence	9	2.0
Total	459	100.0

Notes: Panel A counts papers within overlapping outcome groups. One time or pooled window is not one year of follow-up. One CE paper lacks a category-specific count. Journal results exclude 11 additional welfare-source papers and one author manuscript. Panel B assigns each of the 459 papers to one endpoint category. Attained age, individual follow-up, and policy/cohort event time remain distinct clocks. Coding rules and evidence records are described in Appendix A.2.

Table A2. Interpretation of finite-window evidence and lifetime valuation

<i>A. Interpretation and valuation</i>					
Category			Full sample		Journal sample
Finality or permanence statements			18		17
Completion or window-adequacy arguments only			22		22
Either interpretation category			40		39
Education-based lifetime or career valuation			31		28
Both interpretation and valuation			1		1
At least one category			70		66
Neither detected			389		381
<i>B. Timing of the valued education outcome in 31 papers</i>					
Outcome clock	One window	Multiple windows	Papers	Last endpoint Range (n)	
Attained age	5	1	6	22–54 (5)	
Individual follow-up	9	11	20	1.5–11 (17)	
Education stage only	4	0	4	—	
Institutional event time	—	—	1	5 (1)	
Total	18	12	31		

Notes: Panel A uses 459 papers overall and 447 journal papers. The first two categories are mutually exclusive; valuation can overlap either. Neither detected refers to the reviewed version, not an absence of all long-run discussion. Panel B codes comparable times for the education outcome entering valuation, rather than all outcomes or successive exposed cohorts. One institutional-flow study is separate from individual follow-up. Endpoint ranges use only exact values, with their counts in parentheses; follow-up origins differ across papers. Bounds, pooled intervals, and unaligned stages remain in the sample. The linked 31-paper ledger records sources and coding decisions.

Table B1. Generating scenarios for the observation-design experiment

Scenario	K	b	d	$\Delta H_c(25)$	$\Delta H_c(35)$	NPV
K000	13.3370	0.2575	12.5649	-0.4904	-0.5777	-26.05
K025	13.7876	0.2575	12.5649	-0.1126	-0.1328	-5.99
K050	14.2383	0.2575	12.5649	+0.2652	+0.3121	+14.08
K075	14.6889	0.2575	12.5649	+0.6430	+0.7570	+34.14
K100	15.1395	0.2575	12.5649	+1.0209	+1.2018	+54.20
b000	13.9222	0.1200	12.5649	-1.1271	-0.5345	-25.33
b025	13.9222	0.1903	12.5649	-0.4167	-0.1116	-7.40
b050	13.9222	0.2605	12.5649	+0.0143	+0.0025	+0.23
b075	13.9222	0.3308	12.5649	+0.2553	+0.0323	+4.01
b100	13.9222	0.4011	12.5649	+0.3864	+0.0402	+6.11
d000	13.9222	0.2575	8.0000	+0.0787	+0.0072	+1.26
d025	13.9222	0.2575	13.5000	-0.0295	-0.0027	-0.46
d050	13.9222	0.2575	19.0000	-0.4017	-0.0429	-6.07
d075	13.9222	0.2575	24.5000	-1.2601	-0.1990	-19.24
d100	13.9222	0.2575	30.0000	-2.1886	-0.7111	-40.20

Notes: Each parameter uses the 0, 25, 50, 75, and 100 percent positions of its conditional feasible interval. Scenario names identify the varied parameter and interval position. Only that generating parameter varies; the other two equal the reference values. All three candidate parameters may vary during matching. NPV is in thousands of 2000 dollars per person, under the common signed completion-response rule. Every continuation shares the age-17 baseline level. The mean comparison curve differs slightly from the anchored continuation evaluated at the mean parameter vector.

Table B2. Complete age-25 scans: summary across generating parameters

Generating parameter	Parameter interval	Age-25 effect interval (YS)	Scenarios	Candidate NPV width			Both signs found
				Minimum	Median	Maximum	
K	[13.337, 15.140]	[−0.490, 1.021]	102	18.02	36.73	53.59	53/102
b	[0.120, 0.401]	[−1.127, 0.386]	102	33.21	43.98	58.05	83/102
d	[8.000, 30.000]	[−2.189, 0.079]	102	41.86	52.78	77.62	102/102

Notes: Each row summarizes 101 equally spaced positions in the conditional feasible interval plus the reference value. Only the indicated generating parameter changes; the other two stay at their reference values. All three candidate parameters are free when matching the resulting age-25 effect. Widths are in thousands of 2000 dollars per person. Both signs requires candidates below $-\$0.01$ and above $\$0.01$. Counts summarize a deterministic design and are not path probabilities. The complete 306-row file, including terminal effects, cumulative gaps, generating values, and parameterized extrema, is supplied in the accompanying replication materials.

Table B3. Reference and single-parameter reconstructed earnings NPVs

Policy	Reference	Permanent (K)		Temporary (b)		Temporary (d)	
	NPV	ΔK	NPV	Δb (%)	NPV	Δd	NPV
Florida grant	8.63	0.371	17.53	—	—	—	—
DC tuition aid	8.95	—	—	13.7	9.23	—	—
Adams	-1.07	-0.0244	-1.50	-1.18	-0.37	0.425	-0.31
STBF	0.66	Joint	0.74	—	—	—	—
Wisconsin grant	0.59	0.146	1.25	6.47	0.15	—	—
Kalamazoo	1.10	0.416	1.72	—	—	—	—
Michigan CC	9.45	0.0343	11.76	3.55	3.26	—	—
CUNY Pell	0.56	0.0197	1.23	0.686	0.19	-0.215	0.13
Ohio Pell	2.53	—	—	—	—	—	—
Tennessee HOPE	1.31	0.0575	3.61	2.28	0.43	—	—
Tennessee Pell	-0.30	-0.0188	-0.81	-0.808	-0.10	0.246	-0.06
Georgia HOPE	2.81	0.321	9.39	—	—	—	—
Social Security	2.79	—	—	—	—	—	—
Appropriations: spending	2.70	0.0822	4.96	3.92	0.93	—	—
Early Head Start	n.a.	—	—	—	—	—	—
Texas CC	11.16	0.248	13.99	—	—	—	—
Adult Pell	2.47	0.0815	8.26	—	—	—	—
Appropriations: tuition	0.03	8.23×10^{-4}	0.05	0.0386	0.01	-0.0154	7.16×10^{-3}
HOPE credit	5.95	0.128	12.39	6.05	2.84	—	—
HOPE ind. single: lower	6.44	1.96×10^{-4}	36.06	0.00946	8.04	-0.00414	8.21
HOPE dep. single: lower	-4.14	-2.72×10^{-4}	-16.49	-0.0131	-3.44	0.00575	-3.48
HOPE dep. single: upper	5.06	3.15×10^{-4}	20.18	0.0152	4.20	-0.00665	4.27
HOPE joint: lower	23.75	0.00124	94.72	0.0599	19.73	-0.0263	20.04
HOPE joint: upper	9.72	4.74×10^{-4}	38.76	0.0229	8.08	-0.0100	8.20
Tuition ded. single: lower	6.88	0.0262	17.46	2.02	2.92	—	—
Tuition ded. joint: lower	-1.66	-0.0169	-4.84	-0.611	-0.66	0.183	-0.39
HOPE / Lifetime Learning	-57.97	-0.195	-172.55	—	—	—	—
AOTC single: upper	-1.40	-6.64×10^{-5}	-5.38	-0.00364	-1.09	0.00174	-1.29
AOTC single: lower	31.04	0.00153	118.93	0.0840	24.09	-0.0405	28.69
AOTC joint: upper	-11.54	-4.39×10^{-4}	-44.23	-0.0241	-8.96	0.0115	-10.64
AOTC joint: lower	12.01	4.68×10^{-4}	46.03	0.0257	9.33	-0.0123	11.09
AOTC ind. single: lower	1.86	3.17×10^{-5}	10.03	0.00174	2.18	-8.31×10^{-4}	2.60
AOTC instrument	5.97	9.39×10^{-5}	21.26	0.00390	3.52	-0.00136	2.94
Tennessee Promise	6.46	Joint	n.c.	—	—	—	—
College placement	n.a.	0.0545	n.c.	5.11	n.c.	—	—
Tuition ded. single: upper	0.00	0.000	0.00	0.00	0.00	0.000	0.00
Tuition ded. joint: upper	0.00	0.000	0.00	0.00	0.00	0.000	0.00

Notes: Each pair varies the named parameter family; the other two remain at H^0 values. K changes eventual attainment; b and d effects eventually vanish. ΔK and Δd are in years; Δb is the proportional speed change in percent. STBF jointly changes time enrolled and three credential plateaus; Tennessee jointly changes three credential plateaus. Their outcome-specific changes are saved in the matched-solution data. † marks the same L2 solution also satisfying L3; — denotes no admissible match. NPVs use comparable account units per reference expenditure. n.a.: unavailable reference projection or normalizer; n.c.: no comparable pair, including unresolved source monetary years for EHS and Tennessee. Figure B3 plots the comparable pairs. Appendix B.5 defines the matching procedure. Dep./ind. denote dependent/independent filers; lower/upper denote income thresholds.

Table C1. ACS cohort support and comparison-path fits

Birth cohort	Records	Ages	K	b	d	RMSE
1985	630,409	16–35	13.8926	0.2533	12.5107	0.112
1986	659,595	16–35	13.9037	0.2614	12.6170	0.113
1987	688,856	16–35	13.9704	0.2578	12.5669	0.115

Notes: The full-population ACS extract covers 2000–2022; the selected cohort–age cells occur in 2001–2022. Each cohort contributes all 20 ages. Fits minimize weighted squared differences from person-weighted cell means, using normalized cell population weights within each cohort. The comparison curve averages the three fitted curves with fixed one-third weights. RMSE is in schooling years. Parameter estimates and fit diagnostics are descriptive inputs, not causal policy effects.

Table D1. Source education inputs, clocks, and reconstruction roles

Education input	Effect	SE	Clock / role
<i>STBF: randomized offer, 2012–2014 four-year-college strata</i>			
Time enrolled	0.3648	0.0413	HS+6; joint L2
Any college	0.0072	0.0030	HS+6; joint L2
Any degree (AA+ proxy)	0.0518	0.0150	HS+6; joint L2
Bachelor's degree	0.0813	0.0155	HS+6; joint L2
<i>Adams: eligibility RD, 2005–2006 graduating classes</i>			
Two-year enrollment	0.0070	0.0050	HS+2; conversion
Four-year enrollment	0.0000	0.0070	HS+2; conversion
Two-year completion	0.0000	0.0040	HS+6; conversion
Four-year completion	-0.0250	0.0090	HS+6; conversion
Combined schooling (derived)	-0.0180	0.0240	HS+6; L2
Early schooling (derived)	0.0070	0.0050	HS+2; converted L3
<i>Tennessee Promise: age-window effects of program exposure</i>			
Some college+, ages 18–19	0.03500	0.0040	L3 point target
Some college+, ages 19–20	0.03200	0.0040	L3 point target
Some college+, ages 20–21	0.02500	0.0040	L3 point target
Some college+, ages 21–22	0.02700	0.0030	L3 point target
Some college+, ages 22–23	0.02800	0.0040	L3 point target
Some college+, ages 23–24	0.03500	0.0040	L2
AA+, ages 18–19	0.00200	0.0010	L3 point target
AA+, ages 19–20	0.01400	0.0020	L3 point target
AA+, ages 20–21	0.03100	0.0030	L3 point target
AA+, ages 21–22	0.02600	0.0030	L3 point target
AA+, ages 22–23	0.02900	0.0040	L3 point target
AA+, ages 23–24	0.04000	0.0040	L2
BA+, ages 18–19	-0.00100	0.0000 ^a	L3 point target
BA+, ages 19–20	0.00100	0.0000 ^a	L3 point target
BA+, ages 20–21	0.00001	0.0010	L3 point target
BA+, ages 21–22	-0.00200	0.0020	L3 point target
BA+, ages 22–23	0.01200	0.0030	L3 point target
BA+, ages 23–24	0.02700	0.0040	L2

Notes: Degree effects are in probability units; 0.01 is one percentage point. STBF uses time enrolled separately from its three cumulative credentials; Adams schooling uses retained enrollment/completion multipliers. STBF's deposited input is 0.3648, versus 0.360 printed. Tennessee coefficients and clustered SEs reproduce rounded Table 2 values; its L3 fits these point estimates without SE bands (Appendix D.8). ^a A displayed-zero SE receives a 0.0005 floor in the current 1,000 normal-input repetitions as an imposed reporting-precision scale. No joint confidence region is implied.

Table D2. Source welfare accounts and the path replacement

Component	STBF	Adams	Tennessee Promise
Net earnings NPV	21,150.00	-1,204.04	95,557.69
Reference private nonincome component	32,250.00	2,329.97	2,124.31
Reference pretax public component	32,250.00	1,658.06	14,792.00
L1 WTP	49,170.00	1,364.23	72,837.00
L1 net cost	28,020.00	1,896.35	-10,053.00
Tax / transfer rate	0.2000	0.1979	0.2600

Notes: The table preserves reported L1 amounts, rather than presenting them as reconstructed path values. STBF uses the author nonlinear credential–experience function; the recovered workbook gives earnings NPV of \$21,129.23 rather than the rounded published \$21,150. Its fixed components are \$32,250. Adams transports its source signed income, private payment, and public education-cost functions with completion timing; the reference nonincome components shown here are not frozen residuals for every path. Tennessee uses unscaled public ACS income and a participant-unit bridge of 1/0.2495243, with fixed nonincome components shown above. This bridge comes from the source WTP decomposition and entry effect, not an income-total fit. Its fiscal component is \$14,792 versus \$14,793 rounded in the source. All income functions include their early losses; no extra opportunity cost is deducted. Dollar years and reconstruction qualifications appear in the case appendices.

Table D3. Verified joint paths and paired welfare components

Path	Outcome	K	b	d	$\Delta H(65)$	WTP	Net cost	MVPF
S1	T index	14.3086	0.4268	14.8904	0.0847	38.46	30.70	1.25
	SC+	0.7067	1.4389	19.0818	0.0076			
	AA+	0.5290	2.3078	21.5797	0.0029			
	BA+	0.4556	3.0000	22.1227	0.0451			
S2	T index	14.4236	0.3763	14.9735	0.1997	81.13	20.03	4.05
	SC+	0.7195	0.7400	19.0457	0.0204			
	AA+	0.6860	0.8113	22.7289	0.1600			
	BA+	0.6054	0.8364	22.9004	0.1949			
A1	YS	14.3556	0.3491	14.2621	-0.4790	7.53	14.57	0.52
A2	YS	15.6534	0.1829	12.6675	0.8177	-11.30	-19.64	–
A3	YS	14.7082	0.2762	14.1597	-0.1264	3.00	4.89	0.61
A4	YS	14.7843	0.2554	12.7237	-0.0503	1.82	2.85	0.64
T1	SC+	0.7271	1.0647	19.0777	0.0419	-48.72	32.66	-1.49
	AA+	0.4241	2.2608	21.2173	-0.0479			
	BA+	0.3217	3.0000	21.6230	-0.0406			
T2	SC+	0.7200	3.0000	17.5874	0.0348	152.50	-38.04	$+\infty$
	AA+	0.5728	0.7691	22.1110	0.1008			
	BA+	0.4537	0.8964	22.4814	0.0914			
T3	SC+	0.7203	1.4437	17.8946	0.0351	86.44	-14.83	$+\infty$
	AA+	0.5687	0.8835	22.2619	0.0967			
	BA+	0.4057	1.3628	22.4467	0.0434			
T4	SC+	0.7203	1.4437	17.8946	0.0351	86.44	-14.83	$+\infty$
	AA+	0.5687	0.8835	22.2619	0.0967			
	BA+	0.4057	1.3628	22.4467	0.0434			

Notes: WTP and net cost are thousands of case-specific dollars. S1–S2, A1–A2, and T1–T2 minimize/maximize L2 earnings NPV; A3–A4 do so within exact-match L3. T3–T4 coincide at the Tennessee L3 best-fitting path; they are not distinct bound witnesses. All rows with the same path label form one joint candidate. STBF T index uses an external schooling-shaped timing calibration for the time-enrolled effect; its level is not the author control mean. The original nonlinear earnings function uses separately recovered source mean inputs. Each witness passes the full joint restrictions; parameters are rounded only for display. Age 65 is extrapolated, and Tennessee earnings valuation ends at 64.

Table D4. Tennessee Promise: population matching and continuation limits

Quantity	Some college+	AA+	BA+
Source control mean, ages 23–24	0.6850	0.3800	0.2920
Matched treated mean, ages 23–24	0.7200	0.4200	0.3190
Remaining-completion fraction cap	0.1402	0.2635	0.1978
Implied treated plateau ceiling	0.7593	0.5728	0.4537
Largest saved treated plateau	0.7527	0.5728	0.4537

Notes: The high-school-graduate denominator and prior-year Tennessee residence follow the source population convention. Source control means match levels; cohort fits supply age shapes. Donor completion fractions bound the permitted later attainment under the stated transport assumption. All joint candidates satisfy credential nesting and the current source-window restrictions. The plateau ceiling is not an estimated causal bound.

Table D5. Tennessee Promise: L3 point targets and fitted residuals

Outcome	Age window	Target	Fitted	Residual	Exact L2?
SC+	18–19	3.500	3.677	+0.177	–
SC+	19–20	3.200	2.681	-0.519	–
SC+	20–21	2.500	2.951	+0.451	–
SC+	21–22	2.700	3.335	+0.635	–
SC+	22–23	2.800	3.467	+0.667	–
SC+	23–24	3.500	3.500	+0.000	Yes
AA+	18–19	0.200	0.756	+0.556	–
AA+	19–20	1.400	1.626	+0.226	–
AA+	20–21	3.100	2.518	-0.582	–
AA+	21–22	2.600	2.837	+0.237	–
AA+	22–23	2.900	2.928	+0.028	–
AA+	23–24	4.000	4.000	-0.000	Yes
BA+	18–19	-0.100	-0.027	+0.073	–
BA+	19–20	0.100	-0.076	-0.176	–
BA+	20–21	0.001	-0.141	-0.142	–
BA+	21–22	-0.200	0.028	+0.228	–
BA+	22–23	1.200	1.057	-0.143	–
BA+	23–24	2.700	2.700	+0.000	Yes

Notes: Entries are percentage points. Targets come from Table 2 of Schoonover et al. (2026), retaining the reported precision, including the 0.00001 probability coefficient for BA+ at ages 20–21. Predictions use the same adjacent-age averages and cohort weights. The three latest-window effects are exact. One joint path minimizes equal-weight squared discrepancies subject to the retained support, level, anchor, nesting, and late-completion restrictions. Its eighteen-target RMSE is 0.353 percentage points and maximum absolute residual is 0.667 points. No graphical permutation interval enters this fixed-input fit. Normal-input perturbations are reported separately and do not recover the source permutation distribution.

Table E1. Policy registry, calibration populations, and available L3 paths

ID	Calculation	ACS geography	Birth cohorts	Outcome	L3 paths
E01	Florida access grant	Florida	1982	ys	1
E02	DC tuition assistance	District of Columbia	1981	ys	–
E03	Adams Scholarship	Massachusetts	1987	ys	21
E04	STBF scholarship	Nebraska	1993, 1994, 1995, 1996, 1997	T, C, A, B	–
E05	Wisconsin Scholars Grant	Wisconsin	1990, 1991, 1992	ys	1
E06	Kalamazoo Promise	Michigan	1988	ys	1
E07	Michigan CC tuition	Michigan	1992	ys	115
E08	CUNY Pell grants	New York	1991	ys	37
E09	Ohio Pell grants	Ohio	1981	ys	–
E10	Tennessee HOPE	Tennessee	1989	ys	–
E11	Tennessee Pell eligibility	Tennessee	1987, 1988, 1989, 1990, 1991, 1992	ys	–
E15	Georgia HOPE	Georgia	1975, 1976, 1977, 1978, 1979	ys	–
E16	Social Security student benefits	National	1963	ys	67
E17	College appropriations: spending	National	1983	ys	1
E18	Early Head Start	National	1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999	ba	–
E19	Texas CC tuition	Texas	1987	ys	1
E20	Adult Pell introduction	National	1947	ys	–
E21	College appropriations: tuition	National	1983	ys	1
E22	HOPE tax credit	National	1981	ys	79
E23	HOPE: independent single, start	National	1987, 1988	ys	–
E24	HOPE: dependent single, start	National	1987, 1988	ys	–
E25	HOPE: dependent single, end	National	1987, 1988	ys	–
E26	HOPE: joint, start	National	1987, 1988	ys	–
E27	HOPE: joint, end	National	1987, 1988	ys	–
E28	Tuition deduction: single, start	National	1988	ys	–
E29	Tuition deduction: joint, start	National	1988	ys	–
E30	HOPE and Lifetime Learning	National	1980	ys	–
E31	AOTC: single, end	National	1991, 1992	ys	–
E32	AOTC: single, start	National	1991, 1992	ys	–
E33	AOTC: joint, end	National	1991, 1992	ys	–
E34	AOTC: joint, start	National	1991, 1992	ys	–
E35	AOTC: independent single, start	National	1991, 1992	ys	–
E36	AOTC: simulated instrument	National	1987, 1988, 1989, 1990, 1991, 1992	ys	–
E37	Tennessee Promise	Tennessee	1992, 1993, 1994, 1995, 1996, 1997, 1998	ce, aa, ba	1
E38	Algorithmic college advising	New York	1995	ys	–
E39	Tuition deduction: single, end	National	1988	ys	–
E40	Tuition deduction: joint, end	National	1988	ys	–

Continued on next page

Table E1 (continued)

ID	Calculation	ACS geography	Birth cohorts	Outcome	L3 paths
E41	Cannabis: education channel	RML 14 states	1988, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999	ys	1

Notes: All 38 calculations have numerical path banks; cannabis displays its primary L3. A dash means no additional timing restriction is encoded. L3 requires an additional reported source window under the retained conversion; synthetic duration allocations are excluded. No candidate means a main L3 search with no verified path; it is not a proof of infeasibility. YS denotes a source-specific schooling-equivalent input; STBF jointly uses a time-enrolled index and three cumulative credentials; Tennessee uses joint cumulative SC+, AA+, and BA+ paths. The underlying machine-readable registry additionally records source-study families, source windows, operator weights, source-control qualifications, and information-layer reasons. The main counts retain historical IDs.

Table E2. Source education inputs, observation windows, and priority ranks

Policy	Education input	Source observation window	L1	Own L2	Selected	Layer	Lower	Upper
HOPE: joint, lower kink [f] [†]	Current college attendance	Ages 19–20	1	24	24	L2	−∞	+∞
Tuition deduction: single, lower threshold [†]	Current college attendance	Years 0–4 from expected high-school graduation	1	24	24	L2	−∞	+∞
AOTC: dependent single, lower kink [f] [†]	Current college attendance	Ages 19–20 in 2011	1	24	24	L2	−∞	+∞
AOTC: joint, lower kink [f] [†]	Current college attendance	Ages 19–20 in 2011	1	24	24	L2	−∞	+∞
Tennessee Promise [r]	Some college+, AA+, and BA+	Six overlapping age pairs, 18–19 through 23–24	1	17	1	L3	+∞	+∞
Algorithmic college placement [x]	Earned credits	Pooled follow-up through 2018–2019; duration varies	1	15	15	L2	0.09	+∞
Tuition deduction: single, upper threshold [†]	Current college attendance	Years 0–4 from expected high-school graduation	1	24	24	L2	−∞	+∞
Texas CC tuition	Enrollment and BA completion	Enrollment through HS+6; BA at HS+6 and HS+8	8	6	1	L3	+∞	+∞
Michigan CC tuition	College entry and credentials	Entry within one year; credentials through 2017	9	20	5	L3	13.97	+∞
DC tuition assistance	College entry	Freshman entry within 12 months of high school	10	7	7	L2	8.81	+∞
HOPE: independent single, lower kink [f] [†]	Current college attendance	Ages 19–20	11	24	24	L2	−∞	+∞
HOPE tax credit	Enrollment and persistence	Ages 18–19; first and second college years	12	20	1	L3	+∞	+∞
HOPE: dependent single, upper kink [f] [†]	Current college attendance	Ages 19–20	13	24	24	L2	−∞	+∞
AOTC: simulated instrument [f] [†]	Current college attendance	Age 19 retained; age 22 reported separately	14	24	24	L2	−∞	+∞
Florida access grant	Earned credits	First semester and years 1–4 after high school	15	10	6	L3	10.97	10.97
AOTC: independent single, lower kink [f] [†]	Current college attendance	Ages 19–20 in 2011	16	24	24	L2	−∞	+∞
Social Security student benefits	Completed schooling years	Ages 23 and 28	17	8	8	L3	5.13	5.23
College appropriations: spending	Institution enrollment	Policy year and three lags, 1990–2013	18	20	4	L3	25.23	25.23
Georgia HOPE	Freshman entry by sector	State-year freshman flows, 1988–1997	19	18	18	L2	−2.04	+∞
Ohio Pell grants	College persistence	1999–2000 to 2000–2001	20	14	14	L2	0.18	7.85

Table E2 (continued)

Policy	Education input	Source observation window	L1	Own L2	Selected	Layer	Lower	Upper
HOPE: joint, upper kink [f] [†]	Current college attendance	Ages 19–20	21	24	24	L2	−∞	+∞
Adult Pell introduction	Current college enrollment	Pooled ages 22–35	22	10	10	L2	3.04	4.75
Kalamazoo Promise	Attempted credits	Years 2, 3, and 4 after high school	23	12	9	L3	3.61	3.61
Tennessee HOPE	Two-year college entry	Initial enrollment, 2006–2009 high-school cohorts	24	23	23	L2	−59.36	21.02
STBF scholarship	Time enrolled	Six years after high school	25	12	12	L2	1.25	4.05
Wisconsin Scholars Grant	Earned credits	Semester inputs; cumulative years 2–3 after entry	26	17	11	L3	1.41	1.41
CUNY Pell grants	Earned credits	First and third years after college entry	27	36	21	L3	−5.92	−0.26
College appropriations: tuition	Institution enrollment	Policy year and three lags, 1990–2013	28	21	16	L3	−1.05	−1.05
Tennessee Pell eligibility [†]	College entry by sector	First fall after high school	29	37	37	L2	−∞	2.66
Tuition deduction: joint, upper threshold	Current college attendance	Years 0–4 from expected high-school graduation	30	19	19	L2	−3.86	+∞
Adams Scholarship	Enrollment and AA/BA completion	Enrollment at HS+2; completion at HS+6	31	13	13	L3	0.61	0.64
Early Head Start [r]	BA or higher attainment	Pooled adult attainment, eligible from age 22	32	17	17	L2	−1.40	+∞
AOTC: dependent single, upper kink [f] [†]	Current college attendance	Ages 19–20 in 2011	33	24	24	L2	−∞	+∞
Tuition deduction: joint, lower threshold	Current college attendance	Years 0–4 from expected high-school graduation	34	20	20	L2	−4.29	+∞
HOPE: dependent single, lower kink [f] [†]	Current college attendance	Ages 19–20	35	24	24	L2	−∞	+∞
AOTC: joint, upper kink [f] [†]	Current college attendance	Ages 19–20 in 2011	36	24	24	L2	−∞	+∞
HOPE and Lifetime Learning credits	Current college enrollment	Pooled ages 18–24	37	22	22	L2	−10.75	+∞

Notes: Ranks cover all 37 calculations with L1. [f] denotes a finite tax experiment with an explicit local-linear input conversion, [r] a public-data reconstructed account, and [x] an fiscal-only earnings extension. Finite candidate endpoints are rounded to two decimals before lower-endpoint-first ordering; tied endpoint pairs share a competition rank. Source outcomes and windows need not equal model ages. Unassigned-ratio paths, not their calculations, are omitted from the ordering. Equal endpoint pairs remain tied; bounded monetary components do not guarantee bounded ratios near zero net cost. Cannabis has no whole-policy L1; all 38 remain in Table 3. The restricted 24-case ordering and rounding sensitivities accompany the data. These are conditional display ranks, not social-welfare priorities; finite-path extrema remain candidate based.

Table E3. Reported education timing and its use in reconstruction

Cases	Windows	Source timing evidence	Current reconstruction
Florida access grant	3+	First-semester and years 1–4 credits	Reported timing retained as L3
DC tuition assistance	1	First college entry within 12 months of HS	L2 only; no added timing constraint
Adams Scholarship	3+	Entry, persistence, and BA through year 6	L3: separate early window under stated allocation
STBF scholarship	3+	Semester enrollment; BA in years 4–6	L2 only; no added timing constraint
Wisconsin Scholars Grant	3+	Semester credits; cumulative years 2/3	Reported timing retained as L3
Kalamazoo Promise	3+	Credits in years 2–4; degrees in years 4/6	Reported timing retained as L3
Michigan CC tuition	2	Entry within a year; later credentials	L3: early converted entry component
CUNY Pell grants	2	First-year and three-year credits	Reported timing retained as L3
Ohio Pell grants	1	Entry to next-year persistence	L2 only; no added timing constraint
Tennessee HOPE	1	First post-HS college enrollment	L2 only; no added timing constraint
Tennessee Pell eligibility	1	Initial-fall college-sector choice	L2 only; allocated timing excluded
Georgia HOPE	Other	State-year freshman flows	L2 only; allocated timing excluded
Social Security student benefits	2	Attainment at ages 23 and 28	Reported timing retained as L3
College appropriations	Other	Institution enrollment by policy lag	L3: lag responses under explicit time/population mapping
Early Head Start	1	Pooled adult attainment	L2 only; no added timing constraint
Texas CC tuition	3+	Annual enrollment at HS+0 through HS+6	Reported timing retained as L3
Adult Pell introduction	1	Pooled current adult enrollment	L2 only; no added timing constraint
HOPE tax credit	Other	College-grade and pooled enrollment	L3: separate first-year component under stated mapping
HOPE/AOTC income kinks	1	Contemporaneous credit-kink response	L2 only; no added timing constraint
Tuition-deduction thresholds	3+	Years 0–4 from expected HS graduation	Different operator; L2 in main plot
HOPE and Lifetime Learning credits	1	Pooled current college enrollment	L2 only; no added timing constraint
AOTC: simulated instrument	2	Attendance at ages 19 and 22	Different operator; L2 in main plot
Tennessee Promise	3+	Six age windows, 18–19 to 23–24	Reported timing retained as L3
Algorithmic college placement	2	Initial term; pooled and oldest-cohort follow-up	L2 only; no added timing constraint
Cannabis legalization: education channel	3+	BA completion within 4, 5, 6 years	Reported timing retained as L3; not ranked

Notes: Windows count source ages or follow-up periods, not coefficients. Institutional lags and college-grade contrasts are listed as Other. L3 retains L2 and adds source windows or dynamics under stated outcome, timing, and population mappings. A duration-only split of one input is not L3. Cannabis lacks L1. The timing audit records source locations.

Table E4. Point-input sensitivity: numerical comparisons

Policy	Set	Main	Narrow	Wide	National	Earnings 75%	Earnings 125%	Denser
<i>Direct student financial aid</i>								
Florida access grant	L3	[10.97, 10.97]	[11.12, 11.12]	[10.75, 10.75]	[10.38, 10.38]	[4.90, 4.90]	[44.26, 44.26]	[10.97, 10.97]
DC tuition assistance	L2	[8.81, > 10 ⁶] ^S	— ^S	[−2.57, > 10 ⁶] ^S	[13.89, > 10 ⁶] ^S	[5.33, > 10 ⁶] ^S	[15.10, > 10 ⁶] ^S	[8.81, > 10 ⁶] ^S
Adams Scholarship	L3	[0.61, 0.64]	[0.62, 0.63]	[0.61, 0.64]	[0.62, 0.64]	[0.90, 0.97]	[0.31, 0.40]	[0.61, 0.64]
STBF scholarship	L2	[1.25, 4.05]	[1.27, 2.78]	[1.24, 7.33]	[1.28, 5.81]	[1.19, 2.99]	[1.32, 5.50]	[1.25, 4.05]
Wisconsin Scholars Grant	L3	[1.41, 1.41]	[1.50, 1.50]	[1.27, 1.27]	[1.16, 1.16]	[1.26, 1.26]	[1.56, 1.56]	[1.41, 1.41]
Kalamazoo Promise	L3	[3.61, 3.61]	[3.54, 3.54]	[3.67, 3.67]	[4.01, 4.01]	[2.52, 2.52]	[5.13, 5.13]	[3.61, 3.61]
CUNY Pell grants	L3	[−5.92, −0.26]	[−4.74, −0.85]	[−7.92, 0.48]	[−4.65, 0.60]	[−8.66, 0.09]	[−5.09, −0.55]	[−5.92, −0.26]
Ohio Pell grants	L2	[0.18, 7.85]	[3.22, 7.46]	[−12.66, 8.09]	[1.06, 7.92]	[0.44, 3.83]	[−0.03, 21.24]	[0.18, 7.85]
Tennessee HOPE	L2	[−59.36, 21.02]	[−27.43, 17.84]	[−233.87, 23.72]	[−60.63, 22.35]	[< −10 ⁶ , 6.39]	[−14.31, > 10 ⁶] ^S	[−59.36, 21.02]
Tennessee Pell eligibility	L2	[< −10 ⁶ , 2.66]	[< −10 ⁶ , 2.61]	[< −10 ⁶ , 2.68]	[< −10 ⁶ , 2.66]	[< −10 ⁶ , 1.64]	[< −10 ⁶ , 4.18]	[< −10 ⁶ , 2.66]
Georgia HOPE	L2	[−2.04, > 10 ⁶] ^S	[0.44, > 10 ⁶] ^S	[−4.07, > 10 ⁶] ^S	[−2.46, > 10 ⁶] ^S	[−1.71, 61.09]	[−2.26, > 10 ⁶] ^S	[−2.04, > 10 ⁶] ^S
Social Security student benefits	L3	[5.13, 5.23]	[5.16, 5.21]	[5.09, 5.26]	[5.13, 5.23]	[3.38, 3.43]	[7.97, 8.21]	[5.13, 5.23]
Adult Pell introduction	L2	[3.04, 4.75]	[3.63, 4.74]	[1.61, 4.75]	[3.04, 4.75]	[2.19, 2.78]	[4.15, 8.50]	[3.04, 4.75]
Tennessee Promise	L3	— ^S	— ^S	— ^S	— ^S	— ^S	— ^S	— ^S
<i>Education tax benefits</i>								
HOPE tax credit	L3	— ^S	— ^S	— ^S	— ^S	[252.22, > 10 ⁶] ^S	— ^S	— ^S
HOPE: independent single, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−1000.00, > 10 ⁶] ^S	[−52.08, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: dependent single, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−2.697, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: dependent single, end	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−443.66, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−220.34, > 10 ⁶] ^S	[−161.93, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: joint, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−15.87, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
HOPE: joint, end	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−21.92, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , 1,000] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
Tuition deduction: single, start	L2	[< −10 ⁶ , −8.23] ^S	[< −10 ⁶ , −9.74] ^S	[< −10 ⁶ , −7.51] ^S	[< −10 ⁶ , −8.23] ^S	[< −10 ⁶ , −22.08] ^S	[< −10 ⁶ , −6.22] ^S	[< −10 ⁶ , −8.23] ^S
Tuition deduction: joint, start	L2	[−4.29, > 10 ⁶] ^S	[−4.20, > 10 ⁶] ^S	[−4.36, > 10 ⁶] ^S	[−4.29, > 10 ⁶] ^S	[−4.29, > 10 ⁶] ^S	[−4.29, > 10 ⁶] ^S	[−4.29, > 10 ⁶] ^S
HOPE and Lifetime Learning	L2	[−10.75, > 10 ⁶] ^S	[−10.73, > 10 ⁶] ^S	[−10.77, > 10 ⁶] ^S	[−10.75, > 10 ⁶] ^S	[−36.94, 78.21]	[< −10 ⁶ , −7.59] ^S	[−10.75, > 10 ⁶] ^S
AOTC: single, end	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−133.34, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−25.37, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: single, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−148.48, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: joint, end	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−15.53, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−73.99, > 10 ⁶] ^S	[< −10 ⁶ , 1,000] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: joint, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−100.00, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , 1,000] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: independent single, start	L2	[< −10 ⁶ , > 10 ⁶] ^S	[−19.91, > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S
AOTC: simulated instrument	L2	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[< −10 ⁶ , > 10 ⁶] ^S	[−372.79, > 10 ⁶] ^S	[< −10 ⁶ , 36.36] ^S	[< −10 ⁶ , > 10 ⁶] ^S
Tuition deduction: single, end	L2	[< −10 ⁶ , −6.40] ^S	[< −10 ⁶ , −6.51] ^S	[< −10 ⁶ , −6.31] ^S	[< −10 ⁶ , −6.40] ^S	[< −10 ⁶ , −8.34] ^S	[< −10 ⁶ , −5.72] ^S	[< −10 ⁶ , −6.40] ^S
Tuition deduction: joint, end	L2	[−3.86, > 10 ⁶] ^S	[−3.75, > 10 ⁶] ^S	[−3.93, > 10 ⁶] ^S	[−3.86, > 10 ⁶] ^S	[−3.63, > 10 ⁶] ^S	[−3.98, > 10 ⁶] ^S	[−3.86, > 10 ⁶] ^S
<i>Tuition and public-college funding</i>								
Michigan CC tuition	L3	[13.97, > 10 ⁶] ^S	[23.86, > 10 ⁶] ^S	[7.16, > 10 ⁶] ^S	[7.20, > 10 ⁶] ^S	[5.93, 21.20]	[86.35, > 10 ⁶] ^S	[13.97, > 10 ⁶] ^S
College appropriations: spending	L3	[25.23, 25.23]	[34.88, 34.88]	[16.79, 16.79]	[25.23, 25.23]	[8.05, 8.05]	— ^S	[25.23, 25.23]
Texas CC tuition	L3	— ^S	— ^S	— ^S	— ^S	[14.33, 14.33]	— ^S	— ^S

Table E4 (continued): common sensitivity checks

Policy	Set	Main	Narrow	Wide	National	Earnings 75%	Earnings 125%	Denser
College appropriations: tuition	L3	$[-1.05, -1.05]$	$[-0.65, -0.65]$	$[-1.62, -1.62]$	$[-1.05, -1.05]$	$[-0.62, -0.62]$	$[-1.38, -1.38]$	$[-1.05, -1.05]$
<i>College assessment and course placement</i>								
Algorithmic college placement	L2	$[0.09, > 10^6]^S$	$[0.14, > 10^6]^S$	$[0.07, > 10^6]^S$	$[0.09, > 10^6]^S$	$[0.12, > 10^6]^S$	$[0.07, > 10^6]^S$	$[0.09, > 10^6]^S$
<i>Early childhood services</i>								
Early Head Start	L2	$[-1.40, > 10^6]^S$	$[0.23, 13.26]$	$[-1.49, > 10^6]^S$	$[-1.40, > 10^6]^S$	$[-1.21, 17.35]$	$[-1.57, > 10^6]^S$	$[-1.40, > 10^6]^S$
<i>Education channel of other policies</i>								
Cannabis: education channel	L3	$[-50.08, -50.08]$	$[-48.76, -48.76]$	$[-51.86, -51.86]$	$[-42.74, -42.74]$	–	$[-14.61, -14.61]$	$[-50.08, -50.08]$

Notes: Brackets give the span of positive-cost ratios among verified candidates. Superscript S marks self-financing separately. $> 10^6$ and $< -10^6$ indicate attained finite values beyond those display thresholds, not certified infinite bounds. = means that the main baseline is already national. Narrow and wide multiply the support covariance by 0.5 and 2 and timing deviations by their square roots. Income changes scale the earnings and associated fiscal channels; fixed transfers and education-resource components are held fixed. Each row uses the same point-input specification across columns.

Table E4. Point-input sensitivity: numerical comparisons (continued)

Policy	Specific change	Set	WTP (1,000)	Net cost (1,000)	Positive-cost MVPF
Adams Scholarship	Zero negative income-response template	L3	[1.60, 2.27]	[2.86, 4.89]	[0.46, 0.56]
STBF scholarship	No author-model experience delay	L2	[43.57, 86.16]	[18.77, 29.42]	[1.48, 4.59]
STBF scholarship	Printed time-enrolled input (0.360)	L2	[38.51, 81.19]	[20.01, 30.68]	[1.26, 4.06]
Tennessee Promise	Pooled late-completion allowance	L3	[86.44, 86.44]	[−14.83, −14.83]	− ^s
Tennessee Promise	No source-control level scaling	L3	[86.49, 86.49]	[−14.85, −14.85]	− ^s
HOPE: independent single, start	Credit dose: 0.50 dollars	L2	[−53.78, 101.42]	[−17.37, 9.43]	[< −10 ⁶ , > 10 ⁶] ^s
HOPE: independent single, start	Credit dose: 2 dollars	L2	[−53.74, 101.47]	[−17.38, 9.42]	[< −10 ⁶ , > 10 ⁶] ^s
HOPE: dependent single, start	Credit dose: 0.50 dollars	L2	[−16.27, 30.50]	[−2.77, 1.55]	[< −10 ⁶ , > 10 ⁶] ^s
HOPE: dependent single, start	Credit dose: 2 dollars	L2	[−16.28, 30.48]	[−2.76, 1.55]	[< −10 ⁶ , > 10 ⁶] ^s
HOPE: dependent single, end	Credit dose: 0.50 dollars	L2	[−17.36, 32.61]	[−3.37, 1.86]	[−83.48, > 10 ⁶] ^s
HOPE: dependent single, end	Credit dose: 2 dollars	L2	[−17.34, 32.63]	[−3.37, 1.86]	[< −10 ⁶ , > 10 ⁶] ^s
HOPE: joint, start	Credit dose: 0.50 dollars	L2	[−21.40, 40.34]	[−4.69, 2.53]	[−38.45, > 10 ⁶] ^s
HOPE: joint, start	Credit dose: 2 dollars	L2	[−21.31, 40.46]	[−4.70, 2.52]	[−22.42, > 10 ⁶] ^s
HOPE: joint, end	Credit dose: 0.50 dollars	L2	[−23.38, 44.15]	[−4.88, 2.70]	[< −10 ⁶ , > 10 ⁶] ^s
HOPE: joint, end	Credit dose: 2 dollars	L2	[−23.34, 44.20]	[−4.87, 2.70]	[−42.16, > 10 ⁶] ^s
AOTC: single, end	Credit dose: 0.50 dollars	L2	[−21.85, 43.61]	[−4.95, 2.55]	[−55.67, > 10 ⁶] ^s
AOTC: single, end	Credit dose: 2 dollars	L2	[−21.85, 43.61]	[−4.95, 2.56]	[−48.88, > 10 ⁶] ^s
AOTC: single, start	Credit dose: 0.50 dollars	L2	[−20.82, 41.67]	[−4.43, 2.29]	[< −10 ⁶ , > 10 ⁶] ^s
AOTC: single, start	Credit dose: 2 dollars	L2	[−20.71, 41.82]	[−4.44, 2.29]	[< −10 ⁶ , 297.83] ^s
AOTC: joint, end	Credit dose: 0.50 dollars	L2	[−28.22, 56.53]	[−7.06, 3.65]	[−24.17, > 10 ⁶] ^s
AOTC: joint, end	Credit dose: 2 dollars	L2	[−28.25, 56.47]	[−7.04, 3.65]	[< −10 ⁶ , 39.27] ^s
AOTC: joint, start	Credit dose: 0.50 dollars	L2	[−27.42, 55.02]	[−6.68, 3.46]	[< −10 ⁶ , > 10 ⁶] ^s
AOTC: joint, start	Credit dose: 2 dollars	L2	[−27.37, 55.08]	[−6.69, 3.45]	[< −10 ⁶ , > 10 ⁶] ^s
AOTC: independent single, start	Credit dose: 0.50 dollars	L2	[−91.29, 183.17]	[−36.38, 18.39]	[< −10 ⁶ , > 10 ⁶] ^s
AOTC: independent single, start	Credit dose: 2 dollars	L2	[−91.28, 183.19]	[−36.38, 18.39]	[< −10 ⁶ , 17.72] ^s
AOTC: simulated instrument	Credit dose: 0.50 dollars	L2	[−31.48, 55.32]	[−6.14, 3.47]	[< −10 ⁶ , > 10 ⁶] ^s
AOTC: simulated instrument	Credit dose: 2 dollars	L2	[−31.47, 55.33]	[−6.14, 3.46]	[−19.32, > 10 ⁶] ^s
Early Head Start	FY2001 funded-slot cost proxy	L2	[−38.43, 114.65]	[−0.53, 28.10]	[−1.37, > 10 ⁶] ^s
Cannabis: education channel	All-public-college costs	L3	[−2.53, −2.53]	[0.09, 0.09]	[−28.73, −28.73]
Cannabis: education channel	Community-college costs	L3	[−2.74, −2.74]	[0.13, 0.13]	[−21.38, −21.38]
Cannabis: education channel	Equal state weights	L3	[−2.76, −2.76]	[0.13, 0.13]	[−21.21, −21.21]
Cannabis: education channel	Zero negative income-response template	L3	[−2.55, −2.55]	[0.09, 0.09]	[−27.48, −27.48]

Notes: Monetary components use each registered account unit; EHS and Tennessee are disclosed public-data reconstructions. Each component span is computed across paired paths; separate extrema must not be combined to form an MVPF. Tax-dose rows describe finite credit changes and can be small in absolute dollars. Adams L3 uses an explicit time allocation of the separately reported early entry input. The data file reports unrounded NPV, WTP, cost, ratios, and all L2/L3 results.

Table E5. L3 fixed-point minimum-distance fits

ID	Policy	Targets	RMSE	Max. error	NPV	MVPF
E01	Florida grant	5	0.01024	0.01413	13.84	10.97
E05	Wisconsin grant	9	0.01589	0.03093	3.58	1.41
E06	Kalamazoo Promise	3	0.01245	0.01792	54.20	3.61
E17	College spending	5	0.00111	0.00196	8.58	25.23
E19	Texas tuition	8	0.00388	0.00744	13.12	$+\infty$
E21	Tuition caps	5	0.00214	0.00281	-2.99	-1.05
E37	Tennessee Promise	18	0.35251	0.66733	113.94	$+\infty$
E41	Cannabis: education	3	0.00609	0.00927	-3.42	-50.08

Notes: All eight searches retain one best-fitting path with nonzero additional-input residuals and exact L2 constraints. RMSE includes every encoded target, including the exact retained moment(s); errors are schooling years except Tennessee, which is in percentage points. NPV is thousands of case-specific dollars; price years and policy units differ. The identity weight matrix is fixed within each case. These are conditional fitted values, not ranges from SE screens or certified unique/global solutions. Five other admitted L3 cases retain zero-residual path banks. Tables E6 and D5 report individual discrepancies.

Table E6. L3 schooling-input targets, fitted values, and residuals

ID	Source input	Target	Fitted	Residual	Exact L2?
E01	Year 1 cumulative	0.045750	0.056091	+0.010341	–
E01	Year 2 cumulative	0.111083	0.105024	-0.006060	–
E01	Year 3 cumulative	0.160292	0.146835	-0.013457	–
E01	Year 4 cumulative	0.181917	0.181917	-0.000000	Yes
E01	First semester	0.014792	0.028925	+0.014133	–
E05	Cohort 1 cumulative	0.037500	0.068428	+0.030928	–
E05	Cohort 1: year 1 flow	0.020833	0.025938	+0.005104	–
E05	Cohort 1: year 2 flow	0.008333	0.023708	+0.015375	–
E05	Cohort 1: year 3 flow	0.004167	0.018782	+0.014615	–
E05	Cohorts 2–3: year 1 flow	0.016667	0.037330	+0.020664	–
E05	Cohorts 2–3: year 2 flow	0.041667	0.027938	-0.013729	–
E05	Cohort 2: year 3 flow	0.020833	0.016076	-0.004757	–
E05	Cohorts 2–3 cumulative	0.087500	0.073306	-0.014194	–
E05	Retained weighted aggregate	0.071772	0.071772	-0.000000	Yes
E06	Year 4 cumulative	0.273333	0.273333	+0.000000	Yes
E06	Year 2 cumulative	0.134833	0.122854	-0.011980	–
E06	Year 3 cumulative	0.179542	0.197463	+0.017922	–
E17	Retained four-year sum	0.046013	0.046013	+0.000000	Yes
E17	Lag 0 contribution	0.005041	0.006235	+0.001194	–
E17	Lag 1 contribution	0.013199	0.011239	-0.001960	–
E17	Lag 2 contribution	0.014011	0.013845	-0.000166	–
E17	Lag 3 contribution	0.013762	0.014693	+0.000931	–
E19	Model age 18 cumulative	0.031550	0.038841	+0.007291	–
E19	Model age 19 cumulative	0.077550	0.078264	+0.000714	–
E19	Model age 20 cumulative	0.112750	0.114291	+0.001541	–
E19	Model age 21 cumulative	0.145650	0.144871	-0.000779	–
E19	Model age 22 cumulative	0.172150	0.169401	-0.002749	–
E19	Model age 23 cumulative	0.187250	0.188188	+0.000938	–
E19	Model age 24 cumulative	0.209450	0.202006	-0.007444	–
E19	Model age 26 cumulative	0.218450	0.218450	+0.000000	Yes
E21	Retained four-year sum	0.000461	0.000461	+0.000000	Yes
E21	Lag 0 contribution	-0.000921	0.001154	+0.002075	–
E21	Lag 1 contribution	0.003799	0.000994	-0.002805	–
E21	Lag 2 contribution	0.001785	-0.000126	-0.001910	–
E21	Lag 3 contribution	-0.004202	-0.001562	+0.002640	–
E41	Four-year completion bridge	-0.032000	-0.026976	+0.005024	–
E41	Five-year completion bridge	-0.022000	-0.031273	-0.009273	–
E41	Six-year completion bridge	-0.036000	-0.036000	+0.000000	Yes

Notes: All entries are schooling-equivalent years. Residual equals fitted minus target. The retained L2 moments are exact to numerical precision; additional observations are fixed point targets, not intervals. Display rounding does not change the fitted inputs. The Wisconsin semester and cumulative estimates do not obey an established exact adding-up identity; their reported precision and cohort windows are preserved. The age, population, and conversion bridges are stated in Appendix E.2. No standard error is used to widen acceptance or weight these fits.

Table F1. Willingness to pay and net fiscal cost: point paths and input repetitions

ID	Layer	WTP: point	WTP: repetitions	Net cost: point	Net cost: repetitions
E01	L2	[2.36, 28.08]	[-8.03, 52.16]	[0.99, 1.24]	[-4.08, 7.15]
E01	L3	[11.27, 11.27]	NR	[1.03, 1.03]	NR
E02	L2	[7.45, 49.18]	[2.47, 87.48]	[-3.49, 0.85]	[-12.27, 6.49]
E03	L2	[-11.30, 7.56]	[-762.71, 705.72]	[-19.64, 14.57]	[-220.58, 212.24]
E03	L3	[1.82, 3.00]	[-45.79, 56.43]	[2.85, 4.89]	[-10.11, 18.56]
E04	L2	[38.46, 81.13]	[30.92, 82.06]	[20.03, 30.70]	[19.78, 32.54]
E05	L2	[-10.84, 40.78]	[-26.33, 67.35]	[5.59, 6.87]	[-0.97, 12.94]
E05	L3	[8.98, 8.98]	NR	[6.38, 6.38]	NR
E06	L2	[14.25, 72.47]	[-10.60, 117.97]	[16.05, 16.95]	[5.86, 26.55]
E06	L3	[58.80, 58.80]	NR	[16.29, 16.29]	NR
E07	L2	[-7.57, 20.10]	[-17.91, 39.64]	[-2.71, 1.61]	[-7.46, 3.84]
E07	L3	[1.57, 3.02]	[0.04, 6.30]	[-0.10, 0.11]	[-0.75, 0.52]
E08	L2	[-25.03, 38.02]	[-48.97, 69.47]	[-0.09, 2.76]	[-6.44, 10.87]
E08	L3	[-4.52, -0.25]	NR	[0.76, 0.96]	NR
E09	L2	[0.17, 29.85]	[-1.85, 52.51]	[0.93, 3.80]	[-1.66, 10.04]
E10	L2	[-24.90, 41.94]	[-45.06, 75.14]	[0.42, 2.00]	[-6.32, 10.85]
E11	L2	[-23.23, 35.90]	[-49.35, 67.61]	[-6.82, 13.49]	[-51.50, 46.82]
E15	L2	[-5.51, 43.77]	[-15.11, 85.99]	[-2.22, 2.70]	[-12.24, 8.32]
E16	L2	[35.17, 49.70]	[2.34, 90.95]*	[4.74, 7.61]	[-2.18, 12.88]*
E16	L3	[36.98, 37.39]	NR	[7.14, 7.21]	NR
E17	L2	[-14.20, 36.59]	[-27.11, 65.50]	[-3.29, 2.97]	[-9.99, 6.03]
E17	L3	[7.57, 7.57]	[1.72, 14.73]	[0.30, 0.30]	[-1.06, 1.56]
E18	L2	[-38.43, 114.65]	[-76.78, 114.86]	[-1.10, 27.52]	[-1.14, 34.63]
E19	L2	[5.88, 24.15]	[-2.81, 46.38]	[-1.63, 0.61]	[-6.55, 3.90]
E19	L3	[10.88, 10.88]	[1.99, 22.77]	[-0.08, -0.08]	[-2.37, 2.04]
E20	L2	[0.27, 1.28]	[-0.13, 2.58]	[0.09, 0.27]	[0.01, 0.56]
E21	L2	[-17.26, 31.16]	[-33.61, 59.18]	[-0.94, 2.23]	[-9.07, 6.91]
E21	L3	[-1.21, -1.21]	[-10.37, 6.76]	[1.15, 1.15]	[0.23, 2.67]
E22	L2	[-5.65, 20.51]	[-12.55, 38.78]	[-2.22, 1.29]	[-6.68, 2.96]
E22	L3	[7.67, 9.04]	[0.03, 20.99]	[-0.68, -0.49]	[-3.19, 1.20]
E23	L2	[-53.77, 101.43]	[-97.27, 184.42]	[-17.37, 9.43]	[-38.00, 19.98]
E24	L2	[-16.27, 30.50]	[-30.46, 57.47]	[-2.76, 1.55]	[-9.38, 5.44]
E25	L2	[-17.36, 32.62]	[-32.35, 61.24]	[-3.37, 1.86]	[-10.39, 5.82]
E26	L2	[-21.37, 40.38]	[-39.52, 74.85]	[-4.69, 2.53]	[-12.63, 6.67]
E27	L2	[-23.37, 44.17]	[-42.83, 81.41]	[-4.88, 2.70]	[-13.73, 7.18]
E28	L2	[-9.45, 18.84]	[-18.91, 37.98]	[-3.48, 1.15]	[-8.28, 3.48]
E29	L2	[-30.55, 47.31]	[-60.55, 94.27]	[-9.99, 7.12]	[-20.95, 14.12]
E30	L2	[-12.73, 4.32]	[-26.00, 19.17]	[-0.34, 1.18]	[-2.82, 3.94]
E31	L2	[-21.85, 43.61]	[-40.39, 81.31]	[-4.95, 2.55]	[-13.53, 7.04]
E32	L2	[-20.78, 41.72]	[-38.73, 78.22]	[-4.43, 2.29]	[-12.69, 6.65]
E33	L2	[-28.23, 56.51]	[-51.57, 103.77]	[-7.05, 3.65]	[-17.97, 8.90]

Continued on next page

Table F1 (continued)

ID	Layer	WTP: point	WTP: repetitions	Net cost: point	Net cost: repetitions
E34	L2	[−27.40, 55.04]	[−50.12, 101.16]	[−6.69, 3.45]	[−17.35, 8.56]
E35	L2	[−91.29, 183.18]	[−163.36, 329.98]	[−36.38, 18.39]	[−72.70, 35.95]
E36	L2	[−31.48, 55.32]	[−57.80, 102.28]	[−6.14, 3.47]	[−16.84, 9.45]
E37	L2	[−48.72, 152.50]	[−58.19, 160.81]	[−38.04, 32.66]	[−40.97, 35.97]
E37	L3	[86.44, 86.44]	[53.06, 120.42]	[−14.83, −14.83]	[−26.99, −3.21]
E38	L2	[0.22, 0.22]	[0.22, 0.22]	[−5.12, 2.38]	[−5.86, 2.84]
E39	L2	[−25.86, 41.73]	[−50.69, 82.34]	[−6.74, 4.04]	[−16.12, 9.87]
E40	L2	[−30.22, 49.05]	[−60.77, 99.64]	[−11.39, 7.84]	[−22.99, 14.82]
E41	L2	[−9.83, 13.54]	[−10.49, 14.30]	[−0.18, 0.29]	[−0.21, 0.30]
E41	L3	[−2.35, −2.35]	[−3.08, −1.59]	[0.05, 0.05]	[0.03, 0.07]

Notes: All components are in thousands of the retained case-specific dollars. Monetary years, populations, and policy units differ across cases. Point entries are finite-bank extrema; repetition entries summarize input-draw endpoint quantiles. WTP and cost extrema can use different paths and must not be divided to form an MVPF. Stars mark verified subsets, including 952 L2 draws in E16. E19 L2 has 1,000 verified component repetitions but 999 defined ratios; its new L3 has 1,000 defined ratios. E21 L3 has 1,000 component repetitions and 998 defined ratios. E01, E05, E06, E08, and E16 L3 lack joint draws; E37 L3 uses all 1,000 registered normal-input perturbations with eighteen fixed-rule point-target fits; these are not source permutation draws. All results condition on ACS calibration and the retained accounting. No probability is assigned to a path.

Table F2. Sensitivity to simultaneous marginal input regions without joint draws

ID	Moments	Critical value	WTP	Net cost	Defined MVPF span
E01	5	2.576	[−8.96, 35.01]	[0.84, 1.29]	[−10.68, 27.07]
E05	9	2.773	[−9.92, 46.26]	[5.54, 6.78]	[−1.46, 8.35]
E06	3	2.394	[−12.19, 84.07]	[15.58, 17.11]	[−0.75, 4.99]
E08	2	2.241	[−34.37, 45.54]	[−0.61, 3.15]	[−1.43 × 10 ⁶ , 14.45]
E16	2	2.241	[−6.48, 74.39]	[1.38, 13.66]	[−0.47, 53.78]

Notes: For J encoded source moments, a marginal-normal Bonferroni rectangle uses critical value $\Phi^{-1}(1 - 0.05/(2J))$. This gives an approximate 95 percent simultaneous region for the input means under the marginal normal approximations, without a covariance matrix. For E05 annual credit sums and its aggregate input, marginal SE upper bounds sum the relevant reported SEs. E01 includes its recovered first-semester coefficient. All encoded means, including L2, vary within that rectangle. The table reports extrema among paths found inside it, holding ACS and all accounting prices fixed; these are not certified projections or 95 percent confidence limits for welfare. WTP and net cost use thousands of case-specific dollars. These calculations are separate sensitivities, not bootstrap replacements, and do not enter the main sampling bands or ranking.

Table G1. Published male and female degree effects and standardized education NPVs

Policy	Sex	Effect (pp)	SE (pp)	Paths	NPV span (\$1,000)
Adams Scholarship	Men	-0.8	0.9	2,325	[-15.36, 1.21]
	Women	-1.4	1.0	1,913	[-16.26, -0.61]
STBF scholarship	Men	8.0	–	5,173	[4.22, 12.63]
	Women	8.0	–	5,232	[4.42, 11.95]
Wisconsin Scholars Grant	Men	4.7	–	2,145	[-14.34, 8.28]
	Women	4.7	–	2,108	[-12.21, 3.46]
Kalamazoo Promise	Men	4.1	7.0	5,642	[-3.64, 6.69]
	Women	13.2	7.7	5,436	[8.46, 16.53]
Michigan CC tuition	Men	1.7	1.0	7,737	[-1.42, 6.35]
	Women	1.9	1.1	7,040	[-2.22, 4.80]
Early Head Start	Men	13.0	4.0	1,063	[-5.47, 12.34]
	Women	21.2	5.0	946	[-1.17, 13.96]
Tennessee Promise	Men	1.2	–	2,888	[-2.13, 10.52]
	Women	4.1	–	902	[-1.71, 7.09]

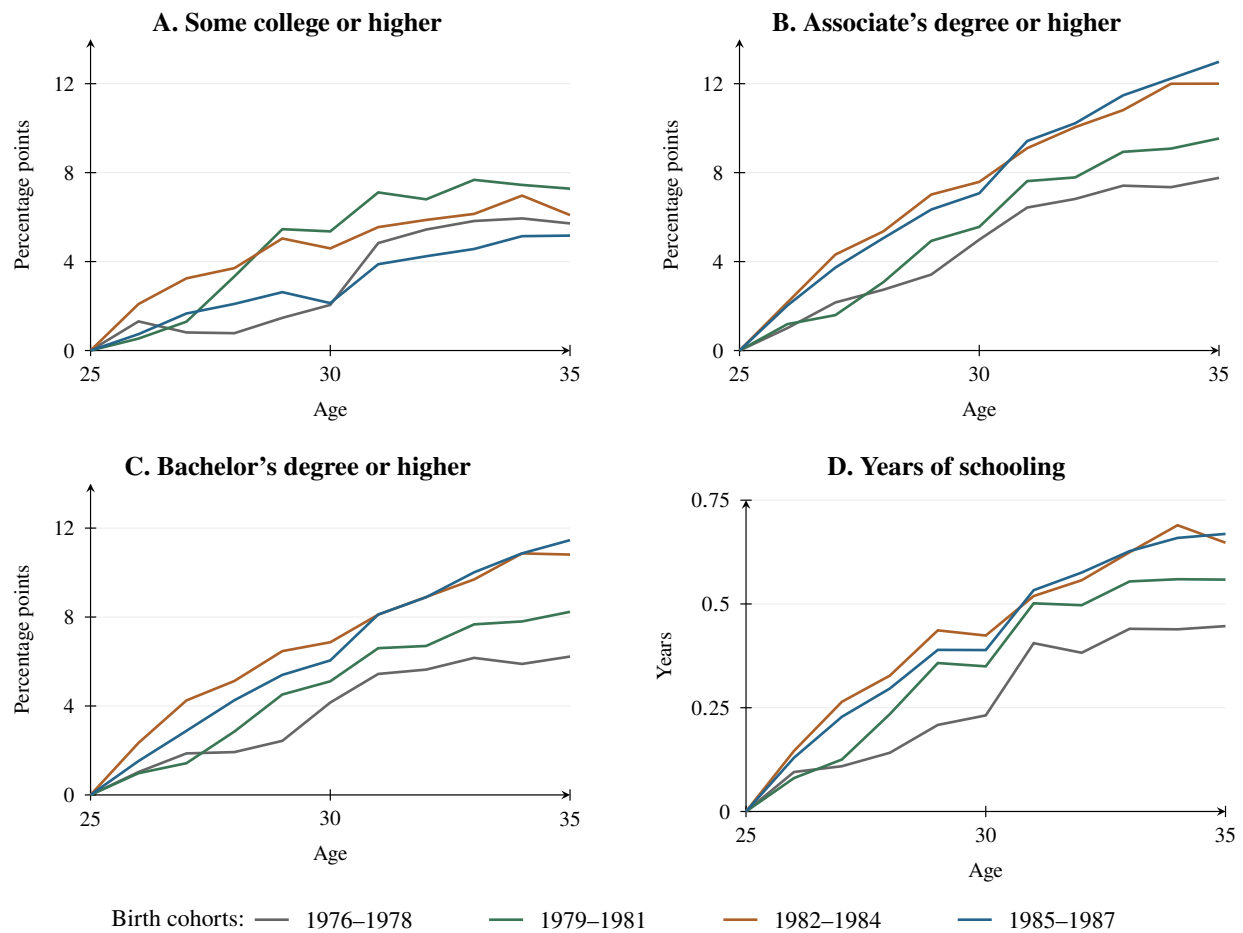
Notes: Effects are the source studies' subgroup point estimates, in percentage points. Source standard errors are reported where available; a dash means that a subgroup SE was not recovered. Tennessee retains a separate auxiliary subgroup interval sensitivity. STBF and Wisconsin use rounded published coefficients; their identical displayed effects do not establish exact equality. The NPV spans allow the terminal effects to vary and therefore combine timing with eventual attainment. They summarize verified finite candidate banks, with no probabilities assigned to paths. All accounts use a common two-schooling-year equivalent per BA increment and the Chapter 3 earnings and forgone-work prices, in 2000 dollars discounted to age 18 at 3 percent. No other education changes, transfers or fiscal costs are included. These are education-channel values, not total-policy MVPFs. Source tables, populations, age mappings and estimands are described in Appendix G.1.

Table G2. Male and female education NPVs at the same terminal effect

Policy	η (pp)	Sex-specific reference curves		Common reference curve	
		Paths (M/F)	$V_f - V_m$ (\$)	Paths (M/F)	$V_f - V_m$ (\$)
Adams Scholarship	-15	32/64	[-449, -249]	43/42	[-101, 35]
	-10	62/64	[-539, -31]	44/46	[-252, 179]
	-5	93/54	[-675, 332]	69/75	[-468, 384]
	0	122/0	—	95/0	—
STBF scholarship	2	0/0	—	0/0	—
	5	71/72	[-585, 733]	72/72	[-664, 664]
	10	66/70	[-626, 559]	67/67	[-587, 587]
	15	0/0	—	0/0	—
Wisconsin Scholars Grant	2	74/83	[-95, 863]	84/84	[-425, 425]
	5	90/0	—	98/98	[-576, 576]
	10	0/0	—	0/0	—
	15	0/0	—	0/0	—
Kalamazoo Promise	2	25/0	—	24/0	—
	5	26/0	—	26/0	—
	10	0/19	—	25/20	[475, 1, 279]
	15	0/18	—	0/19	—
Michigan CC tuition	2	63/65	[-602, 404]	63/64	[-474, 527]
	5	58/60	[-513, 317]	59/60	[-391, 442]
	10	0/0	—	0/0	—
	15	0/0	—	0/0	—
Early Head Start	2	45/43	[37, 208]	49/33	[210, 356]
	5	45/51	[35, 194]	50/44	[210, 347]
	10	41/59	[29, 231]	48/52	[217, 383]
	15	0/0	—	0/53	—
Tennessee Promise	2	21/11	[39, 320]	13/14	[58, 316]
	5	22/11	[46, 277]	13/16	[85, 312]
	10	21/0	—	0/17	—
	15	0/0	—	0/0	—

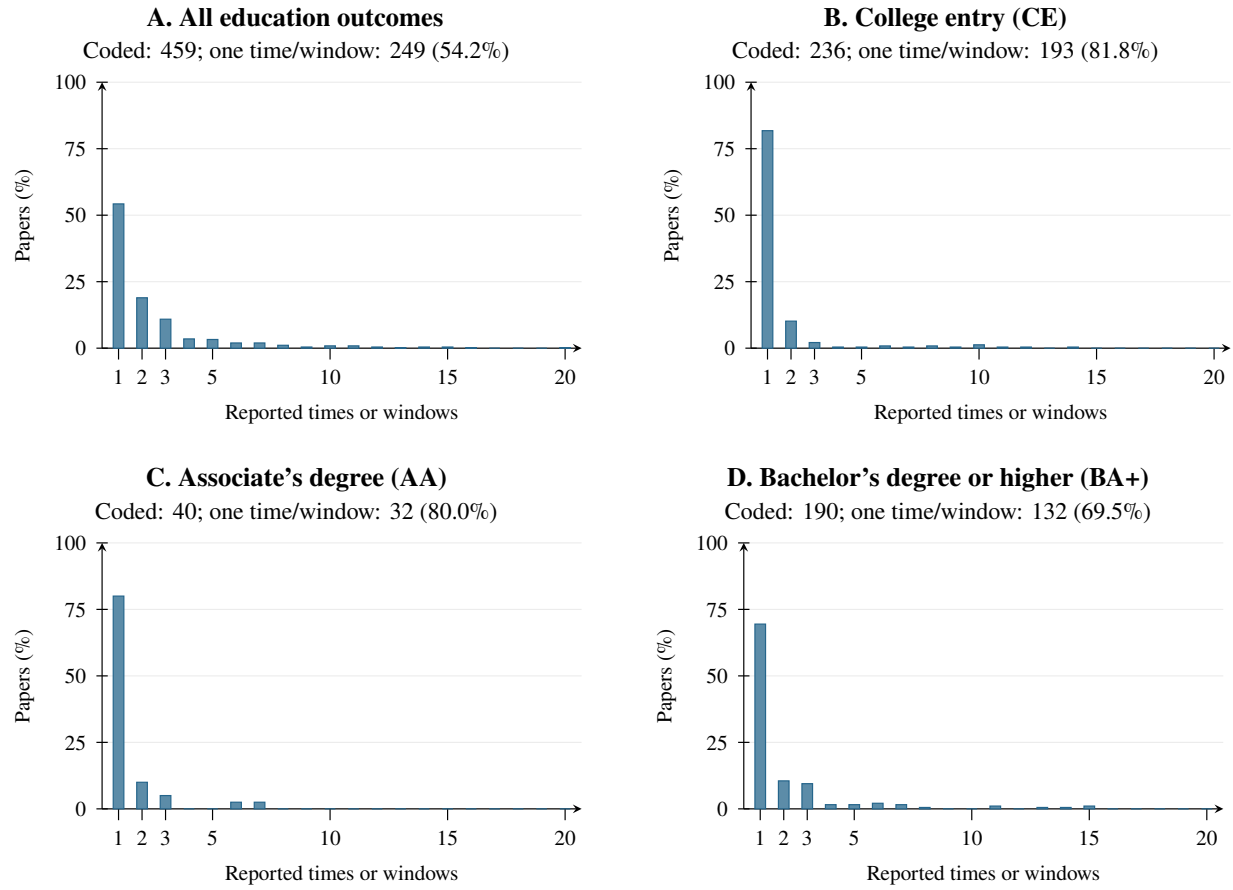
Notes: Each pair matches its own male and female source effects and the same age-65 BA effect η . Prices and exposure-age weights are identical across sexes within a policy. Negative η represents a common terminal reduction, as allowed for Adams. The common-reference sensitivity replaces both reference curves by their equally weighted mean, retaining the same inputs and grid. Entries count verified paths, not observations or repetitions. The displayed dollar span runs from the smallest to the largest female-minus-male NPV difference among the retained pairs. A dash denotes no verified pair at that grid value; finite search failure does not prove incompatibility. Spans are numerical summaries, not confidence intervals or certified global bounds. Tennessee also retains a separate auxiliary subgroup interval sensitivity. Policy populations and estimands differ, so these are within-policy comparisons rather than a cross-policy welfare ranking.

Figure A1. Education growth after age 25 across birth cohorts



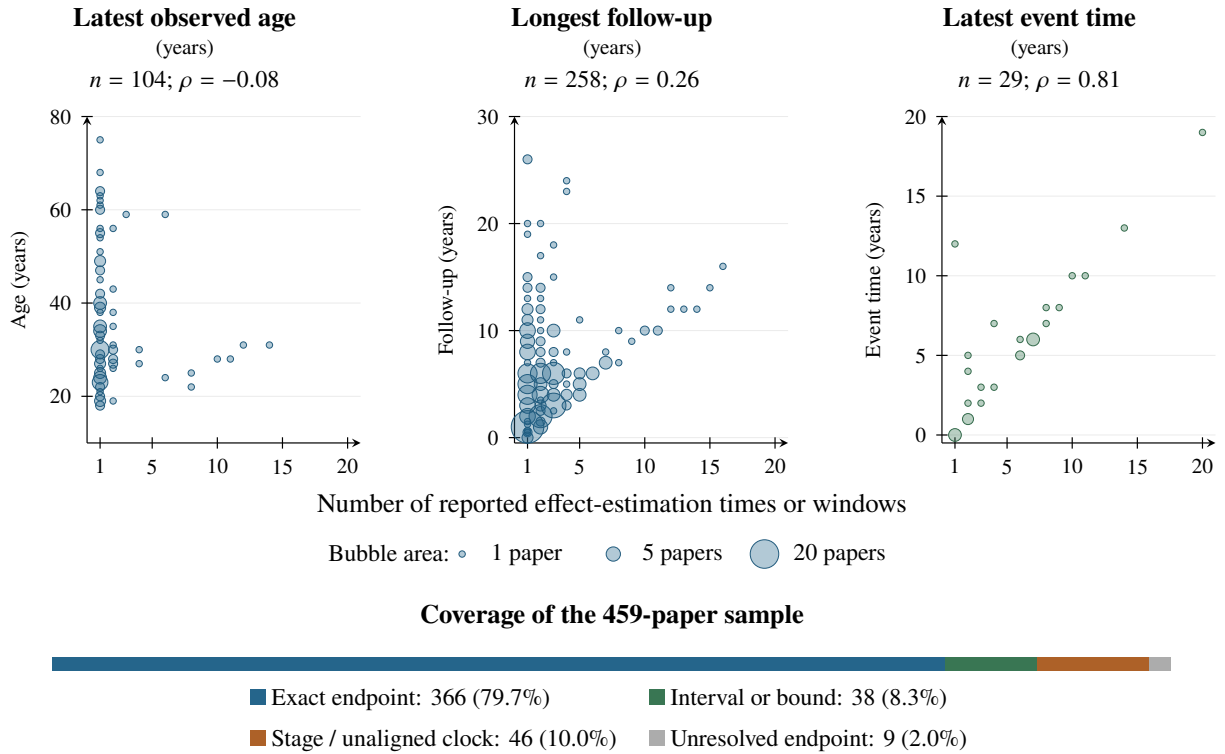
Notes: Each curve measures attainment relative to age 25 for its three-cohort group. ACS 2000–2022 includes all birthplaces and racial and ethnic groups. Person weights apply within cohort–age cells; cohorts have fixed one-third weights. Attainment panels share percentage-point scales; schooling uses years. SC+ is some college or higher; AA+ and BA+ are cumulative thresholds. Appendix A.6 documents the descriptive comparison.

Figure A2. Number of reported effect-estimation times or windows



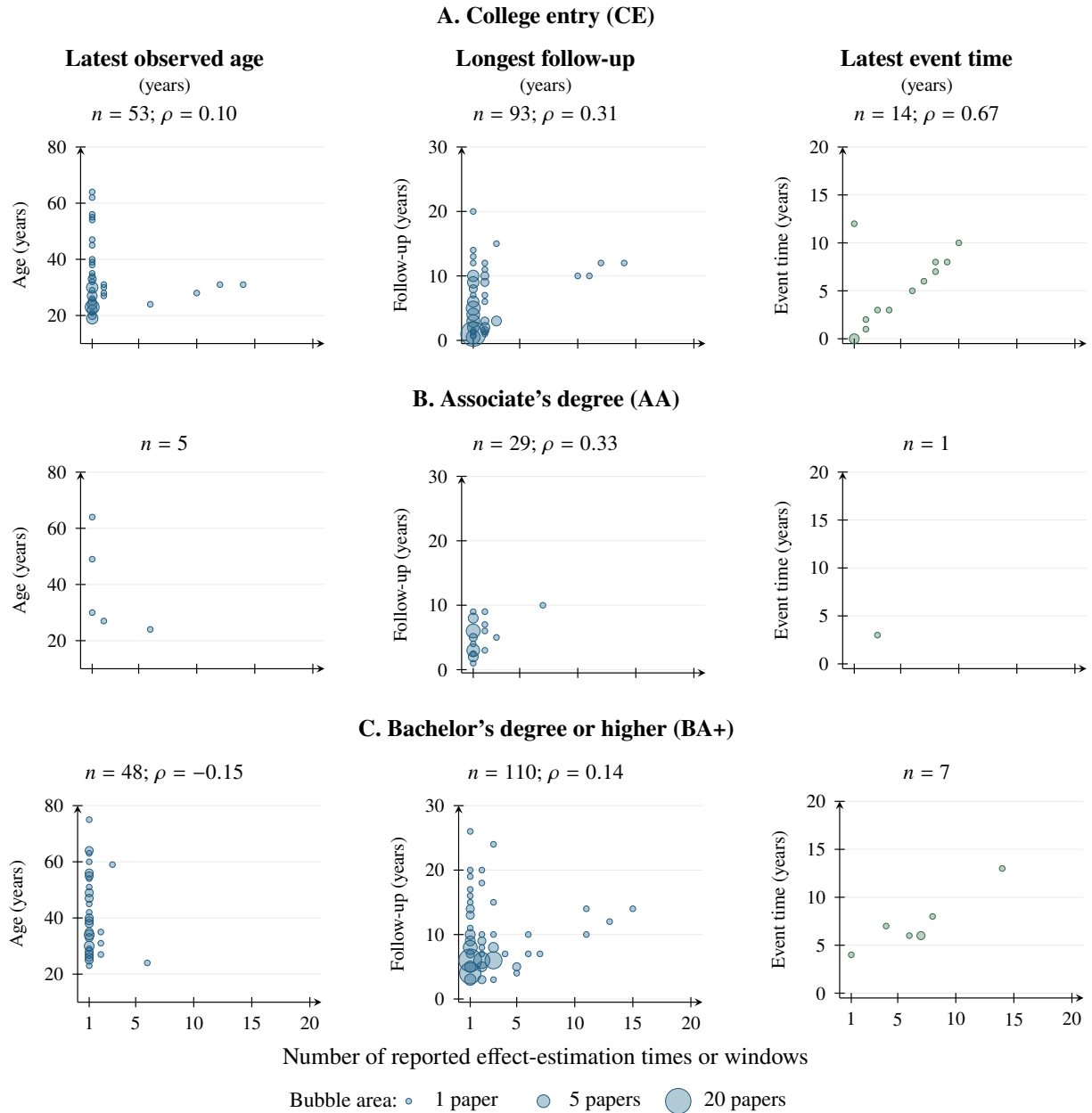
Notes: Percentages use papers with a usable count in each overlapping outcome group; one CE paper is excluded. A point or pooled window counts once; separately reported horizons count separately. Counts do not measure follow-up length or effect persistence. Axes are common. Table A1 gives counts; Appendix A.2 gives coding rules.

Figure A3. Estimation times and observation endpoints: all education outcomes



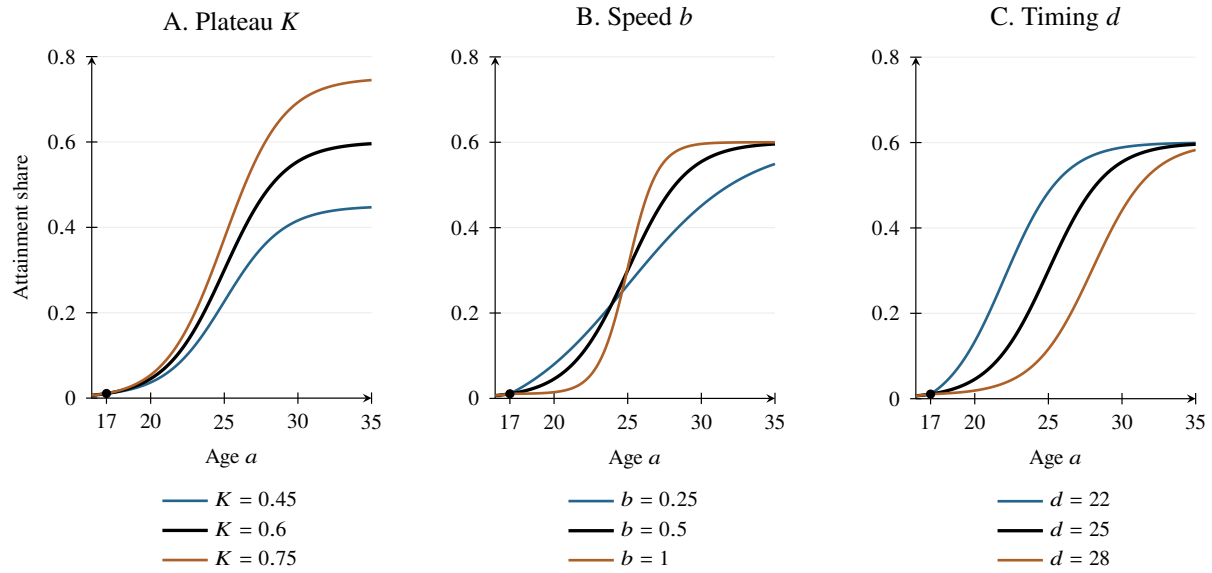
Notes: Bubble area counts papers at each coordinate. Panels separate attained age, individual follow-up, and policy event time, covering 366 distinct papers with overlap across clocks. The coverage bar includes all 459 papers; 93 lack an exact coordinate and receive no imputed endpoint. Correlations are paper-level Spearman correlations. Overall counts and endpoints may summarize different outcomes. Appendix A.2 gives coding rules.

Figure A4. Estimation times and observation endpoints by education outcome



Notes: Rows show college entry, associate-degree outcomes, and BA+ outcomes; columns retain separate clocks. Scales are common within clocks; bubble areas share one count scale. Coverage is 146 CE, 34 AA, and 158 BA+ papers, with overlap. Spearman correlations require at least ten papers and variation in both coordinates. AA preserves each source's exact- or at-least-degree definition.

Figure B1. Parameter changes under a common pre-policy anchor



Notes: One parameter varies per panel; the others remain at $(K, b, d) = (0.60, 0.50, 25)$. All paths share the age-17 reference level. Black, blue, and orange denote reference, smaller, and larger parameter values. K governs the plateau; b and d govern speed and timing conditional on the anchor. In this normalized family, d need not mark half the plateau. Axes show hypothetical attainment shares.

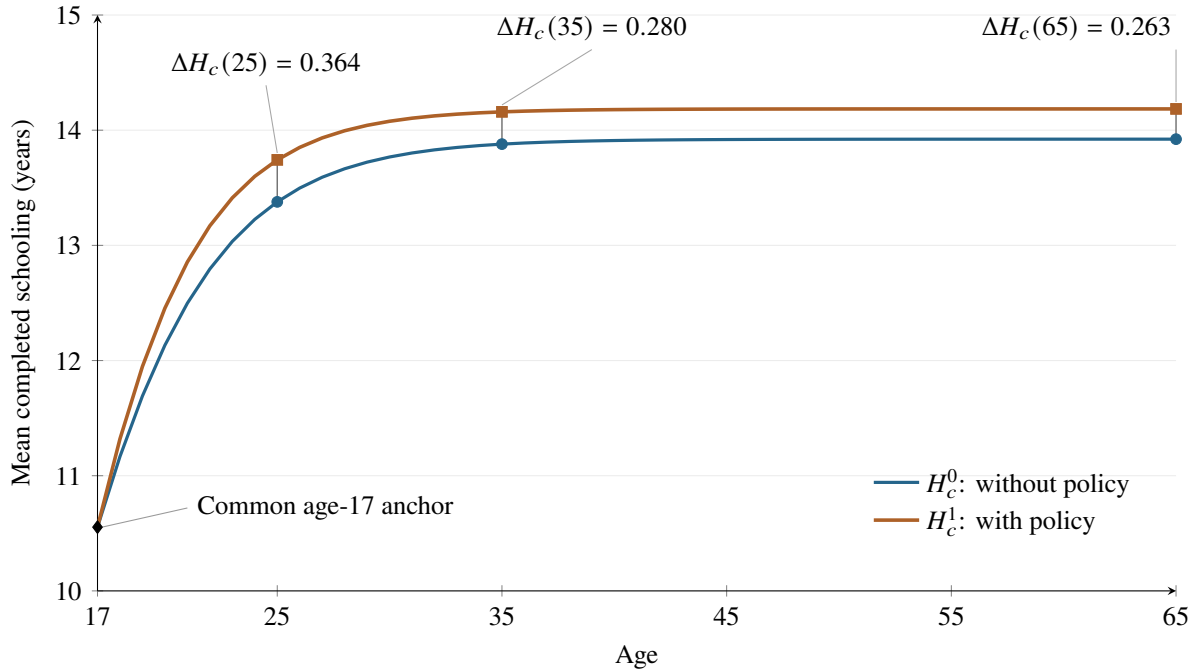


Figure B2. Schooling paths underlying Figure 4

Notes: Blue shows $H_c^0(a)$ and orange shows $H_c^1(a)$ for the same joint-change illustration used in Figure 4. The comparison path averages the fitted ACS profiles for the 1985–1987 birth cohorts with fixed one-third weights. The treated path uses $(\Delta K_c, \Delta b_c, \Delta d_c) = (0.2625, 0.0300, -0.4500)$ and shares the comparison level at age 17. Lines connect the annual values used in the NPV accounts; markers and labels identify the gaps at ages 25, 35, and 65. The early schooling advantage partly narrows as the comparison population catches up, while a positive gap remains at age 65. Baseline values beyond age 35 are extrapolated. The figure displays one stipulated comparison of potential paths.

● L2 match

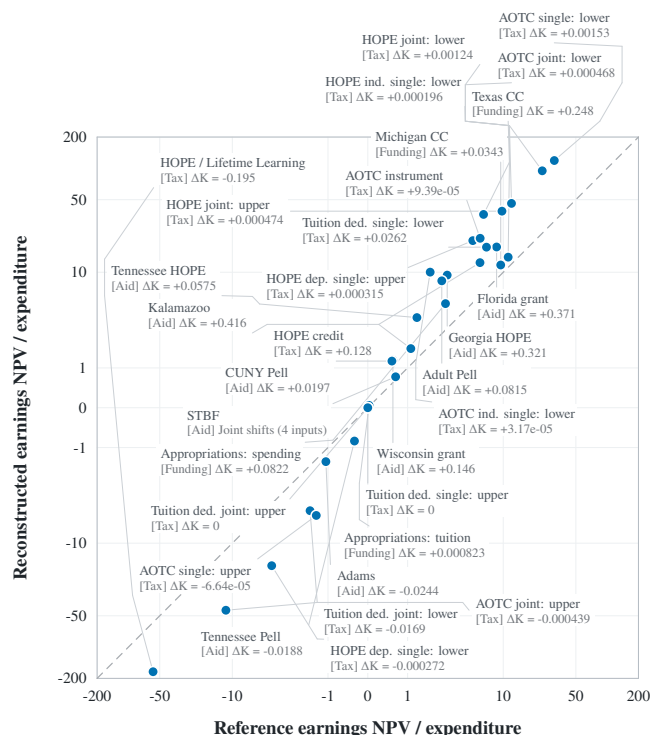
⊙ Same match also passes L3

--- Equality: reconstructed = reference

Earnings NPV / reference expenditure

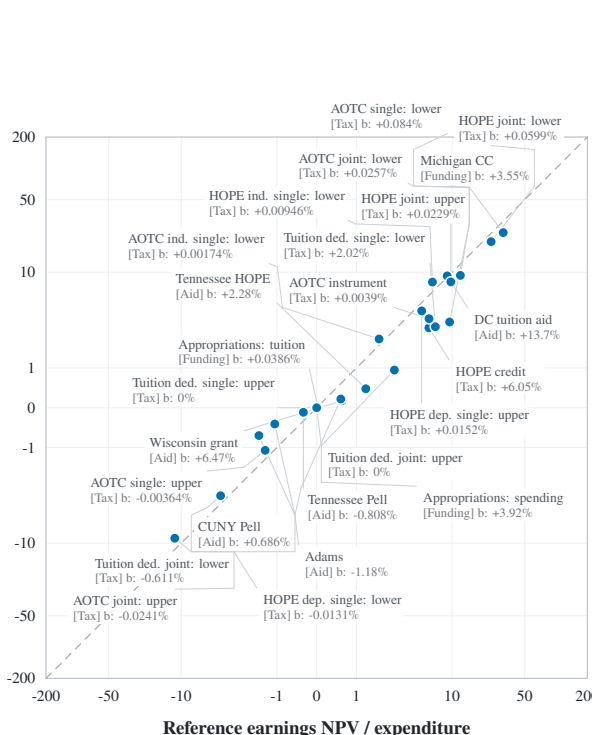
A. Permanent effect (K)

31 paired / 37 reference calculations



B. Temporary effect (b)

25 paired / 37 reference calculations



C. Temporary effect (d)

18 paired / 37 reference calculations

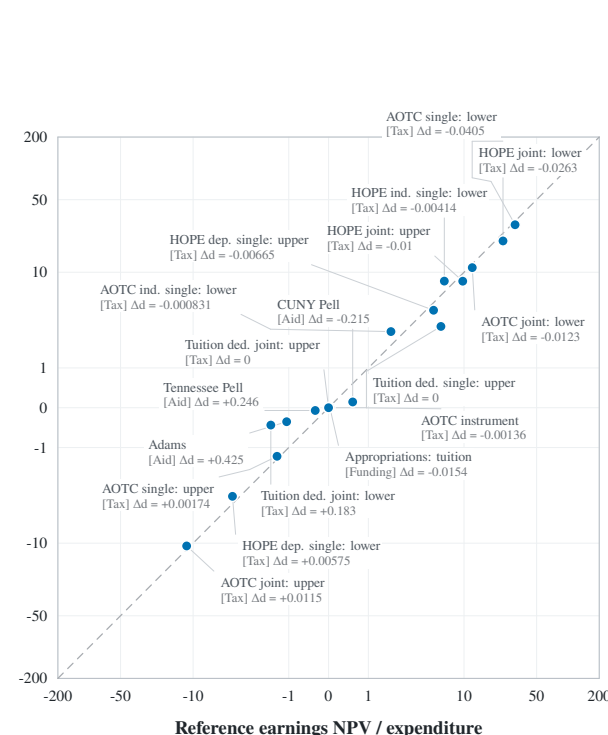


Figure B3. Reference and reconstructed earnings NPVs under permanent and temporary effects

Notes: Both earnings NPVs are divided by the same positive reference expenditure within each calculation, using matched monetary units. Panels use identical asinh scales and square axes; tick labels give actual normalized values. Blue points match L2; red rings mark the same path also satisfying L3. Bracketed labels give the policy types defined in Figure 8. Labels report ΔK in schooling years, proportional b changes in percent, and Δd in years; the other two parameter families stay fixed. STBF uses four joint time and credential inputs; Tennessee Promise uses three credential inputs. Permanent effects change eventual attainment; temporary effects eventually vanish but can retain lifetime value. Early Head Start and Tennessee Promise lack a verified common price vintage for their public reconstruction and source NPV and are excluded from the paired monetary plot. Table B3 gives the values and accounts for omitted pairs. Self-financing policies can have finite earnings NPVs.

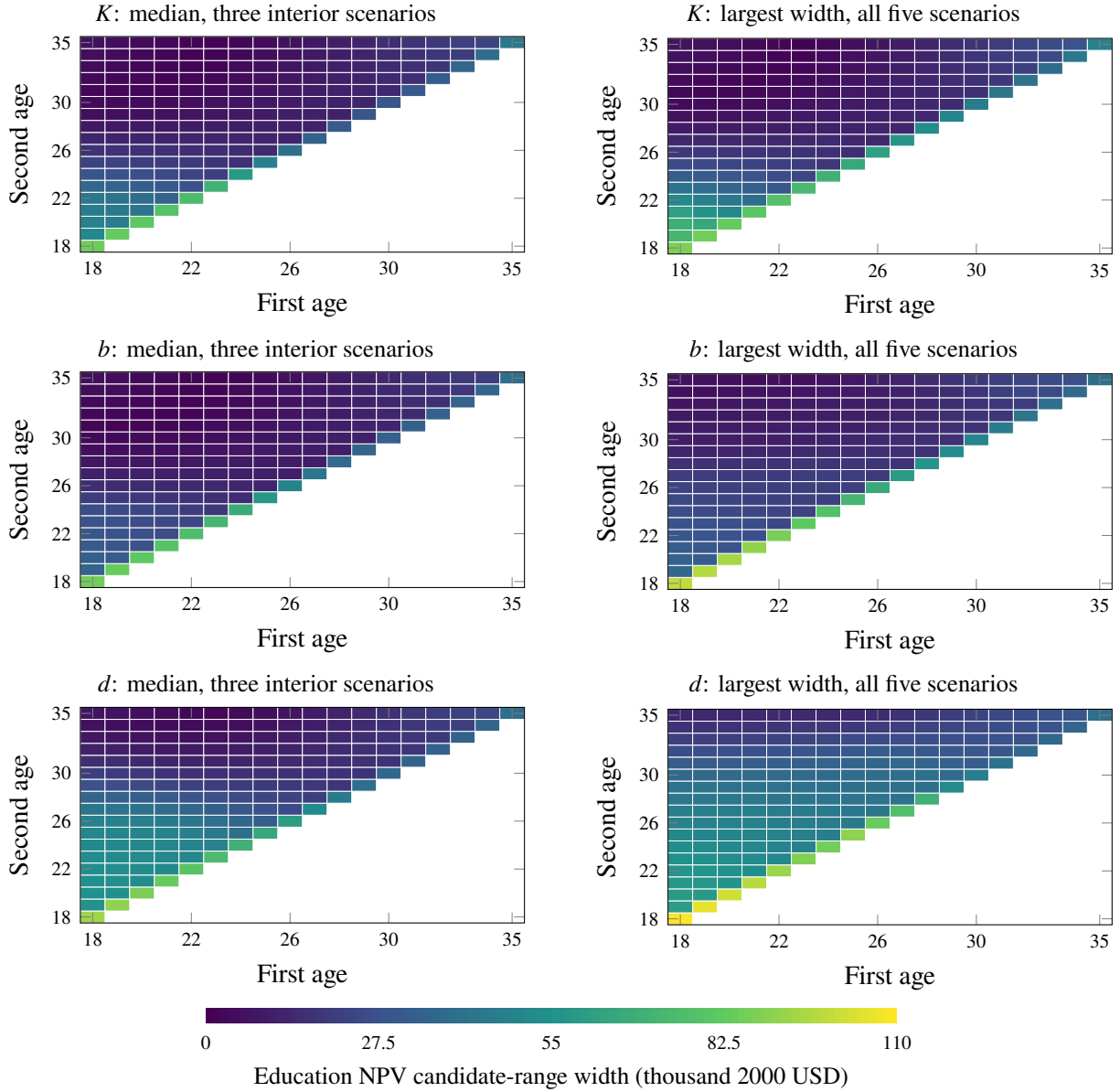


Figure B4. Observation timing and remaining education-NPV ranges

Notes: Diagonal cells represent single-age observations; upper-triangle cells represent two distinct ages. All panels and all four heatmap figures use the same width scale, in thousands of 2000 dollars per person. The widths come from verified numerical candidates under the common baseline, path restrictions, and valuation rule; they are not confidence intervals or certified global bounds. Left panels report medians across the three interior scenarios; right panels report largest widths across all five scenarios.

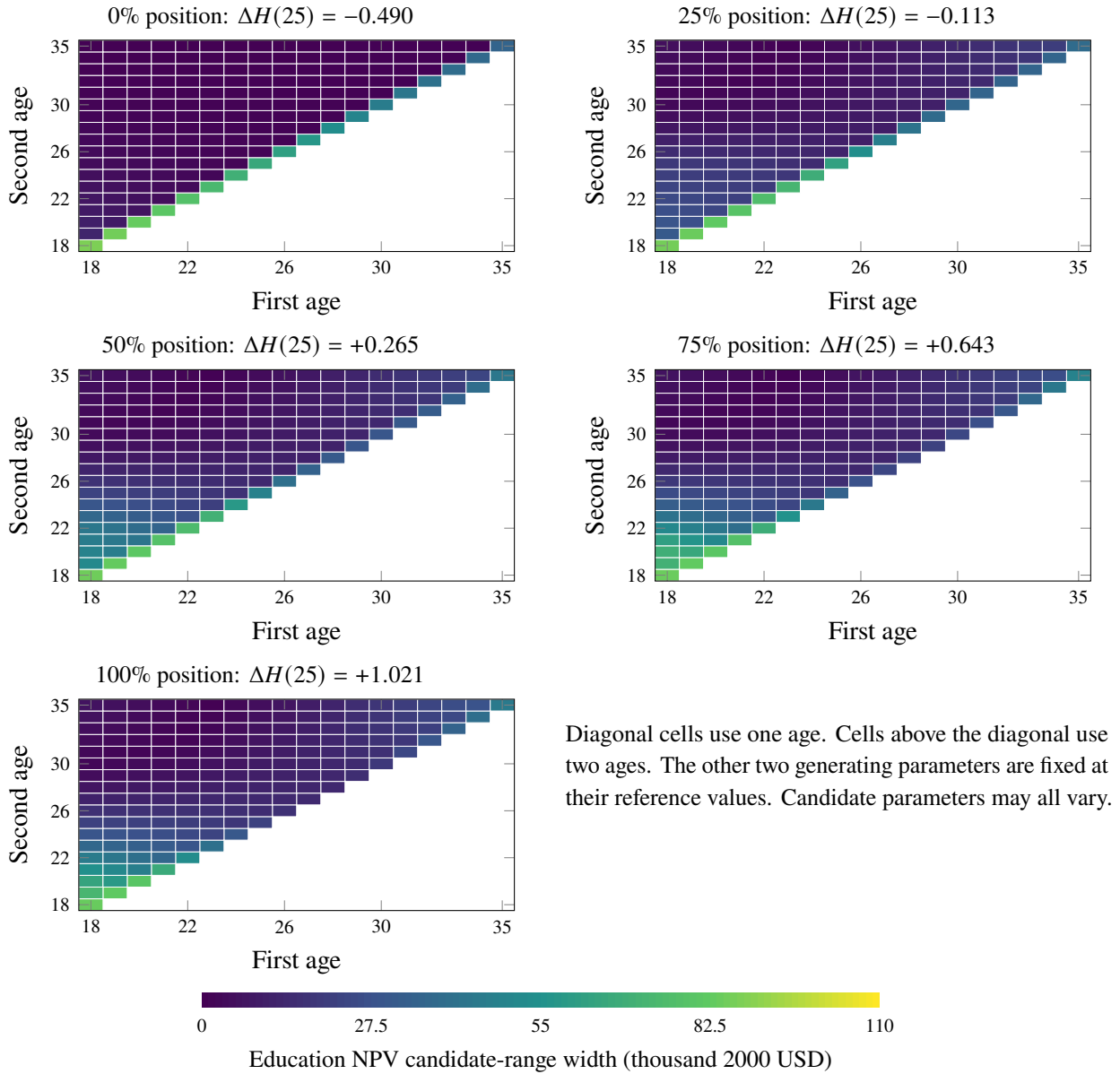


Figure B5. All five plateau scenarios

Notes: Diagonal cells represent single-age observations; upper-triangle cells represent two distinct ages. All panels and all four heatmap figures use the same width scale, in thousands of 2000 dollars per person. The widths come from verified numerical candidates under the common baseline, path restrictions, and valuation rule; they are not confidence intervals or certified global bounds. Each panel fixes one generating scenario at the indicated position of its feasible single-parameter interval. The other two generating parameters remain at their reference values. All three candidate parameters may vary.

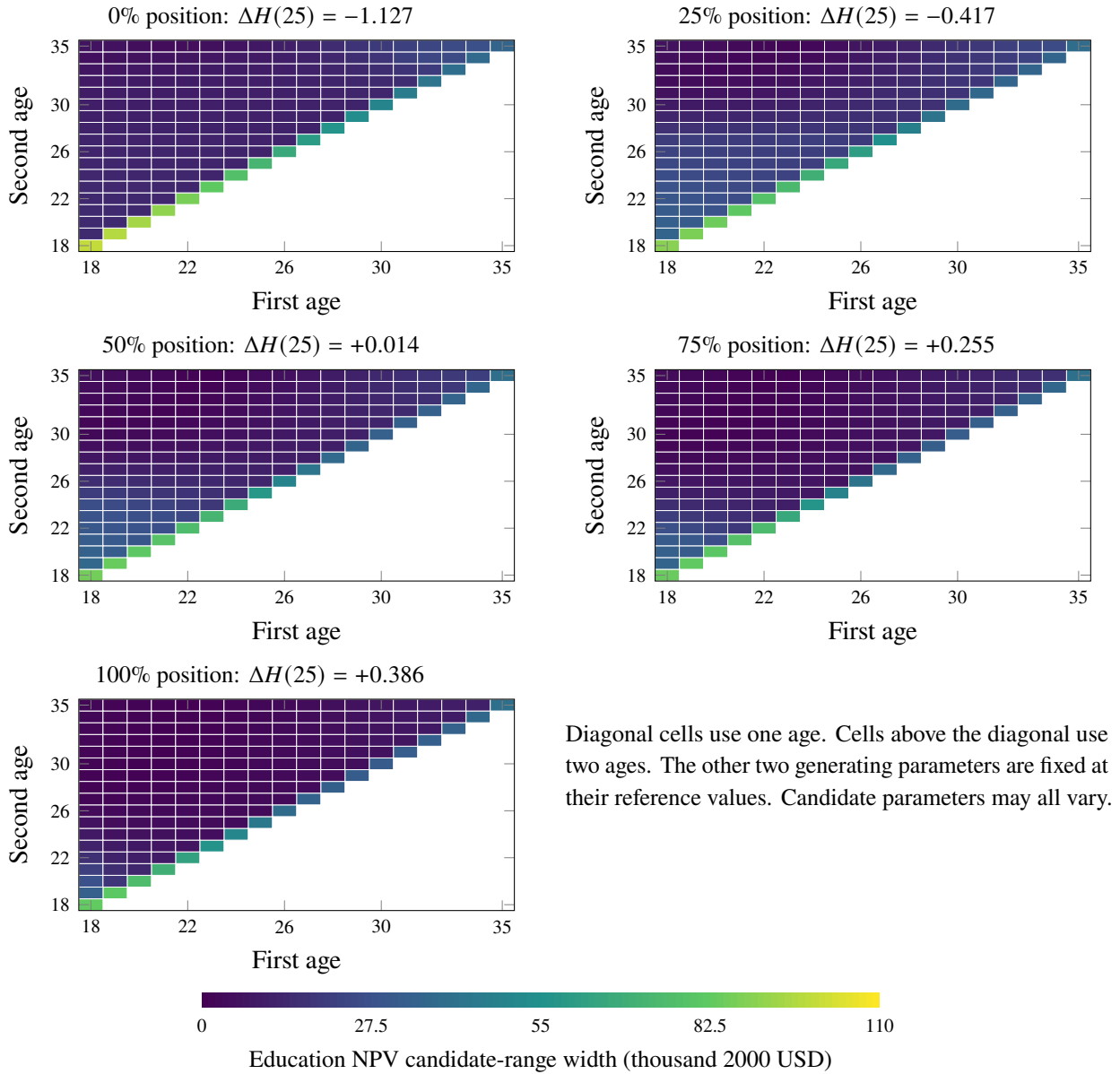


Figure B6. All five speed scenarios

Notes: Diagonal cells represent single-age observations; upper-triangle cells represent two distinct ages. All panels and all four heatmap figures use the same width scale, in thousands of 2000 dollars per person. The widths come from verified numerical candidates under the common baseline, path restrictions, and valuation rule; they are not confidence intervals or certified global bounds. Each panel fixes one generating scenario at the indicated position of its feasible single-parameter interval. The other two generating parameters remain at their reference values. All three candidate parameters may vary.

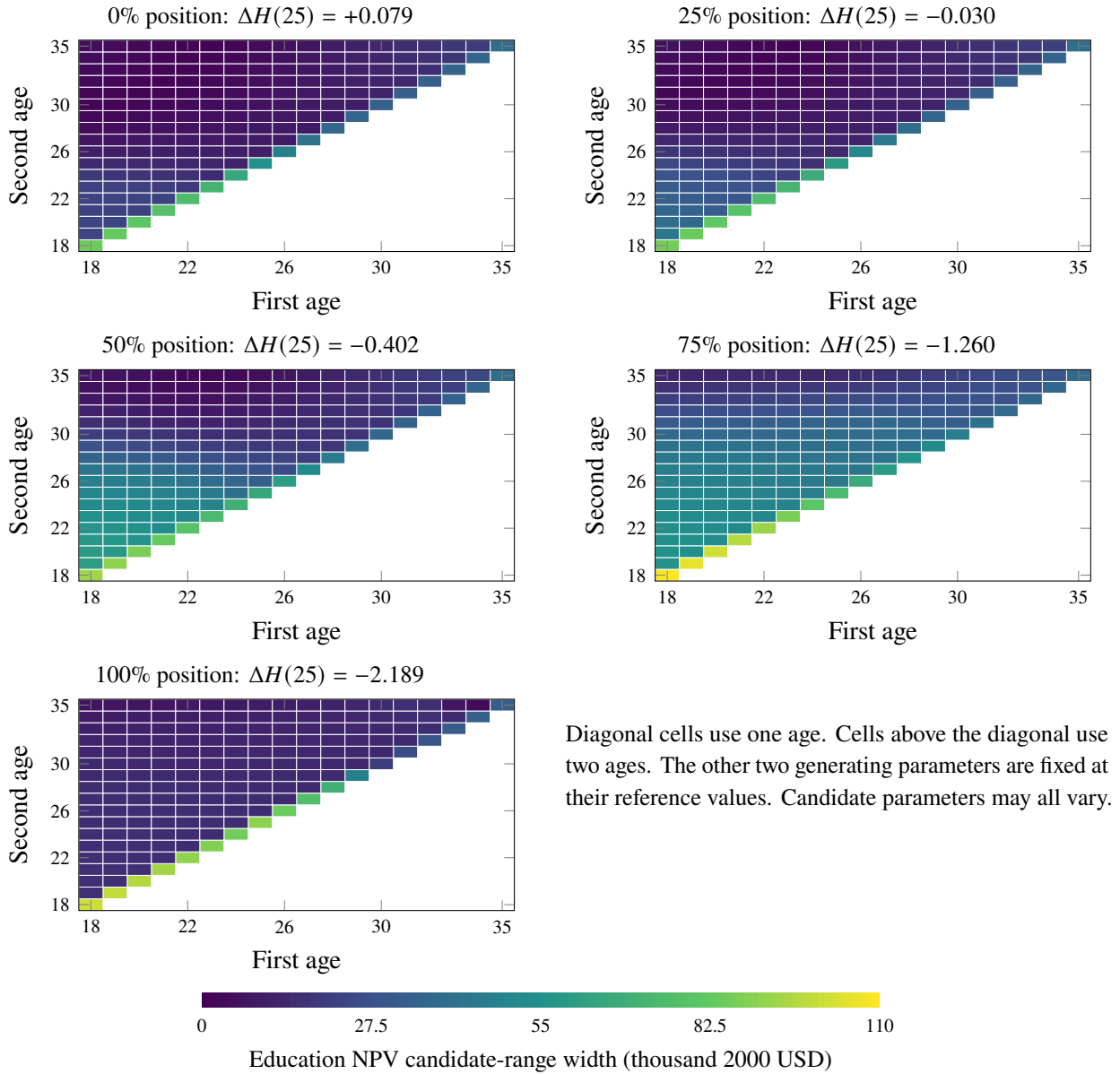


Figure B7. All five timing scenarios

Notes: Diagonal cells represent single-age observations; upper-triangle cells represent two distinct ages. All panels and all four heatmap figures use the same width scale, in thousands of 2000 dollars per person. The widths come from verified numerical candidates under the common baseline, path restrictions, and valuation rule; they are not confidence intervals or certified global bounds. Each panel fixes one generating scenario at the indicated position of its feasible single-parameter interval. The other two generating parameters remain at their reference values. All three candidate parameters may vary.

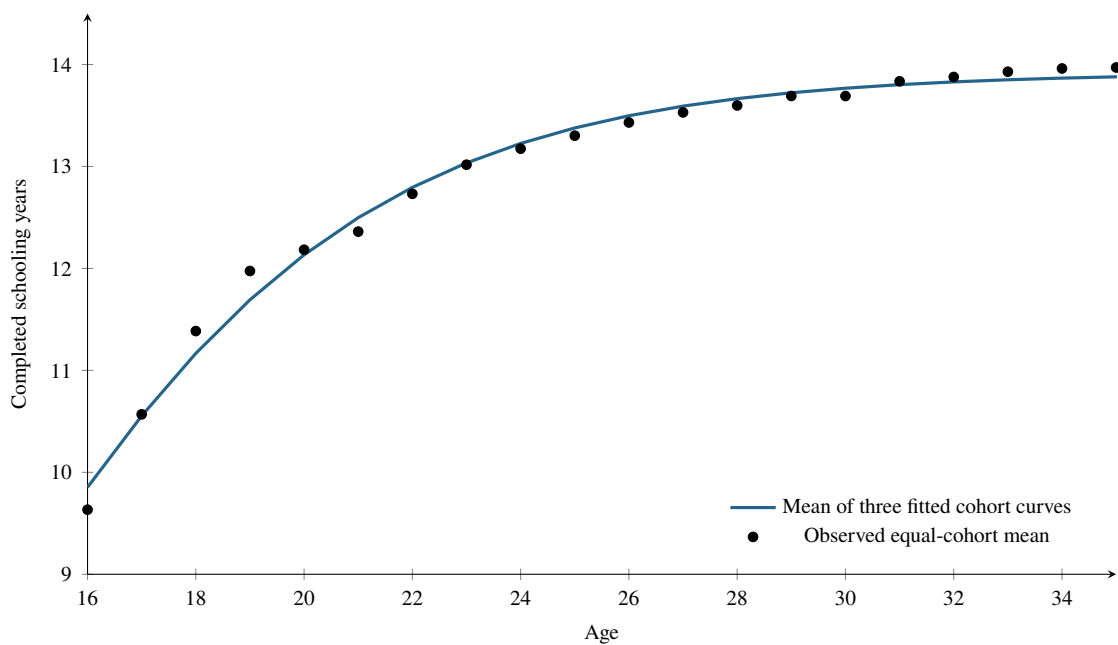
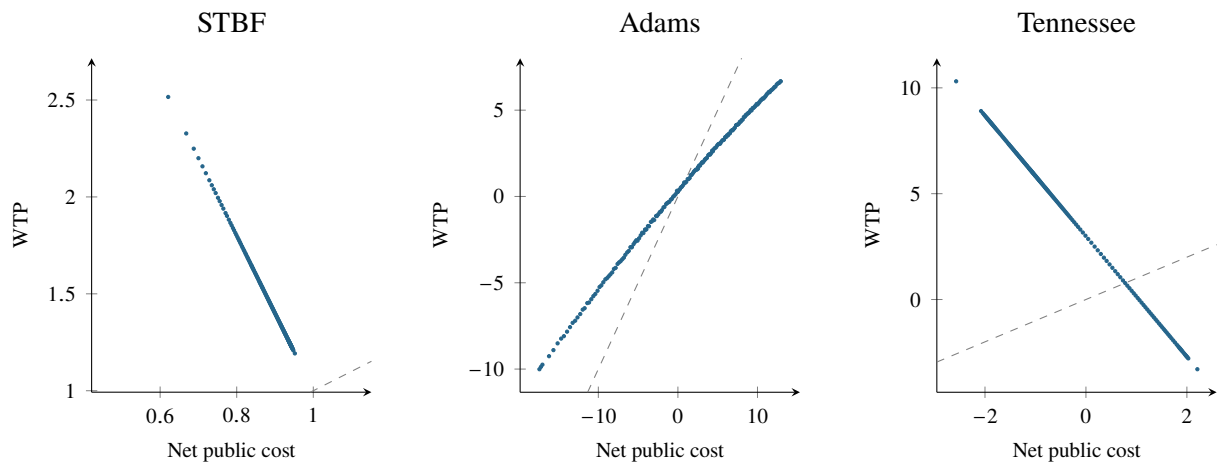


Figure C1. Comparison-path fit: ACS birth cohorts 1985–1987

Notes: The graph covers observed ages 16–35 only. Each cohort is fitted to its person-weighted age means using normalized cell population weights. The three fitted curves and observed cell means receive fixed one-third cohort weights. This in-sample fit diagnostic does not validate extrapolation beyond age 35.

Figure D1. Paired welfare components under the source-specific path mappings



Monetary components are divided by the same fixed reference expenditure within each panel. Points are complete L2 paths, sampled across the earnings range. The dashed line is WTP equal to net public cost. Axis ranges vary by policy; coordinates from different monetary vintages are not dollar comparisons. No separate opportunity cost is deducted from net earnings.

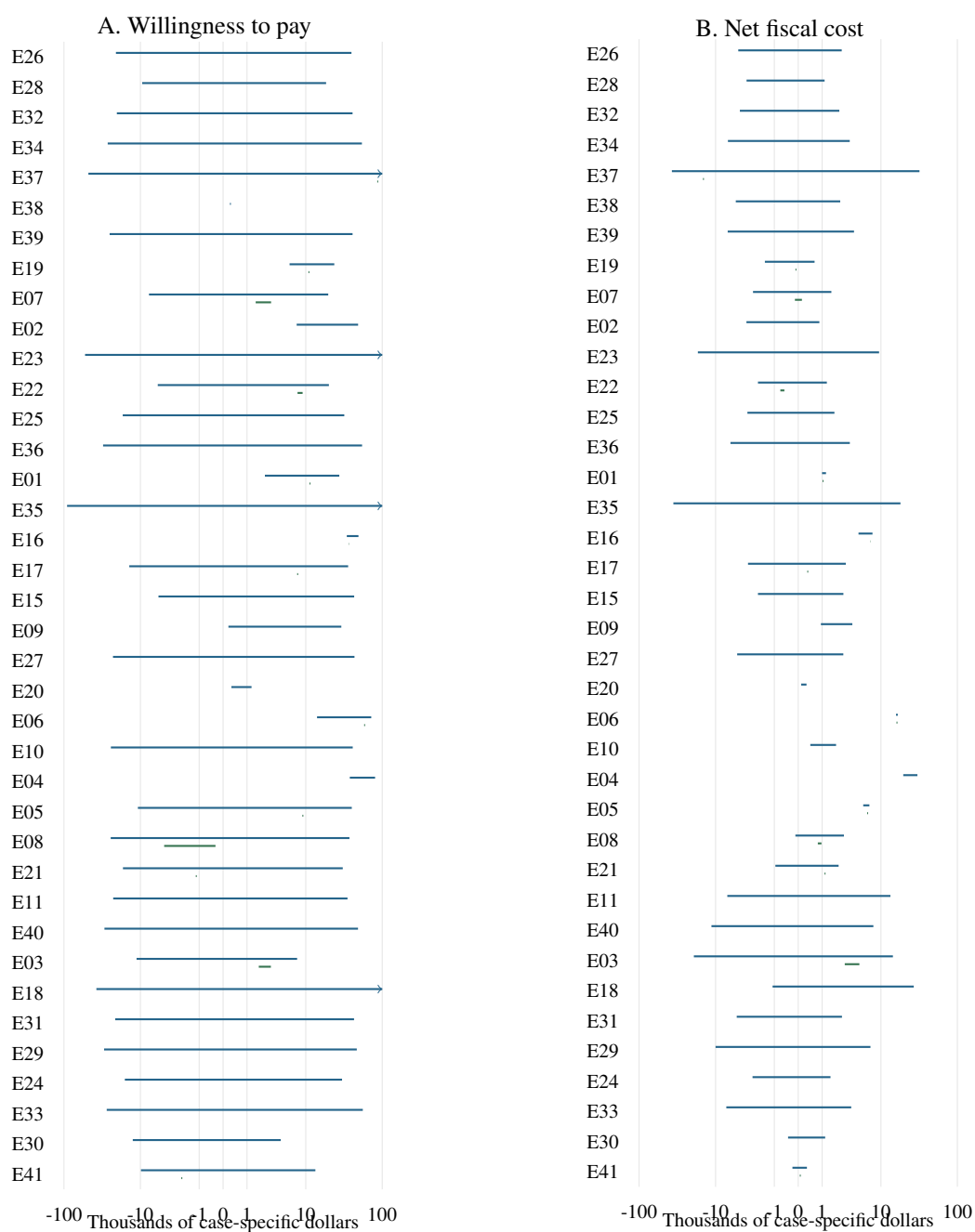


Figure E1. Component spans across all 38 calculations

Notes: Blue spans use L2 and green spans use verified source-timing L3 in Table E1. Crosses denote unsuccessful L3 searches. The axes use the same signed logarithmic transformation and truncate at ± 100 thousand. The component spans are marginal summaries and must not be combined to form an MVPF. Dollar years and policy units differ across cases, so this figure describes within-case continuation sensitivity, not cross-policy dollar rankings.

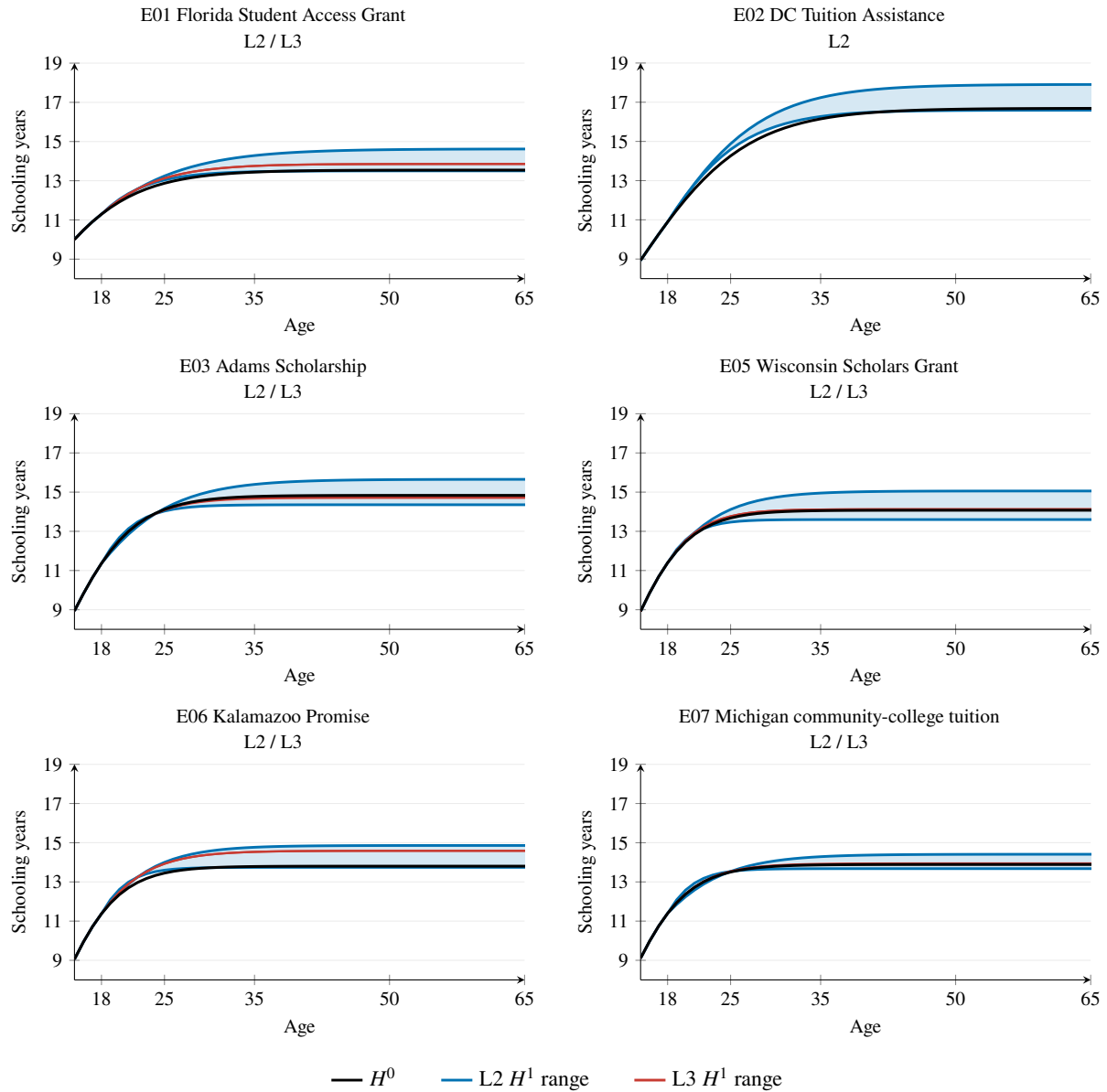


Figure E2. Comparison and treated schooling paths

Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

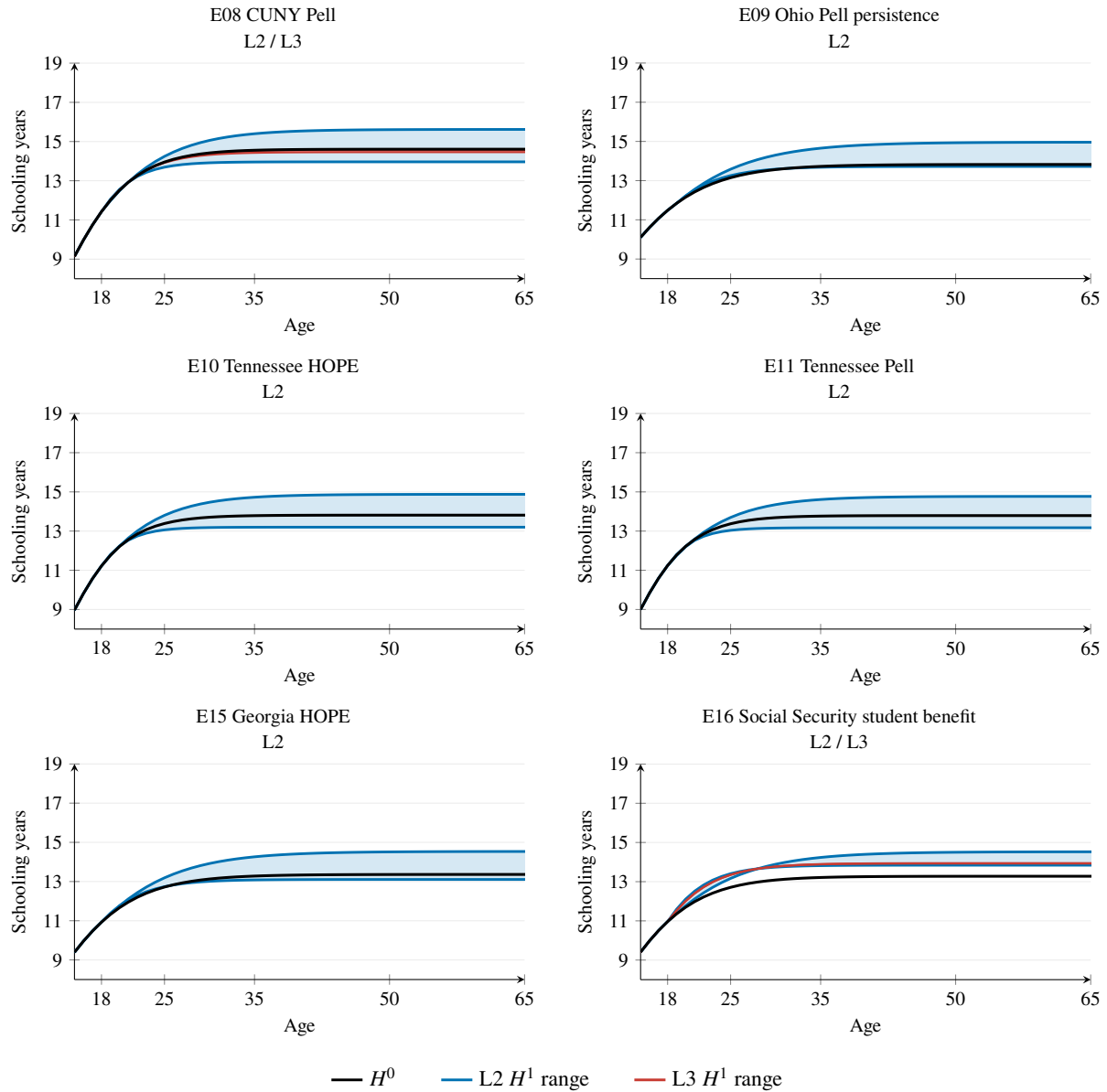


Figure E2. Comparison and treated schooling paths (continued)

Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

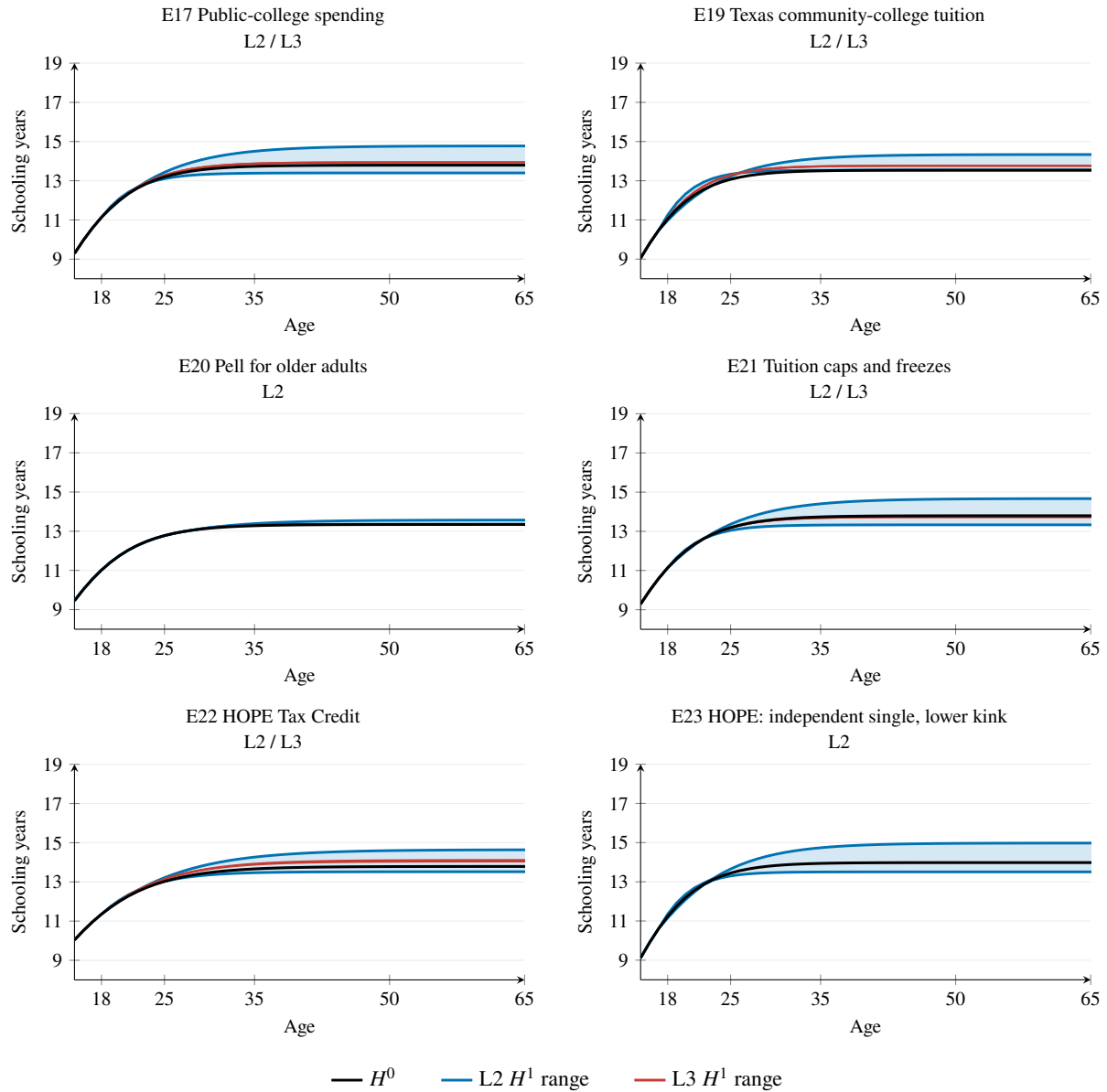


Figure E2. Comparison and treated schooling paths (continued)

Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

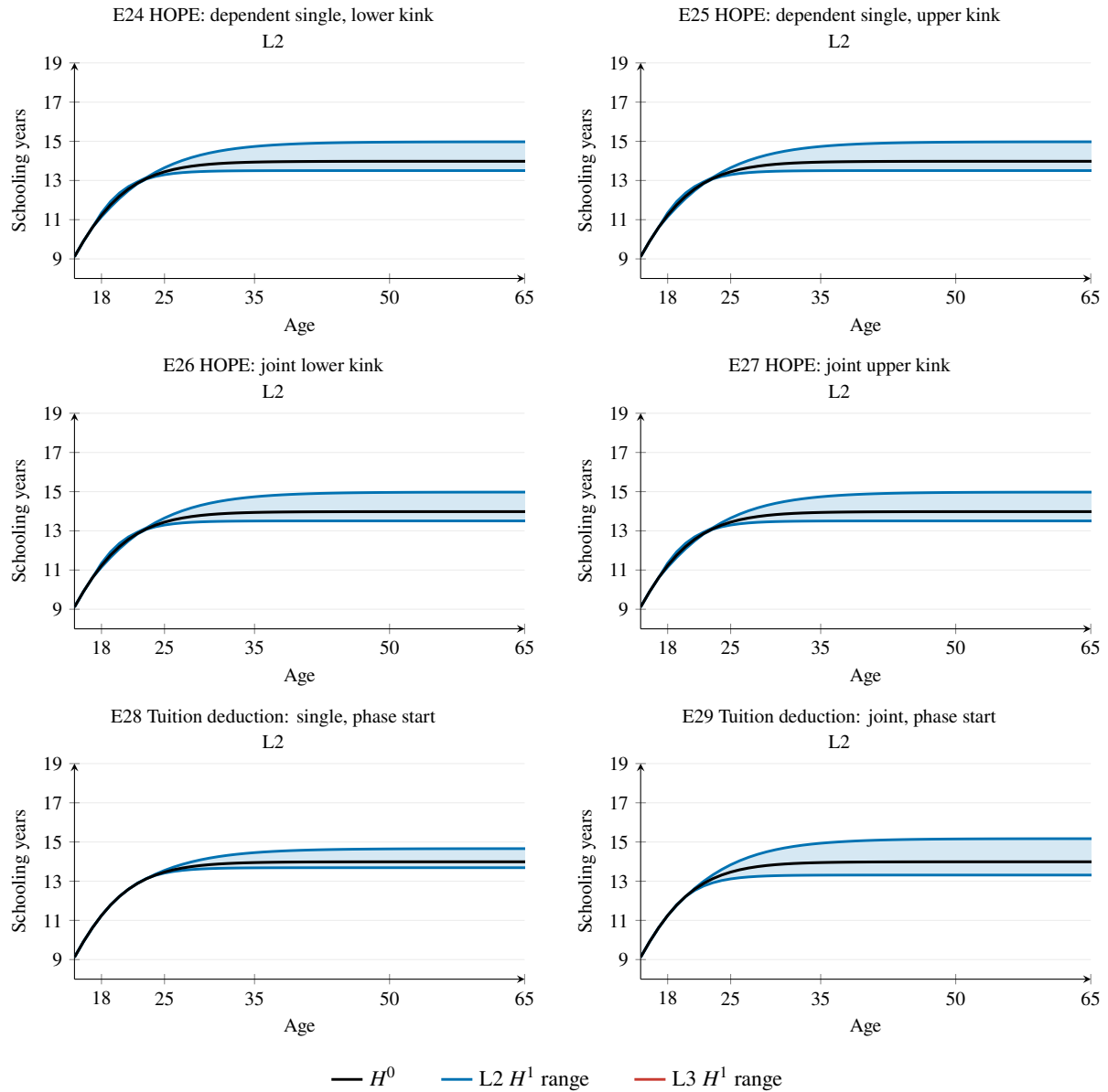


Figure E2. Comparison and treated schooling paths (continued)

Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

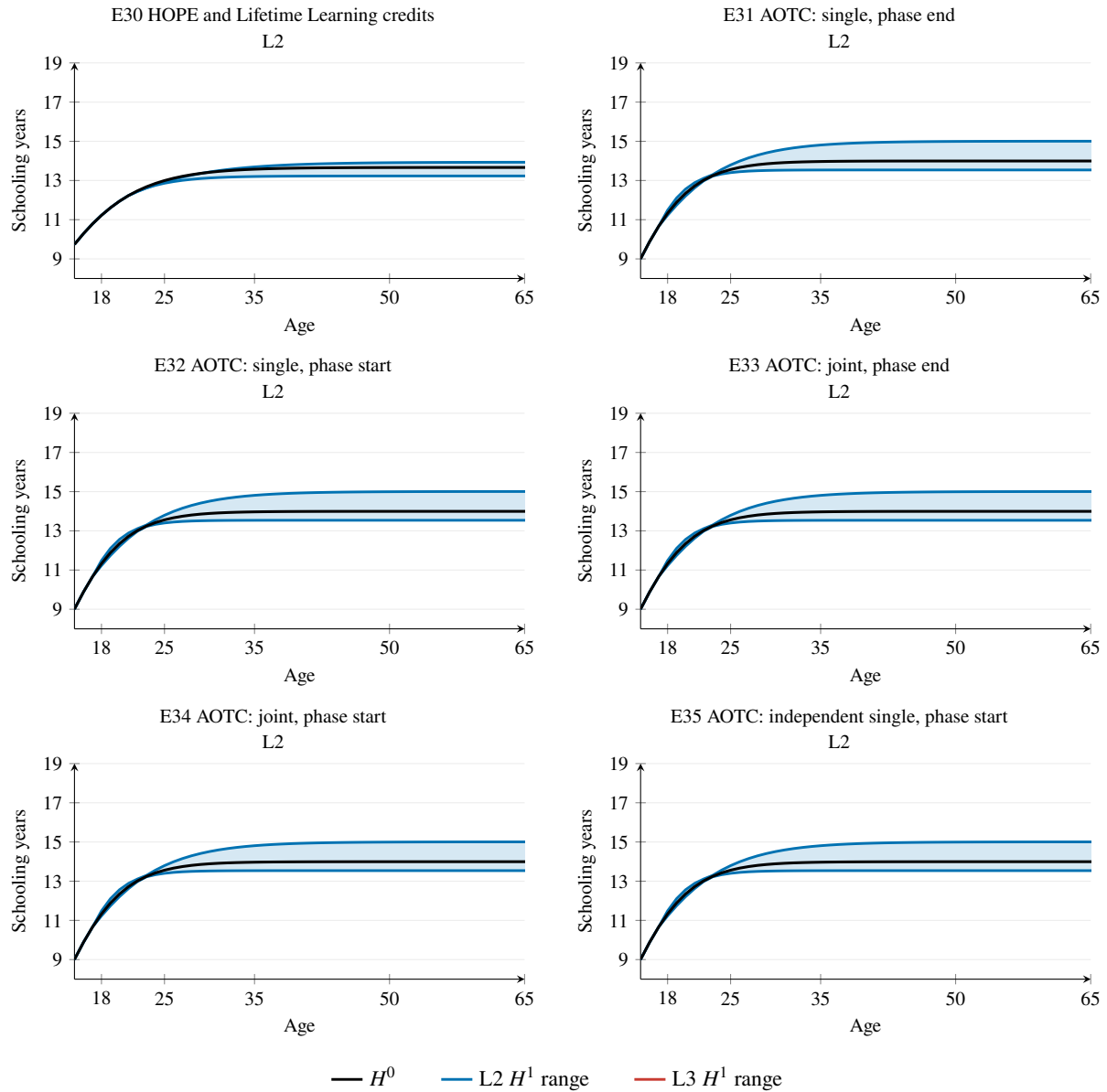


Figure E2. Comparison and treated schooling paths (continued)

Notes: Black is the matched comparison profile H^0 . Blue (L2) and red (L3) boundaries and shading show the pointwise minimum and maximum treated level H^1 among the candidates. An edge may combine different paths across ages. Cannabis shows its primary L3 only. All panels share scales. The ranges summarize the finite search, not confidence bands or path probabilities. Appendix E.2 records source windows and age mappings.

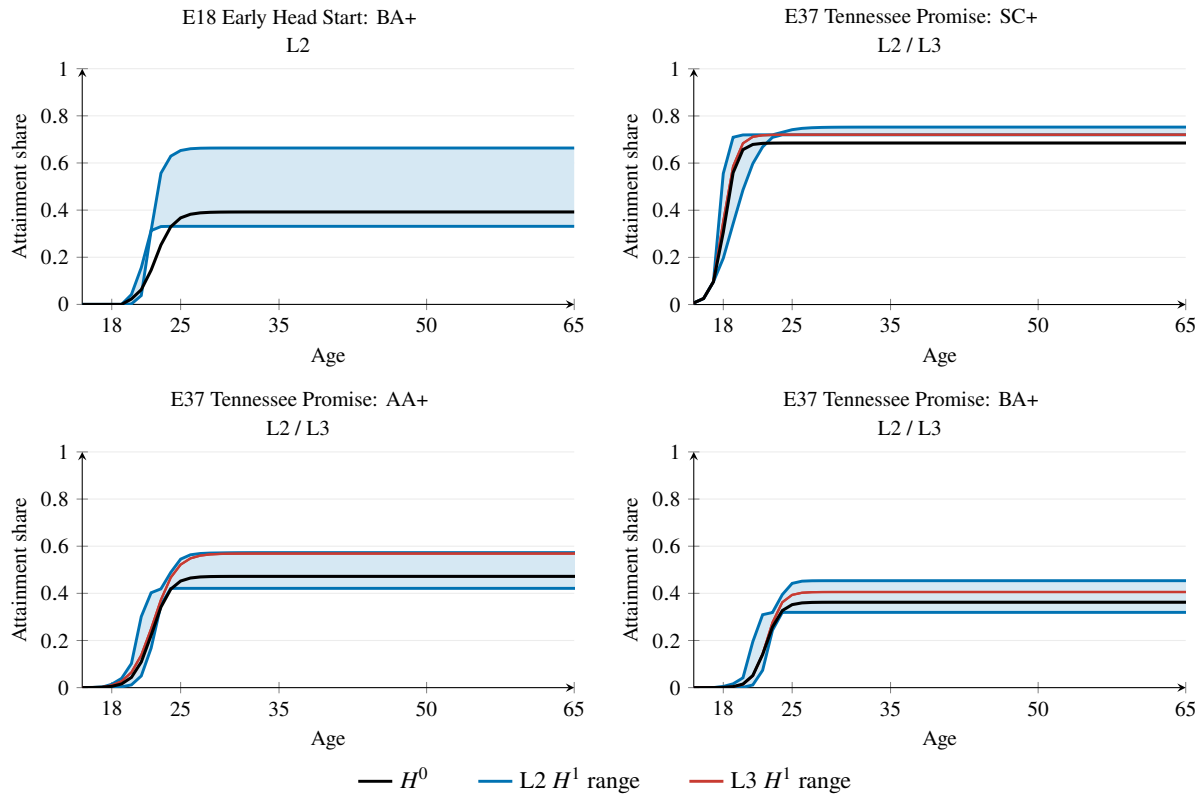
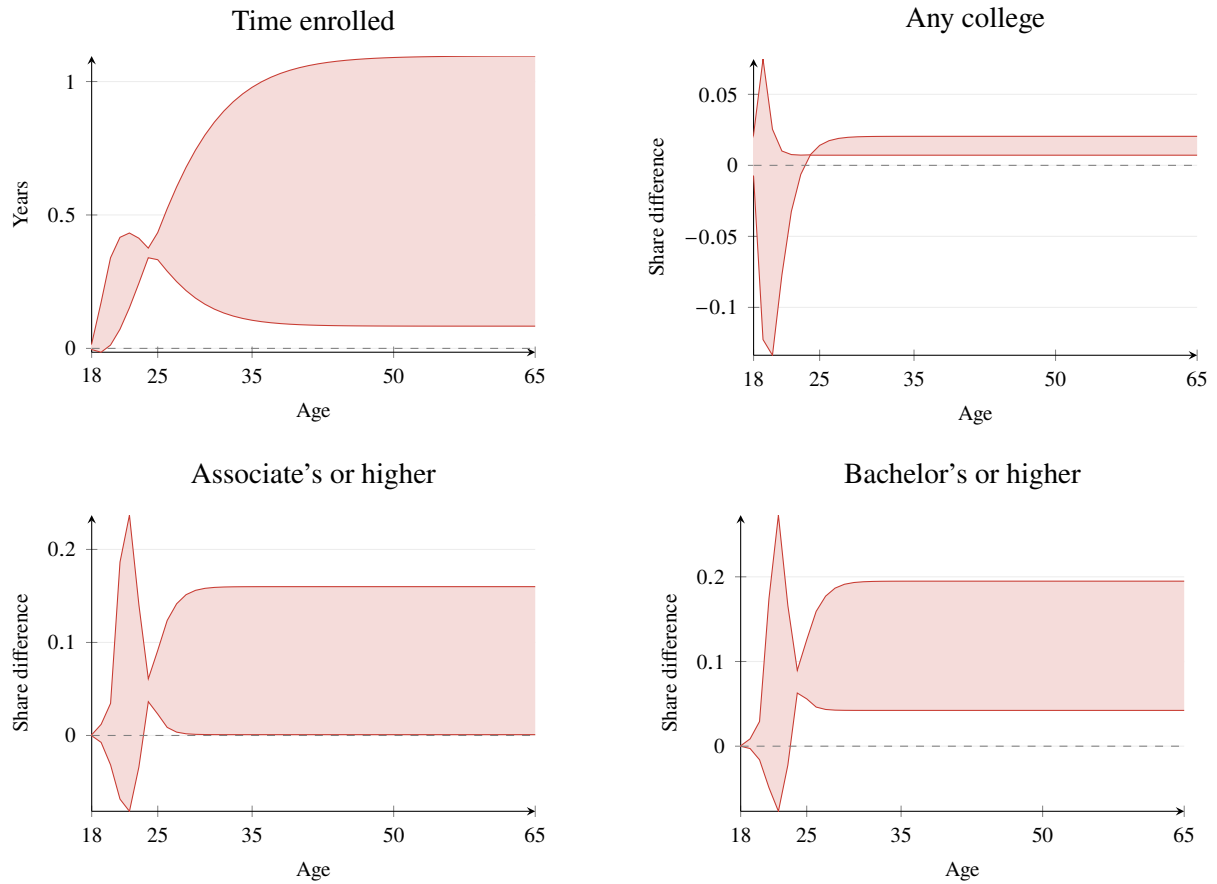


Figure E3. Comparison and treated credential paths

Notes: Black is H^0 ; blue and red boundaries and shading show the pointwise H^1 ranges for L2 and L3. All panels share age and attainment-share scales. Tennessee ranges come from joint candidates satisfying credential nesting and the restrictions in Appendix D.8. Envelope edges may combine different candidates across ages and credentials; they need not form a feasible joint path. These finite-search ranges are not confidence bands or probability distributions. Targets and calibrations differ across the two applications.

Figure E3. Comparison and treated credential paths (continued): STBF effects



Notes: All four six-year inputs are matched jointly using the source age-cohort weights. Ranges show treatment-comparison effects, not treated levels. Credential levels are nested in every candidate, and source mean-input feasibility is checked for the full path. Time enrolled is a separate timing index, not completed schooling. Shaded boundaries can combine different joint candidates and are not confidence bands. The author nonlinear credential-experience model values each complete candidate. No additional observation age supports L3 in this specification.

Figure E4. Point-input sensitivity across the policy collection

Policy calculation and information set	Sensitivity specification						
	Main	Narrow	Wide	National	Earnings 75%	Earnings 125%	Denser
Direct student financial aid							
Florida access grant (L3)							
DC tuition assistance (L2)	S	S	S	S	S	S	S
Adams Scholarship (L3)							
STBF scholarship (L2)							
Wisconsin Scholars Grant (L3)							
Kalamazoo Promise (L3)							
CUNY Pell grants (L3)							
Ohio Pell grants (L2)							
Tennessee HOPE (L2)						S	
Tennessee Pell eligibility (L2)							
Georgia HOPE (L2)	S	S	S	S		S	S
Social Security student benefits (L3)							
Adult Pell introduction (L2)							
Tennessee Promise (L3)	S	S	S	S	S	S	S
Education tax benefits							
HOPE tax credit (L3)	S	S	S	S	S	S	S
HOPE: independent single, start (L2)	S	S	S	S	S	S	S
HOPE: dependent single, start (L2)	S	S	S	S	S	S	S
HOPE: dependent single, end (L2)	S	S	S	S	S	S	S
HOPE: joint, start (L2)	S	S	S	S	S	S	S
HOPE: joint, end (L2)	S	S	S	S	S	S	S
Tuition deduction: single, start (L2)	S	S	S	S	S	S	S
Tuition deduction: joint, start (L2)	S	S	S	S	S	S	S
HOPE and Lifetime Learning (L2)	S	S	S	S		S	S
AOTC: single, end (L2)	S	S	S	S	S	S	S
AOTC: single, start (L2)	S	S	S	S	S	S	S
AOTC: joint, end (L2)	S	S	S	S	S	S	S
AOTC: joint, start (L2)	S	S	S	S	S	S	S
AOTC: independent single, start (L2)	S	S	S	S	S	S	S
AOTC: simulated instrument (L2)	S	S	S	S	S	S	S
Tuition deduction: single, end (L2)	S	S	S	S	S	S	S
Tuition deduction: joint, end (L2)	S	S	S	S	S	S	S
Tuition and public-college funding							
Michigan CC tuition (L3)	S	S	S	S		S	S
College appropriations: spending (L3)						S	
Texas CC tuition (L3)	S	S	S	S		S	S
College appropriations: tuition (L3)							
College assessment and course placement							
Algorithmic college placement (L2)	S	S	S	S	S	S	S
Early childhood services							
Early Head Start (L2)	S		S	S		S	S
Education channel of other policies							
Cannabis: education channel (L3)							

■ Both sides of one ■ Only above one found ■ Only below one found

Notes: Colors refer to positive-cost MVPFs; gray means none was found. S marks a separate self-financing candidate. Equal signs indicate an already national baseline. Each alternative changes one specified assumption, with point inputs fixed. L3 is used where admitted; otherwise L2. Finite searches cannot establish absence of other paths. Table E4 reports values; Appendix E.9 defines the changes.

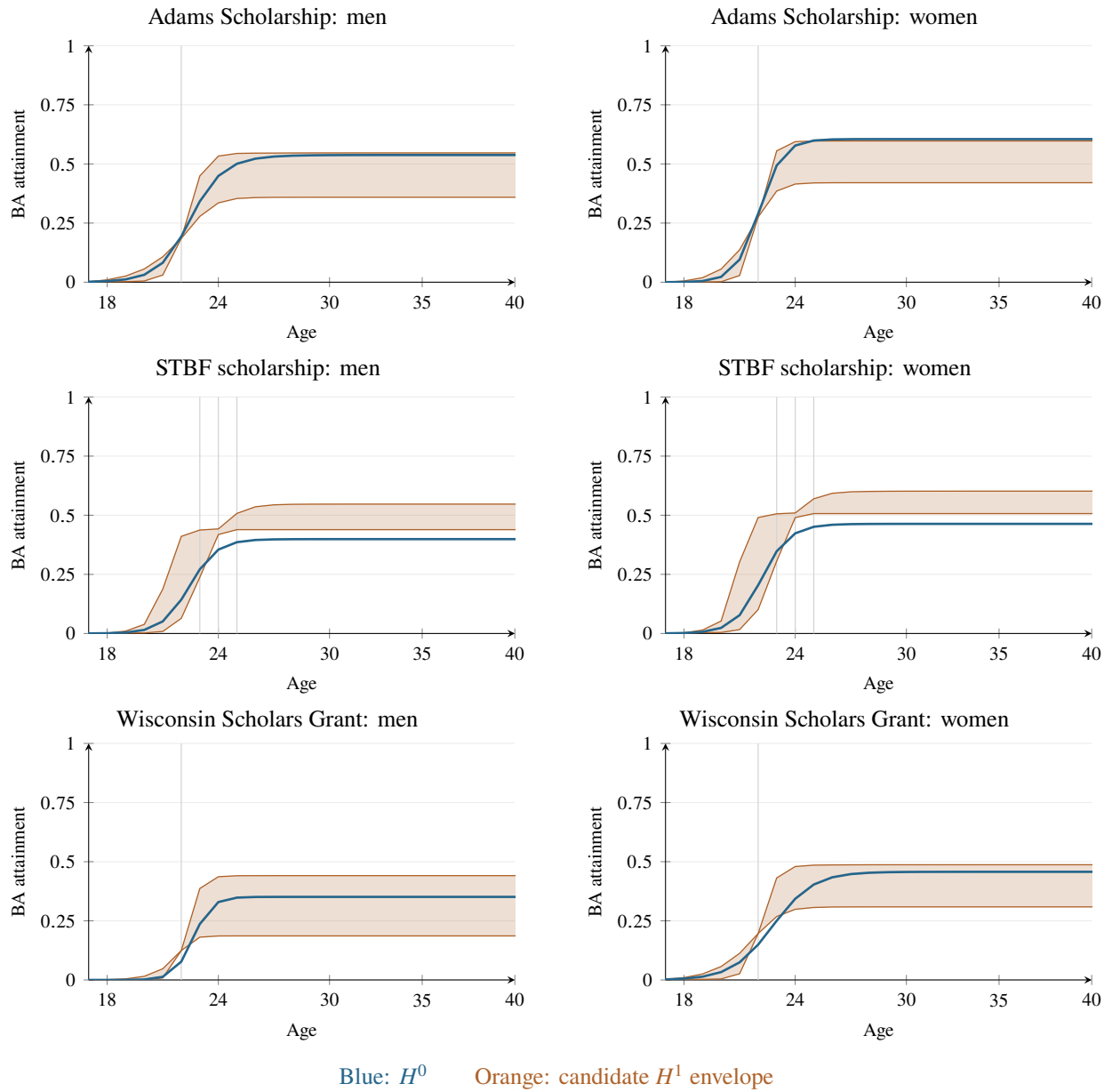


Figure G1. Male and female degree continuations: Adams Scholarship, STBF scholarship, Wisconsin Scholars Grant

Notes: Each panel matches the source study's own subgroup effect and policy-specific ACS geography, cohort weights and population definition. Orange bands are pointwise finite-candidate envelopes, not confidence bands; their edges need not form one complete path. Gray guides mark the ages entering the source moment. When several ages enter, the restriction retains component weights rather than evaluating a midpoint. Both sexes and all panels use the same axes. Curves after age 35 are extrapolations, and recent cohorts can have shorter observed support. Tennessee retains a separate auxiliary subgroup interval sensitivity; Kalamazoo reference levels are rescaled to subgroup control means. Early Head Start uses a full-population reference and the maintained zero-degree-stock approximation through age 19. Appendix G records the source estimands, population transport and valuation assumptions.

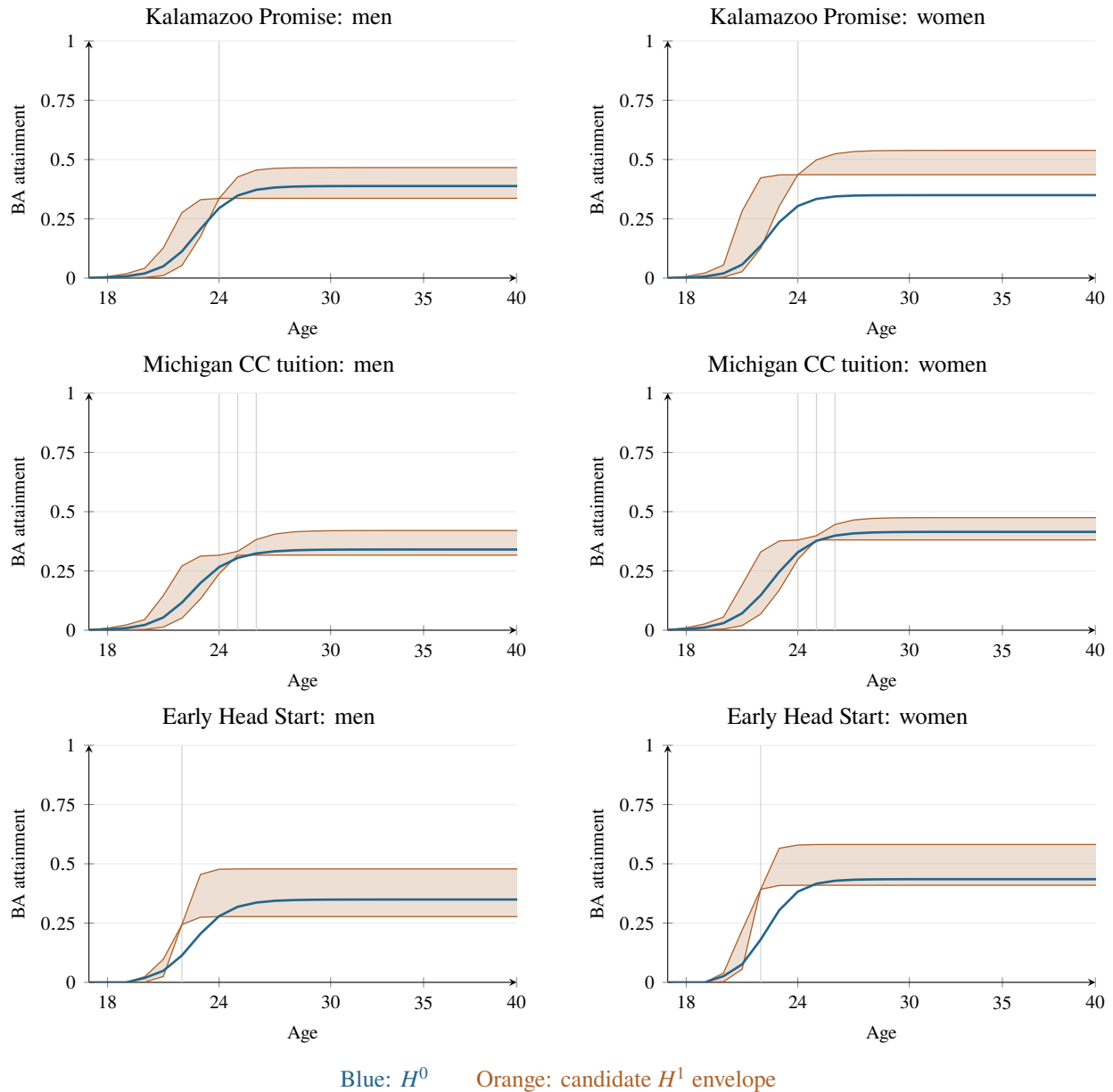


Figure G2. Male and female degree continuations: Kalamazoo Promise, Michigan CC tuition, Early Head Start

Notes: Each panel matches the source study's own subgroup effect and policy-specific ACS geography, cohort weights and population definition. Orange bands are pointwise finite-candidate envelopes, not confidence bands; their edges need not form one complete path. Gray guides mark the ages entering the source moment. When several ages enter, the restriction retains component weights rather than evaluating a midpoint. Both sexes and all panels use the same axes. Curves after age 35 are extrapolations, and recent cohorts can have shorter observed support. Tennessee retains a separate auxiliary subgroup interval sensitivity; Kalamazoo reference levels are rescaled to subgroup control means. Early Head Start uses a full-population reference and the maintained zero-degree-stock approximation through age 19. Appendix G records the source estimands, population transport and valuation assumptions.

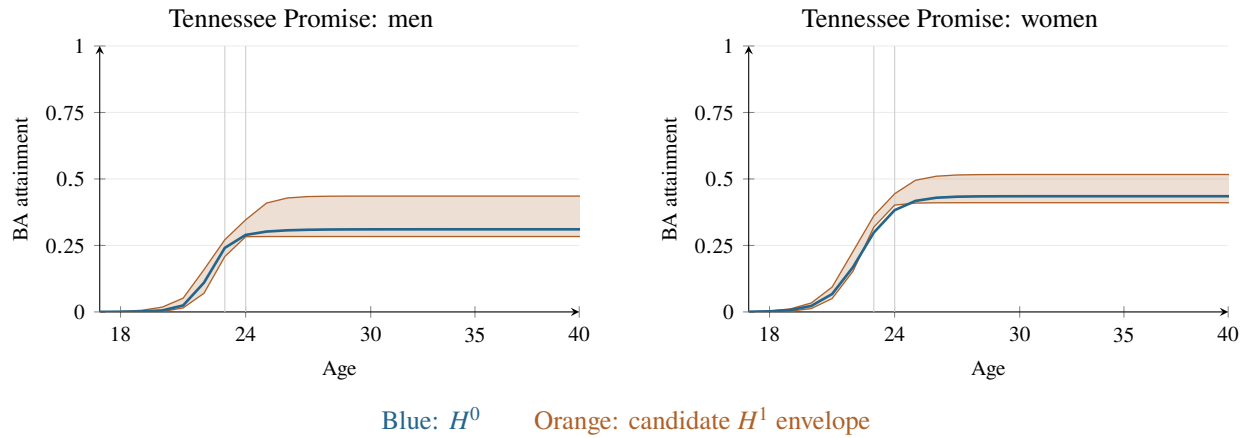


Figure G3. Male and female degree continuations: Tennessee Promise

Notes: Each panel matches the source study's own subgroup effect and policy-specific ACS geography, cohort weights and population definition. Orange bands are pointwise finite-candidate envelopes, not confidence bands; their edges need not form one complete path. Gray guides mark the ages entering the source moment. When several ages enter, the restriction retains component weights rather than evaluating a midpoint. Both sexes and all panels use the same axes. Curves after age 35 are extrapolations, and recent cohorts can have shorter observed support. Tennessee retains a separate auxiliary subgroup interval sensitivity; Kalamazoo reference levels are rescaled to subgroup control means. Early Head Start uses a full-population reference and the maintained zero-degree-stock approximation through age 19. Appendix G records the source estimands, population transport and valuation assumptions.

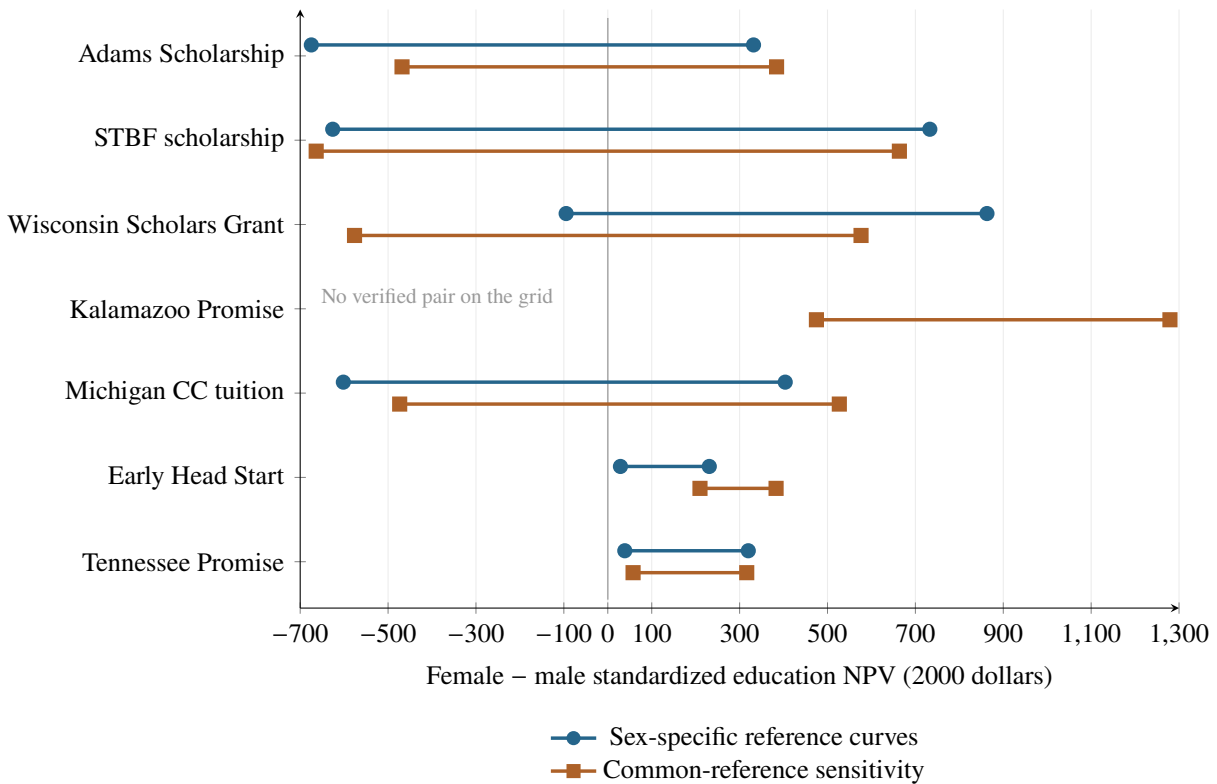


Figure G4. Sex comparisons after fixing terminal education effects and valuation prices

Notes: Each underlying comparison uses two complete paths that match their own source subgroup effects and the same age-65 BA effect. Blue spans use separate sex-specific ACS reference curves; orange spans use their equally weighted mean. Positive differences favor women in the standardized education account. Endpoints summarize the smallest and largest paired differences across the tested terminal grid: -15, -10, -5, 0 percentage points for Adams and 2, 5, 10, 15 for the other policies. The terminal effect is held fixed within each pair; it can differ across pairs in a row. No span is drawn when the saved grid contains no verified pair. Table G2 reports every grid value and path count. The lines summarize endpoints and do not establish that every intermediate value is attained. They are not sampling intervals or probabilities. All values use the common signed seven-year net-income response, a two-schooling-year equivalent per BA increment, and 3 percent discounting to age 18 in 2000 dollars. These accounts exclude other qualifications, transfers, and fiscal costs.